A Lean Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

June 1, 2026

Abstract

We report on a first Lean formalization of the logical corridor from IUT III, Theorem 3.11 to Corollary 3.12. The formalization is deliberately neutral with respect to the correctness of IUT as a proof of the abc conjecture. It isolates the ordered-real comparison required at the end of Corollary 3.12, tracks the finite q^{j^2} and label-quotient phenomena that enter the public dispute, and records exactly which source-facing construction hypotheses are still external.

The current code proves conditional theorems. If the hull/log-volume upper-ray comparison supplies $-|\log(q)| \leq -|\log(\Theta)|$, and if the displayed hypothesis $-|\log(\Theta)| \leq C_\Theta|\log(q)|$ is supplied with $|\log(q)| > 0$, then Lean proves $C_\Theta \geq -1$. If the upper-ray membership is strict, Lean proves $C_\Theta > -1$. In the boundary case $C_\Theta = -1$, Lean proves equality of the q-pilot reading and the Θ-hull boundary, and then classifies the local Frobenioid p_v^N-shift behavior under the actual upper-ray quotient. This does not prove Corollary 3.12 from Mochizuki's published hypotheses; it identifies the remaining mathematical construction problem.

After a source audit against IUT I–IV, Mochizuki's 2026 LeanForm report, and Scholze–Stix, the status is unchanged in kind but sharper in formulation: the final ordered-real implication is formalized, and several collapse-prone comparison levels are separated, but the code still assumes the source construction of the multiradial/HDD/SHE/IPL data that IUT III assigns to Theorem 3.11 and its reorganization.

1 Aim and Sources

The project formalizes a narrow and contested part of Mochizuki's inter-universal Teichmuller theory. The source base is the local copy of IUT I–IV, Mochizuki's 2026 report on formalization, and the Scholze–Stix note. The current target is the portion that Mochizuki's report calls Stage 1:

$$\text{IUT III, Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{IUT III, Corollary 3.12.}$$

The notation “3.11.5” is not a statement in IUT III. It is Mochizuki's 2026 name for the reorganized boundary where the hull-plus-determinant operation is moved before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The formalization is source-facing but not source-complete. It does not construct Hodge theaters, Frobenioids, log-shells, prime strips, or theta links from first principles. Instead, these are represented by typed records with named hypotheses. The point of the current work is to test whether the claimed formal corridor is coherent once every comparison level, label quotient, ordered real line, hull boundary, and quotient collapse is made explicit.

2 Framework and Tooling

The repository is organized as a Lean verification project, not as a prose annotation project. The source papers in `docs/` are treated as the mathematical source of truth. The formalization then moves through three layers:

$$\begin{gathered}\text{source papers}\\ \Downarrow\\ \text{Lean definitions, records, and theorems}\\ \Downarrow\\ \text{this paper draft.}\end{gathered}$$

The paper draft records only the current mathematical state of the formalization. It is not a research log.

The Lean code uses typed interfaces to prevent comparison-level drift. A datum that is merely a representative j^2-coordinate computation has a different type from a balanced full-label datum, and both differ from the final hull/log-volume comparison datum. Likewise, ordered real-line transports, Hodge/SHE preservation records, upper-ray membership, and quotient-collapse audits are separate structures. This is the main anti-collapse mechanism in the framework: a proof cannot silently use a raw coordinate equality where the Corollary 3.12 corridor asks for a hull/log-volume comparison.

The module roles are:

- `IUTStage1SourceCore`: core real-line, pilot, hull, quotient, and C_Θ algebra.
- `IUTStage1Remark312Absorption`: Remark 3.12.2(v) absorption interfaces.
- `IUTStage1IUTIVAlgebra`: Remark 3.12 sign shadows and the conditional IUT IV real-algebra handoff estimates.
- `IUTStage1FiniteLabels`: finite local log-volume endpoints, F_ℓ-label/torsor/sign/cusp-label models, $ZMod$ full labels, theta-root cusp-label source packages, and label-averaged log-volume compatibility.
- `IUTStage1StepX`: procession tensor-packet log-volume, the Step (x) indeterminacy corridor, the theta-hull upper-ray handoff, and local Frobenioid shift diagnostics attached to that boundary.
- `IUTStage1Gaussian`: weighted label averages, $ZMod$ square-weight profiles, Gaussian monoid degree evaluations, and Hodge–Arakelov theta-evaluation source objects.
- `IUTStage1HodgeSHE`: square-weighted full-label transport audits, Hodge/SHE local object data, packet-normalization support, direct-summand local tensor actions, and upper-semi compatibility states.
- `IUTStage1Theorem311`: Theorem 3.11 choice/indeterminacy records, multiradial possible-image records, algorithm-output records, and SHE-alignment source records.
- `IUTStage1StepXI`: Step (xi) hull/determinant and IPL source objects, source obligations, package-audit constructors, and holomorphic hull/determinant source routes.
- `IUTStage1FrobenioidShift`: endpoint-audit surfaces together with the nonarchimedean realified-Frobenioid/log-Kummer, vertical-IQ, and root-of-unity support routes.
- `IUTStage1EndpointAudit`: nested LogVolumeChartAudit refinements around the place-audited multiradial theta-hull endpoint.

- `IUTStage1Source`: remaining public source-package audit surface around the Corollary 3.12 corridor.
- `IUTStage1Experiments`: compiled endpoint checks and diagnostic theorems.

The experiment file is not a substitute for proof. It is a curated surface that forces the important route theorems to elaborate together and exposes Boolean/report endpoints such as whether the present stage settles the dispute. At present that endpoint is deliberately false.

The build discipline is simple. Each mathematical increment is checked by Lean via `lake build Iut.Stage1.IUTStage1Experiments`; the paper is rebuilt from this LaTeX source; and the code is scanned for explicit proof placeholders such as `sorry`, `admit`, and `axiom`. The long-term collaboration plan is compatible with Apodixis: the current typed route endpoints and missing-obligation records give a natural surface for a collaboration tool to show which mathematical interfaces are proved and which remain open.

3 Dispute Interface

In IUT III, Step (x) treats the procession-normalized log-volume as unchanged under (Ind1) and (Ind2), while (Ind3) supplies a one-sided upper-semi comparison. Step (xi) then passes through multiradiality, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot reading in the signed-log upper ray determined by the Θ-pilot:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}.$$

The active source-facing Corollary 3.12 route does not use equality of possible-image regions across an (Ind3) generator. Such an equality exists only in separate quotient bookkeeping interfaces as an explicit stronger hypothesis; the route uses the one-sided upper-semi corridor as its Step (x) input. The current Theorem 3.11 source record now bundles structured IPL/SHE/APT input, coric multiradial output, possible theta-pilot images, the 0-column/1-column pilot distinction, distinct log-shell treatments, and the Hodge-theater history guard. This is still a source interface, not a construction of Theorem 3.11. The IPL layer has been separated into a typed log-volume transport datum: it carries the source and target Hodge-theater sides, identifies the prime-strip link endpoints with the input and output prime strips, records equality of the transported real log-volume, and retains the no-history-identification guard. It is not yet a construction of the full log-link. The current SHE layer now has a synchronization constructor: equality of the canonical one-label Hodge–Arakelov theta-value degree constructs the factored square/full-label preservation package used by the route. This is a Stage 1 source constructor for the available finite-label shadow of SHE, not yet the full simultaneous-holomorphic-expressibility theorem. The hull/determinant layer now has a source constructor attached to the Theorem 3.11 record. It reads the algorithmic certificate and SHE alignment from that record, keeps the record's theta possible-image union tied to the hull endpoint, and derives the determinant-bound comparison $q_{\text{signed}} \leq \theta_{\text{signed}}$. The bridge used by this source data can now be constructed from a displayed common hull containing the target union and a determinant/log-volume bound for that hull; the analytic construction of the hull and determinant remains external source data. The strongest Hodge–Arakelov packet-target route no longer takes the target-side log-volume equality or q-pilot positivity as independent inputs: target equality is derived from the SHE synchronization constructor, and positivity is read from the hull/determinant constructor. The source-side chart calibration is no longer a raw equality input: it is a full-label theta/Hodge calibration certificate, from which equality of the cusp log-volume compatibility records is derived extensionally. The strongest current labelled Step (x) route now consumes a theta-root realified-Frobenioid log-Kummer entry source. The Kummer and mono-analytic forgetting transfers, realization tags, holomorphic-forgetting assertion, and packet-local source alignment are projected from this source object instead of being passed separately to the Corollary 3.12 route. Its labelled $ZMod(\ell)$ capsule family is now constructed as the constant packet-local family over the audited packet local object; the route no longer accepts an arbitrary labelled family or a separate local-object equality for that family. The exact vertical-IQ target has also been retained inside a refined realified Step (x) source, so the route consumes one source object carrying both the realified log-Kummer entry data and the $FMOD$-exact target guard. The refined source now also requires the realified packet target calibration to be exactly the calibration induced by this retained vertical-IQ target. The strongest public experiment route now constructs this refined source from theta-root data, realified Frobenioid transfer data, source alignment, and the vertical-IQ target before entering the Corollary 3.12 corridor. A finite-divisor form of the same route derives the packet-local source alignment from the finite divisor/Frobenioid packet product before constructing the refined exact source. Its public experiment audit endpoint exposes the packet-local/source equality, mono-analytic product equality, divisor-degree normalization, (Ind3)-source equality, and upper-semi inequality used by this construction. For the canonical packet-local labelled family, this exact route no longer requires direct packet-normalization: Hodge labelwise averaging identifies the canonical labelled average with the theta-source average, and the retained vertical-IQ target identifies that average with the packet-normalized capsule value. The final numerical step is elementary:

$$-|\log(q)| \leq -|\log(\Theta)| \leq C_\Theta |\log(q)|, \qquad |\log(q)| > 0 \quad \Longrightarrow \quad C_\Theta \geq -1.$$

Scholze–Stix object to the comparison because the Θ-side data carries j^2-scaled degrees, while a naive identification of ordered real lines appears to remove the room needed for these scalars. The formalization therefore separates four comparison levels:

representative j^2 data
balanced sign-compatible data
averaged cusp-label data
hull/log-volume data.

The final route consumes only the hull/log-volume comparison level. Lean rejects the balanced sign-compatible level as sufficient for the final comparison.

4 Source Audit

The local sources impose the following constraints on the formalization. IUT I constructs the initial Θ-data and the $\Theta^{\pm\mathrm{ell}}$NF-Hodge theaters. It is not merely a notation layer: the paper's central point is that the Hodge theaters separate the additive and multiplicative combinatorial dimensions of a number field. IUT II supplies the Hodge–Arakelov theta evaluation, theta values at ℓ-torsion points, Gaussian Frobenioids, conjugate synchronization, and the multiradial/coric phenomena needed later. The current code models only finite label and Gaussian degree shadows of this material.

IUT III is the direct source for the formal corridor. Step (x) says that procession-normalized log-volumes are invariant under (Ind1) and (Ind2), while (Ind3) becomes an upper-semi inequality after applying log-volume. Step (xi) uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and the 1-column log-Kummer correspondence to put the q-pilot log-volume in the signed-log upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$. Step (xii) warns that replacing global realified Frobenioids by local Frobenioids loses calibration because local objects admit p_v^N-type endomorphisms. Remark 3.12.2 further stresses that the argument depends on weakened data, distinct q/Θ labels, upper-semi indeterminacy, and the asymmetry between the 0-column Θ-pilot and the 1-column q-pilot. The current Step (x) packet source reflects this asymmetry by packaging the 1-column q-pilot non-interference, Kummer/forgetting product preservation, source packet calibration, and packet-normalized target calibration in a single realified-Frobenioid log-Kummer source object. The latest route theorem uses this object directly: it still assumes the finite source object, but it no longer permits the downstream proof to mix unrelated Kummer, forgetting, realization, and packet-local calibration witnesses. The labelled procession average used by this route is also no longer arbitrary: Lean constructs the constant $ZMod(\ell)$-indexed capsule family from the packet local object and proves its normalized value by averaging a constant finite family. The exact target face is now represented by a refined source object rather than by a discarded calibration projection: the object keeps the vertical-IQ target itself and exposes the $FMOD$-exactness predicate together with the packet-normalized target equalities. It also ties the realified packet target calibration to the retained vertical-IQ target, so the exact face cannot be displayed while an unrelated packet calibration is used downstream. This removes direct packet-normalization only for the canonical exact source route; arbitrary labelled families still require their own packet-normalization evidence.

IUT IV uses the resulting log-volume estimate as input to elementary log-volume and Diophantine inequalities. Our IUT IV code is therefore only conditional algebra at present: it records how later estimates consume a Corollary 3.12-shaped bound, but it does not prove that bound from the full arithmetic constructions.

Mochizuki's 2026 report explicitly recommends Stage 1 as the implication IUT III, Theorem 3.11 $\Rightarrow$ Corollary 3.12, with a reorganized intermediate "3.11.5" obtained by moving the hull-plus-determinant operation before the final simultaneous comparison. The same report states that the current official code is skeletal. Our formalization is best read as a more diagnostic Stage 1 interface: it does not replace Stage 2, because it has not constructed Theorem 3.11.

Scholze–Stix argue that, after simplification, the comparison loses the j^2-scaled theta contribution and becomes essentially contentless. The Lean code therefore treats the j^2-representative level, sign quotient, averages, square/full-label preservation, ordered real-line transports, and hull/log-volume endpoint as different types. This confirms that the final Corollary 3.12 endpoint cannot be obtained from the balanced sign-compatible level alone. It does not prove that Mochizuki's source constructions supply the missing higher-level data.

5 Formalized Corridor

The relevant implementation lives in the following Lean modules:

- `Iut/Stage1/IUTStage1SourceCore.lean`
- `Iut/Stage1/IUTStage1Remark312Absorption.lean`
- `Iut/Stage1/IUTStage1IUTIVAlgebra.lean`
- `Iut/Stage1/IUTStage1FiniteLabels.lean`
- `Iut/Stage1/IUTStage1StepX.lean`
- `Iut/Stage1/IUTStage1Gaussian.lean`
- `Iut/Stage1/IUTStage1HodgeSHE.lean`

- `Iut/Stage1/IUTStage1Theorem311.lean`
- `Iut/Stage1/IUTStage1StepXI.lean`
- `Iut/Stage1/IUTStage1FrobenioidShift.lean`
- `Iut/Stage1/IUTStage1EndpointAudit.lean`
- `Iut/Stage1/IUTStage1Source.lean`
- `Iut/Stage1/IUTStage1Experiments.lean`

5.1 Finite label model

The code defines a finite $\mathbb{F}_\ell$-label model, the quotient $|F_\ell|$ consisting of the zero class together with $\mathbb{F}_\ell^\times/\{\pm 1\}$, and the canonical half-range representatives

$$j = 0, \ldots, (\ell - 1)/2.$$

The Stage 1 label model is now connected to the bad-local theta-root data of IUT I, Definition 3.1(e),(f). A new source package consumes the quotient-$\mathbb{Z}$ data of a bad local theta-root model and derives the Stage 1 full label from its canonical coordinate. It also carries the canonical-generator-up-to-sign witness. In the concrete $Z/\ell Z$ quotient case this derived full label is exactly the nonzero canonical full label corresponding to coordinate 1. It proves, for the descended full-label degree model:

$$\deg(\Theta_j) = j^2 \deg(q)$$

for the degree-level Gaussian endpoint, with sign invariance on the full-label quotient. It also proves that $j = 1$ and $j = 2$ give distinct degrees exactly when $\deg(q) \neq 0$. Thus the q^{j^2} phenomenon is present in the model; it is not hidden inside a scalar identification.

The raw coordinate square-weight profile on $\mathbb{Z}/\ell\mathbb{Z}$ is kept as a diagnostic object and is not identified with the descended $|F_\ell|$ theta exponent. The coordinate average, the nonzero-coordinate average, and the absolute-label average are all formalized separately. Lean proves the exact denominator relations and shows that these averages differ strictly for nonzero environment degree. This prevents silently replacing the paper's ℓ-denominator convention by an auxiliary nonzero-only average.

5.2 Indeterminacy and upper-semi route

The Step (x) corridor is represented as:

$$A_0 \quad = \quad A_1 \quad = \quad A_2 \quad \leq U_{\mathrm{Ind3}},$$

where A_0, A_1, A_2 are the before-, after-(Ind1)-, and after-(Ind2)-averaged log-volumes, and U_{Ind3} is the (Ind3) upper-semi target. From this data and the hull/determinant identifications, the code constructs the Step (xi) upper-ray datum:

$$q_{\mathrm{avg}} \leq \Theta_{\mathrm{hull}}.$$

The same route constructs the two q-pilot readings in Step (xi-g) and proves that they are equal to the same fixed real value.

5.3 SHE and Hodge-descent boundary

The current SHE layer is conditional. It has typed records for:

structured SHE square-weight transport
factored square/full-label preservation
Hodge-descent packet transport
nonarchimedean (Ind3) source-target alignment.

Given these inputs, Lean constructs the weighted theta comparison consumed by the final route:

$$q_{\text{signed}} \le \Theta_{\text{signed}}.$$

The nonarchimedean (Ind3) entry, factored square/full-label preservation, and Hodge-descent packet transport now compose directly to this comparison. With the q-pilot positivity condition and the displayed C_Θ-bound, the same composed route instantiates the signed C_Θ endpoint and proves $C_\Theta \ge -1$. The instantiated endpoint also exposes the strict and boundary alternatives: strict source comparison gives $C_\Theta > -1$, while $C_\Theta = -1$ forces equality of the signed q- and Θ-readings and makes the Θ-reading negative. These facts are now also packaged as a single route-level alternative: boundary equality with negative Θ-reading, or strict $C_\Theta > -1$. The code also proves a Gaussian-to-factored-SHE bridge at the present finite degree level. In particular, equality of the source and target environment degrees gives full-label value preservation; nonzero-label target bounds are enough because the zero label has square weight zero. The newest route uses this bridge before applying the nonarchimedean endpoint: Gaussian degree evaluations with preserved environment degree now construct the factored square/full-label preservation package consumed by the nonarchimedean C_Θ dichotomy. This removes one primitive preservation input from that route, but it still assumes the nonarchimedean (Ind3) alignment and the Hodge-descent/SHE source audit. There is also an identity-coordinate specialization. In that case equality of Gaussian degree at the canonical full label 1 implies preservation of the environment degree, so the nonarchimedean C_Θ dichotomy no longer takes environment-degree preservation as an independent hypothesis. The Hodge–Arakelov/Gaussian bridge is now source-facing at the current degree level. A theta-value evaluation source carries the bad-local theta-root label data and a theta-monoid degree d, and constructs the Gaussian degree evaluation

$$\deg_{\text{gau}}(\lambda) = e(\lambda)\, d,$$

where $e(\lambda)$ is the absolute-label theta exponent. The zero label has degree 0, and the canonical label 1 has degree d. Thus equality at the canonical label derives equality of the environment degree and feeds the factored SHE square/full-label obligations. The raw nonarchimedean entry alignment has also been refactored through a source-facing nonarchimedean upper-semi object equipped with 1-column q-pilot log-Kummer data. The upper-semi inequality remains the Step (x) input; the 1-column datum records the q-pilot column and log-Kummer non-interference before converting to the raw real equalities consumed by the current route. For the source side of this compatibility, the code can now use the existing finite tensor-packet chain: Kummer transfer holomorphicF $\to$ holomorphicD and mono-analytic forgetting holomorphicD $\to$ $D^{\vdash}$ preserve product log-volume. Thus the source-side upper-semi equality may be derived from this chain rather than postulated as a bare equality. The newest source package attaches this finite Kummer/forgetting chain to the bad-local theta-root cusp-label source. It carries the canonical nonzero $|F_\ell|$-label and derives the nonarchimedean (Ind3) source equality from product-log-volume preservation along

$$\text{holomorphicF} \to \text{holomorphicD} \to D^{\vdash}.$$

The packet local object, local entry source, mono-analytic product, and holomorphic source product are now packaged as one source-side log-Kummer packet calibration object. The object itself is still supplied source data, but the equality from entry source to (Ind3) source is derived from it together with the Kummer and forgetting maps. The source calibration can now also be constructed from the explicit packet-local/source alignment, the packet-local (Ind3)-source alignment, and the two product-preservation maps. Thus the route no longer has to accept the holomorphic source-product equality as a primitive field when these Step (x) alignments are available. The source and target packet calibrations are now joined by a finite log-Kummer packet-correspondence source carrying the q-pilot column and non-interference guard. Its target face carries the common packet-normalized capsule value, and Lean derives the direct theta-to-entry, theta-to-packet, and entry-to-(Ind3)-target equalities consumed by the upper-semi route. The target calibration no longer contains an independent real representative for this common value: all target equalities are stated directly against the packet-normalized capsule log-volume. This removes one artificial identification layer from the Step (x) interface. The target face has also been refined by a vertical-IQ source object, following IUT III, Proposition 3.10(ii). The theta average, the (Ind3)-target value, and the local entry target are represented as points of one log-Kummer IQ-target whose values agree with the packet-normalized point. The packet-target calibration is then constructed from this vertical-IQ source, so the corresponding theta-to-entry and entry-to-(Ind3)-target equalities are derived from common-target compatibility rather than passed as separate target-calibration inputs. This object is guarded by the exact $FMOD$ side of Proposition 3.10(iii) and Remark 3.10.1: the one-sided $Fmod$ upper-semi mode is classified separately and is not accepted as an exact vertical-IQ target calibration. The local entry target can now also be constructed from the packet-normalized capsule value itself. A packet-normalized entry-target source builds a nonarchimedean upper-semi entry whose target finite log-volume is definitionally the packet-normalized capsule value. The corresponding target calibration therefore no longer needs an independent proof for the entry-target equality; only the theta-average and (Ind3)-target identifications with the same packet-normalized value remain as inputs at this level. The latter can now be reduced further: if the Step (x) target-side (Ind3) alignment gives equality of the theta average with the (Ind3) target, then Lean derives the (Ind3)-target-to-packet-normalized equality from this alignment and the theta-average-to-packet-normalized equality. The packet-correspondence source itself can now be built from this reduced target data together with the source packet/product calibration. Thus the correspondence constructor no longer needs a completed target-calibration record when the target entry and theta-target alignment are available. It can also be built directly from the packet-local source alignments and the same target theta alignment. In this form both the source packet calibration and the target packet calibration are generated before the q-pilot non-interference endpoint is projected. The same reduction has been lifted through the realified-Frobenioid packet source: realified Kummer compatibility still supplies the holomorphicF $\to$ holomorphicD $\to$ $D^{\vdash}$ transfer, while the target face is now generated from the packet-normalized entry target and theta-target alignment. The realified entry source and the theta-root attached realified entry source can also consume the vertical-IQ target source directly; their upper-semi endpoint then derives the theta-to-entry and entry-to-(Ind3)-target equalities through the common log-Kummer target rather than through separate target alignment inputs. This vertical-IQ source now reaches the integrated Hodge/SHE/IPL/hull route: the first-pass C_{Θ} dichotomy can be stated without the ordered-real-line target alignment and without the auxiliary packet-normalized $ZMod(\ell)$ route whenever the exact common IQ-target is supplied. The same route has a source-reduced form: packet-local/source alignment, entry-source/mono-analytic-product alignment, packet-local/(Ind3)-source alignment, and the realified Kummer/forgetting compatibilities construct the source calibration internally before the vertical-IQ target enters the final inequality route. This reduction now also exists at the Step (x) source-object level: an upper-semi packet-normalized entry, packet-local source alignments, realified Kummer/forgetting compatibility, and an exact vertical-IQ target construct one realified log-Kummer entry source before the Hodge/SHE/IPL/hull route is applied. The realified source can now also generate the source packet calibration from packet-local/source alignments before forming the correspondence; consequently the strongest realified Step (x) constructor no longer needs a completed source calibration as input. The first-pass Gaussian route also has a one-correspondence form: it consumes a finite log-Kummer packet-correspondence source and projects the source packet/product calibration and target packet-normalized calibration from that single object before entering the upper-semi corridor. The same correspondence now has a transfer-chain endpoint exposing the holomorphicF $\to$ holomorphicD $\to$ $D^{\vdash}$ product log-volume equalities, the source equality with the (Ind3) real source, the target packet-normalized equalities, and q-pilot non-interference from the same object. The current strongest Step (x) entry source places this correspondence below the realified Frobenioid layer: the selected nonarchimedean upper-semi entry and the realified-Frobenioid packet source are bundled together, and Lean projects the Kummer transfer, mono-analytic forgetting transfer, source calibration, target packet calibration, and q-pilot non-interference guard from that single source object. There is now a reduced constructor for this entry source: entry membership, packet-local/source alignment, packet-local (Ind3)-source alignment, packet-normalized target data, and theta-target alignment construct the realified packet source internally and then derive the upper-semi inequality. The bad-local theta-root label source is now attached to this realified entry source. The corresponding Hodge–Arakelov route consumes one object carrying the canonical nonzero theta-root label, the selected upper-semi entry, and the realified-Frobenioid log-Kummer packet source before entering the C_Θ-dichotomy. This theta-root attached source now has the same source-reduced vertical-IQ constructor: the theta-root label package is combined with the upper-semi packet-normalized entry, packet-local source alignments, realified Kummer/forgetting compatibility, and the exact vertical-IQ target. Lean then projects the canonical-label facts, transfer-chain product equalities, target packet calibrations, q-pilot non-interference, and the Step (x) upper-semi inequality from this single source object. The packet-local source side is now bundled as a named alignment object: packet-local/source equality, entry-source/mono-analytic-product equality, and packet-local/(Ind3)-source equality travel together before constructing the packet source calibration. Conversely, a source packet/product calibration together with the Kummer and mono-analytic forgetting transfers reconstructs this packet-local alignment object; the packet-local/(Ind3)-source equality is derived through the transfer chain rather than supplied as a parallel datum. When the mono-analytic packet product is supplied by the finite divisor/Frobenioid product source, Lean now derives the entry-source/mono-analytic-product equality from the packet-local object identification and the finite divisor realified log-volume computation. This recovery has now been lifted to the realified-Frobenioid packet source: Lean uses the source object's own Kummer and mono-analytic forgetting compatibilities to reconstruct the packet-local source alignment. At the realified Step (x) entry level, the same projection also carries the upper-semi membership of the selected nonarchimedean entry. At the theta-root attached level, the projection also carries the canonical bad-local theta-root generator and nonzero full-label facts. The final Hodge/SHE/IPL/hull route now has an audit form that returns this recovered packet-local source alignment together with the C_Θ dichotomy. The target-only (Ind3) alignment is also derived at route level from the Step (x) packet-normalized theta source and the exact vertical-IQ target, with the $FMOD$-exactness guard retained. An audit form of this route exposes the derived equality between the Hodge-descent theta average and the (Ind3) target value. At the packet-correspondence layer, this source-alignment object and the exact vertical-IQ target now construct one log-Kummer correspondence source. The realified-Frobenioid packet source uses this correspondence-level constructor, so source and target Step (x) calibrations enter through a single packet source before the theta-root entry source is formed. The same correspondence-level reduction is now available for the packet-normalized ordered target: the named source-alignment object constructs the source face before the theta-average and target (Ind3) alignments construct the target face. The theta-root attached entry source now also projects this packet correspondence, so the canonical bad-local label facts and the full Step (x) transfer/target calibration endpoint are exposed from one source object. The Hodge/SHE/IPL/hull route has the same realified endpoint: its alignment record supplies the Hodge–Arakelov Gaussian comparison, canonical-one preservation, q-pilot positivity, and theta/Hodge log-volume calibration, while the Step (x) data enters through the single theta-root attached realified source. The route now also has a source-reduced endpoint: the theta-root label package, entry membership, packet-local source alignments, packet-normalized target alignment, and realified Frobenioid compatibility construct the Step (x) source inside the theorem before proving the same $C_\Theta = -1$ boundary alternative or strict $C_\Theta > -1$ conclusion. For the vertical-IQ form, the theorem now constructs the theta-root attached realified Step (x) source object directly from this source-alignment object and the exact IQ-target before applying the Hodge/SHE/IPL/hull endpoint. At the non-theta-root realified entry level, the corresponding source-aligned vertical-IQ endpoint now exposes the $FMOD$-exactness guard itself, together with Kummer/forgetting product preservation, target packet normalization, q-pilot non-interference, and the upper-semi inequality. The same exactness exposure has been lifted through the bad-local theta-root source: the canonical theta-root label data, exact vertical-IQ condition, packet-normalized target equalities, and Step (x) upper-semi inequality now appear in one theta-root attached endpoint. The packet-normalized theta-source calibration is now also consumed through a single Step (x) alignment object: the constant $ZMod(\ell)$ packet-normalized route and the proof that it has the same theta source as the Hodge-descent route are bundled before Lean derives the theta-average-to-packet-normalized equality used by the final corridor. The ordered-real-line form now has the same source-aligned boundary: the packet-local source-alignment object and realified Kummer/forgetting compatibility construct the source calibration internally before the matched theta-target alignment is used. This has been narrowed further to a target-only (Ind3) alignment: source-side packet-to-(Ind3) information is taken from the packet-local source-alignment object, while only the theta-average-to-target equality remains as ordered-real input. That target-only alignment is now constructible from Step (x) packet-normalized theta-source alignment together with an exact vertical-IQ target: Lean derives both sides through the common packet-normalized value and keeps the $FMOD$-exactness guard visible. The corresponding realified-source audit removes the independent packet-local source-alignment parameter: the alignment is recovered from the theta-root attached realified log-Kummer entry source, while the exact vertical-IQ target supplies the target-side equality and $FMOD$ guard. The stronger realified-source audit also removes the separate packet-normalized theta-source route at this boundary: theta packet normalization is projected from the same realified log-Kummer entry source, and the vertical-IQ target supplies only the (Ind3)-target packet-normalized equality. The procession-normalized log-volume layer now has a direct finite $\mathbb{F}_\ell$-labelled source: capsule log-volumes are indexed by $ZMod(\ell)$, the total is the finite sum over all labels, the normalized value is this sum divided by ℓ, and Lean proves invariance of the average under label translation. At this finite-label boundary, the (Ind1)/(Ind2) preservation step no longer has to be represented as pointwise equality of labelled summands: Lean proves preservation of the normalized average by reindexing the $ZMod(\ell)$-sum, after which a pointwise upper bound on the twice-translated family gives the one-sided (Ind3) bound for the original average. The target-side packet calibration can now consume this labelled Step (x) average directly: the equality $\theta_{\text{avg}} = \text{packetNormalized}$ is derived by composing θ_{avg} with the labelled normalized average and then with the audited packet-normalized capsule value. This dependency is tagged separately from a bare direct real-line identification. The same labelled-average dependency has been lifted into the realified Frobenioid packet source constructor, so the realified Step (x) packet source no longer needs a raw packet-normalized theta-average equality at this boundary. It is now also lifted through the theta-root attached Step (x) entry source and through the Hodge/SHE/IPL/hull route: the route may consume the labelled average and its equality with the Hodge-descent theta average directly, rather than a constant packet-normalized $ZMod(\ell)$ route object. The same route now has a source-aligned form: the source packet/product calibration is constructed from packet-local source alignment and the Kummer/forgetting transfers, while the target packet calibration is constructed from the labelled procession average. A further route removes the raw theta-source-to-labelled-average real equality: the labelled capsule source is aligned labelwise with the theta-source $ZMod(\ell)$-average, and Lean derives equality of the two averages from the finite averaging formula. The labelled-average-to-packet-normalized equality is now also derived in the source layer: if each labelled capsule has the displayed local object, Lean proves that the labelled average is that local object's finite log-volume, and direct packet-normalization then identifies it with the packet-normalized capsule value. The labelled route can now replace its remaining target-side ordered-real alignment by an exact vertical-IQ target: Lean derives the (Ind3)-target equality from the same packet-normalized value used by the labelled theta-source alignment. In the finite-divisor variant, the source packet-local alignment is now constructed from the finite divisor/Frobenioid tensor packet product before the direct-labelled vertical-IQ route is applied. The labelwise equality between the labelled capsule source and the Hodge-descent theta-source $ZMod(\ell)$-average is now derived from local Hodge-descent transport: each theta-source label is identified with the packet local object's finite log-volume, as is each labelled capsule value. The vertical-IQ route can now also be instantiated with a finite divisor/Frobenioid packet product on the mono-analytic side. In this form the route constructs the theta-root attached realified log-Kummer entry source from the finite divisor packet product and the exact vertical-IQ target before applying the Hodge/SHE/IPL/hull endpoint. The packet-local source alignment is now recovered inside that source object, not passed as a separate route input. The Remark 3.9.5 hull boundary has also been lowered from an abstract intersection-closed predicate to a product-hull source that matches the paper's $\lambda \cdot O$ description. The declaration

`IUTStage1Remark395ProductHullSystemSource`

carries a family of product hulls, their log-volumes, and for each region a distinguished parameter whose product hull is the intersection of all product hulls containing that region. From this data Lean constructs the ordinary `IUTStage1Remark395HolomorphicHullSystem`; the experiment endpoint proves that $\phi(P)$ is exactly this selected product hull, contains P, is minimal among product hulls containing P, and has the stated product-hull log-volume. There is now a direct-product constructor for this boundary,

`IUTStage1Remark395DirectProductHullSystemSource`.

Here a product hull is a literal coordinate product and the selected intersection parameter is the family of coordinate projections of P. Lean proves that this coordinate-projection product is the intersection of all product hulls containing P, then projects to `IUTStage1Remark395ProductHullSystemSource`. Thus this layer no longer supplies the product-hull intersection law as an independent field. This product-hull source now has a principal scalar-multiple refinement,

`IUTStage1Remark395PrincipalProductHullSystemSource`.

It records the local integer/direct-product region O, scalar maps $\lambda \cdot (-)$, nonzero-parameter witnesses, and principal hulls of the source-paper form $\lambda \cdot O$. It then derives the previous

`IUTStage1Remark395ProductHullSystemSource`. Thus the hull boundary can distinguish a genuinely Remark 3.9.5(i)-shaped product hull from an arbitrary family of subsets before the analytic construction of the scalars is completed. The supplied principal-minimality law has now been narrowed further by

`IUTStage1Remark395DirectProductPrincipalHullSystemSource`.

This constructor starts from the direct-product hull theorem and a scalar presentation of each principal region. Its remaining source equality is the pointwise identification $\lambda \cdot O = \prod_v H_v(\lambda)$; once that is available, Lean derives the principal intersection/minimality law from the coordinate-projection product-hull theorem. Thus the principal layer no longer has to restate the whole "smallest $\lambda \cdot O$ containing P" property when its scalar images have already been matched to direct-product hulls. This equality boundary has now been factored once more by

`IUTStage1Remark395ScalarImageDirectProductHullSource`.

Instead of accepting the extensional equality $\lambda \cdot O = \prod_v H_v(\lambda)$, this source records the two directional scalar-image laws: scalar multiples of local-integer points land in the corresponding direct-product hull, and every direct-product-hull point has a scalar preimage in O. Lean proves the set equality from these two laws and then constructs the direct-product principal hull source above. Thus the next valuation/local-ring layer can target concrete image and preimage statements rather than a monolithic equality of subsets. The product-level image/preimage laws have now been pushed down to coordinate factors by

`IUTStage1Remark395CoordinateScalarImageDirectProductHullSource`.

Here O is the literal product of local integer factors O_v, and $\lambda \cdot (-)$ is assembled coordinatewise from local scalar maps $\lambda_v \cdot (-)$. The source only supplies the one-place assertions $\lambda_v O_v \subseteq H_v(\lambda)$ and the corresponding local preimage statement for points of $H_v(\lambda)$. Lean combines these coordinate witnesses by dependent choice to prove the global scalar-image landing and preimage laws, then constructs the scalar-image direct-product hull source. Thus the next finite-extension valuation step can work one local factor at a time. There is now a stricter scalar-parameter version of this coordinate boundary:

`IUTStage1Remark395ScalarParameterDirectProductHullSource`.

This source no longer asks for local image/preimage laws for arbitrary subsets. Instead each coordinate has an actual scalar-parameter type and a named local region $H_v(\lambda_v)$ satisfying the source-shaped equality

$$H_v(\lambda_v) = \lambda_v \cdot O_v.$$

Lean derives the product landing law, product preimage law, and equality between the principal scalar-image hull and the direct-product hull for the product scalar parameter. This is the current lowest scalar-image boundary before the finite-extension valuation model supplies the concrete local parameters and proves the equality $H_v(\lambda_v) = \lambda_v O_v$. The principal hull presentation is now tied to the valuation-ball local-ring cover. The declaration

`IUTStage1Remark395PrincipalValuationBallProductHullCoverSource`

carries a valuation-ball factor-calibrated Haar tensor-packet cover together with a principal product-hull source, identifies the valuation cover's hull system with the principal hull system, and identifies

O with the anchor valuation-ball direct-product cell. Its endpoint proves that the selected principal hull $\lambda \cdot O$ is the same subset as the valuation-ball direct-product cell union, that its log-volume is the calibrated cell sum, and that the same valuation cover still projects to the adjusted Ob3/Ob5 determinant source. This does not construct the analytic scalar parameters from the source papers, but it removes a previous separation between the $\lambda \cdot O$ hull notation and the nonarchimedean local-ring/Haar cover used by the determinant route. There is now also a product-carrier version of this valuation-ball bridge,

`IUTStage1Remark395DirectProductPrincipalValuationBallProductHullCoverSource`.

Its valuation cover lives on an actual dependent product, and its principal hull source is the direct-product-backed scalar source above. The endpoint therefore identifies the selected $\lambda{\cdot}O$ hull first with the coordinate/direct-product hull selected by the projection construction, then with the valuation-ball direct-product cell union, and finally carries the same adjusted Ob3/Ob5 determinant normalization handoff. The remaining scalar obligation is now just the per-parameter analytic equality between the source paper's scalar image $\lambda \cdot O$ and the corresponding valuation/direct product hull. This valuation-ball bridge has also been lowered to the scalar-image source:

`IUTStage1Remark395ScalarImageValuationBallProductHullCoverSource`.

It consumes the two directional scalar-image laws, constructs the direct-product principal hull source internally, and then projects through the same valuation-ball determinant route. Its endpoint therefore keeps the valuation cover, calibrated cell-sum log-volume, and adjusted Ob3/Ob5 handoff, but the scalar trust boundary exposed at this product-carrier level is no longer a subset equality. It is precisely the landing and preimage pair that the next finite-extension valuation/local-ring construction must prove. The coordinate-local version of this same valuation bridge is now exposed as

`IUTStage1Remark395CoordinateScalarImageValuationBallProductHullCoverSource`.

It starts with the coordinate source above, constructs the scalar-image valuation cover internally, and keeps the selected principal hull identified with both the direct-product hull and the valuation-ball direct-product cell union. Consequently the adjusted Ob3/Ob5 determinant handoff can now be reviewed at a boundary whose remaining scalar hypotheses are one-place statements for the local factors O_v and $H_v(\lambda)$, rather than global statements about $\lambda \cdot O$. The selected scalar-parameter version of this valuation bridge is now exposed as

`IUTStage1Remark395SelectedScalarParameterValuationBallProductHullCoverSource`.

It combines the scalar-parameter source with the valuation-ball cover by choosing the product scalar parameter $(\lambda_v)_v$ whose local regions $H_v(\lambda_v)$ are exactly the coordinate projections of the current possible-image union. Lean then proves that the selected scalar image $\lambda \cdot O$ is the direct-product hull selected by the holomorphic hull operator and hence the valuation-ball direct-product cell union used by the adjusted Ob3/Ob5 determinant source. This leaves the next finite-extension valuation task in its sharpest current form: construct the actual local parameters λ_v and prove $H_v(\lambda_v) = \lambda_v O_v$ from the local IUT source data. This scalar-parameter source has now been lowered once more to actual local multiplication by nonzero elements:

`IUTStage1Remark395NonzeroScalarMultiplicationDirectProductHullSource`.

Here the parameter type at a place v is the subtype of nonzero local scalars, the scalar action is the local multiplication map $x \mapsto \lambda_v x$, and the region $H_v(\lambda_v)$ is defined as the image of O_v under this map. Consequently the equality

$$H_v(\lambda_v) = \lambda_v O_v$$

is no longer a supplied field at this boundary; it is the defining construction from which Lean derives the product landing, product preimage, and principal-hull/direct-product equality. The valuation-ball determinant handoff has the corresponding nonzero-scalar form

`IUTStage1Remark395NonzeroScalarMultiplicationValuationBallProductHullCoverSource`.

Its endpoint verifies that the selected nonzero scalar image is still the holomorphic family hull and the valuation-ball direct-product cell union used by the adjusted Ob3/Ob5 determinant source. The remaining local-field task is therefore more specific: identify the source-paper scalar parameters as nonzero elements of the local factors and connect the local integer factors O_v to the valuation-ring/unit-ball construction already present in the Haar normalization branch. This second connection has now been formalized at the product-cover boundary by

```
IUTStage1Remark395ValuationUnitBallNonzeroScalarMultiplication
  ProductHullCoverSource.
```

This source keeps the valuation-cover place index explicit and records a coordinate-to-place selector. At the anchor chart it requires each selected compact-open valuation ball to be the unit ball $\mathcal{O}_v = \{x \mid \|x\|_v \le 1\}$, proves that the anchor local factor is the realized ring of integers, derives the global equality

$$O = \prod_v O_v = \text{anchor valuation direct-product cell},$$

and then constructs the previous nonzero-scalar valuation cover. Thus the $\lambda_v O_v$ hull now enters the adjusted Ob3/Ob5 determinant route through valuation-unit-ball provenance, not through a separately supplied global anchor-cell equality. What remains is to construct the coordinate-to-place selector and the nonzero scalars from the source-paper Frobenioid/Kummer local data rather than pass them as source fields. This valuation-ball provenance has now been lifted to the finite-divisor vertical-IQ endpoint. The declaration

```
IUTStage1Remark395PrincipalValuationBallBackedConstructorBacked
  ConstructedHullDeterminantFiniteDivisorVerticalIQSource
```

requires the constructor-backed adjusted Ob3/Ob5 source to be the record-canonical form of the adjusted determinant source projected from the principal valuation-ball cover. From this equality Lean derives the principal hull-operator equality, the equality between the valuation-ball possible-image union and the Theorem 3.11 possible-image union, and the identity between the selected route hull and the valuation-ball direct-product cell union. The public endpoint

```
remark395PrincipalValuationBallBackedConstructorBackedConstructed
  Ob3Ob5AdjustedHullDeterminantFiniteDivisorVerticalIQ
  _cThetaDichotomyWithPrincipalValuationBallAudit
```

then returns this valuation-ball audit next to the same finite-divisor vertical-IQ C_Θ dichotomy. There is also a canonical-scale specialization of this endpoint:

```
remark395PrincipalValuationBallBackedConstructorBackedConstructed
  Ob3Ob5AdjustedHullDeterminantFiniteDivisorVerticalIQ
  _canonicalCThetaScaleDichotomyWithPrincipalValuationBallAudit.
```

Here the scale is not supplied as an external real number. It is the local source-attached value

$$C_{\Theta}^{\mathrm{can}} = \frac{\theta_{\mathrm{signed}}}{-q_{\mathrm{signed}}},$$

and the bound

$$\theta_{\mathrm{signed}} \leq C_{\Theta}^{\mathrm{can}} \cdot (-q_{\mathrm{signed}})$$

is derived in Lean from q-pilot positivity by cancellation. The returned `PrincipalValuationBallBackedCanonicalCThetaScaleAudit` keeps both the valuation-ball record audit and the constructor-backed hull/determinant bridge audit in the statement, so this endpoint no longer takes `thetaSigned_le_cTheta_absLogQ` as a hypothesis at this local source boundary. It is still a local canonical scale for the present source package, not yet a construction or identification of Mochizuki's global C_{Θ}-constant from the full IUT papers. The valuation-unit-ball/nonzero-scalar product source now has the analogous finite-divisor boundary wrapper. The new source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalarBacked
  ConstructorBackedConstructedHullDeterminantFiniteDivisor
  VerticalIQSource
```

contains the principal valuation-ball finite-divisor source, the literal local product scalar source, and an explicit carrier equivalence from the product carrier to the target real-line carrier. Its transport audit records that the product local-integer region, selected nonzero scalar image, and valuation-unit-ball direct-product cover all transport to the same principal $\lambda_v O_v$ hull. The public experiment endpoint

```
remark395ValuationUnitBallNonzeroScalarBackedConstructorBacked
  ConstructedOb3Ob5AdjustedHullDeterminantFiniteDivisor
  VerticalIQ_canonicalCThetaScaleDichotomyWithTransportAudit
```

therefore exposes the same canonical-scale vertical-IQ dichotomy together with the product-carrier/unit-ball transport provenance. The remaining source-facing gap has moved to constructing this carrier equivalence, the coordinate-to-place selector, and the nonzero scalars from the actual Frobenioid/Kummer local data. This transport boundary has now been sharpened once more. The reduced source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  FiniteDivisorTransportSynchronizationSource
```

does not ask separately for the transported valuation-cover equality. It only records the carrier equivalence, the transport of the local-integer region, and the transport of the selected nonzero scalar image to the principal hull. Lean then derives

$$e\big(\text{valuation unit-ball direct-product cover}\big) = \text{principal } \lambda_v O_v \text{ hull}$$

from the product-cover theorem

```
selectedScalarImageHull_eq
  _valuationBallDirectProductCellUnion.
```

The experiment endpoint

```
remark395ValuationUnitBallNonzeroScalarTransportSynchronized
  ConstructorBackedConstructedOb3Ob5AdjustedHullDeterminant
  FiniteDivisorVerticalIQ_canonicalCThetaScaleDichotomy
  WithTransportAudit
```

therefore reaches the same canonical-scale vertical-IQ dichotomy with one fewer transported equality at the trust boundary. The carrier side of the same route has also been separated into a charted source. Instead of supplying an arbitrary equivalence from the local product carrier to the target real-line copy, the new

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  TargetCarrierChartSource
```

records a product-to-real coordinate, a real-to-product realization, and the two inverse laws. Lean derives the carrier equivalence

$$((d : \delta) \to A_d) \simeq \texttt{Point target}$$

from this chart, and the charted finite-divisor synchronization endpoint then forwards to the reduced transport-synchronization route. Thus the current trust boundary has moved from an unstructured carrier equivalence to the construction of the product real-coordinate chart itself. The remaining source-facing task is to build that chart from the actual Frobenioid/Kummer local valuation data, together with the coordinate-to-place selector and nonzero scalar preimage laws. For the finite-divisor endpoint itself, this carrier requirement has now been weakened once more to the actual data used by the comparison: a forward transport into the target real-line copy. The source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  TargetPointTransportSource
```

keeps the same valuation-unit-ball/nonzero-scalar product source and principal valuation-ball source, but replaces the equivalence or inverse chart by a map

$$((d : \delta) \to A_d) \longrightarrow \texttt{Point target}.$$

The transported valuation-cover equality is again derived from the selected scalar-image/product-cover theorem, and the public endpoint reaches the same canonical-scale vertical-IQ dichotomy while returning a `TargetPointTransportAudit`. Thus inverse reconstruction of the local product carrier is no longer part of the finite-divisor trust boundary. The remaining source-facing construction is the forward target log-coordinate map itself, with proofs that it sends the local integer region and selected nonzero scalar image to the corresponding principal $\lambda_v O_v$ regions. This has now been sharpened to a coordinate-derived target transport source. The new

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  CoordinateTargetTransportSource
```

does not accept an arbitrary map to `Point target`. It takes a real coordinate map

$$((d : \delta) \to A_d) \longrightarrow \mathbb{R}$$

together with coordinate descriptions of the target principal local-integer region and selected principal hull. Lean proves the elementary wrapper lemma that the induced map $x \mapsto \{\texttt{coord} := f(x)\}$ has the required point-level image if and only if the coordinate image has the required real-region image. The coordinate-derived endpoint then forwards through `TargetPointTransportSource`. Thus the remaining transport obligation is now stated at the real log-coordinate level: construct the coordinate map from the local valuation/Kummer data and prove the two coordinate image equalities.

The selected-hull half of this obligation has now been pushed one layer closer to Remark 3.9.5(i). The source-core theorem

`selectedScalarImageHull_eq_selectedNonzeroScalar_image`

proves that the selected product hull is literally the image $\lambda \cdot O$ of the local integer product region under the selected coordinatewise nonzero scalar multiplication. The new

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`SelectedImageCoordinateTargetTransportSource`

therefore asks for the target coordinate image of this explicit $\lambda \cdot O$ region, not for the already-packaged `selectedScalarImageHull` image. Lean rewrites through the source-core equality and then applies the coordinate-derived endpoint. The remaining selected-hull transport payload is now the real-coordinate image law for an explicit nonzero-scalar multiplication image. The local-integer half has now been pushed down in the same way. The valuation-unit-ball product source already proves

$$\texttt{localIntegerRegion} = \texttt{valuationCover.directProductCell valuationCover.anchor},$$

i.e., the local integer product is the anchor product of valuation unit balls $O = \prod_v O_v$. The new

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`ValuationAnchorCoordinateTargetTransportSource`

therefore states the local-integer coordinate image for this explicit anchor valuation-cell product and states the selected-hull coordinate image for the explicit $\lambda \cdot O$ image. Lean derives the previous selected-image coordinate transport source by rewriting through these two source equalities. Thus both coordinate-image transport obligations are now attached to concrete Remark 3.9.5(i) regions O and λO, rather than to packaged endpoint regions. The target coordinate map itself has now been restricted one step further. The new

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`FiniteLocalLogCoordinateTargetTransportSource`

does not accept an arbitrary real-valued coordinate functional on the source product carrier. It carries local readings

$$\ell_v : ((d : \delta) \to A_d) \longrightarrow \mathbb{R}$$

for the finite valuation-place set and defines the target coordinate by

$$x \longmapsto \sum_v \ell_v(x).$$

Lean then projects this finite local log-coordinate source to the valuation-anchor coordinate transport source. The remaining image laws are therefore the two explicit statements

$$\Big(\sum_v \ell_v\Big)(O) = R_O, \qquad \Big(\sum_v \ell_v\Big)(\lambda O) = R_{\lambda O},$$

where R_O and $R_{\lambda O}$ are the real-coordinate descriptions of the principal local-integer region and selected principal hull on the target real-line copy. Thus the arbitrary target-coordinate-map freedom has been removed from this boundary; the next mathematical task is to construct the local log-coordinate readings from the valuation/Kummer model and prove these two image equalities. The scalar finite-sum image laws have now been reduced to a product-image statement for the tuple of local coordinates. The helper

`finiteLocalLogCoordinateSumRegion`

defines the real region obtained from local coordinate regions $R_v \subseteq \mathbb{R}$ by

$$\Big\{ \sum_v r_v \mid r_v \in R_v \Big\}.$$

The theorem

`finiteLocalLogCoordinateSumRegion.image_sum_eq_of_vector_image`

proves that if

$$x \longmapsto (v \mapsto \ell_v(x))$$

maps a source region onto the product of the local coordinate regions, then $x \mapsto \sum_v \ell_v(x)$ maps the same source region onto the corresponding sum region. The new

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  LocalLogCoordinateProductImageTargetTransportSource
```

therefore supplies the vector-valued image laws for O and λO; Lean derives the finite-sum transport source and then enters the same valuation-anchor finite-divisor endpoint. The remaining source-facing problem is now the local product-coordinate theorem for $(\ell_v)_v(O)$ and $(\ell_v)_v(\lambda O)$, rather than an unstructured image theorem for the scalar sum. That product-coordinate theorem has now been split into its two source-facing halves. The lemma

`localLogCoordinateVector_image_eq_product_of_landing_and_preimage`

proves the vector-valued product image equality from coordinatewise landing

$$x \in S \Rightarrow \ell_v(x) \in R_v \quad \text{for all } v$$

and simultaneous realization

$$(r_v)_v \in \prod_v R_v \Rightarrow \exists x \in S,\ \ell_v(x) = r_v \quad \text{for all } v.$$

The corresponding

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  LocalLogCoordinatePreimageTargetTransportSource
```

now supplies these landing and simultaneous-preimage laws for O and λO; Lean derives the product-image source, then the finite-sum source, and finally the valuation-anchor endpoint. Thus the remaining mathematical construction is the actual local analytic/Kummer realizer for compatible tuples of valuation-place log coordinates, not a supplied vector-valued image equality. The simultaneous-preimage half has now been lowered once more to a direct product assembly theorem. The generic lemma

`simultaneousLocalLogCoordinatePreimage_of_placewise_factor_preimages` states that if each placewise coordinate value $r_v \in R_v$ has a local factor preimage b_v, and if admissible choices $(b_v)_v$ assemble to a source point whose local log-coordinate readings are exactly the factor coordinates, then the whole tuple $(r_v)_v$ has a simultaneous preimage. The source record

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  LocalLogCoordinatePlacewisePreimageTargetTransportSource
```

uses this lemma for both the anchor region O and the selected scalar image λO. Thus the remaining preimage payload is no longer a global existential statement over the product carrier; it is the placewise local scalar-image/preimage theorem together with the direct-product assembly law that identifies the assembled point with the prescribed log-coordinate tuple. The coordinatewise landing half has also been lowered to local factor data. The lemma

```
localLogCoordinate_lands_of_placewise_factor_landing
```

proves landing in $\prod_v R_v$ from a point-to-factor projection, membership of that factor in the appropriate local source region, agreement of the source local coordinate with the factor coordinate, and the local factor landing theorem. The new

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  LocalLogCoordinateFactorwiseTargetTransportSource
```

uses this landing lemma together with the placewise-preimage assembly theorem, so the public route now derives both sides of the vector-valued product-image equality from factorwise local data. The projected factorwise refinement further removes the abstract point-to-factor and assembly maps from this boundary. A valuation-place/product index equivalence $e : \gamma_{\mathrm{loc}} \simeq \delta$ identifies each local place with an actual coordinate of the dependent product $\prod_{d:\delta} A_d$. The source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  LocalLogCoordinateProjectedFactorwiseTargetTransportSource
```

sets the local factor carrier at v to $A_{e(v)}$, takes projection to be evaluation at $e(v)$, and takes assembly to be the standard dependent-product reindexing equivalence `Equiv.piCongrLeft`. The corresponding public endpoint therefore routes to Corollary 3.12 after proving the factorwise transport laws from product evaluation and reindexing. The remaining local-log payload at this boundary is the source-facing image description of the coordinate regions O_v and $\lambda_v O_v$ under the chosen local logarithmic coordinate, together with the valuation-place/product-index identification. This image-description boundary has now been lowered once more. The source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  ValuationLogImageProjectedFactorwiseTargetTransportSource
```

defines the local real regions themselves as

$$\ell_v(O_v), \qquad \ell_v(\lambda_v O_v),$$

where ℓ_v is the selected local valuation/log-coordinate map. The previous projected-factorwise source is obtained by projection, and the image equalities for O_v and $\lambda_v O_v$ are now definitional theorems. Thus this boundary no longer asks for separate local coordinate-image laws; at that level, the remaining inputs were the actual construction of the local maps ℓ_v from the valuation/Kummer model and the global finite-sum identifications of the target real-line principal regions. The finite-sum target boundary has now been lowered another step. The source

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`ValuationLogFiniteSumTargetPointSource`

defines the target point of a local product element x by the coordinate

$$\sum_v \ell_v(x_v).$$

It carries the typed compatibility saying that the source's coordinate-place map agrees with the chosen valuation-place/product-index equivalence, then proves internally that the coordinate images of $O = \prod_v O_v$ and $\lambda O = \prod_v \lambda_v O_v$ are exactly the finite-sum regions $\sum_v \ell_v(O_v)$ and $\sum_v \ell_v(\lambda_v O_v)$. The remaining synchronization at this lower boundary is point-level transport of these two source regions to the principal finite-divisor target regions; the real-line finite-sum equalities are no longer independent fields of the exact-theta constructor. This boundary has now been lowered again by

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`CoordinatePlaceValuationLogFiniteSumTargetPointSource`.

Here the valuation-cover coordinate-place map is represented source-side by an equivalence $\delta \simeq \gamma_{\mathrm{local}}$. Lean derives the previous finite-sum synchronization law

$$\texttt{coordinatePlace}(e^{-1}(v)) = v$$

from the source equality $\texttt{coordinatePlace} = e$. Thus the remaining coordinate-index work is no longer an arbitrary target-boundary compatibility; it is the construction of this equivalence from the actual local valuation indexing. The local coordinate itself has now been lowered to a Haar log-volume source. The new local source

`IUTStage1Remark395LocalFactorHaarLogCoordinateSource`

assigns to a local factor x_v a local compact-open region U_{v,x_v} and defines

$$\ell_v(x_v) = \mu_{\log,v}(U_{v,x_v})$$

using the normalized finite-extension Haar log-volume source already present in the valuation/Haar branch. The corresponding finite-sum source

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`HaarLogCoordinateFiniteSumTargetPointSource`

therefore projects to the coordinate-place finite-sum route with ℓ_v no longer a bare function. Lean proves that the O_v and $\lambda_v O_v$ coordinate regions are images under normalized Haar log-volume of the assigned local regions. This has now been strengthened to a compact-open Haar source. The local declaration

`IUTStage1Remark395LocalFactorCompactOpenHaarLogCoordinateSource`

uses the finite-extension compact-open topology/Haar normalization object, requires every assigned U_{v,x_v} to satisfy the compact-open predicate, and then projects to the normalized Haar log-coordinate source. The finite product declaration

`IUTStage1Remark395ValuationUnitBallNonzeroScalar`
`CompactOpenHaarLogCoordinateFiniteSumTargetPointSource`

threads this compact-open provenance through the same target-point finite-sum endpoint and exposes the compact-open Haar route at the experiment surface. This has now been lowered one more step to centered valuation balls. The local declaration

`IUTStage1Remark395LocalFactorCenteredValuationBallHaarLogCoordinateSource`

takes a proper ultrametric local factor together with the valuation-ball Haar normalization source, defines

$$U_{v,x_v} := \overline{B}(x_v, r_v),$$

where r_v is the compact-open radius of the valuation-ball Haar source, and proves that this closed ball is open by the ultrametric inequality and compact by properness. Thus the compact-open predicate is no longer supplied at this boundary; it is constructed from the valuation-ball topology. The finite product source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPointSource
```

threads these centered balls through the same finite-sum target map and projects to the compact-open Haar route at the experiment surface. This local boundary has now been lowered to the valued-field integer source. The declaration

`IUTStage1Remark395LocalFactorValuedFieldIntegerCenteredValuationBallHaarLogCoordinateSource`

starts from a valued-field integer proper-ultrametric Haar normalization source: the local ring is the actual valued-field integer subring $\mathcal{O}_K$, the valuation topology is the valued-field ultrametric topology, and the centered coordinate is still

$$x_v \longmapsto \mu_{\log}(\overline{B}(x_v, r_v)).$$

Lean constructs the proper-ultrametric valuation-ball Haar source used by the centered-ball layer and then constructs the centered local coordinate source. The finite product source

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  ValuedFieldIntegerCenteredValuationBallHaarLogCoordinate
  FiniteSumTargetPointSource
```

threads these valued-field integer local coordinates through the same target finite sum and exposes the corresponding experiment route and Step (xi) constructor. The same boundary has now been specialized to the concrete base local field $\mathbb{Q}_p$. The local declaration

`IUTStage1Remark395LocalFactorPadicCenteredValuationBallHaarLogCoordinateSource`

starts from the p-adic proper-ultrametric Haar source: the integer ring is identified with $\mathbb{Z}_p$, the residue prime is p, the normalization degree is 1, and the uniformizer action is multiplication by p. It projects to the valued-field integer centered-ball source without adding a new abstract local-ring field. It now also projects to the finite-extension centered-ball source for $K = \mathbb{Q}_p$, proving that the same closed-ball local region and normalized Haar log-coordinate are obtained after passing through the degree-one finite-extension normalization route. The finite-product declaration

```
IUTStage1Remark395ValuationUnitBallNonzeroScalar
  PadicCenteredValuationBallHaarLogCoordinate
  FiniteSumTargetPointSource
```

uses valuation places as the product coordinates and assigns to each place v an actual local carrier $\mathbb{Q}_{p_v}$. Its target coordinate is definitionally

$$(x_v)_v \longmapsto \sum_v \mu_{\log}\big(\overline{B}(x_v, r_v)\big),$$

and the experiment surface now reaches the same topology/Haar Ob5–Ob7 canonical-scale finite-divisor route through this p-adic source. The Milestone 2 hull layer now contains a direct formalization of Remark 3.9.5(ii)–(iii)'s hull-map skeleton and approximant family. The hull map $\phi(P)$ is represented by a closure operator; Lean exposes the Remark 3.9.5(ii) laws

$$\phi(H) = H, \qquad P \subseteq \phi(P), \qquad P_1 \subseteq P_2 \Rightarrow \phi(P_1) \subseteq \phi(P_2),$$

together with idempotence $\phi(\phi(P)) = \phi(P)$. For a region P, an approximant record carries a hull H, proves $H \subseteq \phi(P)$, and proves

$$\mu_{\log}(P) \leq \mu_{\log}(H) \leq \mu_{\log}(\phi(P)).$$

For the Corollary 3.12 possible-image union, the public endpoints now also return $P \subseteq \phi(P)$, each $P_i \subseteq \phi(P)$, q-pilot containment in $\phi(P)$, and $\phi(\phi(P)) = \phi(P)$. The $\Xi(P)$-part of Remark 3.9.5(iii) is now represented at the same finite level: an exact hull approximant is a $\Phi(P)$-approximant H with $\mu_{\log}(H) = \mu_{\log}(P)$. When P is already closed under the hull operator, Lean constructs the canonical exact approximant. The possible-image Step (xi) source can now also be built from such an exact approximant and a weighted determinant source; in that route Lean derives $\mu_{\log}(P) = \mu_{\log}(H) = \det_{\log}$, rather than only the inequality $\mu_{\log}(P) \leq \mu_{\log}(H)$. Remark 3.9.5(iv)'s statement $\phi(P) \in \Phi(P)$, hence $H_{\Phi(P)} = \phi(P)$, is now represented by a finite approximant-family object. Every member of the family is a $\Phi(P)$-approximant inside $\phi(P)$, and if one distinguished member is the canonical hull, Lean proves that the family union and its log-volume are exactly those of $\phi(P)$. The corresponding $\Xi(P)$-family statement is also represented. Each member is an exact approximant with log-volume $\mu_{\log}(P)$; if the family contains the canonical hull, its union is $\phi(P)$. In the special case where P is already closed under hull formation, the canonical one-point $\Xi(P)$-family has union $P = \phi(P)$ and log-volume $\mu_{\log}(P)$, matching Remark 3.9.5(iv)'s closed-region case. For possible-image families, the exact $\Xi(P_B)$ boundary has now been sharpened: `IUTStage1Remark395PossibleImageExactXiFamilySource` does not accept an arbitrary exact family, but only the source calibration

$$\mu_{\log}(\phi(P_B)) = \mu_{\log}(P_B).$$

From this equality Lean constructs the canonical exact approximant and a $\Xi(P_B)$-family over either the one-point index or the same index type as a chosen $\Phi(P_B)$-family. At the possible-image-family level, this exact-approximant family now also interfaces with the Ob5 upper-semi quotient: if a $\Xi(P_B)$-family is available and one possible image is nonempty, Lean proves

$$H_{\Xi(P_B)} = \phi(P_B)$$

and proves that the quotient image of $H_{\Xi(P_B)}$ is the same collapsed point as the quotient image of that nonempty possible image. The same source-facing family now has an explicit Ob6 log-volume endpoint: for a $\Phi(P_B)$-family and an exact $\Xi(P_B)$-family either supplied directly or constructed from $\mu_{\log}(\phi(P_B)) = \mu_{\log}(P_B)$, Lean proves

$$H_{\Phi(P_B)} = \phi(P_B) = H_{\Xi(P_B)}$$

and records the inequalities

$$\mu_{\log}(P_B) \leq \mu_{\log}(H_{\Phi(P_B)}), \qquad \mu_{\log}(P_B) \leq \mu_{\log}(H_{\Xi(P_B)}),$$

together with the corresponding inequalities for each selected possible image P_β. Thus Remark 3.9.5(Ob6)'s warning that estimates must pass through hull-approximant unions, rather than through the raw possible-image region alone, is now a named checked source endpoint. The next source-facing checkpoint is also present. The Ob7 log-Kummer compatibility source attaches a constructed hull/determinant bridge to a finite $F^{\times\mu}$-prime-strip lift. Its endpoint proves that the q-region log-volume is bounded by the retained prime-strip global log-volume, while also recording the prime/place square, local realified-Frobenioid log-volume compatibility, and Frobenius-like unit-character compatibility. This keeps the comparison object in the Frobenioid/log-Kummer language required by Remark 3.9.5(Ob7), instead of collapsing immediately to a bare real number. The canonical hull $\phi(P)$ is itself an approximant, and every approximant is fixed by hull formation. This is the compactness guard emphasized in the paper: the indeterminacy in choosing a hull approximant remains inside $\phi(P)$. The bounded-family version of Remark 3.9.5(v)–(vi) is now represented by a family-hull quotient source. For a family P_β, the code forms the hull of the union, uses it as the collapsed subset in the upper-semi quotient, and proves that any two nonempty possible regions have the same quotient image. This quotient source is now connected directly to the typed Theorem 3.11 indeterminacy layer. The declaration

```
IUTStage1MultiradialThetaImages
  .remark395Ob5QuotientEndpoint_of_related
```

starts from a multiradial possible-image package with its explicit (Ind1), (Ind2), (Ind3) quotient relation, forms the corresponding Remark 3.9.5 family P_B, and proves that related choices have equal possible-image regions and equal upper-semi quotient images after collapsing $\phi(P_B)$. The experiment surface exposes the same bridge as

```
multiradialThetaImages_remark395IndeterminacyOb5Endpoint.
```

This Theorem 3.11 bridge now has a strengthened Ob5–Ob7 endpoint,

```
IUTStage1MultiradialThetaImages
  .remark395Ob5Ob6Ob7Endpoint_of_related
```

with experiment wrapper

```
multiradialThetaImages
  _remark395IndeterminacyOb5Ob6Ob7Endpoint.
```

The endpoint forms the same Remark 3.9.5 possible-image family from the typed (Ind1), (Ind2), (Ind3) quotient, proves the Ob5 quotient collapse for related choices, threads the family through the Ob6 $H_{\Phi(P_B)}$ and $H_{\Xi(P_B)}$ log-volume inequalities, and then attaches the Ob7 $F^{\times\mu}$-prime-strip source. Two synchronization equalities require the Ob7 hull/determinant bridge to use the same possible-image family and the same holomorphic hull operator, so this audit does not allow an unrelated prime-strip log-volume comparison to masquerade as the Theorem 3.11 possible-image comparison. As before, equality across an (Ind3) generator is not inferred from the one-sided Step (x) upper-semi inequality; it is precisely the explicit image-invariance field of the typed multiradial quotient package. The map statement in Remark 3.9.5(v) is now lifted to this bounded-family source: any set map from a nonempty possible region P_i into another possible region P_j induces, after quotienting by the family hull, the same image as the quotient map on P_i itself. The "formal empty set" extension in the same remark is represented by a finite formal-intersection system: a family of nonempty subsets of the collapsed subset S. Lean proves that every component has quotient image $\{S\}$, that all component images agree, and that component maps induce the same collapsed image after quotienting. Remark 3.9.5(vi)'s expected-equivalence statement is also represented for the bounded-family hull: for arbitrary nonempty regions A, B, their quotient images are both the collapsed point if and only if $A \subseteq \phi(P_B)$ and $B \subseteq \phi(P_B)$. The Ob5 compatibility of this quotient formalism with $\det^{\otimes M}(\phi(P_B))$ and $\mu_{\log}(\phi(P_B))$ is now represented at finite log-volume level: the same family hull that collapses the possible regions is identified with the weighted determinant log-volume, and the associated tensor-power normalization recovers this family-hull log-volume. Lean now also derives $\mu_{\log}(\bigcup_\beta P_\beta) \leq \det_{\log}$ from hull monotonicity and this determinant identification. The public Ob5 endpoint exposes these facts together with the expected-equivalence condition: for nonempty regions A, B, both quotient images are the collapsed family-hull point if and only if $A, B \subseteq \phi(P_B)$. The Ob9 log-volume passage is now represented at the same finite hull level. A realified semi-simplification correspondence between two hull shadows sends closed hulls to closed hulls, has a two-sided inverse on closed hulls, and preserves hull log-volume. Applied to the bounded-family hull, Lean proves that the transported hull still has log-volume $\det_{\log}$. This is the coarse log-volume content of Ob9; it is not yet a construction of the full log-link or its arbitrary-iterate indeterminacy. The Corollary 3.12 possible-image approximant source now projects the same estimate directly, so the upper-ray route can read $\mu_{\log}(P) \leq \det_{\log}$ from that source object. Remark 3.9.6 is represented by a finite product-formula theorem: if the log-link preserves the local log-volume at every place and the source and target conversion ratios agree, then the weighted global arithmetic degree sums agree. This has also been specialized to the Corollary 3.12 possible theta-image family, so the possible-image hull used by the upper-ray route and the Ob5 family-hull quotient/determinant source are definitionally the same hull. The canonical specialization now exposes this family-union-to-determinant estimate at its endpoint. The Corollary 3.12 possible-image approximant source projects to this family-hull quotient source, so the possible-image family used in the upper-ray route also carries the corresponding choice-independence theorem. Since the q-pilot region is contained in the chosen approximant and the approximant lies in the same family hull, the code also proves that a nonempty q-pilot region has the same family-hull quotient image as any nonempty possible theta-image choice. The same approximant now feeds the Step (xi) upper-ray constructor. If the q-pilot region lies in an approximant H, and if $\mu_{\log}(H)$ is the determinant-normalized log-volume, then Lean derives

$$\mu_{\log}(q) \leq \mu_{\log}(H) \leq \mu_{\log}(\phi(P)), \qquad \mu_{\log}(q) \leq \det_{\text{norm}}.$$

Thus this part of the corridor no longer has to use only the canonical hull itself; it can use any certified hull-approximant inside $\phi(P)$. This has now been specialized to the possible-image union in Corollary 3.12. One source object carries the family of possible theta images, the union P, a certified approximant $H \in \Phi(P)$, $q \subseteq H$, and determinant normalization of $\mu_{\log}(H)$. The Step (x)-to-Step (xi) comparison is now expressed by a named handoff proposition over this fixed corridor and hull source. Thus the two numerical identifications $\mu_{\log}(H) = \text{Ind3}_{\text{upper}}$ and $\mu_{\log}(q) = \overline{\mu}_{\log}(q)_{\text{pre-Ind}}$ are no longer loose parameters of the endpoint; they are the explicit source obligation required before the Step (xi-f) upper-ray datum is formed. Lean then derives the whole chain

$$\mu_{\log}(P) \leq \mu_{\log}(H) \leq \mu_{\log}(\phi(P)), \qquad \mu_{\log}(q) \leq \mu_{\log}(H) \leq \det_{\text{norm}}.$$

The determinant side has also been sharpened at finite log-volume level. Following Remark 3.9.5(vii), (Ob3-1)–(Ob3-3), a localization summand now carries a rank > 1, a bundle log-volume, a structure-sheaf adjustment, and a positive weight. A determinant source forms the finite weighted sum of the adjusted summands, then derives the positive tensor-power log-volume and the normalized determinant log-volume by definition. This replaces supplied tensor-power and normalized values for this new route, but it remains a log-volume skeleton rather than a construction of arithmetic vector bundles. The localization input itself has been refined one step further: a finite direct-summand source computes the bundle log-volume as a sum of direct summand log-volumes before projecting to the localization summand interface. Thus the weighted determinant can now be built from finite direct-summand localization sources. The Ob3-1-2 structure-sheaf adjustment is also named explicitly. A structure-sheaf adjusted source carries

$$L_{\mathrm{adj}} = \sum_{\gamma} L_{\gamma} - L_{\mathcal{O}(-)}$$

and proves that its weighted value is the same adjusted summand consumed by the determinant source. Thus the current determinant route separates direct summands, subtraction of the $\mathcal{O}(-)$ contribution, positive weighting, tensor-power formation, and normalization as distinct Lean interfaces. The possible-image approximant route can now consume this weighted determinant source directly. The equality between the approximant log-volume $\mu_{\log}(H)$ and the weighted normalized determinant value is now carried by a named Ob3-3 compatibility certificate. After this certificate is supplied, the old determinant record is only a projection used by the common upper-ray interface. At the package-ledger boundary, the hull/determinant source data can now be constructed from a concrete common hull K, a proof $P_{\mathrm{out}} \subseteq K$, and a determinant/log-volume estimate $\mu_{\log}(K) \leq \theta_{\mathrm{signed}}$. The resulting Lean endpoint returns target-union containment, the determinant bound, all-target bounds, and equality between the package's named hull/determinant bridge and this constructed bridge. Thus the remaining source obligation is the construction of K and its bound, not an opaque common-bound bridge. A more source-shaped constructor specializes K to $\phi(P_{\mathrm{out}})$ for an abstract hull operator ϕ. Its endpoint proves that the bridge hull is definitionally this hull of the target union, and reduces the source input to the determinant/log-volume estimate for $\phi(P_{\mathrm{out}})$ plus equality with the package's named bridge. The Set-based holomorphic-hull shadow now induces the Region-level hull operator and Region measure used by this package bridge when the underlying type is the point set of a real-line copy. The endpoint identifies the Region hull with the original point-set hull, proves region containment, and recovers monotonicity of the induced log-volume. At the package level, this induced hull operator can now be used directly: if the charted package measure is identified with the induced Region measure and the point-set hull of P_{out} satisfies the determinant bound, then the hull/determinant source data is constructed from the holomorphic-hull shadow. The remaining bridge equality is explicit, so the package cannot display one hull construction while proving with another. The source package's theta-pilot possible-image record now also feeds the canonical hull approximant directly: Lean proves that its set-theoretic union is exactly the package target union before applying the determinant comparison. The possible-image approximant source now feeds this package bridge in the canonical case. If its possible-image union is the package target union, its chosen approximant is the canonical hull, the package measure is the induced Region measure, and the normalized determinant value is at most θ_{signed}, then Lean constructs the same hull/determinant source data. No general containment of an arbitrary approximant is assumed. This has been lifted to the Theorem 3.11 source constructor: the approximant union may be tied to the source record's theta possible-image union, and the record then supplies the equality with the package target union. Thus the constructor no longer needs prebuilt hull/determinant source data in this canonical case. The same constructor now accepts the finite weighted determinant source from Remark 3.9.5(vii), (Ob3), directly. The determinant bound is stated for the weighted determinant log-volume, and Lean derives the normalized determinant bound consumed by the canonical hull bridge. A further specialization constructs the approximant as the canonical hull $\phi(P)$ itself. In the current route the q-pilot input may be containment in the possible-image union $P = \bigcup_i P_i$; Lean then derives q-pilot containment in $\phi(P)$ from hull extensivity. Thus direct containment in $\phi(P)$ is no longer a primitive input for this canonical possible-image specialization. Remark 3.9.5(vii), (Ob4), is represented at the same finite level: a naive Frobenius/tensor-power log-volume object records the passage $L \mapsto ML$ for $M > 0$, and Lean proves that normalized log-volume recovers L. The weighted determinant source projects to this object, so the M-th tensor-power copy does not add a further log-volume ambiguity in the current corridor. A comparison object also proves that two positive tensor-power presentations with the same base log-volume have the same normalized log-volume, even when the tensor degrees differ. This has now been upgraded to the Ob4-3 family form: `IUTStage1TensorPowerTwistFamily` records a reference tensor-power presentation and an arbitrary family of reconstructed twisted presentations over the same base log-volume. Lean proves that every twist has the reference normalized log-volume, that any two twists have equal normalized log-volume, and that all normalized log-volume bounds are equivalent across the twist family. This is the finite log-volume reading of the source-paper statement that tensor-power-twist indeterminacies have no substantive effect on the estimates used in Corollary 3.12. The package hull/determinant bridge now has an Ob4-shaped input form: a bound on the normalized tensor-power log-volume of the weighted determinant source is converted by Lean to the determinant bound used by the canonical hull bridge. In the canonical-hull form, this Ob4 input also derives the approximant-equals-$\phi(P)$ witness internally. At the package boundary the same canonical Ob4 route can now start from q-pilot containment in the possible-image union P; Lean derives containment in $\phi(P)$ before constructing the hull/determinant bridge. The same endpoint is exposed at the Theorem 3.11 source-record level, so the possible-image union used by the canonical hull is the record's multiradial theta-image union before being identified with the package target union. For arbitrary possible-image families that are later identified with the Theorem 3.11 record, the constructor now has the same source shape: start from q-pilot containment in the displayed possible-image union and derive containment in its canonical hull internally. The record-native specialization now has the same $q \subseteq P_{\mathrm{rec}}$ input form: the hull containment needed by Step (xi) is derived from the record's own possible-image union. A record-native specialization removes the arbitrary possible-image-family input: the family is $i \mapsto P_i$ from the Theorem 3.11 source record, and Lean proves that its set-theoretic union is the record's theta-image union. Below the packet layer, the code now has a first realified Frobenioid degree source. A finite Frobenioid object carries a divisor degree, a unit log-volume, and their realified log-volume. A finite divisor-product source now records a tensor-packet product over base valuations whose packet factors are the real readings of the corresponding divisor-prime contributions plus base-local unit contributions; when the unit terms sum to the divisor source's unit log-volume, Lean derives that the packet product log-volume is the realified divisor log-volume. A finite divisor-monoid source computes this divisor degree from prime multiplicities and prime degrees, matching the finite shadow of the divisor monoid in IUT I, Example 3.5. The same source now has a divisor-level tensor product that adds multiplicities and realified log-volumes when the prime-degree data agree; its projection to the degree-object layer agrees with the existing degree-object tensor product. It also has the Ob4 naive Frobenius tensor-power operation: multiplying the divisor multiplicities and unit log-volume by M constructs a new finite Frobenioid object whose divisor degree and realified log-volume are proved to be M times the original values. The older realified degree-object layer now has the same Ob4 operation, and the projection from the finite divisor-monoid source to that degree-object layer is proved to commute with tensor-power formation. The theta-version of IUT I, Example 3.5(ii), is now represented at this finite level: the same divisor degree is read through a formal $\log(\Theta)$-generator, and normalization of this generator to 1 recovers the ordinary finite divisor source. The Ob4 tensor-power operation also lifts to this theta copy, where Lean proves that the theta divisor log-volume is multiplied by M. The D-, or log-shell, version of Example 3.5(iii) is represented similarly: a distinguished local Frobenius/log-shell reading is divided by the extension degree before it is applied to the finite divisor degree. Tensor-powering this copy leaves the normalized Frobenius reading fixed and multiplies the resulting log-shell divisor log-volume by M. The three finite copies now have a compatibility package: if they share the same base divisor source and the theta/log-shell readings are normalized, then their log-volume readings are equal. This compatibility package is now stable under simultaneous Ob4 tensor-powering: the ordinary, theta, and log-shell copies are tensor-powered over the same object, the base identifications and normalizations are preserved, and all three finite log-volume readings are multiplied by M. The environment/Gaussian realified Frobenioid comparison of IUT II, Corollary 4.6(v), is represented at the same finite level by an evaluation package between two such compatible copy systems. The package stores only the ordinary divisor-source identification; Lean derives equality of the ordinary, theta, and log-shell log-volume readings across the environment and Gaussian sides from that identification and the copy-compatibility lemmas. The comparison is also stable under simultaneous Ob4 tensor-powering: the ordinary-source equality is preserved, the theta/log-shell comparison is rederived on the tensor-powered evaluation, and the two ordinary realified log-volume readings are multiplied by M. The same comparison is now connected to the finite local-global restriction layer, reflecting the F-prime-strip formulation following Corollary 4.6(v): if the environment and Gaussian global objects are the degree projections of their ordinary divisor sources and their local restrictions agree pointwise, then Lean derives equality of the global realified log-volumes and of all local restricted log-volume readings. The prime-index part of the same F-prime-strip statement is now explicit: finite prime-label types for C_{env} and C_{gau} come with bijections to the common valuation set V, and the evaluation map between prime labels is required to commute with these bijections. Lean then transports the local compatibility theorem along this prime-place square, deriving equality of restricted and local realified log-volumes at evaluated primes. The next functorial step toward the $\Theta^{\times\mu}$- and $\Theta^{\times\mu}_{\text{gau}}$-links of Corollary 4.7(iii) is also represented at finite level. A lift to an $F^{\times\mu}$-prime-strip stores an environment-side unit character; the Gaussian-side unit character is defined by transport along the prime-evaluation equivalence. Thus Lean proves unit-character compatibility at evaluated primes rather than accepting it as an independent bridge field. The coric $F^{\vdash\times\mu}$ invariance of Corollary 4.7(iv) is represented by a common unit character on the valuation set V. Once the environment unit character is the pullback of this coric character along $Prime(C_{\text{env}}) \simeq V$, Lean derives the corresponding Gaussian equality from the prime-evaluation square. The coric global realified Frobenioid compatibility of Corollary 4.7(v) is represented by a positive real scale on finite local-global Frobenioid collections. This is the finite shadow of the $R_{>0}$-orbit language: given scaled equality of the global realified log-volume and of the restricted local prime readings, Lean derives the scaled local-object and local-prime log-volume equalities from the existing local-global collection laws. The Frobenius-picture graph of Corollary 4.7(vi) is represented by the directed successor relation $n \to n+1$ on $\mathbb{Z}$. Lean proves that translations preserve this oriented graph and that any map exchanging the adjacent vertices 0 and 1 cannot preserve directed edges. This captures the finite order-theoretic content of the paper's statement that the chain has translation symmetry but not arbitrary permutation symmetry. Proposition 3.10(ii)'s column log-Kummer correspondence is now represented at the same finite level. For each vertical coordinate $m \in \mathbb{Z}$, the source carries a packet of group elements acting on one common IQ-carrier labelled by $n, \circ$. Translation in the column transports group elements, and Lean proves that the transported element acts by the same endomorphism on the common carrier. This records the translation-invariance of the log-Kummer action; it does not yet construct the underlying number fields. The realified-Frobenioid part of Proposition 3.10(iii) is represented at degree level by a family of realified Frobenioid objects indexed by $m \in \mathbb{Z}$. The log-link translation law preserves the extracted realified log-volume, including adjacent

$m \mapsto m \pm 1$ shifts. The associated etale-picture core of Corollary 4.11(i),(ii) is represented by a common $D^{\vdash}$-core projection for all integer-labelled spokes. Lean proves that this projection is constant across successor steps and is unchanged by arbitrary permutations of the integer-labelled spokes, capturing the finite mono-analytic core/permutation-symmetry contrast with the Frobenius-picture chain. The value-group restriction discussed in Remark 4.10.3 is exposed as a combined finite-label endpoint: for a Gaussian degree evaluation, the canonical full label $j = 1$ has degree equal to the environment q-degree, the canonical square weight at $j = 1$ is 1, and the singleton restriction to $j = 1$ is not stable under translation by 1. Thus the identity/pivotal specialization and the failure of compatibility with the full label symmetry are checked in the same typed statement. A finite tensor-product operation on these degree objects adds divisor degrees and unit log-volumes, and Lean proves that the realified log-volume of the tensor product is the sum of the two realified log-volumes. This operation is also lifted to the tensor-packet source layer: if an existing tensor-packet product source is certified to carry the tensor-product Frobenioid degree, Lean derives additivity of its packet product log-volume. A Frobenioid Kummer-compatibility certificate then projects to the coric tensor-packet transfer by deriving equality of product log-volumes from equality of realified Frobenioid log-volumes, while retaining the Hodge-theater history guard. This is still a finite degree-level shadow, not a construction of Mochizuki's Frobenioids, but it moves one equality source below the packet-correspondence interface. The same degree layer now has finite Frobenioid morphisms. A morphism records the divisor-degree shift and unit-log-volume shift between two realified objects; Lean derives the corresponding shift in realified log-volume and proves identity and composition laws at this degree/log-volume level. This models the additive content of the pull-back and structure morphisms appearing in Remark 3.9.5(ix), (cQ3). These morphisms now transport across Ob4 tensor-powering: the source and target degree objects are tensor-powered, and the divisor and unit shifts of the induced morphism are multiplied by M. The Ob8/Ob9 vertical-shift issue is now attached directly to the Step (xi-g) handoff. A log-Kummer vertical-shift calibration pairs the Step (x)-to-hull upper-ray source with the global realified Frobenioid calibration. Lean proves that the calibrated value is the hull boundary used by the upper ray, that both q-pilot computations are bounded by this calibrated value, and that a shifted local value coincides with the calibrated value only when the shift term vanishes. Remark 3.9.5(ix)'s distinction between (cQ3),(cQ4) and (iQ5) is now represented by a finite closed-loop guard. The guard ties the hull/determinant log-volume obtained from the (sQ3) route and the calibrated log-Kummer log-volume obtained from the (sQ4) route to the same Step (xi) boundary. Lean then proves that the two values coincide, while the direct shifted local log-volume agrees with this boundary exactly in the zero-shift case. This does not construct $F^{\times\mu}$-prime-strips; it records the present formal boundary between the source-faithful route and the log-volume-only quotient. The global-to-local realified restriction from IUT I, Example 3.5 is also encoded at degree level: the restricted global prime log-volume is $[K_v : (F_{\mathrm{mod}})_v]^{-1}$ times the local prime log-volume, and Lean proves that multiplying by the extension degree recovers the local prime log-volume. The same restricted value now normalizes the local p_v^N-shift source: the local shift is expressed with this restricted global prime step, and the automorphism-or-zero-step criterion is restated in terms of the restricted realified prime log-volume. Remark 3.9.5(ix)'s local-global collection language is represented at the same finite degree level. A collection contains a local realified Frobenioid object for each v, one global realified Frobenioid object, and localization restriction factors. Lean proves that each local log-volume is the restricted global value and that multiplying by the corresponding extension degree recovers the global log-volume. This is a finite log-volume shadow of the local-global category, not a construction of the category itself. The collection and its restriction factors now transport coherently under Ob4 tensor-powering: local objects, the global object, restricted prime log-volumes, and local prime log-volumes are all multiplied by M, while extension-degree rescaling remains valid. This local-global collection now also backs the finite cQ3/cQ4 closed-loop guard: if the collection's global realified log-volume is the cQ3 hull/determinant boundary, then Lean proves that every localized value rescales to both the cQ3 boundary and the cQ4 calibrated log-Kummer boundary. Thus the Step (xi) boundary can be read from the local-global collection rather than from a bare real number alone. The local-global collection has also been refined by structure morphisms in the finite Frobenioid degree category. For each v, a degree morphism from the local object to the global object records divisor and unit shifts. Lean proves that the resulting morphism shift is exactly the additive form of the localization rescaling. This models the degree/log-volume content of the "structure poly-morphism" language in Remark 3.9.5(ix), (cQ3). The structure-morphism collection now transports through Ob4 tensor-powering: the underlying local-global collection is tensor-powered, and every structure morphism acquires divisor and unit shifts multiplied by M. The same finite category now has compatible morphisms between two such local-global structure collections. A compatible morphism consists of local morphisms at each v, one global morphism, and a commutativity condition: the local-then-structure path and the structure-then-global path have the same divisor and unit shifts. Lean derives equality of the total realified log-volume shift along these two paths. The same compatible morphism now transports through Ob4 tensor-powering: the global morphism, all local morphisms, and the two paths in the compatibility square acquire shifts multiplied by M, and the square remains commutative. IUT I, Example 5.1 and Remark 5.1.1 distinguish the diagonal Frobenioid $F_{\langle J \rangle}$ from the product of underlying categories F_J: the former carries constant distributions in j, while the latter allows arbitrary distributions. The code now models this finite realified log-volume distinction. A diagonal distribution has total log-volume $|J|$ times its common value, with the realified restriction factor propagated through this sum; a distribution with two unequal coordinates is proved not to come from a diagonal constant distribution. IUT I, Example 5.4(vii) and Remark 5.4.2 are represented by a finite pivotal distribution object: a chosen pivotal index supplies the common diagonal log-volume, the summed interpretation gives total $|J|$ times this pivotal value, and the averaged interpretation recovers the pivotal value itself. IUT I, Definition 5.5 is now represented at finite log-volume level by an NF-bridge mode object: the bridge target is either the summed capsule log-volume or the normalized averaged capsule log-volume. For the same capsule distribution, Lean proves that the summed bridge target is $|J|$ times the averaged bridge target; for a pivotal distribution, these specialize to $|J|$ times the pivotal value and to the pivotal value itself. The adjacent Θ-bridge and ΘNF-gluing condition of Definition 5.5(iii) are formalized at the same finite level: the Θ side reads the common capsule total, and a compatibility record requires the NF- and Θ-bridge morphisms to use the same index bijection and the same capsule distribution. Lean then proves that a summed NF target equals the Θ-target, while an averaged NF target becomes the Θ-target after multiplication by $|J|$. Corollary 5.6(iii) is represented by a finite gluing torsor: once compatible NF and Θ bridge halves have been fixed, the gluing parameter is modeled by the additive F_ℓ-torsor on $\mathbb{Z}/\ell\mathbb{Z}$. Lean proves zero action, additivity of the action, existence and uniqueness of the translation between two gluing parameters, and preservation of the shared index-bijection and capsule data. This layer is now connected to the nonarchimedean packet correspondence: a realified-Frobenioid log-Kummer packet source projects the Frobenioid compatibility data to the Kummer $F \to D$ transfer and to the mono-analytic forgetting transfer, and then feeds the existing packet correspondence endpoint. Thus the Step (x) source equality used by the route can be obtained from realified Frobenioid log-volume compatibility plus the remaining packet source/target calibrations. IUT III, Remark 3.11.4 further says that the root-of-unity and cyclotomic indeterminacies arising from forgetting Frobenius-like labels are invisible for the objects under consideration. The present code formalizes the finite sign-unit part of this assertion: a unit in the subgroup $\{1, -1\} \subset (\mathbb{Z}/\ell\mathbb{Z})^\times$ preserves the zero/nonzero full cusp label, every full-label log-volume expression, and the averaged Gaussian log-volume. This is not yet a construction of the full cyclotomic rigidity isomorphism;

it is the finite-label component that the current Stage 1 interface can prove without an additional Frobenioid/cyclotome layer. The same finite-label layer now also proves the corresponding exclusion: an affine coordinate change $j \mapsto t + aj$ preserves the full cusp label if and only if $t = 0$ and a is represented by such a sign-unit invisibility certificate. For nonzero Gaussian environment degree, the same classification is equivalent to pointwise preservation of the Gaussian degree profile. Thus the current code has a finite analogue of the "no further indeterminacies" assertion of Remark 3.11.4(ii). It also separates this cyclotomic invisibility from the stronger square-weight preservation used by the factored SHE layer: a root-unit invisibility action always preserves the full-label map, but it preserves the raw representative-square profile, equivalently the combined square/full-label preservation package, if and only if the root unit is the identity. The integrated route now composes the Hodge–Arakelov theta-value sources with this packet-target nonarchimedean source. The resulting theorem derives the same boundary/strict C_Θ dichotomy from source-facing Gaussian evaluation data and source-facing upper-semi data, rather than from two unrelated collections of raw equalities. For the target side, the strongest route now uses a target-only Step (x) alignment

$$\Theta_{\mathrm{avg}} = U_{\mathrm{Ind3}}$$

together with calibration of the local entry target against U_{Ind3}. The direct equality $\Theta_{\mathrm{avg}} = \mathrm{entry}_{\mathrm{target}}$ is then derived. When the theta-source average is identified with the packet-normalized capsule log-volume, this target alignment is itself derived from calibration of U_{Ind3} with the same packet-normalized value. The local entry target may now also be calibrated against this packet-normalized value, so the equality $\mathrm{entry}_{\mathrm{target}} = U_{\mathrm{Ind3}}$ is derived rather than supplied directly in the strongest route.

5.4 Step (xi-f) algebra

The final real algebra is fully formalized. In informal notation, the main theorem is:

Theorem 1 (Formal Step (xi-f))**.** *Assume $q < 0$, $q \le \theta$, and $\theta \le C(-q)$. Then $C \ge -1$. If $q < \theta$, then $C > -1$. If $C = -1$, then $q = \theta$ and $\theta < 0$.*

This theorem is instantiated by the Step (x)-to-Step (xi) route whenever the hull/log-volume upper-ray comparison and the displayed C_Θ-bound are available. It is also instantiated by the current nonarchimedean (Ind3)-plus-factored-SHE route, subject to the same positivity and displayed-bound hypotheses. Thus the remaining gap is not the final ordered-real implication, including its strict and boundary branches, but the construction of the source hypotheses consumed by that implication.

6 Boundary Quotient Experiment

The most developed diagnostic concerns Step (xii), where local Frobenioid p_v^N-ambiguity is compared with the globally calibrated log-volume. The formal model records a local shift

$$L_N = L_0 + N\delta, \qquad N \in \mathbb{Z},\ \delta \in \mathbb{R},$$

and a global calibration that fixes exponent 0. Lean proves

$$L_N = L_0 \iff N\delta = 0,$$

and, if $\delta \neq 0$,

$$L_N = L_M \iff N = M.$$

At the boundary branch $C_\Theta = -1$, the formal corridor identifies

$$q_{\text{pilot}} = \Theta_{\text{hull}} = L_0.$$

The local shift model is now source-facing: a local Frobenioid log-shell submodule carries a p_v^N-action whose log-volume effect is $L_0 \mapsto L_0 + N\delta$. The local action records whether the p_v^N-endomorphism is an automorphism, and in the current model this is equivalent to $N = 0$. The global realified Frobenioid calibration package selects exponent 0, hence its calibrated value is L_0. If the local endomorphism is not an automorphism and $\delta \neq 0$, Lean proves that the local shifted value cannot equal the globally calibrated value. The same calibration can now be built from the restriction-normalized local source, so the local step is the restricted global prime log-volume rather than an independent real parameter. For the actual upper-ray quotient that collapses $\{x \mid x \leq \Theta_{\text{hull}}\}$, Lean proves the exact classification

$$[L_N] = [q_{\text{pilot}}] \quad \Longleftrightarrow \quad N\delta \leq 0,$$

and the complementary separation criterion

$$[L_N] \neq [q_{\text{pilot}}] \quad \Longleftrightarrow \quad N\delta > 0.$$

If $\delta > 0$, this becomes:

$$[L_N] = [q_{\text{pilot}}] \Longleftrightarrow N \leq 0, \qquad [L_N] \neq [q_{\text{pilot}}] \Longleftrightarrow 0 < N.$$

This is not a proof for either side of the Mochizuki–Scholze–Stix dispute. It is a precise reading of one simplified quotient mechanism inside the current formal corridor. It shows that quotient collapse can identify unequal underlying log-volumes if they lie in the collapsed upper ray, while positive local shifts remain separated. The decisive question is therefore not the ordered-real algebra; it is whether the source IUT constructions actually supply the required calibrated hull/log-volume comparison without making an illicit identification of histories or comparison levels.

7 Current Results

The project currently proves the following source-facing results in Lean:

1. The finite q^{j^2} degree rule, sign quotient, half-range representatives, zero/nonzero split, and average formulas are formalized.
2. The bad-local theta-root quotient-$\mathbb{Z}$ package now feeds the Stage 1 $|F_\ell|$ full-label model through its canonical coordinate; the concrete $Z/\ell Z$ case yields the canonical nonzero full label.
3. A single label-independent scalar cannot absorb all representative j^2-degrees when $\deg(q) \neq 0$.
4. The raw representative branch does not descend to the sign quotient, while the balanced full-label Gaussian branch does.
5. The Step (x) averaged-log-volume corridor feeds a Step (xi) upper-ray comparison once determinant/hull identifications are supplied.
6. A nonarchimedean (Ind3) entry with factored square/full-label preservation now directly supplies the final weighted-theta route and the signed $C_\Theta \geq -1$ endpoint, once q-positivity and the displayed C_Θ-bound are supplied.

7. The same nonarchimedean route now exposes the strict branch $C_\Theta > -1$ and the boundary consequence $C_\Theta = -1 \Rightarrow q_{\mathrm{signed}} = \Theta_{\mathrm{signed}} < 0$. It also packages these as one dichotomy at the route level.

8. Gaussian degree evaluations with preserved environment degree now feed directly into this nonarchimedean C_Θ dichotomy, constructing the factored square/full-label SHE package internally.

9. In the identity-coordinate Gaussian route, equality at the canonical full label 1 supplies the environment-degree preservation needed by the same nonarchimedean C_Θ dichotomy.

10. The Hodge–Arakelov theta-value evaluation source now constructs the Gaussian degree evaluation from a theta-monoid degree and the bad-local theta-root label source; canonical-label preservation feeds the factored SHE square/full-label obligations.

11. The same finite Hodge–Arakelov layer now has an audited theta evaluation wrapper matching the current IUT II shadow: it keeps the theta-root source, bad-local quotient identification, Gaussian degree evaluation, full-label compatibility, canonical $j = 1$ behavior, and the canonical square-weight profile as one source-facing object. A pair of these objects constructs the SHE synchronization source and can also construct the finite Hodge/SHE transport source once the allowed-forgetting witness is given.

12. A nonarchimedean upper-semi object equipped with 1-column q-pilot log-Kummer data now supplies the raw nonarchimedean (Ind3) entry alignment used by the canonical Gaussian route.

13. The source-side equality in that compatibility can be derived from product-log-volume preservation along the finite Kummer-plus-forgetting tensor-packet chain.

14. A theta-root attached Kummer/forgetting source package now combines the canonical bad-local $|F_\ell|$-label source with a source-side packet/product calibration object and that finite chain, deriving the nonarchimedean source equality.

15. Source-side and target-side packet calibrations are now joined by one nonarchimedean log-Kummer packet-correspondence source carrying the q-pilot non-interference guard. Its projections derive the theta-to-packet, theta-to-entry, entry-to-(Ind3)-target, and packet-normalized target equalities used by the route; the target calibration now refers directly to the packet-normalized capsule value rather than to a separate real representative.

16. The strongest Gaussian first-pass route can now consume this single packet-correspondence source and project both its source and target faces, instead of taking source packet equalities and target calibration as separate route inputs.

17. A realified-Frobenioid log-Kummer entry source now bundles the selected upper-semi inclusion with the realified packet source; the strongest Step (x) route projects Kummer transfer, mono-analytic forgetting, source/target packet calibration, and q-pilot non-interference from this single source.

18. A theta-root attached realified entry source now combines the canonical bad-local theta-root label package with that realified Step (x) source, and the Hodge–Arakelov route can consume this attached source directly.

19. The Hodge/SHE/IPL/hull route now also has a realified endpoint: route alignment supplies the Hodge–Arakelov and q-pilot data, and Step (x) enters through the theta-root attached realified source.

20. Remark 3.9.5(iii)'s hull-approximant family is now represented directly: an approximant lies inside the canonical hull $\phi(P)$, has log-volume between P and $\phi(P)$, and is fixed by hull formation.

21. The bounded-family upper-semi quotient of Remark 3.9.5(v)–(vi) is represented: each possible region lies in the hull of the family union, and any two nonempty possible regions have identical images in the quotient that collapses this family hull.

22. The Ob5 determinant/log-volume compatibility of this family quotient is represented at finite level: the same $\phi(P_B)$ quotient source is paired with a weighted determinant certificate, and Lean proves that the determinant log-volume and normalized tensor-power log-volume are the family-hull log-volume.

23. This Ob5 compatibility is specialized to the Corollary 3.12 possible theta-image family: the upper-ray source hull and the family quotient determinant/log-volume source share the same $\phi(P_B)$.

24. The canonical possible-image Ob5 endpoint now returns the family-union-to-determinant estimate, not only the quotient collapse and determinant equality.

25. The Corollary 3.12 possible-image approximant source now projects to this bounded-family quotient source, yielding the same quotient-image equality for nonempty possible theta-image choices.

26. In that same source, q-pilot containment in the chosen approximant implies q-pilot containment in the family hull; if the q-pilot region is nonempty, its quotient image agrees with the quotient image of any nonempty possible theta-image choice.

27. Such a certified hull-approximant now also constructs the Step (xi) upper-ray datum: if the q-pilot region is contained in the approximant and the approximant log-volume is determinant-normalized, then Lean derives the q-pilot bound by the approximant, by the canonical hull, and by the determinant log-volume.

28. The approximant route is now specialized to the Corollary 3.12 possible-image union. A single source object binds the possible theta-image family, its union, the chosen hull-approximant, q-pilot containment in that approximant, and determinant normalization; the experiment endpoint returns the union/approximant/hull and q-pilot/determinant inequalities.

29. The determinant input can now be produced from a finite weighted localization source: adjusted vector-bundle summand log-volumes are summed, the positive tensor-power value is derived, and normalization is proved to return the determinant log-volume before projecting to the determinant interface.

30. Each localization summand can now itself be produced from a finite direct-summand source: the bundle log-volume is the sum of direct-summand log-volumes, then the structure-sheaf adjustment and positive weight produce the adjusted localization contribution.

31. The Ob3-1-2 structure-sheaf adjusted source makes the $\mathcal{O}(-)$ subtraction explicit before projection to the localization summand. A corresponding determinant constructor proves that summing these weighted adjusted sources gives the determinant log-volume used by the route.

32. The possible-image hull-approximant endpoint now has a weighted determinant-source constructor. The route consumes the weighted summand source and projects to the common upper-ray interface only after a named Ob3-3 compatibility certificate identifies the approximant log-volume with the weighted normalized determinant value.

33. The weighted determinant route now has a canonical-hull specialization: the approximant is constructed as $\phi(P)$, q-pilot containment is stated directly in this hull, and the approximant-equals-hull proof is derived by Lean rather than supplied as a separate route input.

34. The Ob4 tensor-power passage is now represented at log-volume level: M-th tensor-power formation multiplies by $M > 0$, and normalization recovers the base log-volume. The weighted determinant source projects to this tensor-power object, and two tensor-power presentations with the same base log-volume have equal normalized log-volume. The new `IUTStage1TensorPowerTwistFamily` extends this from a pairwise comparison to a family of Ob4-3 tensor-power-twist reconstructions: all twists share the reference normalized log-volume, and normalized log-volume bounds are invariant under replacing one twist by another.

35. The bounded-family hull/determinant source now proves the direct estimate from possible-image union to determinant log-volume: $\mu_{\log}(\bigcup_\beta P_\beta) \leq \det_{\log}$.

36. The Corollary 3.12 possible-image approximant source now exposes the same possible-image-union-to-determinant estimate directly.

37. The all-in-one target-charted Hodge/IPL determinant possible-image route now has a named Step (xi) comparison-chain audit, `targetChartedHodgeIPLDeterminantPossibleImageRoute_stepXIComparisonChainAudit`. This audit derives, at the route boundary, the chain

$$\mu_{\log}(q_{\text{pilot}}) \leq \mu_{\log}(P_B) \leq \mu_{\log}(\phi(P_B)) = \det_{\text{norm}} = \det_{\log} \leq \theta_{\text{signed}},$$

together with $q_{\text{signed}} \leq \theta_{\text{signed}}$. Thus the public route no longer exposes only the final signed comparison; it also records the possible-image containment and determinant-normalization chain that produces it.

38. This comparison-chain audit is now threaded through the finite-divisor vertical-IQ C_Θ endpoint as `targetChartedHodgeIPLDeterminantPossibleImageRouteFiniteDivisorVerticalIQ_stepXIComp` The endpoint returns the same Step (xi) log-volume chain next to the conditional Corollary 3.12 dichotomy, so the finite Step (x) boundary can be audited through the constructed possible-image/determinant comparison rather than only through a coarse Gaussian-to-Step (xi) summary.

39. The package-level hull/determinant bridge can now be built from the Ob4-normalized tensor-power bound associated to a weighted determinant source. The conversion to the determinant bound is a Lean theorem, not a separate bridge assumption.

40. A first realified Frobenioid degree layer is present: equality of realified Frobenioid log-volumes now derives the product-log-volume equality used by the tensor-packet coric transfer, with theater histories still kept distinct. The degree object also has a finite tensor-product operation:

divisor degree, unit log-volume, and realified log-volume are proved to add. A tensor-packet source certified by this Frobenioid product inherits product-log-volume additivity.

41. The realified Frobenioid degree object can now be constructed from a finite divisor-monoid source: Lean computes the divisor degree as $\sum_p m_p \deg(p)$ before adding the unit log-volume.

42. A finite divisor-product source now turns a base-valuation tensor-packet product into a realified Frobenioid packet source when every packet factor is the real reading of its divisor-prime contribution plus a base-local unit contribution, and the unit contributions sum to the divisor source's unit log-volume. The equality between packet product log-volume and realified Frobenioid log-volume is derived from the finite sum, not supplied as a separate field.

43. The Step (x) packet-local source alignment can now consume this finite divisor-product source: after identifying the packet-local object with the local entry source and the divisor realified log-volume, Lean derives the entry-source/mono-analytic-product equality used by the log-Kummer source calibration.

44. The procession-normalized Step (x) average now has an $\mathbb{F}_\ell$-labelled source object: Lean computes the total as $\sum_{j \in ZMod(\ell)} v_j$, the normalized value as $\ell^{-1} \sum_j v_j$, and proves this average is unchanged by translating all labels.

45. The corresponding finite-label indeterminacy endpoint proves that two successive label translations preserve the normalized average by finite reindexing; an upper bound on the twice-translated summands then yields the (Ind3) upper bound for the original average.

46. A target-side Step (x) calibration constructor now derives $\theta_{\text{avg}} = \text{packetNormalized}$ from the $\mathbb{F}_\ell$-labelled procession average, instead of taking this equality as an unclassified direct input.

47. The realified Frobenioid packet source has the same labelled-average constructor: Kummer/forgetting compatibility, source calibration, and labelled Step (x) average construct the full packet source and expose the ordinary log-Kummer packet correspondence endpoint.

48. The labelled-average constructor now reaches the theta-root attached Step (x) entry source and the Hodge/SHE/IPL/hull route. The route derives the packet-normalized theta calibration by composing theta-source-to-labelled-average with labelled-average-to-packet-normalized equality.

49. The labelled-average Hodge/SHE/IPL/hull route now has a source-aligned form: packet-local source alignment and Kummer/forgetting transfers construct the source calibration internally.

50. The theta-source-to-labelled-average equality can now be derived from labelwise agreement between the capsule-labelled source and the Hodge-descent theta-source $ZMod(\ell)$-average.

51. The labelled-average-to-packet-normalized equality can now be derived from labelled capsule local-object equality plus direct packet-normalization; the integrated route exposes this stricter source boundary.

52. The direct-labelled route now has an exact vertical-IQ target form: the target-side (Ind3) equality is derived through the common packet-normalized value, not passed as ordered-real alignment data.

53. The direct-labelled vertical-IQ route now has a finite-divisor form: source packet-local alignment is constructed from finite divisor/Frobenioid product data rather than supplied as a separate alignment record.

54. The Hodge direct-labelled finite-divisor route can now consume the all-in-one target-charted Hodge/IPL/determinant possible-image route source directly: the source-derived Hodge/SHE/IPL/hull bridge is projected from that route source before the labelled Step (x) boundary is entered.

55. The direct-labelled and finite-divisor routes no longer require labelwise equality with the theta-source $ZMod(\ell)$-average as route input; it is derived from Hodge-descent label-local-object transport.

56. The theta-root attached Step (x) source can now be constructed from the same finite divisor-product source and an exact vertical-IQ target. The endpoint exposes the canonical bad-local theta-root label, Kummer/forgetting product equalities, finite divisor product equality, target packet normalization, q-pilot non-interference, and the upper-semi bound.

57. The finite divisor-monoid source now has its own tensor product: multiplicities and unit log-volumes add, and Lean proves additivity of the induced divisor degree and realified log-volume.

58. The same finite divisor-monoid source now has an Ob4-style naive Frobenius tensor-power operation: multiplicities and unit log-volume are multiplied by M, and Lean proves the induced divisor degree and realified log-volume are multiplied by M.

59. The older realified degree-object layer now has the same Ob4 tensor-power operation, and the projection from a finite divisor-monoid source to this layer commutes with tensor-power formation.

60. The projection of this divisor-level tensor product to the realified degree-object layer agrees with the existing degree-object tensor product, preventing these two Frobenioid shadows from drifting apart.

61. The theta-version of the finite divisor source is now represented: a formal $\log(\Theta)$-generator scales the divisor degree, normalization to 1 recovers the ordinary source, and tensor products preserve additivity when the theta generators match. The Ob4 tensor-power operation also lifts to this copy, multiplying the theta divisor log-volume by M.

62. The D/log-shell version is also represented at finite level: Lean normalizes a distinguished local Frobenius log-volume by the extension degree, recovers the ordinary source when this normalized value is 1, and proves tensor-product additivity when normalized values match. The Ob4 tensor-power operation preserves the normalized Frobenius reading and multiplies the log-shell divisor log-volume by M.

63. The ordinary, theta, and log-shell divisor copies now have a compatibility package: shared base divisor data plus normalized readings imply equality of all three finite log-volume readings. Simultaneous Ob4 tensor-powering of the three copies preserves this compatibility package and multiplies all three finite log-volume readings by M.

64. The finite environment/Gaussian realified Frobenioid comparison attached to IUT II, Corollary 4.6(v), is now encoded as an evaluation package between two compatible copy systems. The only stored bridge is equality of the ordinary divisor source; Lean derives equality of the

ordinary, theta, and log-shell readings across the two sides. Simultaneous Ob4 tensor-powering preserves the stored ordinary-source equality, rederives the theta/log-shell comparison, and multiplies the two ordinary realified log-volume readings by M.

65. This evaluation package is now tied to the existing finite local-global restriction layer: if the two global objects are the degree projections of the ordinary environment/Gaussian divisor sources and the local restrictions agree at every v, then Lean derives equality of the global realified log-volume and of the pointwise local restricted log-volume readings.

66. The prime-index skeleton of the corresponding F-prime-strip evaluation is now explicit: finite prime-label types for C_{env} and C_{gau} carry bijections to V, and the evaluation equivalence between prime labels must commute with these bijections. Lean transports the local compatibility theorem along this square to obtain equality at evaluated primes.

67. The finite $F^{\times\mu}$-prime-strip lift used by the $\Theta^{\times\mu}$- and $\Theta^{\times\mu}_{\mathrm{gau}}$-links is now represented by transport of a unit character along the prime-evaluation equivalence. The Gaussian unit character is constructed from the environment unit character, and Lean proves compatibility at evaluated primes.

68. The coric $F^{\vdash\times\mu}$ invariant of Corollary 4.7(iv) is now represented at finite level: a common unit character on V pulls back to the environment side, and Lean derives the Gaussian-side pullback equality from the prime-evaluation square.

69. The coric global realified Frobenioid compatibility of Corollary 4.7(v) is now represented by a positive real scale on finite local-global Frobenioid collections. From scaled global and restricted-local readings, Lean derives the scaled local-object and local-prime log-volume readings.

70. The Frobenius-picture graph of Corollary 4.7(vi) is now represented as the directed successor graph on $\mathbb{Z}$. Lean proves translation invariance and rules out edge preservation by any map that swaps adjacent vertices 0 and 1.

71. The etale-picture $D^{\vdash}$-core of Corollary 4.11(i),(ii) is now represented as a common core projection for all integer-labelled spokes. Lean proves successor-step constancy and invariance under arbitrary permutations of the spoke labels.

72. The value-group restriction of Remark 4.10.3 is now a single endpoint: the Gaussian degree at $j = 1$ equals the environment q-degree, the square weight at $j = 1$ is 1, and the singleton-1 restriction is not translation-invariant.

73. The realified Frobenioid degree layer now has finite morphisms: divisor and unit shifts determine the target realified log-volume, identity has zero shift, and composition adds both shifts. These morphisms are stable under Ob4 tensor-powering, with divisor and unit shifts multiplied by M.

74. The Ob8/Ob9 vertical-shift guard is now tied to the Step (xi-g) two-computation handoff: global realified Frobenioid calibration selects the hull upper-ray boundary, both q-pilot readings are bounded by it, and shifted local coincidence is equivalent to a zero shift term.

75. The cQ3/cQ4 portion of Remark 3.9.5(ix) now has a finite closed-loop guard: the hull/determinant value and the calibrated log-Kummer value are proved equal as the same Step (xi) boundary, while the direct log-volume-only shifted value agrees with that boundary only under the zero-shift condition.

76. The global-to-local realified Frobenioid restriction factor of IUT I, Example 3.5 is formalized as inverse-extension-degree scaling of the local prime log-volume, with the degree-normalization identity proved in Lean. This restricted value is connected to the local p_v^N-shift source used in the Step (xii) diagnostic, and global calibration can now be constructed from this restriction-normalized source.

77. A finite local-global realified Frobenioid collection now models the degree/log-volume content of Remark 3.9.5(ix): each local object is a restricted global value, and extension-degree rescaling recovers the global realified log-volume. The collection and its restriction factors are stable under Ob4 tensor-powering, with the realified readings multiplied by M.

78. The cQ3/cQ4 closed-loop guard can now be backed by such a local-global collection: the global collection value is identified with the cQ3 hull/determinant boundary, and every localized value rescales to both the cQ3 and cQ4 Step (xi) boundaries.

79. The local-global collection now carries finite structure morphisms from local objects to the global determinant object. The morphism divisor/unit shift is proved equal to the additive localization rescaling term. The structure-morphism collection is stable under Ob4 tensor-powering, with every structure-morphism shift multiplied by M.

80. Compatible morphisms between local-global structure collections are now represented: local-then-structure and structure-then-global paths have equal divisor/unit shifts and hence equal total realified log-volume shift. This compatibility is preserved by Ob4 tensor-powering of the source and target local-global structure collections.

81. The diagonal/arbitrary distribution distinction of IUT I, Example 5.1 and Remark 5.1.1 is formalized at finite realified log-volume level: diagonal distributions are constant in j, their total is $|J|$ times the common restricted value, and a two-coordinate mismatch cannot be represented as diagonal.

82. The pivotal distribution of IUT I, Example 5.4(vii), is modeled by a chosen index in a diagonal distribution; Lean proves the summed log-volume formula and the normalized averaged formula of Remark 5.4.2.

83. The finite NF-bridge log-volume model of IUT I, Definition 5.5, separates summed and averaged bridge targets and proves their $|J|$ scaling relation for a common capsule distribution.

84. The finite ΘNF-bridge compatibility model records the same-index-bijection and same-capsule requirements of Definition 5.5(iii), and proves the corresponding Θ-target equalities for summed and averaged NF modes.

85. The finite gluing set of Corollary 5.6(iii) is modeled as the additive F_ℓ-torsor on $\mathbb{Z}/\ell\mathbb{Z}$, with compatibility data preserved under gluing translation.

86. The realified Frobenioid layer now feeds the nonarchimedean log-Kummer packet source: Kummer and mono-analytic forgetting transfers are projected from Frobenioid compatibility before the packet correspondence endpoint is applied.

87. A packet-normalized Step (x) entry-target source now constructs the target side of a nonarchimedean upper-semi entry from the capsule-family normalized log-volume. The target-calibration constructor derived from this source removes the separate entry-target-to-packet-normalized equality.

88. The same target-calibration layer can now derive the (Ind3)-target-to-packet-normalized equality from theta-target alignment and theta-average packet normalization, instead of accepting that equality directly.

89. A vertical-IQ target source now models the exact common-target side of the log-Kummer correspondence in Proposition 3.10(ii). It derives the theta-average, (Ind3)-target, and entry-target packet-normalized equalities used by the packet target calibration.

90. The vertical-IQ source carries an $FMOD$-exactness guard, reflecting Remark 3.10.1's distinction between exact $FMOD$ Frobenioid compatibility and one-sided $Fmod$ upper semi-compatibility.

91. The exact vertical-IQ realified entry source now enforces that the realified packet target calibration is the calibration induced by the retained exact vertical-IQ target.

92. The canonical exact Step (x) experiment route now constructs this refined source internally from theta-root data, realified Frobenioid transfer data, packet-local source alignment, and the exact vertical-IQ target.

93. A finite-divisor canonical exact route now derives the packet-local source alignment from the finite divisor/Frobenioid packet product before constructing the exact source used by the Corollary 3.12 corridor.

94. Its compatibility and audit forms now construct the same finite-divisor source object first; the packet-local alignment and audit equalities are then projected from that object rather than from a separate alignment endpoint.

95. The strongest finite-divisor canonical exact route consumes this finite-divisor packet data as one source object: finite divisor packet, mono-analytic realization, packet-local/source equality, divisor realified-degree equality, and (Ind3)-source equality are projected from that object rather than supplied as unrelated route arguments.

96. The public experiment audit form of this finite-divisor exact route exposes the source equality, mono-analytic product equality, divisor-degree normalization, (Ind3)-source equality, and upper-semi inequality used in that construction.

97. The source-aligned realified Step (x) entry endpoint now exposes this $FMOD$-exactness guard at the same level as the Kummer/forgetting transfer chain, target packet-normalized equalities, q-pilot non-interference, and upper-semi inequality.

98. The theta-root attached source-aligned Step (x) endpoint now carries the same $FMOD$-exactness guard together with the canonical bad-local theta-root label facts.

99. The log-Kummer packet-correspondence source can now be constructed from source packet/product calibration plus the packet-normalized entry target and theta-target alignment. The q-pilot non-interference, source equality, and target equality endpoint is then recovered from this constructed correspondence.

100. The packet-correspondence endpoint now exposes the full finite Step (x) transfer chain: Kummer product preservation, mono-analytic forgetting product preservation, source equality with the (Ind3) real source, target packet-normalized equalities, and q-pilot non-interference.

101. The source packet/product calibration itself now has a constructor from packet-local/source alignment, packet-local (Ind3)-source alignment, and Kummer/forgetting product-log-volume preservation. The source holomorphic-product equality is derived by this constructor.

102. A calibrated source packet/product face can now be projected back to the packet-local source-alignment object once the Kummer and mono-analytic forgetting transfers are supplied; the packet-local (Ind3)-source equality is reconstructed through that transfer chain.

103. The same projection is now available from a realified-Frobenioid log-Kummer packet source itself: its internal Kummer and forgetting compatibilities supply the transfer chain needed to recover the packet-local source-alignment object.

104. The realified Step (x) entry source now projects the same packet-local source alignment while retaining the selected entry's upper-semi membership.

105. The theta-root attached realified Step (x) entry source now projects the same packet-local source alignment while retaining the canonical generator and nonzero full-label facts.

106. The Hodge/SHE/IPL/hull route has a source-audit endpoint: from one theta-root attached realified Step (x) source, Lean returns the recovered packet-local alignment equalities together with the C_Θ dichotomy.

107. The source-aligned Hodge/SHE/IPL/hull route now derives its target-only (Ind3) alignment from the Step (x) packet-normalized theta source and the exact vertical-IQ target source, retaining the $FMOD$-exactness guard at the route boundary.

108. The audit form of this vertical-IQ route also exposes the derived equality from the Hodge-descent theta average to the (Ind3) target value, before returning the C_Θ dichotomy.

109. The packet-correspondence constructor has a stronger alignment form: packet-local source alignment, packet-local (Ind3)-source alignment, packet-normalized entry target, and theta-target alignment generate both packet calibrations before projecting the q-pilot endpoint.

110. The realified-Frobenioid log-Kummer packet source has the same reduced target-data constructor: Frobenioid compatibility yields the Kummer and forgetting transfers, while packet-normalized target data is generated from the entry-target source and theta-target alignment.

111. The vertical-IQ target source has been lifted through the realified-Frobenioid entry source and the theta-root attached entry source, so their upper-semi endpoints no longer need independent target alignment equalities when common-target log-Kummer data is available.

112. The integrated Hodge/SHE/IPL/hull C_Θ-dichotomy route now has a vertical-IQ form: exact common-target log-Kummer data supplies the target equalities used by Step (x) before the route enters the final Corollary 3.12 real inequality.

113. The integrated vertical-IQ route also has a source-reduced form: source packet calibration is constructed internally from packet-local alignments and realified Kummer/forgetting compatibility.

114. The same source-reduced vertical-IQ data now constructs the realified Step (x) log-Kummer entry source itself, before any final C_Θ-dichotomy theorem is invoked.

115. The realified packet source now also has a source-reduced constructor: packet-local/source alignment and packet-local (Ind3)-source alignment generate the source packet calibration using the Frobenioid-derived Kummer and forgetting transfers.

116. The realified Step (x) entry source has the same reduced boundary: entry membership plus source/target alignment data constructs the packet source internally and projects q-pilot non-interference, source equality, target equality, and the upper-semi inequality.

117. The same realified packet source now exposes a combined calibration endpoint: it proves Kummer and forgetting product-log-volume preservation, packet-local equality with the entry source, entry-source equality with the (Ind3) source, packet-normalized equality for the theta average, (Ind3) target, and entry target, and q-pilot non-interference.

118. The theta-root attached realified entry source now has the same source-reduced vertical-IQ constructor: the upper-semi packet entry, packet-local source alignments, realified Kummer/forgetting compatibility, and exact vertical-IQ target construct one source object that projects the canonical nonzero theta-root label facts and the realified Step (x) upper-semi endpoint.

119. The packet-correspondence source now has the same source-aligned vertical-IQ constructor: a named packet-local source-alignment object and an exact vertical-IQ target source construct one correspondence object before the realified-Frobenioid layer is applied.

120. The packet-correspondence source also has a source-aligned packet-normalized ordered-target constructor: the source face is generated from the named packet-local alignment object, while the target face is generated from the packet-normalized entry and theta-target alignment.

121. The theta-root attached realified entry source now projects the underlying packet correspondence endpoint, combining canonical bad-local label facts with transfer-chain equalities, packet target calibrations, q-pilot non-interference, and the upper-semi bound.

122. The Hodge/SHE/IPL/hull route now consumes this reduced theta-root Step (x) data directly: the theorem constructs the realified source internally from a named packet-local source-alignment object and exact vertical-IQ target, then concludes the current C_Θ dichotomy.

123. The same route has an ordered-real-line form: the packet-source and theta-target (Ind3) real equalities are extracted from one matched-scale ordered transport alignment, so they are no longer accepted as unrelated route inputs at this boundary.

124. The ordered packet-normalized route now has a source-aligned form: packet-local source alignment plus realified Kummer/forgetting compatibility construct the source calibration internally before the route enters the theta-root attached Step (x) source.

125. The same source-aligned ordered route now has a target-only alignment form: the source-side (Ind3) equality is supplied by the source alignment object, and the route consumes only the theta-target (Ind3) equality as ordered-real input.

126. The target-only (Ind3) alignment can now be constructed from a Step (x) packet-normalized theta-source alignment and an exact vertical-IQ target source.

127. The realified-source vertical-IQ audit recovers the packet-local source-alignment object from the theta-root attached realified log-Kummer entry source, so this route boundary no longer accepts that source alignment as an independent input.

128. The stronger realified-source vertical-IQ audit also projects theta-average packet normalization from that same source, removing the separate Step (x) packet-normalized theta-source route from the audit boundary.

129. The finite-divisor vertical-IQ route now constructs this theta-root attached realified source internally from the finite divisor/Frobenioid packet product, then invokes the realified-source audit. Thus the integrated route no longer reconstructs a packet-local alignment independently of the source object.

130. In the ordered form, the packet-normalized theta calibration can now be obtained from a constant $ZMod(\ell)$ packet-normalized route. The theorem requires equality of the Hodge-descent theta source with the packet-normalized theta source, preventing a packet-normalization audit for one theta source from being used for another.

131. The same source equality is now bundled with the packet-normalized route as a Step (x) theta-source alignment, and the Hodge/SHE/IPL/hull endpoint consumes that aligned object instead of separate route/equality inputs.

132. The selected packet-normalized nonarchimedean entry is now bundled with its membership in the audited upper-semi inclusion family. The route consumes this single source object rather than a constructed entry and separate membership proof.

133. The source side of that Step (x) entry can now be supplied by a named packet/product source calibration. The route no longer receives the packet-local/source and source/mono-analytic-product equalities separately; they are projected from the calibration object before the realified Frobenioid source is constructed.

134. Below that calibration, the source/mono-analytic-product equality can now be derived from the finite divisor/Frobenioid packet product source whenever the packet-local object is identified with the entry source and the finite divisor realified log-volume.

135. The Hodge/SHE/IPL/hull vertical-IQ route has a finite-divisor form: it consumes the finite divisor packet source directly and constructs the packet-local source alignment internally before deriving the $C_\Theta = -1$ boundary alternative or strict $C_\Theta > -1$ conclusion.

136. At the entry-source level, the finite-divisor vertical-IQ constructor now first projects the full packet-source calibration from the finite-divisor packet source; the packet-local alignment is no longer the primary constructor at this boundary.

137. The finite-divisor packet source now also constructs the packet correspondence source itself: source calibration, vertical-IQ target calibration, and q-pilot non-interference are bundled before the realified packet and entry sources are formed.

138. The same finite-source packet-correspondence boundary now covers the entry-target/theta-target branch: the packet-normalized entry target supplies the target face while the finite-divisor packet source supplies the source face.

139. The ordered Step (x) route itself now has the finite-source boundary: the finite-divisor packet source supplies the source alignment before the ordered (Ind3) target comparison enters the Hodge/SHE/IPL/hull endpoint.

140. The theta-target branch of the ordered Step (x) route now has the same finite-source boundary: the finite-divisor packet source supplies the source alignment, while the only remaining ordered target input is the theta-target (Ind3) alignment.

141. The first-pass Corollary 3.12 experiment endpoint now has a finite Step (x) source form: the upper-semi packet-normalized entry, finite-divisor packet source, realified Kummer/forgetting compatibilities, and exact vertical-IQ target construct the realified log-Kummer entry source before the route enters the existing C_Θ dichotomy theorem.

142. The public finite-divisor vertical-IQ experiment surface now includes the source-object form and an audit endpoint, so the Step (x) source equalities are checked as projections of one finite-divisor/Kummer source.

143. The integrated Hodge/SHE/IPL/hull finite-divisor vertical-IQ route now has the same source-object boundary. Its compatibility form constructs the finite-divisor packet-local source and delegates to the source-object route.

144. The direct-labelled finite-divisor vertical-IQ route and its Hodge-derived labelwise specialization now use this source-object boundary as well; the finite-divisor compatibility forms no longer build an independent packet-local source alignment in their final proof path.

145. The theta-root attached realified Step (x) finite-divisor constructor now has the same finite-source form. The finite-divisor compatibility endpoint constructs this source object first and then exposes the theta-root label, Kummer/forgetting, product, target-normalization, and upper-semi facts as projections from it.

146. The ordered Step (x) vertical-IQ target route now has the same finite-source boundary. Its finite-divisor compatibility form first bundles the finite divisor/Frobenioid data, then derives the source alignment used by the ordered target route from that source object.

147. The labelled-average Step (x) route has the same finite-source boundary: finite $ZMod$-averaging supplies packet normalization, while the finite-divisor packet source supplies the source-side packet-local alignment.

148. The direct-normalized labelled-average route has the same source boundary: direct packet normalization derives the labelled average, and the finite-divisor packet source supplies the Step (x) source alignment.

149. The theta-labelled target and vertical-IQ route now have the same finite-source boundary: the labelled finite average supplies the theta-source calibration, the exact vertical-IQ target supplies the target alignment, and the finite-divisor packet source supplies the source alignment.

150. The finite sign-unit/root-of-unity part of Remark 3.11.4 is formalized: sign-unit action preserves full cusp labels, full-label log-volume readings, Gaussian degrees, and averaged Gaussian log-volume.

151. The same finite layer classifies the allowed unit-affine label changes: full-label preservation is equivalent to zero translation plus a sign-unit invisibility certificate; under nonzero Gaussian environment degree this is also equivalent to pointwise Gaussian-degree preservation.

152. The root-unit action preserves the raw representative-square profile, equivalently the factored square/full-label package, only for the identity unit; root-of-unity invisibility is therefore not silently promoted to SHE square-weight preservation.

153. The current integrated route composes Hodge–Arakelov theta-value sources with the packet-target nonarchimedean source to obtain the same $C_\Theta = -1$ boundary alternative or strict $C_\Theta > -1$ conclusion.

154. A Theorem 3.11 multiradial source record now packages structured IPL/SHE/APT, coric output, possible theta images, pilot-column asymmetry, distinct log-shell treatments, and the no-history-identification guard.

155. The IPL source layer now has an explicit log-volume transport record with source and target Hodge-theater sides, prime-strip endpoint compatibility, log-volume preservation, and no-history-identification.

156. The SHE source layer now derives the current square/full-label preservation package from Hodge–Arakelov canonical-one synchronization, exposing pointwise representative comparison, transported-average equality, and the retained Hodge-history separation.

157. The Step (xi) hull/determinant layer now has a Theorem 3.11 source constructor: source obligations are derived from the source record where possible, the record's theta possible-image union is tied to the hull endpoint, and the determinant bound yields $q_{\text{signed}} \le \theta_{\text{signed}}$.

158. The theta-pilot possible-image family of the source package now feeds the canonical holomorphic-hull approximant directly: Lean identifies its set-theoretic union with the package target union before the determinant comparison is applied.

159. The package-level hull/determinant source bridge can now be constructed from a concrete common hull containing the target union and a determinant/log-volume bound for that hull. The experiment endpoint returns target-union containment, the determinant bound, all-target bounds, and equality with the package's named hull/determinant bridge.

160. The same source bridge has a hull-operator specialization: the common hull is $\phi(P_{\text{out}})$ for the target-union region P_{out}, and the endpoint records this equality before deriving containment, determinant, and all-target bounds.

161. The Set-based holomorphic-hull/log-volume shadow now induces a Region-level hull operator and Region measure on real-line points. The experiment endpoint proves exact agreement with the original point-set hull, containment, and induced log-volume monotonicity.

162. The package-level hull/determinant bridge can now be constructed from that holomorphic-hull shadow when the package measure is identified with the induced Region measure and the point-set hull satisfies the determinant bound.

163. The possible-image approximant source can now construct the package hull/determinant bridge in the canonical case: the possible-image union must be the package target union, the approximant must be the canonical hull, and the normalized determinant value must be bounded by θ_{signed}.

164. The Theorem 3.11 hull/determinant constructor can now be built directly from the source record plus this canonical approximant data, with the record's possible-image union supplying the bridge to the package target union.

165. The same Theorem 3.11 constructor can now be built from a canonical possible-image approximant backed by the finite weighted determinant source of Remark 3.9.5(vii), (Ob3). The endpoint exposes the equality between the approximant log-volume and the weighted determinant log-volume.

166. The canonical weighted determinant constructor now has the stronger canonical-hull input form: the approximant is definitionally $\phi(P)$, q-pilot containment is imposed in this hull, and the endpoint exposes the derived approximant-region equality.

167. The canonical possible-image route can now instead consume $q \subseteq P = \bigcup_i P_i$. The containment $q \subseteq \phi(P)$ used by the upper-ray comparison is derived from hull extensivity, so the source interface is closer to the Step (xi) possible-image construction.

168. The canonical-hull route now has the same Ob4 tensor-power input form: a normalized tensor-power bound is converted to the determinant bound, while Lean derives the approximant-equals-$\phi(P)$ witness from the canonical hull constructor.

169. The package-level canonical Ob4 route now has a $q \subseteq P$ source form: bridge construction, determinant normalization, target-union containment, and the signed-volume hull bound are all exposed after Lean derives $q \subseteq \phi(P)$.

170. This canonical-hull Ob4 endpoint is also public for the Theorem 3.11 source constructor: the source record's theta possible-image union is the hull input before the package target-union identification is applied.

171. The arbitrary-family Theorem 3.11 Ob4 constructor now also has a $q \subseteq P$ source form. The family is then identified with the record's theta possible-image union, and Lean derives the hull containment consumed by the package-level source data.

172. The record-native Theorem 3.11 Ob4 route can now consume $q \subseteq P_{\mathrm{rec}}$, where P_{rec} is the record's own possible-image union; Lean derives $q \subseteq \phi(P_{\mathrm{rec}})$ before constructing the final hull/determinant source constructor.

173. A record-native specialization now derives that theta-image union from the Theorem 3.11 record's own possible-image family, instead of accepting an arbitrary image family plus a separate union equality.

174. The Step (x)-to-Step (xi) upper-ray bridge can now be built directly from a certified possible-image hull approximant. The determinant normalization is read from the approximant/determinant compatibility, rather than supplied as a separate equality at the bridge boundary.

175. The Step (x)-to-Step (xi) hull comparison is now a named handoff proposition over a fixed procession corridor and fixed hull source; from this handoff Lean derives q-pilot upper-ray membership and determinant normalization at the Step (xi-f) boundary.

176. The same upper-ray bridge now has a log-Kummer vertical-shift calibration endpoint: the globally calibrated Frobenioid log-volume is the hull boundary used in Step (xi-g), not an independently chosen real value.

177. The Step (xi) endpoint now exposes the finite cQ3/cQ4 closed loop: hull/determinant passage and log-Kummer calibration produce the same boundary, and the shifted local log-volume route is equivalent to a zero-shift specialization.

178. A Hodge/SHE/IPL/hull packet-target route now derives the target-side log-volume equality from SHE synchronization and reads q-positivity from the hull constructor; the source-side theta/Hodge equality is derived from a full-label calibration certificate in the route-alignment source record.

179. The Hodge/SHE/IPL/hull alignment is now also exposed through a source-derived bridge object. This object keeps the Theorem 3.11 multiradial record, the IPL log-volume transport, the finite Hodge–Arakelov SHE transport, the Step (xi) hull/determinant constructor, and the theta/Hodge calibration certificate as separate source components, then constructs the route-level log-volume alignment internally. The finite transport records the source and target Hodge-theater sides, finite label transport, full-label log-volume preservation, transported-average preservation, and the present no-history-identification guard.

180. The finite Hodge/SHE transport now has a source-boundary layer: Lean constructs it from a Hodge-theater descent bridge, a typed allowed-forgetting witness, the current Hodge–Arakelov synchronization source, and the finite full-label preservation data. The witness records the source theater, target theater, descent operation, permission proof, and history-separation guard, separating the history/forgetting obligation from the downstream transport object.

181. The allowed-forgetting witness now has a canonical history-separated finite-stage form. Instead of treating the permission predicate as an unrelated route input, the public Hodge–Arakelov endpoint can take it to be the history-separation condition of the Theorem 3.11 Hodge-theater descent bridge carried by the source record. This removes a loose external Prop from the public corridor while preserving the trust boundary: the full source-paper construction of the Hodge-theater forgetting operation is still not claimed.

182. The IPL log-volume transport now has a source construction from the same finite Hodge/SHE transport plus a Theorem 3.11 IPL-link construction source. The construction builds the input-prime-strip link with source and target endpoints definitionally equal to the recorded input and output prime strips, then aligns the resulting IPL datum with the pre-ledger certificate. Its Hodge-theater sides are projected from the finite transport, and its log-volume preservation equality is the transported average preservation supplied by the finite Hodge/SHE layer.

183. The public IPL boundary also has a certificate-pinned link-source form. This form consumes the pre-ledger certificate's own IPL datum together with just the endpoint equalities saying that its link source and target are the input and output prime strips. Thus the strongest finite-divisor route no longer has to carry a separate constructed IPL-datum alignment when the certificate datum itself already satisfies the Theorem 3.11 endpoint shape.

184. The Step (xi) holomorphic-hull/determinant route now has a source boundary wrapper around the record-native Ob4 constructor. The wrapper keeps the holomorphic hull shadow, $q \subseteq P_{\text{rec}}$ containment, weighted determinant source, tensor-power bound, bridge equality, positivity, and normalization as separate auditable fields, then constructs the existing Theorem 3.11 hull/determinant constructor and exposes $q_{\text{signed}} \leq \theta_{\text{signed}}$.

185. The strongest Hodge–Arakelov/history-separated route can now consume that Step (xi) holomorphic-hull/determinant source as a single object. This keeps the q-pilot containment and determinant payload intact, but no longer expands the record-native hull fields at the public Hodge–Arakelov boundary.

186. The Step (xi) source boundary now also has a choice-linked q-pilot form: a named Theorem 3.11 possible-output choice carries $q \subseteq P_{\text{choice}}$, and Lean derives $q \subseteq P_{\text{rec}}$ by the possible-image union inclusion. Thus the strongest Hodge–Arakelov/history-separated endpoint has a variant whose q-pilot containment is not an independent public field.

187. That choice-linked Step (xi) source now projects to the existing upper-ray and two-computation corridor. From the canonical hull and weighted-determinant compatibility, Lean constructs the $R_{\leq -|\log(\Theta)|}$-style upper-ray datum, proves membership of the q-pilot log-volume in this ray, derives the determinant upper bound for that q-pilot log-volume, and records equality of the input-prime-strip and output-hull q-pilot readings.

188. The choice-linked Step (xi) source now has a stricter exact-theta form. Instead of carrying the normalized tensor-power upper bound as a separate field, it carries the equality identifying θ_{signed} with the normalized Ob4 tensor-power log-volume of the weighted determinant source. Lean then derives the old tensor-power bound, projects back to the existing choice-linked hull source, and exposes a public Hodge–Arakelov/history-separated finite-divisor endpoint for this stronger source shape.

189. That exact-theta Step (xi) source now has a side-conditioned form as well. The q-pilot positivity and source normalization proofs are no longer two loose fields in this variant; they are projected from $IUTStage1SourceSideConditions$, the same side-condition boundary used when promoting source data to the Stage 1 ledger.

190. The exact-theta Step (xi) source now also has a hull/det-data-backed form. In this variant, the equality with the package's named hull/determinant bridge is no longer carried directly. Instead the source consumes $IUTStage1SourceHullDetData$, projects the package bridge equality already stored in that object, and then identifies the stored bridge data with the record-canonical Step (xi) hull/tensor-power construction.

191. The same boundary now has a record-native bounded-family hull form. This new source object specializes the Ob5 bounded-family hull/determinant source to the Theorem 3.11 possible-image family, so the family union P_{rec}, its hull $\phi(P_{\text{rec}})$, the weighted determinant source, and the normalized Ob4 tensor-power log-volume are tied to one record-level object. The strengthened endpoint assumes $\theta_{\text{signed}} = \mu_{\log}(\phi(P_{\text{rec}}))$; Lean then derives the older exact tensor-power equality and the tensor-power bound from the Ob5 hull/determinant compatibility.

192. The record-native bounded-family source now projects back to the generic Ob5 bounded-family hull/determinant source. Thus the record-level possible-image family reuses the existing quotient-collapse theorem, possible-region containment in the family hull, determinant/log-volume comparison, and normalized tensor-power endpoint instead of duplicating a weaker record-specific summary.

193. The history-separated Hodge–Arakelov, certificate-pinned IPL, finite-divisor vertical-IQ endpoint now consumes this family-hull-backed source directly. The endpoint constructs the source-derived Hodge/SHE/IPL/hull bridge from this source, so the public route no longer has to present the exact tensor-power equality as the Step (xi) source-facing input. The experiment module exposes the same route as the audit theorem `familyHullBackedFiniteDivisorVerticalIQ_cThetaDichotomy`.

194. The family-hull-backed endpoint now also has a constructed-IPL form at the same boundary. The source-derived bridge is built directly from the input-prime-strip

link construction source, so the bridge audit retains the equality between the package certificate's IPL datum and the constructed datum, before applying the same finite-divisor vertical-IQ route. The experiment module exposes this as `constructionFamilyHullBackedFiniteDivisorVerticalIQ_cThetaDichotomy`.

195. The same family-hull Step (xi) source now has an obligations-backed form. It consumes one $IUTStage1SourceHullDetObligations$ object and projects the package hull/determinant data, q-pilot positivity, and source normalization from that object, rather than carrying them as separate Step (xi) fields. The experiment module exposes this stronger audit route as `obligationsBackedFamilyHullFiniteDivisorVerticalIQ_cThetaDichotomy`.

196. The obligations-backed family-hull route now also has the constructed IPL form. Thus the public boundary simultaneously keeps the input-prime-strip link construction visible and projects the Step (xi) hull/determinant data, q-pilot positivity, and normalization from the single obligations object. The experiment module exposes this as the corresponding constructed-IPL, obligations-backed family-hull dichotomy theorem.

197. The exact theta/family-hull identification now also has a Hodge-calibrated source form. Instead of supplying `thetaSigned = familyHullLogVolume` directly, the source object factors the equality through the Hodge–Arakelov theta-monoid degree, and the experiment module exposes this route as `hodgeCalibratedFamilyHullFiniteDivisorVerticalIQ_cThetaDichotomy`.

198. The Hodge-calibrated family-hull route now has a constructed-IPL form as well. It keeps the input-prime-strip link construction visible while the Step (xi) source still factors the exact theta/family-hull equality through the Hodge–Arakelov theta-monoid degree.

199. This Hodge-calibrated route now has a charted refinement. The source object calibrates the transported pre-ledger theta chart coordinate to the Hodge–Arakelov theta-monoid degree, and Lean derives the package `thetaSigned` equality from the existing chart theorem. The experiment module exposes this route as `chartedHodgeCalibratedFamilyHullFiniteDivisorVerticalIQ_cThetaDichotomy`.

200. The charted Hodge-calibrated route now also has the constructed-IPL form. The public boundary keeps the input-prime-strip link construction visible while retaining the chart-derived theta/Hodge scalar equality.

201. The charted Hodge/family-hull calibration now has a determinant-factored form. The source object calibrates the Hodge–Arakelov theta-monoid degree to the normalized determinant log-volume of the record-native family hull, and Lean derives the family-hull equality from the Ob5 hull/determinant compatibility. The experiment module exposes this route as `determinantChartedHodgeFamilyHullFiniteDivisorVerticalIQ_cThetaDichotomy`.

202. The determinant-factored route now has a summand-factored source form. The source object calibrates the Hodge–Arakelov theta-monoid degree to the finite sum of adjusted localization summands of the weighted determinant source; Lean derives the determinant and normalized determinant equalities before entering the same finite-divisor endpoint. The experiment module exposes this route as `summandChartedHodgeFamilyHullFiniteDivisorVerticalIQ_cThetaDichotomy`.

203. The summand-factored route now also has a source-calibrated Hodge–Arakelov input form. The source Hodge–Arakelov evaluation carries the full-label theta/Hodge log-volume calibration and projects the previous `IUTStage1SourceThetaHodgeLogVolumeCalibration` internally. The experiment module exposes this boundary as the source-calibrated variant of the summand-charted finite-divisor theorem.

204. The same source-calibrated summand route now has a constructed-IPL form. Instead of receiving the certificate-pinned IPL-link source and its endpoint equalities directly, the boundary consumes the Theorem 3.11 input-prime-strip link construction source and projects the link source internally before entering the finite-divisor vertical-IQ endpoint.

205. The constructed-IPL, source-calibrated summand route now also has a synchronized Hodge–Arakelov input form. The source object carries the calibrated source evaluation, target evaluation, and canonical one-label degree preservation together, so the public finite-divisor boundary no longer receives the Hodge/SHE synchronization equality separately.

206. The synchronized summand route now has a target-charted Step (xi) source form. Instead of receiving the charted-theta/source-Hodge scalar equality as an immediate summand-calibration field, the source reads the transported theta chart as the target Hodge–Arakelov canonical one-label degree and derives the source theta-monoid degree equality from the synchronized canonical-one preservation.

207. The target-charted summand source now has a possible-image q-pilot form. The q-pilot region is fixed to the chosen Theorem 3.11 possible-image region, so the public Step (xi) source no longer carries a separate q-region and $q \subseteq P_{\text{choice}}$ proof.

208. The target-charted possible-image summand route now has a Hodge-layer charted synchronization form. The target theta chart/canonical-one equality is projected from a target-charted Hodge–Arakelov synchronization object, while the Step (xi) source retains only the determinant summand equality and record-native possible-image hull data.

209. The target-charted possible-image route now factors the Hodge theta/summand comparison through a Hodge/determinant family-hull payload. This payload packages the record-native family-hull determinant source, the package measure calibration, and the Hodge theta-monoid/summand equality; the possible-image Step (xi) source then projects these fields instead of exposing them separately. The source now also exposes the converted side-conditioned possible-image hull/determinant endpoint before the Hodge/IPL wrapper is introduced. The experiment surface exposes the payload endpoint, the determinant possible-image endpoint, this side-conditioned hull endpoint, the constructed-IPL bridge audit, and the finite-divisor vertical-IQ C_Θ-dichotomy route.

210. The synchronized target-charted possible-image summand/family-hull source now reduces to the constructor-built Step (xi) boundary. The named audit records the fixed Theorem 3.11 possible-image q-region, target chart/source theta-monoid equality, finite adjusted-summand equality, theta/family-hull equality, obligations bridge equality, q-positivity, normalization, and the raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. A product-hull companion audit now carries the Remark 3.9.5 smallest-product-hull provenance through this same synchronized source whenever its family-hull shadow is induced by a product-hull system.

211. The target-charted Hodge/IPL side now has a packaged construction source. It binds the target-charted Hodge–Arakelov synchronization to the constructed Theorem 3.11 input-prime-

strip link, projects the finite Hodge/SHE transport, and then projects the IPL log-volume transport from that same finite transport. The determinant possible-image bridge and finite-divisor vertical-IQ endpoint can now consume this single Hodge/IPL payload instead of separate Hodge synchronization and constructed-IPL arguments. The construction source now also exposes the history-separated finite Hodge/SHE transport-source endpoint and the constructed IPL log-volume transport endpoint itself. The all-in-one route projects these finer audits rather than reaching through the finite construction source directly.

212. The strongest target-charted finite-divisor corridor now has an all-in-one Step (xi) route source. It binds the packaged Hodge/IPL construction source to the determinant possible-image payload, and the experiment surface exposes the route-source endpoint, the Hodge–Arakelov theta-evaluation/Gaussian endpoint, the history-separated Hodge–Arakelov finite Hodge/SHE transport-source endpoint, the finite Hodge/SHE transport endpoint, the constructed IPL log-volume transport-source endpoint, the constructed Hodge/SHE/IPL endpoint, the side-conditioned possible-image hull/determinant endpoint, the holomorphic-hull/log-volume endpoint, the canonical hull/tensor-power bridge endpoint, the source-boundary raw comparison endpoint, source-owned bridge endpoint, the bridge-level raw source-derived comparison endpoint, and finite-divisor packet-source and finite-divisor C_Θ-dichotomy endpoints from this single object. It now also exposes a finite-source Step (x) boundary audit before the external C_Θ numeric bound: the route-owned bridge data, raw source comparison, packet-local finite-source equalities, and vertical-IQ target equalities are all projected from the bundled route/source objects. The same route now has an exact vertical-IQ finite-source form: the realified log-Kummer Step (x) source retains the exact vertical-IQ target and transfer data, and the route projects its Kummer/forgetting compatibilities and $FMOD$-exact target internally. This exact finite route is now also bundled as a single source package joining the Step (xi) route source, upper-semi entry, finite packet-local source, and exact vertical-IQ realified Step (x) source before the C_Θ bound is applied. The bundle also exposes the realified Step (x) correspondence audit: theta-root canonical-label data, Kummer/Frobenioid product preservation, mono-analytic forgetting, q-pilot non-interference, and the upper-semi inequality are projected from the retained exact source. A compact source-derived corridor audit now records one load-bearing fact from each stage of the strongest chain: theta root, history-separated Hodge/SHE, constructed IPL, possible-image hull containment, raw comparison, exact vertical-IQ, non-interference, and Step (x) upper-semi control. A companion Gaussian-front audit starts one step earlier, at the Hodge–Arakelov theta-evaluation source: canonical theta labels, Gaussian zero degree, square-weighted degree formula, full-label log-volume compatibility, and target chart comparison are tied to the same corridor facts.

213. The finite-divisor vertical-IQ Hodge/SHE/IPL/hull endpoint now has a source-derived form: it consumes this separated bridge object instead of a prebuilt route alignment before constructing the same finite-divisor Step (x) source and proving the $C_\Theta = -1$ boundary alternative or strict $C_\Theta > -1$ conclusion.

214. The same finite-divisor vertical-IQ endpoint now also has the milestone-facing constructor form: it consumes the finite Hodge/SHE transport source, a Theorem 3.11 IPL-link construction source, record-native Step (xi) hull data, and the theta/Hodge calibration certificate directly. The Step (xi) inputs are the record's possible-image union, $q \subseteq P_{\mathrm{rec}}$, the holomorphic hull shadow, weighted determinant source, tensor-power bound, and bridge equality. The endpoint constructs the IPL log-volume transport, the hull/determinant source, and the

source-derived Hodge/SHE/IPL/hull bridge internally, then applies the finite-divisor vertical-IQ route.

215. The constructor-backed finite-divisor endpoint now preserves the Remark 3.9.5 product-hull provenance. A product-hull-backed wrapper records that the measured hull operator is induced by the product-hull system, and the audit proves selected-product-hull equality, q-pilot containment, minimality among product hulls containing P_{rec}, product-hull log-volume equality, and the same raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison before returning the downstream C_Θ dichotomy. The same wrapper now also exposes the Ob5 quotient/determinant compatibility at this product-hull boundary: for any nonempty comparison possible image, the q-choice possible image and comparison possible image have the same collapsed upper-semi quotient image modulo the selected product hull, and the selected hull log-volume is identified with the normalized determinant, determinant log-volume, and normalized tensor-power value. The Ob5 scalar companion is now explicit as well: after quotienting the real line by the singleton $\{\mu_{\log}(\phi(P_B))\}$, the normalized determinant scalar, ordinary determinant scalar, normalized tensor-power scalar, and family-hull log-volume scalar all have the same collapsed image. This records the Remark 3.9.5(Ob5) assertion that the upper-semi formalism applies compatibly to $\phi(P_B)$, $\det^{\otimes M}(\phi(P_B))$, and $\mu_{\log}(\phi(P_B))$. Finally, the product-hull wrapper now projects to the source-facing possible-image family package: the Theorem 3.11 family P_B, its canonical $\Phi(P_B)$ approximant, and any supplied exact $\Xi(P_B)$ family are audited at the same boundary, with $H_{\Xi(P_B)}$ identified with the selected product hull. The Ob6 hull-approximant log-volume audit is also exposed at this boundary: $H_{\Phi(P_B)}$ and $H_{\Xi(P_B)}$ are both identified with the selected product hull, while the raw family union, comparison possible image, and q-pilot region are estimated by the corresponding hull-approximant log-volumes before the normalized determinant equality is invoked. The same boundary now carries an exact-theta bridge audit: the package value θ_{signed} is identified with the Remark 3.9.5 family-hull log-volume, the normalized tensor-power/determinant upper bound is derived from this equality, and the q-pilot log-volume comparison is threaded through the selected product hull into θ_{signed}.

216. The milestone-facing endpoint has a Hodge–Arakelov source form as well: it consumes source and target Hodge–Arakelov theta-evaluation sources, canonical one-label degree preservation, and a typed allowed-forgetting witness, then constructs the finite Hodge/SHE transport internally. In this form the canonical square-weight profile and theta-root source are read directly from the Hodge–Arakelov source side before the endpoint applies the same Theorem 3.11 IPL, record-native Step (xi), and finite-divisor vertical-IQ route.

217. A stronger Hodge–Arakelov endpoint removes even that explicit allowed-forgetting argument: it builds the finite Hodge/SHE transport with the record's history-separated descent-bridge permission, then follows the same IPL, record-native Step (xi), and finite-divisor vertical-IQ path.

218. The strongest Hodge–Arakelov/history-separated endpoint can now use the certificate-pinned IPL-link source. Its public assumptions no longer include an arbitrary finite Hodge/SHE transport, a free allowed-forgetting predicate, a free square-weight profile, a separate theta-root source, a constructed-IPL-datum alignment, or the raw record-native Step (xi) hull/determinant field list.

219. The target-side theta-to-entry equality is no longer a direct input in the strongest route. It is derived from a Step (x) theta-target alignment and calibration of the entry target with the upper-semi target value.

220. The Step (x) theta-target alignment can itself be derived from packet-normalized theta-source equality and calibration of the upper-semi target with the packet-normalized value.

221. The entry-target calibration can likewise be expressed by equality to the packet-normalized value, deriving equality with the upper-semi target.

222. The Step (xi-f) real algebra proves $C_\Theta \geq -1$, strictness under strict upper-ray membership, and equality constraints in the boundary case.

223. The final route rejects the balanced sign-compatible comparison level as sufficient; it requires hull/log-volume comparison data.

224. The local Frobenioid shift model is now sourced by a p_v^N log-volume action package and a global realified calibration package; a non-automorphic local shift with nonzero step is separated from the calibrated value, and the upper-ray quotient experiment gives the exact threshold for collapse versus separation in the boundary upper ray.

225. Several downstream IUT IV numerical estimates are formalized as ordered-real algebra conditional on the Corollary 3.12 input.

The first-pass experiment report evaluates the route-theorem, finite Gaussian, collapse-obstruction, upper-ray, and IUT IV algebra flags to `true`. The flag `disputeSettledByCurrentStage` remains `false`.

The current code base is large enough to distinguish three meanings of "available". First, pure ordered-real consequences are proved outright. Second, route theorems prove that certain named source-facing records imply the desired endpoint. Third, the records themselves remain obligations when they stand for Hodge-theater, Frobenioid, log-Kummer, hull, determinant, or multiradial constructions not yet formalized from IUT I–III. Only the first two meanings are currently achieved.

8 What Is Not Yet Proved

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The following mathematics is still external to the code:

1. initial Θ-data and its arithmetic hypotheses from IUT I;
2. Hodge theaters, Frobenioids, log-shells, prime strips, and theta links;
3. the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
4. the construction of the actual (Ind1), (Ind2), (Ind3) data in Theorem 3.11;
5. the genuine holomorphic hull and determinant constructions used in Step (xi), beyond the present source-data constructor and ordered-real upper-ray model;
6. the full proof that the typed SHE and Hodge-descent transport obligations follow from Mochizuki's source constructions, beyond the current Hodge–Arakelov canonical-one synchronization shadow;
7. the construction of the input prime-strip link (IPL), simultaneous holomorphic expressibility (SHE), and algorithmic parallel transport (APT) properties as actual consequences of Theorem 3.11 rather than records;

8. the arithmetic-divisor, height, log-different, and log-conductor constructions behind the IUT IV Diophantine applications.

9 Implementation Audit

The code contains no `sorry`, `admit`, or Lean axioms in the Stage 1 corridor. The current risk is not hidden unsoundness of this form, but mathematical incompleteness: several records are deliberately named obligation interfaces. These interfaces are acceptable only as temporary boundaries. The next phase must replace them by constructions from the source papers or mark them as unresolved.

For the Source-Derived Step (xi) Bridge milestone, the checked public endpoint is

```
Iut.Stage1.Experiments.boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ.
```

Its direct audit companion is

```
Iut.Stage1.Experiments.boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ
  _withDirectMilestoneAudit.
```

Both declarations have been added to the `checkdecls` list. A kernel `#print axioms` check for these two declarations, the explicit constructed-IPL alias, and the underlying namespaced constructed-IPL endpoint reports only `propext`, `Classical.choice`, and `Quot.sound`; in particular, no `sorryAx` appears. Thus the milestone is kernel-clean at the usual Lean classical/quotient trust boundary.

The Remark 3.9.5 holomorphic-hull/determinant milestone has now started at the source-object level. The new declaration `IUTStage1Remark395HolomorphicHullOperator` records the source-facing ϕ-operator laws directly: closed hulls are fixed, $P \subseteq \phi(P)$, inclusions are preserved, $\phi(\phi(P)) = \phi(P)$, and log-volume is monotone. It projects to the older closure-operator shadow used by the existing bridge code, but the public source object no longer begins with that shadow. This has now been backed by the earlier construction in Remark 3.9.5(ii): `IUTStage1Remark395HolomorphicHullSystem` defines $\phi(P)$ as the intersection of all hulls that contain P. Its only source closure input is that this intersection is again a hull; Lean then derives P1–P3, idempotence, and $\mu_{\log}(P) \leq \mu_{\log}(\phi(P))$, before projecting to the existing ϕ-operator interface. The possible-image family layer has also been made source-facing: `IUTStage1Remark395PossibleImageFamilySource` records the family $P_B = \{P_\beta\}_{\beta \in B}$, its union, the canonical hull $\phi(P_B)$, the $\Phi(P_B)$ and $\Xi(P_B)$ approximant types projected from this ϕ-operator, and the Remark 3.9.5(Ob5) upper-semi quotient collapse. The experiment-surface declaration

```
Iut.Stage1.Experiments.remark395PossibleImageFamilySource_endpoint
```

proves that the family union and each possible image lie in $\phi(P_B)$, that the canonical Φ-approximant is $\phi(P_B)$, and that any two nonempty possible regions have the same quotient image after collapsing $\phi(P_B)$. The exact-approximant companion

```
Iut.Stage1.Experiments.remark395PossibleImageFamilySource
  _XiFamilyOb5Endpoint
```

proves the same Ob5 collapse for $H_{\Xi(P_B)}$, while recording that each chosen exact approximant has log-volume $\mu_{\log}(P_B)$. This supplied-family endpoint is now accompanied by the more source-local constructor

```
Iut.Stage1.Experiments.remark395PossibleImageExactXiFamilySource
  _toXiFamilyOb5Endpoint
```

and its Ob6 companion. These endpoints start only from the exactness calibration $\mu_{\log}(\phi(P_B)) = \mu_{\log}(P_B)$, build the canonical exact $\Xi(P_B)$-family, and then recover the same quotient collapse and log-volume comparison. Thus this part of the corridor now records the specific mathematical equality needed to make $\phi(P_B)$ exact, instead of treating the whole $\Xi(P_B)$-family as primitive. There is now also an `ofHullSystem` constructor and experiment endpoint: starting from the intersection-defined hull system and a family $\{P_\beta\}$, Lean constructs the possible-image source and proves that the canonical hull is precisely the system's $\phi(\bigcup_\beta P_\beta)$, which is a hull by the intersection-closure input. Thus the Ob5 quotient collapse for possible images is already downstream of the minimal-hull construction, not merely of a supplied ϕ-operator. The product-hull source now has a direct-product constructor as well: `IUTStage1Remark395DirectProductHullSystemSource` defines product hulls as coordinate products, defines the selected hull parameter by coordinate projections of the region, and proves the "smallest product hull" equality before projecting to the existing product-hull and holomorphic-hull systems. The determinant side now has a source-facing Ob3/Ob4 package as well:

`IUTStage1Remark395Ob3Ob4AdjustedDeterminantSource`.

It records localization summands, the structure-sheaf adjustment $O(-)$, weighted adjusted log-volumes, the positive tensor power M, and the normalized determinant log-volume before projecting to the older weighted determinant skeleton. The experiment-surface theorem

```
Iut.Stage1.Experiments.remark395Ob3Ob4AdjustedDeterminantSource
  _endpoint
```

checks the adjusted-summand formula, determinant summation, tensor-power formula, and normalized determinant equality. The constructed Step (xi) source also has a constructor `ofOb3Ob4AdjustedDeterminantSource` whose endpoint verifies that this source-facing Ob3/Ob4 package is the determinant source used by the existing hull/determinant corridor. This determinant package now has a lower source-facing constructor layer. The new objects

`IUTStage1LocalizedArithmeticVectorBundle`,
`IUTStage1StructureSheafAdjustedLocalizedVectorBundleSource`,
`IUTStage1Remark395Ob3Ob4LocalizedVectorBundleDeterminantSource`

separate the local-ring-labelled vector-bundle localization of Step (xi-d) from the later $O(-)$ structure-sheaf adjustment and determinant/tensor-power normalization. Their experiment endpoints prove that the localized rank-> 1 direct-summand log-volume projects to the adjusted localization source, and that the resulting Ob3/Ob4 determinant source has the same weighted summand formula and normalized determinant equality as the older adjusted determinant package. The Ob3/Ob5 determinant-compatibility input has also been separated from the low-level certificate:

`IUTStage1Remark395Ob3Ob5DeterminantCompatibilitySource`.

This object states, at the possible-image-family level, that $\mu_{\log}(\phi(P_B))$ is the normalized weighted-determinant log-volume. Lean derives the older `IUTStage1HullApproximantWeightedDeterminantCompatibility` certificate from this equality and then derives the ordinary determinant equality from the Ob3/Ob4 normalization theorem. The record-canonical Step (xi) specialization `IUTStage1Remark395RecordOb3Ob5DeterminantCompatibilitySource` fixes P_B to the Theorem 3.11 possible-image family. Both the generic and record-canonical Ob3/Ob5 source objects now project to the bounded-family hull/determinant source used by the downstream exact-theta Step (xi) routes: the endpoints `remark395Ob3Ob5DeterminantCompatibilitySource_boundedFamilyEndpoint` and `remark395RecordOb3Ob5DeterminantCompatibilitySource_boundedFamilyEndpoint` verify the family union, canonical hull, determinant log-volume equality, and normalized tensor-power equality after this projection. The generic bridge from Ob3/Ob5 compatibility to the Step (xi) log-volume chain now also has an exact-theta constructor:

```
remark395HullDeterminantBridgeSource
  _ofOb3Ob5CompatibilitySourceOfThetaEqFamilyHullLogVolume
  _endpoint.
```

This route no longer takes the Ob4 tensor-power bound as an independent input. Given $q \subseteq \bigcup_\beta P_\beta$ and the source calibration $\theta_{\text{signed}} = \mu_{\log}(\phi(P_B))$, Lean derives the tensor-power bound from the bounded-family hull/determinant source and then proves

$$\mu_{\log}(q) \leq \det_{\text{norm}} \leq \theta_{\text{signed}}.$$

The record-canonical Ob3/Ob5 source now also has a public constructor endpoint from the Ob3/Ob4 adjusted determinant source. Given the remaining source-facing equality

$$\mu_{\log}(\phi(P_B)) = \det_{\text{norm}}(P_B),$$

Lean constructs the record Ob3/Ob5 compatibility object, projects the Ob3/Ob4 localization-summand and tensor-power formulas, derives equality with the ordinary determinant log-volume, and then projects the resulting object to the bounded-family Ob5 hull source. The experiment surface exposes this as

```
remark395RecordOb3Ob5DeterminantCompatibilitySource
  _ofAdjustedDeterminantSource_endpoint.
```

The remaining equality has now been pushed one level closer to Remark 3.9.5, (vii), (Ob3-1)–(Ob3-3). The new source-core object

```
IUTStage1Remark395Ob3Ob5DeterminantCompatibilitySource
  .IUTStage1Remark395Ob3Ob5AdjustedDeterminantLogVolumeSource
```

does not assume equality with the normalized determinant directly. Instead it records the finite Ob3-3 summand formula

$$\mu_{\log}(\phi(P_B)) = \sum_\beta \text{adj}_\beta,$$

where the summands are the Ob3/Ob4 structure-sheaf-adjusted localization contributions. Lean then derives equality with the determinant log-volume and with the normalized determinant log-volume from the adjusted determinant source. The record-canonical specialization fixes P_B to the Theorem 3.11 possible-image family and exposes

```
remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  _endpoint.
```

Thus the current open analytic/log-volume input is no longer $\mu_{\log}(\phi(P_B)) = \det_{\text{norm}}$ as a monolithic assertion, but the more source-facing finite summand identity $\mu_{\log}(\phi(P_B)) = \sum_\beta \text{adj}_\beta$. This identity can now also be stated at the localized vector-bundle level: the constructor

```
IUTStage1Remark395Ob3Ob5AdjustedDeterminantLogVolumeSource
  .ofLocalizedVectorBundleDeterminantSource
```

takes the equality with the sum of local-ring-labelled adjusted vector-bundle summands and derives the adjusted determinant source, determinant log-volume equality, and normalized Ob3/Ob5 compatibility from it. The equality with the localized adjusted sum has now been factored once more. The source declaration

```
IUTStage1Remark395LocalizedHullVectorBundle
  DecompositionSource
```

records a finite decomposition of the canonical family hull into localized hull-regions together with two source-facing facts: the hull log-volume is the sum of these localized region log-volumes, and each localized region log-volume is the corresponding adjusted localized vector-bundle summand. Lean derives the Ob3-3 summand identity from these two facts and then constructs the same Ob3/Ob5 adjusted determinant log-volume source. The localized-region containment has now been pushed one level lower. The source declaration

```
IUTStage1Remark395LocalizedHullCover
  VectorBundleSource
```

starts from the intersection-defined Remark 3.9.5 hull system and a finite cover

$$\phi(P_B) = \bigcup_{\beta} H_\beta$$

by localized hull-regions. From this cover identity, Lean proves $H_\beta \subseteq \phi(P_B)$ for each β, derives the family-hull log-volume sum from the supplied cover-additivity theorem, and then constructs the localized hull/vector-bundle decomposition source described above. The pointwise summand calibration has now been factored as well. A local source object

```
IUTStage1LocalizedHullRegion
  VectorBundleCalibrationSource
```

attaches a single localized hull-region H_β to its structure-sheaf-adjusted localized arithmetic vector bundle and records the source equality

$$\mu_{\log}(H_\beta) = w_\beta \cdot \big(L_{\beta,\mathrm{bundle}} - L_{\beta,\mathcal{O}(-)}\big).$$

Lean then identifies the right hand side with the weighted adjusted localized vector-bundle summand. The aggregate declaration

```
IUTStage1Remark395CalibratedLocalized
  HullCoverVectorBundleSource
```

assembles these calibrated local objects, constructs the finite localized determinant source from them, and then projects to the localized hull-cover source. Thus the cover-level pointwise equality $\mu_{\log}(H_\beta) = \mathrm{adj}_\beta$ is no longer a bare field. The finite cover-additivity input has also been separated from this aggregate object. The declaration

```
IUTStage1FiniteAdditiveHullLogVolumeSource
```

records the source-paper law that a finite pairwise-disjoint family of localized hull-regions satisfies

$$\mu_{\log}\left(\bigcup_{\beta} H_\beta\right) = \sum_{\beta} \mu_{\log}(H_\beta).$$

The refined aggregate source

```
IUTStage1Remark395FiniteAdditive
  CalibratedLocalizedHullCoverVectorBundleSource
```

assumes pairwise disjointness of the localized regions and this finite additivity law, then derives the exact cover-sum equality consumed by the calibrated cover source. The pairwise-disjointness input has now been pushed one level lower as well. The source declaration

```
IUTStage1LocalizedHullRegionOwnershipSource
```

records a single owner map from hull points to optional localized indices and states that each localized hull-region is the corresponding owner fiber. From this fiber description Lean proves pairwise disjointness of distinct localized regions. The aggregate declaration

```
IUTStage1Remark395OwnedFiniteAdditive
  CalibratedLocalizedHullCoverVectorBundleSource
```

therefore constructs the finite-additive calibrated cover source without taking localized-region disjointness as a bare field. The cover identity itself has now been factored through the same owner map. The declaration

```
IUTStage1Remark395OwnerTotalFiniteAdditive
  CalibratedLocalizedHullCoverVectorBundleSource
```

assumes that every point of the family hull $\phi(P_B)$ has a localized owner and that each owner fiber remains inside $\phi(P_B)$. Together with the fiber description of H_β, Lean derives $\phi(P_B) = \bigcup_\beta H_\beta$ and then projects to the owner-fiber finite-additive cover route. The owner map itself has now been constructed from a lower family-hull indexed source. The declaration

```
IUTStage1Remark395FamilyHullIndexedFiniteAdditive
  CalibratedLocalizedHullCoverVectorBundleSource
```

takes as primitive data an index function on the subtype of points of $\phi(P_B)$, together with the assertion that H_β is the corresponding index fiber. Lean defines the ambient owner map to be this index on $\phi(P_B)$ and `none` off $\phi(P_B)$, then derives owner-totality, fiber containment in $\phi(P_B)$, and the owner-total cover source. The indexing source has now been pushed down to a direct-product localization source. The declaration

```
IUTStage1Remark395DirectProduct
  LocalizedHullCoverVectorBundleSource
```

represents H_β as the finite direct-product cell cut out by all local factor regions at the localization places. Its separation hypothesis is local: two distinct indices are separated by at least one disjoint local factor. From this Lean proves pairwise disjointness of the direct-product cells, uses the cover equality

$$\phi(P_B) = \bigcup_\beta H_\beta$$

to choose the unique cell containing a hull point, and then constructs the family-hull indexed source rather than taking the index map as primitive. The finite-additivity boundary has also been localized to this same direct-product cover. The declaration

```
IUTStage1Remark395DirectProductAdditive
  LocalizedHullCoverVectorBundleSource
```

does not assume a universal finite-additive law for arbitrary region families. Instead it records the cover-specific Step (xi) equality

$$\mu_{\log}\left(\bigcup_{\beta} H_\beta\right) = \sum_{\beta} \mu_{\log}(H_\beta)$$

for the direct-product localization cells. Lean transports this equality across the identification of H_β with the calibrated localized regions, derives the calibrated cover source, and proves $\mu_{\log}(\phi(P_B))$ equals the sum of the adjusted Ob3/Ob4 localization summands. This cover-specific equality has now been derived from a lower finite local-factor volume source. The declaration

```
IUTStage1Remark395LocalFactorVolume
  DirectProductLocalizedHullCoverVectorBundleSource
```

assigns a log-volume contribution to each local factor of each direct-product cell. Its source laws compute

$$\mu_{\log}(H_\beta) = \sum_{v} \mu_{\beta,v}, \qquad \mu_{\log}\left(\bigcup_{\beta} H_\beta\right) = \sum_{\beta}\sum_{v} \mu_{\beta,v}.$$

Lean then derives the direct-product cover additivity by finite-sum congruence and projects to the cover-specific additive source. Thus the previous cover-additivity assumption is now reduced to the local-factor log-volume computation for the localized arithmetic-vector-bundle cells. The local-factor volumes are now also tied to the Ob3 vector-bundle data. The declaration

```
IUTStage1Remark395VectorBundleLocalFactor
  DirectProductLocalizedHullCoverSource
```

defines the local factor $\mu_{\beta,v}$ to be the direct-summand log-volume of the localized arithmetic vector bundle attached to β at the place v. Its source laws compute each H_β from the associated bundle log-volume and compute the cover from the resulting finite bundle sum. Lean then derives the previous local-factor source and proves that the local-factor sum agrees with the weighted adjusted Ob3/Ob4 summand already used by the determinant package. Thus the local-factor layer is no longer an independent real-valued family; it is read from the localized vector-bundle summands. The factor regions themselves have now been pushed down one more source layer. The declaration

```
IUTStage1Remark395LocalRingChartedVectorBundle
  DirectProductHullCoverSource
```

introduces a local-ring chart for each factor of each direct-product localization cell. The chart records the local-ring label and the region that represents the corresponding local factor. Its source laws identify the chart local ring with the local-ring label of the localized arithmetic vector bundle, identify each chart log-volume with the associated direct-summand log-volume, compute each H_β as the sum of these chart-factor log-volumes, and compute the cover as the finite sum of charted cell log-volumes. Lean then derives the vector-bundle local-factor source. Thus the regions entering the direct-product cover are no longer an arbitrary family parallel to the bundle data; they are read from local-ring localization charts tied to the same localized vector bundles. The chart/bundle matching has also been isolated at the individual factor level. The new

```
IUTStage1LocalRingVectorBundleFactorCalibrationSource
```

records, for a fixed (β, v), that the chart lies over the same local ring as the localized arithmetic vector bundle and that the chart region's log-volume is the corresponding direct-summand log-volume. The aggregate

`IUTStage1Remark395CalibratedLocalRingCharted`
`VectorBundleHullCoverSource`

then builds the previous charted direct-product source from these per-factor calibrations, together with the remaining cell factorization and cover additivity laws. This separates the local-ring/direct-summand calibration obligation from the finite direct-product geometry obligation. The cover additivity law has now been separated from the calibrated charted source as well. The declaration

`IUTStage1Remark395FiniteAdditiveCalibrated`
`LocalRingChartedVectorBundleHullCoverSource`

keeps the same per-factor calibrations and cell-factorization law, but replaces the charted cover log-volume equality by a finite-additive hull log-volume source. Distinct direct-product cells are proved disjoint from the local-factor separation condition, and Lean applies finite additivity to derive

$$\mu_{\log}\left(\bigcup_{\beta} H_\beta\right) = \sum_{\beta} \mu_{\log}(H_\beta).$$

Thus the cover equality consumed by the charted vector-bundle route is no longer an independent field of that route. The per-cell direct-product equality has now been factored as well. The new source declaration

`IUTStage1DirectProductCellLogVolumeSource`

records the Step (xi-d) tensor/direct-product log-volume law for one finite localization cell:

$$\mu_{\log}\left(\bigcap_{v} H_{\beta,v}\right) = \sum_{v} \mu_{\log}(H_{\beta,v}).$$

The aggregate

`IUTStage1Remark395ProductLogVolumeFiniteAdditive`
`CalibratedLocalRingChartedVectorBundleHullCoverSource`

therefore no longer takes the cell-factorization equality as a bare field. It derives the equality for each H_β from the single-cell product log-volume source, proves disjointness from local-factor separation, applies finite additivity to the direct-product cover, and then projects to the previous finite-additive charted source. This source has now been lowered once more to a tensor/direct-sum datum for the actual charted cells. The declaration

`IUTStage1FiniteTensorDirectSumLogVolumeSource`

records a finite tensor-product region, its identification with the corresponding direct-product cell, the local factor log-volumes, and the finite direct-sum formula for the tensor-product log-volume. From these data, Lean derives the single-cell law

$$\mu_{\log}(H_\beta) = \sum_{v} \mu_{\log}(H_{\beta,v}).$$

The aggregate

```
IUTStage1Remark395TensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
```

assigns such a tensor/direct-sum source to each charted localization cell, derives the cell log-volume equalities, proves the same pairwise-disjoint cover additivity, and projects to the finite-additive calibrated charted cover source. The tensor-cell source has now been lowered to compact-open local factor data. The declaration

```
IUTStage1LocalCompactOpenTensorFactorSource
```

records the local ring label, compact-open local factor region, and normalized local log-volume for one nonarchimedean tensor factor. A second declaration,

```
IUTStage1CompactOpenTensorRegionDirectSumSource
```

records the tensor-product region obtained from a finite family of these compact-open factors, identifies it with the simultaneous local-factor cell, and proves the finite direct-sum log-volume formula. Lean then constructs `IUTStage1FiniteTensorDirectSumLogVolumeSource` from this compact-open direct-sum datum, including the chart-region transport needed when the compact-open factor regions are identified with the already charted local factor regions. At the packet level,

```
IUTStage1Remark395CompactOpenTensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
```

requires one such compact-open tensor direct-sum construction for each charted localization cell, proves that the compact-open factor regions and local-ring labels match the charted local factors, constructs the tensor-packet source, and then derives the same finite-additive cover equality and bundle-log-volume projection. The compact-open tensor factor source has now been lowered to a nonarchimedean local normalized log-volume datum. The declaration

```
IUTStage1NonarchimedeanLocalCompactOpenLogVolumeSource
```

records a local carrier, local-ring label, residue prime, finite-extension degree, integral structure, compact-open subset, uniformizer-scaled subset, and their realization as regions in the current real-line copy. It includes the IUT IV Proposition 1.4 normalization laws needed in the present corridor: the integral structure has normalized log-volume zero, the normalized log-volume is the degree-normalized raw log-volume, and multiplication by a uniformizer shifts the normalized log-volume by $-\log(p)$. Lean constructs `IUTStage1LocalCompactOpenTensorFactorSource` from this datum. The finite direct-sum layer

```
IUTStage1NonarchimedeanLocalTensorRegionDirectSumSource
```

then constructs the compact-open tensor-region source from a finite family of these local nonarchimedean sources and the direct-sum tensor-region formula. Finally,

```
IUTStage1Remark395NonarchimedeanLocalTensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
```

uses one such local tensor source for each charted localization cell, verifies the local compact-open regions and local-ring labels against the charted factors, constructs the compact-open tensor-packet source, and inherits the finite-additive cover and bundle-log-volume conclusions. The local nonarchimedean source has now been lowered once more to the finite-extension Haar normalization boundary of IUT IV, Proposition 1.4. The declaration

`IUTStage1FiniteExtensionHaarCompactOpenLogVolumeSource`

records the raw Haar/log-volume function on local compact-open subsets of a finite extension of $\mathbb{Q}_p$, its degree-normalized version, the realization of local subsets as real-line regions, the integral structure $\mathcal{O}_K$, and the uniformizer-scaled compact-open subset. Its endpoint checks the source-paper normalizations

$$\mu_{\log}(\mathcal{O}_K) = 0, \qquad \mu_{\log}(p \cdot U) = \mu_{\log}(U) - \log(p),$$

and constructs `IUTStage1NonarchimedeanLocalCompactOpenLogVolumeSource`. The declaration

`IUTStage1FiniteExtensionTensorProductDirectSumDecompositionSource`

records the IUT IV direct-sum decomposition step for finite tensor products over $\mathbb{Z}_p$: the tensor-product region is identified with the simultaneous local compact-open cell, and its normalized log-volume is the finite sum of the degree-normalized Haar log-volumes of the direct summands. This source constructs the nonarchimedean local tensor-region direct-sum source. At the packet level,

```
IUTStage1Remark395FiniteExtensionHaarTensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
```

assigns such a finite-extension Haar tensor decomposition to each charted localization cell and then constructs the nonarchimedean local tensor-packet source used by the Step (xi) cover route. The finite-extension Haar source has now been lowered to compact-open topology provenance. The declaration

`IUTStage1FiniteExtensionCompactOpenTopologyHaarNormalizationSource`

records the compact-open predicate on the local carrier, closure of compact-opens under finite intersections, the ring-of-integers compact-open, the chosen compact-open subset, a uniformizer action preserving compact-opens, the raw Haar measure, and the raw and degree-normalized Haar log-volume functions. Its endpoint verifies compact-open provenance for $\mathcal{O}_K$, U, and $p \cdot U$, checks $\log(\mu_{\mathrm{Haar}}(U))$ at the raw level, and constructs `IUTStage1FiniteExtensionHaarCompactOpenLogVolumeSource`. The tensor layer

`IUTStage1CompactOpenTopologyTensorProductDirectSumSource`

constructs the finite-extension tensor-product direct-sum source from these compact-open topology factors. Finally,

```
IUTStage1Remark395CompactOpenTopologyHaarTensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
```

feeds compact-open topology/Haar normalization data for each charted cell into the finite-extension Haar tensor-packet source and hence into the same finite-additive Step (xi) cover route. The local compact-open topology source has also been lowered one step further to a Haar-measure-backed source. The declaration

`IUTStage1AdditiveHaarCompactOpenNormalizationSource` uses mathlib's additive Haar-measure class on the local additive carrier, records openness and compactness for the ring of integers and selected compact-open subset, normalizes the ring of integers to Haar mass 1, and records the finite-extension uniformizer scaling law for real Haar mass. Lean then derives the raw logarithmic shift by $-[K : \mathbb{Q}_p] \log(p)$, divides by the finite-extension degree, and obtains the normalized IUT IV shift $\mu_{\log}(pU) = \mu_{\log}(U) - \log(p)$. The companion declaration

`IUTStage1AdditiveHaarTensorProductDirectSumSource`

feeds these Haar-backed local factors into the compact-open tensor/direct-sum source, so the tensor summation layer now has a route whose local inputs are actual additive Haar measures rather than arbitrary real-valued Haar functions. There is now a valuation-ball refinement below this Haar source. The declaration

`IUTStage1ProperUltrametricValuationBallTopologySource`

now starts from a proper ultrametric metric topology on the local additive carrier. The ultrametric inequality proves openness of positive-radius closed balls, while properness supplies compactness by the closed-ball compactness theorem. Thus the compact-open valuation-ball theorem is constructed from two recognizable local topological hypotheses rather than from separate open/compact fields. The companion declaration

`IUTStage1ProperUltrametricValuationBallAdditiveHaarNormalizationSource`

feeds this proper-ultrametric source into the ultrametric Haar route. The local ring input has also been sharpened from an abstract unit-ball label to the valued-field integer subring supplied by `mathlib`. The declaration

`IUTStage1ValuedFieldIntegerUnitBallSource`

works over a nontrivially normed ultrametric field and uses `Valued.integer.mem_iff` to prove

$$\mathcal{O}_K = \{x \mid \|x\| \leq 1\} = \overline{B}(0, 1).$$

It then identifies this valued-field integer subring with the radius-one ball in the proper-ultrametric valuation-ball topology. The companion declaration

`IUTStage1ValuedFieldIntegerProperUltrametric`
`HaarNormalizationSource`

uses $\mathcal{O}_K$ itself as the local-ring label and transports the Haar mass-one normalization from $\mathcal{O}_K$ to the radius-one valuation ball before projecting into the proper-ultrametric Haar normalization route. There is now a base local-field specialization of this statement. The declaration

`IUTStage1PadicIntegerUnitBallSource`

works over $\mathbb{Q}_p$, identifies $\mathcal{O}_{\mathbb{Q}_p}$ with `PadicInt.subring` p, and proves that its underlying set is the image of the coercion $\mathbb{Z}_p \hookrightarrow \mathbb{Q}_p$. This also records the local-ring instance on $\mathbb{Z}_p$, so the ring-of-integers object is no longer only the abstract valued-field subring. The base residue quotient has now also been made concrete. The declaration

`IUTStage1PadicIntegerResidueQuotientSource` uses the mathlib reduction map `PadicInt.toZMod` : $\mathbb{Z}_p \to \mathbb{Z}/p\mathbb{Z}$, proves that its additive kernel is the maximal ideal of $\mathbb{Z}_p$, and applies the first isomorphism theorem to identify

$$\mathbb{Z}_p/\ker(\texttt{toZMod}) \simeq_{\text{add}} \mathbb{Z}/p\mathbb{Z}.$$

Transporting the $\mathbb{Z}/p\mathbb{Z}$-vector-space structure across this additive equivalence gives a one-dimensional residue module, and Lean proves

$$\#(\mathbb{Z}_p/\ker(\texttt{toZMod})) = p = p^{\operatorname{finrank}_{\mathbb{Q}_p}\mathbb{Q}_p}.$$

The same source now also constructs

$$\texttt{Fin}(p^{[\mathbb{Q}_p:\mathbb{Q}_p]}) \simeq \mathbb{Z}_p/\ker(\texttt{toZMod})$$

and finite-index representatives whose quotient classes are definitionally the indexed residue classes. This upgrades the base-field residue quotient from a cardinality result to the representative data needed by the coset partition layer, without a supplied finite quotient equivalence. This has now been threaded into the base local coset/Haar boundary. The declaration

```
IUTStage1PadicBaseFieldResidueQuotientDilationCoset
  HaarCharacterNormalizationSource
```

constructs the additive equivalence

$$\mathbb{Z}_p \simeq_{\text{add}} \mathcal{O}_{\mathbb{Q}_p},$$

proves that the transported $\ker(\texttt{toZMod})$ is exactly the subgroup $p\mathcal{O}_{\mathbb{Q}_p}$, applies the first isomorphism theorem to build

$$\mathcal{O}_{\mathbb{Q}_p}/p\mathcal{O}_{\mathbb{Q}_p} \simeq_{\text{add}} \mathbb{Z}_p/\ker(\texttt{toZMod}),$$

and then supplies the finite quotient equivalence and representatives consumed by the quotient-coset partition/Haar-character source. Thus the base-field coset decomposition used to derive $\mu(p\mathcal{O}_{\mathbb{Q}_p}) = p^{-1}$ now starts from the concrete `PadicInt.toZMod` residue quotient rather than from a supplied quotient package. The companion declaration

```
IUTStage1PadicProperUltrametricHaarNormalizationSource
```

fixes the residue prime to p, the finite-extension degree to 1, and the uniformizer scale map to $x \mapsto px$, then projects to the valued-field integer Haar route. The Haar mass-one and p-scaling law are still carried as analytic local-field inputs, but their ambient field, local ring, residue prime, degree, and uniformizer map are now the actual p-adic ones. This base local-field source now also projects into the finite-extension Haar boundary itself by viewing $\mathbb{Q}_p/\mathbb{Q}_p$ as a finite extension. Lean constructs

```
IUTStage1PadicProperUltrametricHaarNormalizationSource.
  toPadicFiniteExtensionProperUltrametricHaarNormalizationSource
```

and proves that its valued-field integer projection agrees with the direct $\mathbb{Q}_p$ valued-field Haar source. Thus the finite-extension degree appearing in the normalization is not an independent parameter at the base place: it is $\operatorname{finrank}_{\mathbb{Q}_p}(\mathbb{Q}_p) = 1$, with the same base-prime dilation $x \mapsto px$. The declaration

```
IUTStage1UltrametricValuationBallTopologySource
```

starts from an ultrametric metric topology on the local additive carrier and proves that every positive-radius closed valuation ball $\{x \mid d(x, 0) \leq r\}$, $r > 0$, is open. It still carries compactness of these balls as the local compactness/properness input, but the openness part of the compact-open assertion is now derived from the nonarchimedean triangle inequality rather than supplied independently. The companion declaration

`IUTStage1UltrametricValuationBallAdditiveHaarNormalizationSource`

projects this ultrametric source into the positive-radius topology-backed Haar route. The declaration

`IUTStage1PositiveRadiusValuationBallTopologySource`

isolates the local-field topological theorem that every positive-radius valuation ball $\{x \mid \|x\|_v \leq r\}$, $r > 0$, is compact-open. The topology-backed declaration

`IUTStage1ValuationBallTopologyAdditiveHaarNormalizationSource`

then derives the unit ball and selected ball compact-open predicates from this single positive-radius theorem before projecting to the older additive Haar normalization route. Thus the source boundary has moved from four separate open/compact fields for two balls to one uniform valuation-ball compact-open theorem. The declaration

`IUTStage1ValuationBallAdditiveHaarNormalizationSource`

records a nonarchimedean valuation-norm function on the local carrier and defines the ring of integers as the unit ball $\{x \mid \|x\|_v \leq 1\}$. The selected local compact-open subset is likewise a valuation ball $\{x \mid \|x\|_v \leq r\}$ for a positive radius r, and the uniformizer operation is the image under a pointwise scale map. Lean derives the compact-open predicate as `IsOpen` $\wedge$ `IsCompact`, proves closure under finite intersections in a T_2 space, and constructs the additive Haar normalization source from the valuation-ball data. The companion tensor source

`IUTStage1ValuationBallTensorProductDirectSumSource`

feeds valuation-ball local factors into the additive Haar tensor/direct-sum source. The valuation-ball route now also reaches the charted Step (xi) cover boundary itself. The new declaration

`IUTStage1Remark395ValuationBallHaarTensorPacketFiniteAdditive`
`CalibratedLocalRingChartedVectorBundleHullCoverSource`

assigns to each charted localization cell a valuation-ball tensor/direct-sum source, proves that every local charted region and local-ring label is realized by the corresponding valuation-ball factor, and then constructs the existing compact-open topology/Haar tensor-packet cover source. Its endpoint exposes the valuation-ball identities for $\mathcal{O}_K$ and U, compact-openness of the selected valuation balls, the finite direct-product cell log-volume formula, the finite-additive cover sum, and the inherited bundle-log-volume projection. Thus the current IUT IV local-volume corridor now proceeds from valuation balls to additive Haar normalization, then to compact-open topology, finite-extension Haar log-volume, tensor/direct-sum summation, and finally the charted Step (xi) cover route. The local chart calibration in this valuation-ball route has now been pushed down another level. The per-factor declaration

`IUTStage1ValuationBallVectorBundleFactorCalibrationSource` constructs the local-ring chart from a valuation-ball Haar factor itself: the chart local ring is the valuation-ball factor's local ring, and the chart region is the realized selected valuation ball. Its source law identifies the localized vector-bundle direct summand with the normalized Haar log-volume of that valuation ball, after which Lean derives the ordinary chart/direct-summand calibration source. The packet-level declaration

```
IUTStage1Remark395ValuationBallFactorCalibratedHaarTensorPacket
  FiniteAdditiveCalibratedLocalRingChartedVectorBundleHullCoverSource
```

uses these per-factor valuation-ball calibrations to build the previous valuation-ball charted-cover source. Thus the factor chart is no longer a separate object merely matched to valuation-ball data; at this boundary it is constructed from the valuation-ball compact-open region. This same factor-calibrated valuation-ball source now also projects into the newer finite-cover-additive/tensor-measure-backed cover corridor. The projection uses the broad finite-additive law only to construct the selected family-local finite-cover certificate, then Lean derives the tensor-cell cover sum

$$\mu_{\log}\Big(\bigcup_{\beta} H_\beta\Big) = \sum_{\beta} \mu_{\log}^{\text{tensor}}(H_\beta)$$

and reads the adjusted Ob3/Ob5 determinant equalities from that constructed finite-cover source. This connects the valuation-ball Haar factor route to the current narrower cover-audit path without treating the tensor-cover theorem as an independent payload. The record-canonical valuation-ball handoff has also been tightened at the source-core boundary. The theorem

```
IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  .ofValuationBallFactorCalibratedHaarTensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
  _toSourceCoreAdjustedLogVolumeSource_eq
```

proves that, once the valuation-ball possible-image family is synchronized with the Theorem 3.11 record family, the record-canonical Ob3/Ob5 source forgets back to exactly the valuation-ball adjusted determinant log-volume source. Downstream principal valuation-ball finite-divisor sources can therefore use this equality as a derived synchronization, rather than carrying the valuation-ball Ob3/Ob5 equality as an unrelated payload. This summand-calibrated source now feeds the milestone-facing constructed Step (xi) source directly. The new constructor

```
IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  .toConstructedHolomorphicHullDeterminantSource
```

first derives the record Ob3/Ob5 compatibility object from the finite summand formula, then uses that derived compatibility to build `IUTStage1Remark395ConstructedHolomorphicHullDeterminantSource`. The experiment endpoint

```
remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  _toConstructedHolomorphicHullDeterminantSource_endpoint
```

returns the finite-summand equality, the determinant and normalized determinant identifications, and the chain

$$\mu_{\log}(q) \le \det_{\log}(P_B) \le \Theta_{\text{signed}},$$

so the constructed Step (xi) route no longer receives the canonical hull–determinant compatibility as an independent input in this constructor. This constructed-source layer now also has an exact-theta refinement,

```
IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  .toConstructedHolomorphicHullDeterminantSource
  OfThetaEqFamilyHullLogVolume.
```

For an arbitrary q-region $q \subseteq P_{\mathrm{rec}}$ contained in the record possible-image union, this constructor derives the Ob4 normalized tensor-power bound from

$$\theta_{\mathrm{signed}} = \mu_{\log}(\phi(P_B)),$$

using the bounded-family hull/determinant source projected from the Ob3/Ob5 compatibility. Its public endpoint records the exact-theta equality, the derived tensor-power inequality, the adjusted-summand-to-determinant identifications, the two-step hull/determinant chain, and $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$. Thus the general constructed Step (xi) source has the same no-free-tensor-bound interface that was previously only available after choosing a possible-image q-region; the package measure calibration and the record-canonical bridge equality remain explicit source obligations. There is now also an obligations-backed form of this arbitrary-q exact-theta constructor,

```
IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  .toConstructedHolomorphicHullDeterminantSource
  OfThetaEqFamilyHullLogVolumeFromObligations.
```

It projects the package hull/determinant bridge equality, q-pilot positivity, and source normalization from one `IUTStage1SourceHullDetObligations` object. The endpoint still exposes the synchronization identifying that obligations-backed bridge datum with the record-canonical Remark 3.9.5 hull/tensor-power construction for the chosen q-region. Thus, at the general constructed-source level, the remaining explicit inputs are the measure calibration, the exact-theta family-hull identity, q-containment in the record possible-image union, and this single bridge-synchronization statement. The measured Ob3/Ob5 adjusted source now removes that separate measure calibration input as well. Its constructor

```
IUTStage1MeasureCalibratedRemark395RecordOb3Ob5Adjusted
  DeterminantLogVolumeSource.toConstructedHolomorphicHull
  DeterminantSourceOfThetaEqFamilyHullLogVolumeFromObligations
```

projects the package measure equality from the measured family-hull source, derives the Ob4 tensor-power bound from $\theta_{\mathrm{signed}} = \mu_{\log}(\phi(P_B))$, and keeps only the q-containment, exact-theta identity, obligations object, and record-canonical bridge synchronization as explicit parameters. There is also a selected-possible-image specialization,

```
IUTStage1MeasureCalibratedRemark395RecordOb3Ob5Adjusted
  DeterminantLogVolumeSource.toPossibleImageConstructed
  HolomorphicHullDeterminantSourceOfThetaEqFamilyHullLogVolume
  FromObligations.
```

Here the q-region is fixed to `recordThetaPossibleImage record qChoice`, and Lean derives its containment in the Theorem 3.11 possible-image union from the record's choice-indexed union constructor rather than receiving a separate q-containment proof. The same summand source now also feeds the constructor-built possible-image Step (xi) source used by the finite-divisor-facing corridor. The constructor

`IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource`
`.toPossibleImageConstructorBuiltHolomorphicHullDeterminantSource`

fixes the q-pilot region to a chosen Theorem 3.11 possible image, derives its record-union containment, derives the Ob3/Ob5 compatibility from the finite summand formula, and then builds `IUTStage1PossibleImageConstructorBuiltHolomorphicHullDeterminantSource`. Its experiment endpoint

`remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource`
`_toPossibleImageConstructorBuiltHolomorphicHullDeterminantSource_endpoint`

therefore exposes the constructor-built source endpoint, the Step (xi) bridge-inequality audit, the derived compatibility object, and the summand-to-determinant equalities at one review boundary. This removes one more independent compatibility certificate from the public constructor-built route, although the package measure calibration and record-canonical hull/determinant bridge equality are still explicit inputs. The finite-summand source now also has an exact-theta family-hull constructor,

`IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource`
`.toPossibleImageFamilyHullExactThetaHullDetObligationsBackedSource`.

Here the record Ob3/Ob5 family hull is still derived from the adjusted-summand identity, but the Ob4 tensor-power bound used in the record-canonical Step (xi) bridge is no longer a separate route input. It is derived from the exact-theta source equality

$$\theta_{\text{signed}} = \mu_{\log}(\phi(P_B)).$$

The corresponding experiment endpoint exposes the selected possible-image q-region, the package hull/determinant bridge, q-positivity, normalization, the derived weighted determinant source, the adjusted-summand-to-determinant equalities, the derived tensor-power bound, and $q_{\text{signed}} \leq \theta_{\text{signed}}$. The measured adjusted-summand source

`IUTStage1MeasureCalibratedRemark395RecordOb3Ob5`
`AdjustedDeterminantLogVolumeSource`

packages the same finite Ob3–3 adjusted-summand construction together with the package measure equality for the derived record family hull. Its exact-theta endpoint first projects the measured family-hull source and then invokes the possible-image constructor above. Thus, in this route, the measure calibration $\mu = \mu_{\log}(\phi(\cup_B P_B))$ is no longer a loose endpoint argument; it is part of the source-facing adjusted-summand object. This adjusted source is now threaded through the target-charted Hodge Step (xi) layer as well. The declarations

`IUTStage1TargetChartedHodgeMeasureCalibrated`
`AdjustedSummandFamilyHullSource`,
`IUTStage1TargetChartedHodgeMeasureCalibrated`
`AdjustedPossibleImageHullDetObligationsBackedSource`

project to the older measured determinant routes only internally. Their experiment endpoints keep the finite adjusted-summand identity

$$\mu_{\log}(\phi(P_B)) = \sum_{\gamma} (\text{localization}_{\gamma} - \text{structure sheaf}_{\gamma}) = \det_{\log}$$

visible next to the Hodge theta-degree calibration, the selected Theorem 3.11 possible-image q-region, and the obligations bridge equality. The adjusted-summand payload has also been moved into the constructor-backed Step (xi) route. The source

```
IUTStage1TargetChartedHodgeConstructorBacked
  MeasureCalibratedAdjustedPossibleImageHullDetSource
```

uses the record-canonical Theorem 3.11 hull/tensor-power constructor to generate the possible-image hull/determinant obligations, while its determinant input is projected from the measured adjusted-summand family-hull source. Its endpoint exposes, in one checked proposition, the selected q-region, the measure calibration, the two finite adjusted-summand equalities

$$\mu_{\log}(\phi(P_B)) = \sum_{\gamma}(\text{localization}_\gamma - \text{structure sheaf}_\gamma) = \det_{\log},$$

the equality between generated obligations and the constructor hull/determinant data, the record-canonical bridge equality, q-positivity, normalization, and $q_{\text{signed}} \leq \theta_{\text{signed}}$. The all-in-one route

```
IUTStage1TargetChartedHodgeIPLConstructorBacked
  MeasureCalibratedAdjustedPossibleImageRouteSource
```

then combines this constructor-backed adjusted source with the Hodge/IPL construction. Its audit projects internally to the constructor-backed measured route, but keeps the adjusted-summand Step (xi) audit as public provenance. The actual log-volume bridge has now been separated as a named source object:

`IUTStage1Remark395HullDeterminantBridgeSource`.

It takes q-region containment in the possible-image union, Ob3/Ob5 compatibility between $\phi(P_B)$ and the determinant-normalized log-volume, and the Ob4 normalized tensor-power bound by θ_{signed}, then proves the chain

$$\mu_{\log}(q_{\text{region}}) \leq \mu_{\log}(\phi(P_B)) = \overset{\text{norm}}{\det_{\log}} = \det_{\log} \leq \theta_{\text{signed}}.$$

The experiment-surface theorem `remark395HullDeterminantBridgeSource_endpoint` exposes this chain. The direct two-step normalized form is also exposed as `remark395HullDeterminantBridgeSource_normalizedBridge`, which records

$$\mu_{\log}(q_{\text{region}}) \leq \det_{\text{norm}} \leq \theta_{\text{signed}}.$$

The Step (xi) specialization `IUTStage1Remark395RecordHullDeterminantBridgeSource` fixes the family to the Theorem 3.11 possible images and the bound to `package.preLedger.thetaSigned`; its endpoint verifies the record-canonical q-to-determinant-to-theta comparison, while `remark395RecordHullDeterminantBridgeSource_normalizedBridge` exposes the corresponding record-canonical normalized bridge. The record bridge constructors from Ob3/Ob5 compatibility and from the adjusted Ob3-3 summand identity now expose this normalized two-step chain in their endpoint statements, so a reviewer no longer has to pass through a separate determinant-log endpoint to see the $\det_{\text{norm}}$ comparison. The constructed summand-charted Hodge/family-hull source now exposes this same comparison as a separate subgoal-5 audit:

`remark395SummandCalibratedStepXILogVolumeChainAudit`.

This endpoint starts from the selected Theorem 3.11 possible-image q-region, uses the record-family hull containment and monotonicity, applies the Ob3/Ob5 family-hull/determinant equality, and then uses the finite summand calibration to derive $\det_{\log} = \theta_{\text{signed}}$. Thus the theorem returns the explicit Lean chain

$$\mu_{\log}(q_{\text{region}}) \leq \det_{\log} \leq \theta_{\text{signed}},$$

together with the determinant summand-sum equality and the raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. The same synchronized finite-divisor boundary now also has an Ob5 quotient refinement:

```
synchronizedTargetChartedPossibleImageSummandHodgeFamilyHull
  IPLConstructionFiniteDivisorVerticalIQ_boundaryEndpoint
  WithStepXIOb5QuotientBoundaryAudit.
```

This audit keeps the no-C_Θ finite Step (x) packet and vertical-IQ equalities, the summand-calibrated Step (xi) chain $\mu_{\log}(q_{\text{region}}) \leq \det_{\log} \leq \theta_{\text{signed}}$, the synchronized reduction to the constructor-built Step (xi) source, and the Ob1–Ob5 quotient bridge for a second nonempty Theorem 3.11 possible-image choice. Thus the strongest current summand route exposes both the numerical log-volume chain and the Ob5 collapsed quotient images before the final C_Θ estimate is supplied. The product-hull Ob5–Ob7 route now has the analogous canonical-scale boundary

```
remark395ProductHullBackedConstructorBackedConstructed
  Ob3Ob5AdjustedHullDeterminantFiniteDivisorVerticalIQ
  _canonicalCThetaScaleDichotomy
  WithOb5Ob6Ob7SynchronizedAuditOfQPilotRegionNonempty.
```

This endpoint keeps the synchronized Ob5 quotient collapse, Ob6 Φ/Ξ-approximant comparison, and Ob7 prime-strip log-Kummer compatibility in the returned audit, but replaces the external numeric `thetaSigned_le_cTheta_absLogQ` input by the local scale

$$C_{\Theta,\text{can}} = \frac{\theta_{\text{signed}}}{-q_{\text{signed}}}.$$

Lean derives the corresponding bound from q-pilot positivity and then applies the same finite-divisor vertical-IQ route at this scale. As above, this is a source-attached local scale, not an identification with the global C_Θ-constant of the IUT papers. The same synchronized finite-divisor boundary now has a remaining-payload variant:

```
synchronizedTargetChartedPossibleImageSummandHodgeFamilyHull
  IPLConstructionFiniteDivisorVerticalIQ_boundaryEndpoint
  WithStepXIOb5QuotientRemainingPayloadBoundaryAudit.
```

This audit adds the constructor-built `ConstructorBuiltRemainingPayloadAudit` to the quotient-refined boundary. Consequently, the finite Step (x) packet/vertical-IQ equalities, the summand-derived chain $\mu_{\log}(q_{\text{region}}) \leq \det_{\log} \leq \theta_{\text{signed}}$, the Ob5 quotient independence, and the still-open Step (xi) source-paper payload now sit in one review object. The constructed Step (xi) source also has `ofRecordHullDeterminantBridgeSource`, so the bridge payload may now be passed as one source-level mathematical object instead of as unrelated compatibility and tensor-bound fields. There is now a sharper constructor `ofRecordOb3Ob5CompatibilitySource` that builds the constructed Step (xi) source from the record Ob3/Ob5 source equality plus the remaining Ob4 tensor-bound, measure-calibration, bridge-equality, positivity, and normalization data. The corresponding Step (xi) source

```
IUTStage1Remark395ConstructedHolomorphicHullDeterminantSource
```

combines this ϕ-operator with the Theorem 3.11 possible-image union, q-pilot containment in that union, the finite Ob3/Ob4 weighted determinant source, tensor-power normalization, and the record-canonical hull/determinant bridge equality. Lean proves inside this source object that the possible-image union, each possible image, and the q-pilot region lie in $\phi(P)$, and that the source object projects the possible-image family to the Ob5 family-hull/determinant log-volume source. In particular, Lean now also proves the family-hull log-volume equality with the weighted determinant and the family-union log-volume bound by that determinant before proving

$$\mu_{\log}(q_{\text{region}}) \leq \underset{\log}{\overset{\text{norm}}{\det}} \leq \theta_{\text{signed}}.$$

The experiment-surface theorem

```
Iut.Stage1.Experiments.boundarySignedEqualityOrStrictCTheta_from
  _remark395ConstructedHullDeterminantFiniteDivisorVerticalIQ
```

threads this constructed Remark 3.9.5 source through the finite-divisor vertical-IQ corridor. This ordinary form still accepts a chosen C_Θ and the corresponding numeric estimate as explicit inputs. The same source now has a canonical-scale companion

```
Iut.Stage1.Experiments.boundarySignedEqualityOrStrictCTheta_from
  _remark395ConstructedHullDeterminantFiniteDivisorVerticalIQ
  _canonicalCThetaScale,
```

where the constructed Remark 3.9.5 source object defines the scale as $\theta_{\text{signed}}/(-q_{\text{signed}})$. The source-level audit `ConstructedCanonicalCThetaScaleAudit` records the

$$\mu_{\log}(q_{\text{region}}) \leq \underset{\log}{\overset{\text{norm}}{\det}} \leq \theta_{\text{signed}} \leq C_{\Theta,\text{can}} \cdot (-q_{\text{signed}})$$

chain and proves the final ordered-real dichotomy without an external `thetaSigned_le_cTheta_absLogQ` hypothesis for this local scale. The new experiment-surface target

```
Iut.Stage1.Experiments.remark395ConstructedHolomorphic
  HullDeterminantSource_canonicalCThetaScaleChain
```

exposes this chain directly from the constructed source object, while `remark395ConstructedHolomorphicHullDeterminantSource_canonicalCThetaScaleAudit` exposes the full audit record. The source equality $\mu_{\log}(\phi(P_B)) = \det_{\log}^{\text{norm}}$, the Ob4 tensor-bound input, and the equality with the older algorithmic hull/determinant bridge remain source obligations rather than full arithmetic vector-bundle constructions, but the Ob3/Ob5 certificate and ordered log-volume comparison are now derived from named source-level objects.

The endpoint type `Stage1Comparison` is not a source proof: its comparison field is the Corollary-3.12-shaped conclusion, and the corresponding theorem is only a projection. The source-facing route must construct this endpoint from the ledger. The ledger now records the opaque output certification and ties the Θ-common bound used in the final inequality to `chartedContainer.apply certificate`, so the displayed charted container cannot be paired with an unrelated convenient bound. Throughout the current corridor, "upper ray" means the signed-log region $\{x \mid x \leq b\}$, not the ordinary order-theoretic ray $\{x \mid b \leq x\}$.

The largest engineering issue has been reduced to the remaining organization of the endpoint-route families. Eight mechanical splits have been made: `IUTStage1IUTIVAlgebra.lean` now carries the Remark 3.12 sign shadows and the conditional IUT IV real-algebra handoff estimates, while `IUTStage1FiniteLabels.lean` carries the finite-label and theta-root cusp-label source layer, and `IUTStage1StepX.lean` carries the Step (x) upper-ray corridor together with its boundary Frobenioid-shift diagnostics. The Gaussian and Hodge–Arakelov theta-evaluation shadow layer now lives in `IUTStage1Gaussian.lean`, and the Hodge/SHE support layer now lives in `IUTStage1HodgeSHE.lean`. The Theorem 3.11 source-choice and multiradial possible-image layer now lives in `IUTStage1Theorem311.lean`. The Step (xi) hull/determinant and IPL source layer now lives in `IUTStage1StepXI.lean`. The first endpoint audit family and nonarchimedean realified-Frobenioid/log-Kummer support layer now live in `IUTStage1FrobenioidShift.lean`; the original `IUTStage1Source.lean` file has now also been split into `IUTStage1EndpointAudit.lean`, for the nested LogVolumeChartAudit refinements, and `IUTStage1Source.lean`, for the public source-package audit surface. Low-risk future refactors are:

Structured endpoint route families	smaller route modules,
Generated dependency/audit report	separate checked target.

These refactors should be mechanical only: move declarations, preserve names where possible, and rebuild after each move.

10 Next Work

The next mathematical milestone is to construct, rather than assume, more of the source-facing SHE and Hodge-descent preservation data. The immediate route is:

$$\begin{array}{c} q^{j^2} \text{ finite Gaussian endpoint with pivotal restriction} \\ \Downarrow \\ \text{Gaussian and splitting monoid construction} \\ \Downarrow \\ \text{factored square/full-label preservation} \\ \Downarrow \\ \text{Hodge/SHE transport audit} \\ \Downarrow \\ \text{hull/log-volume comparison} \\ \Downarrow \\ C_\Theta \geq -1. \end{array}$$

The project should continue by replacing the present record fields for factored SHE preservation and Hodge-descent source-target alignment with the corresponding constructions from IUT I–III. The first source-derived bridge layer has now been added: the final finite-divisor vertical-IQ route can consume separated IPL, SHE, hull/determinant, and theta/Hodge calibration source components instead of an opaque route-alignment argument. The SHE component has been refined to a finite Hodge/SHE transport package that exposes the two theater sides, the finite label equivalence, full-label log-volume preservation, transported-average preservation, and the current history-separation guard. It now also has a typed source boundary that carries the Hodge-theater descent bridge and a typed allowed-forgetting witness separately from the synchronization-derived finite preservation data. The remaining work is to make each of those components itself follow from the relevant source-paper constructions, rather than supplying them as records. The IPL component can now be constructed at the public finite-divisor vertical-IQ endpoint from the finite Hodge/SHE transport together with a Theorem 3.11 IPL-link construction source. The endpoint no longer receives a free-standing IPL datum, source/target endpoint equalities, or a separate IPL log-volume transport source: the constructed input-prime-strip link gives the endpoint equalities definitionally, the source object aligns that constructed datum with the pre-ledger certificate, and log-volume preservation is read from the finite Hodge/SHE transport. There is now also a certificate-pinned variant of this boundary. It reads the IPL datum directly from the pre-ledger certificate and asks only for the two endpoint equalities of that certificate link. Consequently the public Hodge–Arakelov/history-separated route may avoid the stronger certificate-equals-constructed-IPL-datum assumption. The remaining source-paper task for this layer is to construct those certificate endpoint equalities, and eventually the input-prime-strip link itself, from the Theorem 3.11 IPL machinery. At the next source boundary, the same public endpoint can now receive a synchronized Hodge–Arakelov source object. This object carries the calibrated source evaluation, target evaluation, and canonical one-label degree preservation together, and it projects the finite Hodge/SHE transport used by the route. The allowed-forgetting predicate is fixed to the history-separation condition of the record's Hodge-theater descent bridge, so the public route no longer needs to be handed the finite transport record, a free square-weight profile, a separate theta-root source, a separate source/target evaluation synchronization equality, or a free permission predicate. These are constructed or projected from the synchronized Hodge–Arakelov and Theorem 3.11 source data. The actual Hodge-theater forgetting operation and the deeper construction of the theta-evaluation sources remain open source-paper obligations. The summand-charted Step (xi) input also has a synchronized target-charted form: the transported theta chart is calibrated against the target Hodge–Arakelov canonical one-label degree, and Lean derives the charted-theta/source-Hodge equality using the synchronized canonical-one preservation before projecting to the previous summand source. Thus the public route no longer has to receive that charted source-Hodge scalar equality as a primitive Step (xi) summand-calibration field. The same source now has a possible-image q-pilot refinement: the q-region is definitionally the chosen Theorem 3.11 possible-image region, and the $q \subseteq P_{\text{choice}}$ containment is projected by reflexive membership rather than supplied as an additional Step (xi) field. The experiment surface exposes this refinement as an audit endpoint before the finite-divisor route is applied: it reports the definitional q-region equality, the target-charted Hodge/summand identities, the hull-bridge equality, the sign and normalization side conditions, and the derived $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. This source now also projects to the generic possible-image side-conditioned holomorphic-hull/determinant source. In that projection, the Ob4 normalized tensor-power bound is derived from the Hodge/family-hull calibration, the record-canonical hull/determinant bridge equality is derived from the package hull-det obligations, and the q-pilot side conditions are read from the same obligations object. The experiment surface therefore has a raw constructed-IPL endpoint for this summand/family-hull route, proving $q_{\text{signed}} \leq \theta_{\text{signed}}$ before the later C_Θ numeric bound is invoked. The summand/family-hull q-comparison payload is now also promoted to a named audit proposition, bundling the possible-image q-pilot choice, determinant summand formula, family-hull log-volume identification, obligation-backed hull/determinant bridge equality, and raw signed comparison. The source-derived bridge constructor itself now has the same specialized input form: it consumes the calibrated Hodge–Arakelov synchronization, the constructed IPL source, and the synchronized target-charted possible-image summand/family-hull Step (xi) source, then performs the generic hull-source projection internally. The experiment surface exposes a bridge audit endpoint for this constructor, so the constructed bridge no longer has to be reviewed via an intermediate generic holomorphic-hull/determinant source. The finite-divisor vertical-IQ C_Θ endpoint now has the same direct bridge form: the public Step (xi) input is the synchronized target-charted possible-image summand source, and the boundary theorem builds the source-derived Hodge/SHE/IPL/hull bridge internally before entering the generic finite-divisor route. Thus this endpoint no longer has to reach the bridge through the older target-charted/summand projection chain. The same summand/family-hull route now has a no-C_Θ finite-divisor boundary returning the named q-comparison audit together with the finite Step (x) packet equalities and vertical-IQ target equalities, separating the summand Step (xi)/Step (x) provenance from the later numeric estimate. The target chart comparison can now be moved one layer earlier. A target-charted Hodge–Arakelov synchronization records that the transported pre-ledger theta chart is the target Hodge–Arakelov canonical one-label degree; Lean then derives the source theta-monoid reading from the synchronized canonical-one preservation. The associated possible-image summand Step (xi) source no longer carries this target chart/canonical-one equality as a hull-side calibration field; it keeps only the determinant-side finite summand equality plus the record-native possible-image hull data. The bridge audit and finite-divisor C_Θ-endpoint expose this refined source boundary. This refined target-charted route now also has a no-C_Θ finite-divisor boundary: it returns a named bridge audit, including the constructed IPL-datum checks, target-chart synchronization, log-volume preservation, raw signed comparison, and history-separation guard, together with the finite Step (x) packet equalities and vertical-IQ target equalities. The target-charted possible-image route now has a determinant-payload refinement. A dedicated Hodge/determinant family-hull source packages the record-native family-hull determinant source, the equality between the package measure and the hull log-volume measure, and the Hodge theta-monoid equality with the finite sum of adjusted determinant summands. The possible-image Step (xi) source consumes this single payload and projects the earlier family-hull, measure-calibration, and summand-equality fields internally before entering the same side-conditioned hull/determinant bridge. The bridge audit now has a determinant-refined raw finite-divisor boundary as well: it reuses the target-charted summand bridge audit, adds the family-hull/determinant log-volume equality, and carries the finite Step (x) packet and vertical-IQ target equalities before the C_Θ estimate is invoked. The experiment surface therefore reports the determinant log-volume equality in addition to the target theta chart, Hodge theta-monoid, possible-image q-region, hull-det bridge, and raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. The finite-divisor vertical-IQ experiment endpoint now exposes this determinant payload as the strongest public Step (xi) boundary in this corridor. The Hodge/IPL side of this route has also been bundled one layer earlier. A target-charted Hodge–Arakelov IPL-construction source now contains the target-charted Hodge synchronization and the constructed Theorem 3.11 input-prime-strip link. From these data Lean constructs the finite Hodge/SHE transport source, reads the IPL log-volume transport from that finite transport, and preserves the target/source log-volume equality without a separate transport equality field. The determinant possible-image bridge and finite-divisor vertical-IQ experiment endpoints now consume this single Hodge/IPL payload together with the Step (xi) determinant possible-image source. This packaged route now also has a no-C_Θ finite-divisor boundary that returns the packaged-source bridge audit together with the finite Step (x) packet equalities, vertical-IQ target equalities, and raw signed comparison before the numeric estimate is supplied. The Hodge/IPL construction source itself now exposes the history-separated finite Hodge/SHE transport-source audit: the source and target Hodge–Arakelov value sources, record-native forgetting permission, pointwise representative comparison level, identity coordinate equivalence, and Hodge-theater separation are visible before the constructed-IPL layer is formed. The strongest public target-charted route now packages these two sides into one source object before the finite Step (x) data are introduced. This all-in-one route source exposes the target theta chart comparison, constructed IPL-datum equality, possible-image q-region, determinant summand equality, family-hull/determinant equality, package hull-det bridge equality, log-volume transport preservation, positivity, normalization, and raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison as projections from a single source-facing payload. It now also has a divisor-packet no-C_Θ boundary returning a named all-in-one route bridge audit, the finite Step (x) packet equalities, the vertical-IQ target equalities, and the raw signed comparison; thus the source-derived Step (xi) bridge is reviewed as a single route object before the finite packet source is constructed. The same named route audit now also travels through the exact vertical-IQ finite-source boundary, alongside the log-Kummer vertical-IQ calibration-source equality, the finite-source equalities, and the exact target equalities. The same route now also projects explicitly to the possible-image side-conditioned holomorphic hull/determinant source, exposing the chosen Theorem 3.11 possible-image q-region, containment in the record possible-image union, the Ob4/tensor-power determinant bound, positivity, normalization, and the raw signed comparison before the finite Step (x) vertical-IQ data or the separate C_Θ numeric bound enter. On the Hodge/SHE side, the same route now exposes the finite transport audit separately: the structured-SHE flag, source/target Hodge theaters, coordinate transport, full-label log-volume preservation, transported-average preservation, and history-separation guard are route-level projections. On the Hodge/IPL side, the all-in-one route also projects directly to the finite Hodge/SHE plus constructed-IPL source object and exposes the target theta chart comparison, certificate-pinned IPL datum, constructed-IPL datum, log-volume preservation, allowed forgetful transport, and Hodge-theater history separation as route-level audits. The packaged target-charted Hodge–Arakelov/IPL source now exposes both the history-separated finite Hodge/SHE transport-source endpoint and the constructed IPL log-volume transport endpoint itself, including the source/target Hodge–Arakelov value sources, record-native forgetting permission, pointwise representative comparison level, identity coordinate equivalence, certificate-pinned IPL datum, input/output prime-strip link endpoints, finite Hodge-theater source and target sides, log-volume preservation, allowed forgetful transport, and history separation; the all-in-one route projects these source-level endpoints. Upstream of these transport audits, the route now also exposes the Hodge–Arakelov theta-evaluation layer: the canonical theta-root generator and full label, zero and canonical Gaussian degrees, the finite q^{j^2} square-weight degree formula, source full-label theta calibration, and the target chart/canonical-one comparison are all route-level projections. Between this Gaussian layer and the finite transport object, the route now exposes the factored SHE preservation constructor itself: canonical one-label preservation for the source and target Hodge–Arakelov theta evaluations constructs identity coordinate transport, coordinate-square preservation, full-label map and value preservation, transported-average preservation, and the Hodge-theater history-separation guard. The next route projection pins the finite Hodge/SHE transport source itself to the history-separated constructor from these source and target Hodge–Arakelov theta-evaluation sources: the source/target value sources, history-separated forgetful permission, pointwise representative comparison level, identity coordinate equivalence, and Hodge-theater history separation are exposed before the constructed-IPL layer is used. The constructed IPL log-volume transport source is now exposed at the same route boundary: the certificate-pinned IPL datum, constructed datum, input/output prime-strip link endpoints, finite Hodge-theater source and target sides, log-volume preservation, allowed forgetful transport, and history separation are all projected before the Step (xi) hull/determinant source enters. The same route boundary now also has a constructor-identity endpoint, `constructedIPLTransportSourceEqConstructedEndpoint`, which states that the projected IPL/log-volume source is exactly `ofTheorem311IPLLinkConstructionSource` applied to the finite Hodge/SHE transport and the carried input-prime-strip-link construction. The same boundary now records the provenance of the IPL log-volume equality: the IPL source log-volume is the source average of the finite Hodge/SHE transport audit, the IPL target log-volume is the transported target average, and their equality is the transported-average preservation theorem from the finite Hodge/SHE layer. The same constructed-IPL boundary now also exposes the choice-wise input-prime-strip links generated by the Theorem 3.11 link construction: for every possible output choice, the constructed link has source equal to the route's input prime strip and target equal to that choice's prime strip. This is exposed by the experiment suffix `thetaMonoidMatchedConstructedIPLChoiceLinkEndpoint`, giving a finite choice-indexed shadow of the IPL condition before Step (xi) hull data is used. The bundled exact vertical-IQ route now carries this same finite-average provenance together with the route-owned IPL datum, raw Step (xi) comparison, log-Kummer vertical-IQ target calibration, Frobenioid precision flag, and Step (x) upper-semi inequality before the external C_Θ numeric bound is used. The same bundled exact-route boundary now also carries the Remark 3.9.5(Ob5) family-hull guard: for two nonempty Theorem 3.11 possible images, both regions lie in the record family hull and have the same upper-semi quotient image, while the determinant/log-volume identifications, raw signed comparison, log-Kummer non-interference, and Step (x) upper-semi bound are projected from the same source package. This Ob5 guard is now also threaded through the Gaussian-front audit: the canonical Hodge–Arakelov theta label, zero Gaussian degree, square-weight formula, two arbitrary nonempty possible images, their common family-hull quotient image, and the exact Step (x) log-Kummer/upper-semi facts appear in one endpoint. The same all-in-one route source now feeds the Hodge direct-labelled finite-divisor vertical-IQ endpoint directly: instead of passing the generic source-derived bridge at the public Step (x) boundary, Lean projects the bridge from the target-charted Hodge/IPL/determinant possible-image route and then applies the Hodge-derived labelled-average and finite-divisor packet-local construction. On the Step (xi) side, the target-charted Hodge/determinant possible-image source now exposes the converted side-conditioned hull/determinant endpoint itself: the chosen Theorem 3.11 possible image, its containment in the record possible-image union, the tensor-power bound, q-positivity, normalization, and raw comparison are visible before the Hodge/IPL route source is assembled. The all-in-one route projects this source-level endpoint. The route also exposes the canonical hull/tensor-power bridge equality itself: the common-container hull/determinant bridge is identified with the record-canonical Theorem 3.11 possible-image hull data, weighted determinant, measure calibration, and tensor-power bound before the finite Step (x) vertical-IQ source or the external C_Θ numeric bound enter. This canonical bridge audit is now available at the target-charted Hodge/determinant Step (xi) source boundary itself; the all-in-one route endpoint projects it from that source. The same route now also exposes the source-paper-shaped Step (xi) passage from Theorem 3.11 possible images to the holomorphic hull and then to the log-volume bound: the endpoint records certification, the SHE datum, equality of the theta-pilot possible-image union with the record union, containment of that union in the applied holomorphic hull, the θ_{signed} log-volume bound, and the resulting raw comparison $q_{\text{signed}} \leq \theta_{\text{signed}}$. The route-level theorem is now a projection of the reusable possible-image Step (xi) source-level holomorphic-hull/log-volume endpoint; this endpoint is also available at the target-charted Hodge/determinant possible-image source boundary before the Hodge/IPL construction wraps it into the all-in-one route. The same route now also exposes the determinant summand/family-hull payload as its own audit: the target theta-chart readings, Hodge theta-monoid summand formula, family-hull/determinant log-volume equality, package measure-to-hull-log-volume calibration, tensor-power normalization, determinant tensor bound, and raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison are projected at the all-in-one route boundary before finite Step (x) or the external C_Θ bound enter. The same all-in-one route now exposes the raw comparison at the source boundary before the generic bridge is assembled: the certificate-pinned IPL datum, preserved IPL log-volume, source Hodge–Arakelov value source, chosen possible-image q-region, and $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison are all projected from the route source. The bridge-level raw comparison endpoint now uses this source-boundary comparison for its final inequality while still auditing the assembled bridge's IPL and Hodge/SHE projections. The same all-in-one route now has a direct Gaussian-to-Step (xi) audit endpoint: the theta-root canonical label and square-weighted Gaussian degree formula, the finite Hodge/SHE history-separated transport, the constructed IPL log-volume preservation, the constructed choice-wise input-prime-strip links, the chosen possible-image q-region with its union containment, the determinant summand calibration, the package hull/determinant bridge datum, and the raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison are projected from one route source before any finite Step (x) vertical-IQ input enters. This audit has also been promoted to a named route-source proposition, and the finite-divisor vertical-IQ boundary now has an audited endpoint that returns this Gaussian-to-Step (xi) payload together with the C_Θ dichotomy. Consequently the public finite-divisor route exposes the constructed Hodge–Arakelov/SHE/IPL/hull chain used for the raw comparison at the same boundary where the Step (x) packet data and external numeric bound enter. The finite-source form of this boundary now also has a no-C_Θ audited endpoint: before the ordered-real dichotomy is invoked, it returns the named Gaussian-to-Step (xi) audit together with the route-owned bridge, raw comparison, finite-source packet equalities, and vertical-IQ target equalities. The bundled finite exact vertical-IQ source now has the same audited C_Θ boundary: one source package returns the named Gaussian-to-Step (xi) route audit, the exact vertical-IQ Frobenioid-isomorphism precision flag, and the final dichotomy. This moves the strongest finite bad-local corridor's public endpoint closer to the intended source-derived chain while still leaving the external numeric C_Θ estimate explicit. The same bundled source now also exposes the corresponding raw audit endpoint before the C_Θ estimate is supplied: it returns the named Gaussian-to-Step (xi) audit, exact vertical-IQ precision, target log-Kummer calibration, the raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison, q-pilot log-Kummer non-interference, and the Step (x) upper-semi bound. This separates the source-derived Step (xi)/Step (x) corridor from the still-open numerical estimate in the public theorem surface. The same route also exposes the Remark 3.9.5(Ob5) bounded-family quotient mechanism: for any two nonempty Theorem 3.11 possible images, the endpoint proves that both lie in the record possible-image family hull and have the same image under the upper-semi quotient by that hull. The audit carries the determinant log-volume identification, the family-union-to-family-hull inequality, the normalized tensor-power equality, and the raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison, so the "do not choose a convenient possible image as canonical" guard is visible before the finite Step (x) vertical-IQ source and the external C_Θ bound enter. The route source now owns the assembled source-derived bridge via a `toSourceDerivedBridge` projection, and the finite-divisor vertical IQ theorem consumes that route-owned bridge instead of spelling out the generic bridge constructor at the Step (x) boundary. This separates the Step (xi) comparison and bridge assembly from the later finite Step (x) and C_Θ boundary. Its finite Step (x) boundary now also has a packet-source form: the public all-in-one route can consume a `NonarchimedeanFiniteDivisorPacketLocalSource` directly, so the divisor packet, mono-analytic theater, and packet-local log-volume equalities are audited inside the named Step (x) source object rather than repeated as loose endpoint arguments. The same public route now exposes the finite-source Step (x) boundary before applying the C_Θ numeric bound: the endpoint records the route-owned IPL datum and log-volume preservation, the raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison, the packet-local source log-volume chain, and the vertical-IQ target identifications from the supplied finite-source and target-source records. A tighter exact vertical-IQ variant now consumes a `NonarchimedeanThetaRootExactVerticalIQRealifiedEntrySource` instead of a separate target-source record: the endpoint projects the retained log-Kummer vertical-IQ calibration source, the Kummer/forgetting transfer guards, and the exact target before applying the same C_Θ boundary. The current strongest public endpoint packages these data into one bundled exact-route source object, so the route, upper-semi entry, finite Step (x) source, and exact vertical-IQ source are audited as projections before the external numeric bound enters. Its realified Step (x) correspondence endpoint now exposes the theta-root canonical generator, nonzero canonical label, Kummer/forgetting log-volume preservation, q-pilot log-Kummer non-interference, and the local-object upper-semi bound from the same bundled exact source. Its source-derived corridor endpoint additionally records the history-separated Hodge/SHE transport guard, certificate-pinned IPL datum, IPL log-volume preservation, possible-image hull containment, $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$, exact vertical-$IQ$ calibration, and Step (x) non-interference in one audit statement. The Gaussian-front corridor endpoint extends this by projecting the Hodge–Arakelov theta-evaluation source's canonical label, Gaussian degree normalization, square-weight formula, full-label compatibility, and target chart comparison before entering the same Hodge/SHE, IPL, hull, and Step (x) chain. A determinant-refined Gaussian corridor endpoint now keeps the Step (xi) determinant payload visible in this same bundled exact route: after the Hodge–Arakelov Gaussian data it projects the target-chart/source theta-monoid comparison, finite adjusted-summand formula, family-hull determinant log-volume equality, package measure calibration, determinant tensor bound, raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison, exact vertical-IQ calibration, q-pilot log-Kummer non-interference, and the Step (x) upper-semi estimate. The direct-labelled Step (x) finite-divisor route now also has a source-derived Hodge/SHE/IPL/hull form. Instead of taking the older route-alignment record, a free square-weight profile, and a separate theta-root source, this boundary projects the route alignment and canonical square-weight profile from the source-derived bridge and reads the theta-root package from the source Hodge–Arakelov side of the finite Hodge/SHE transport. It still constructs the finite-divisor packet-local source before entering the exact vertical-IQ target route. The source-calibrated vertical-IQ route now has this projection form as well: the log-Kummer packet source-calibration remains explicit, while the route alignment, canonical square-weight profile, and theta-root package are obtained from the source-derived bridge before the theta-root realified Step (x) source is constructed. The earlier realified-entry vertical-IQ audit endpoint has also been given the same source-derived projection boundary. Its public theorem still takes the realified Step (x) source and the exact vertical-IQ target as inputs, because those are the objects whose target-side $FMOD$ and packet-local audit facts are being reported, but the Hodge/SHE/IPL/hull alignment and canonical square-weight profile are now read from the source-derived bridge rather than supplied independently. The theta-root packet-local entry route has the same source-derived form. The entry membership, Kummer/forgetting compatibilities, packet-local source/mono-analytic equalities, and the two (Ind3) real equalities remain visible Step (x) inputs, while the theta-root source, canonical square-weight profile, and Hodge/SHE/IPL/hull route alignment are projected from the source-derived bridge. The ordered-real-line packet-local variant now factors through this same source-derived entry route: the ordered alignment supplies the packet-source and theta-target (Ind3) equalities, while the bridge supplies the route alignment, canonical profile, and theta-root source. The constant-$ZMod_l$ packet-normalized variant has the same projection boundary. The packet route still supplies the theta-average normalization after the theta-source guard is applied, and the ordered alignment still supplies the (Ind3) real equalities, but the Hodge/SHE/IPL/hull alignment, canonical profile, and theta-root source are projected from the source-derived bridge. The upper-semi $ZMod_l$ entry specialization now follows this same source-derived route: the upper-semi source packages the chosen nonarchimedean entry together with its membership proof, and the bridge supplies the route alignment, canonical profile, and theta-root source. The source-calibrated $ZMod_l$ route has also been projected through the source-derived bridge. The packet/source calibration object remains the Step (x) input, but the Hodge/SHE/IPL/hull alignment, canonical profile, and theta-root source are no longer separate public arguments. The labelled-average refinement of this source-calibrated route has the same projection boundary. The finite $ZMod_l$ labelled procession average and its packet-normalized comparison remain visible Step (x) data, while the route alignment, canonical profile, and theta-root source are supplied by the source-derived bridge. The source-aligned labelled-average route has also been given this bridge projection. Its packet-local source-alignment object remains the Step (x) input, while the source-derived bridge supplies the Hodge/SHE/IPL/hull alignment, canonical profile, and theta-root source. The finite-source labelled-average route now follows the same pattern. The finite divisor packet source still supplies the packet-local source alignment, while the source-derived bridge supplies the route alignment, canonical profile, and theta-root source before entering the source-aligned labelled-average boundary. The finite-divisor labelled-average compatibility route now has the same projection form. Its divisor/Frobenioid packet data is still bundled into the finite Step (x) source object, while the source-derived bridge supplies the route alignment, canonical profile, and theta-root source. The theta-labelled source-aligned variant now has the same bridge boundary. The labelled capsule average alignment still derives the theta-source average from finite labelwise averaging, while the source-derived bridge supplies the route alignment, canonical profile, and theta-root source. The named source-aligned vertical-IQ route now has the same projection boundary: the packet-local source-alignment object remains the Step (x) input, but the route alignment, canonical square-weight profile, and theta-root package are obtained from the source-derived Hodge/SHE/IPL/hull bridge. The packet-local vertical-IQ compatibility route has the same source-derived bridge form. This variant keeps the packet-local Kummer/forgetting and (Ind3) source alignments explicit, but obtains the Hodge/SHE/IPL/hull alignment, canonical square-weight profile, and theta-root source by projection from the separated source-derived bridge. It now also has the all-in-one target-charted Hodge/IPL/determinant possible-image source form: the public packet-local route consumes that source object directly and projects the source-derived bridge internally before entering the packet-local vertical-IQ comparison. The corresponding no-C_Θ packet-local endpoint is now exposed as well: it returns the all-in-one source-derived bridge audit, the exact vertical-IQ target audit, and the three packet-local Step (x) source equalities before the external numeric bound is introduced. A choice-linked variant of this no-C_Θ packet-local endpoint now exposes the constructed IPL choice links at the same boundary: for every possible output choice, the constructed link has source equal to the bridge's input prime strip and target equal to that choice's prime strip, before the raw signed comparison and exact vertical-IQ target audit are consumed. The experiment surface exposes this review target with the suffix `choiceLinkedBoundaryEndpoint`. The ordinary finite-divisor Step (x) route now has the same projection boundary: the finite-divisor packet-local source is still constructed from the divisor/Frobenioid packet data, while the Hodge/SHE/IPL/hull alignment, canonical square-weight profile, and theta-root package are read from the source-derived bridge. This gives the experiment surface a finite-divisor vertical-IQ theorem whose public route inputs no longer include the older alignment/profile/theta-root triple. At the Step (xi) boundary, the strongest public Hodge–Arakelov route can now consume the holomorphic-hull/determinant source object directly. This is a review-surface improvement rather than a closure of Step (xi): the source object still contains the holomorphic hull shadow, weighted determinant source, tensor-power bound, bridge equality, positivity, and normalization. The *Lean* source now also names this trust boundary explicitly as a constructor-built Step (xi) boundary audit: it records the selected possible-image q-region, the internally built Theorem 3.11 hull/determinant constructor and obligations object, the record-canonical hull/tensor bridge equality, the tensor-power determinant bound, q-positivity, normalization, and the raw signed comparison in one review target. The corresponding canonical-one constructed-IPL finite-divisor endpoint exposes this named audit before entering the packet/vertical-IQ boundary. The stronger charted Hodge-calibrated family-hull source now reduces to this constructor-built boundary as well. In this reduction, the possible-image q-region is fixed by the Theorem 3.11 record, θ_{signed} is read through the transported pre-ledger theta chart and the Hodge–Arakelov theta-monoid degree, the tensor-power bound is derived from the record-family-hull log-volume calibration, and the bridge datum, q-positivity, and normalization are projected from one hull/determinant obligations object. This gives the constructor-built audit a stronger source-facing predecessor rather than treating its bound and side conditions as free fields. The synchronized target-charted possible-image summand/family-hull source now feeds the same constructor-built boundary through a named reduction audit. This adds the target-chart/source theta-monoid comparison and the finite adjusted summand equality to the review surface before the data is forgotten to the charted Hodge/family-hull constructor audit. The record-hull finite-divisor endpoint can now consume the constructor-built possible-image Step (xi) source object directly. Thus the operation identifiers, selected possible-image q-region, determinant source, Ob3/Ob5 compatibility, measure calibration, tensor-power bound, hull/determinant bridge equality, q-positivity, and normalization are projected from one constructor source before the packet-local vertical-IQ comparison is invoked. The experiment wrapper `possibleImageConstructorBuiltRecordHullFiniteDivisorVerticalIQ_cThetaDichotomy` exposes this as the public C_Θ-dichotomy surface. The $q \subseteq P_{\mathrm{rec}}$ containment now has a stronger choice-linked source form, where $q \subseteq P_{\mathrm{choice}}$ is supplied and $q \subseteq P_{\mathrm{rec}}$ is derived from the possible-image union. This choice-linked source also projects to the formal Step (xi-e/f/g) upper-ray record: the q-pilot log-volume is shown to lie in the upper ray, to be bounded by the determinant log-volume, and to have matching input/output q-pilot readings. A stricter exact-theta variant now replaces the free tensor-power-bound field by the equality $\theta_{\mathrm{signed}} = \mathrm{NormVol}(\mathrm{Ob4Det})$, from which Lean derives the bound used by the older constructor. The side-conditioned exact-theta variant additionally projects q-pilot positivity and normalization from the existing Stage 1 side-condition object, rather than carrying these as independent Step (xi) fields. A further hull/det-data-backed variant projects the package bridge equality from the existing hull/determinant source-data object and then records only the compatibility of that stored bridge with the record-canonical Step (xi) construction. The exact-theta data-backed source now also has a possible-image-native form: its q-pilot region is definitionally the chosen Theorem 3.11 possible image, so $q \subseteq P_{\mathrm{choice}}$ is projected by reflexive membership before the same data-backed source is constructed. The experiment surface exposes both this source audit endpoint and the corresponding finite-divisor vertical-IQ C_Θ route. The newest record-native bounded-family variant replaces the loose exact tensor-power equality by the source-shaped assertion that θ_{signed} is the log-volume of the Theorem 3.11 possible-image family hull; the normalized tensor-power equality is then derived from the Ob5 determinant compatibility. That variant is now wired into the finite-divisor vertical-IQ endpoint chain through a dedicated source-derived bridge constructor, rather than through the older backed exact-theta source. The same bounded-family data-backed source now has a possible-image-native form: the q-pilot region is definitionally the selected Theorem 3.11 possible image, while the Ob5-shaped equality $\theta_{\mathrm{signed}} = \mathrm{Vol}_{\log}(H_{\mathrm{fam}})$ remains the source datum. The constructed-IPL finite-divisor endpoint and the experiment surface expose this stronger family-hull route. This route is now visible at the source-derived bridge boundary as well: a calibrated Hodge–Arakelov synchronization supplies the finite Hodge/SHE transport and theta/Hodge calibration, the constructed-IPL source supplies the input-prime-strip link, and the possible-image family-hull source supplies the fixed q-region and Ob5 log-volume comparison before the finite-divisor vertical-IQ theorem is applied. The same possible-image family-hull route now has an obligations-backed source form: the package hull/determinant bridge data, q-pilot positivity, and normalization are projected from one `IUTStage1SourceHullDetObligations` object, while the q-region remains definitionally the chosen Theorem 3.11 possible image. The experiment surface exposes both the bridge audit and the calibrated finite-divisor C_Θ-dichotomy for this obligations-backed route. A further Hodge-calibrated possible-image form removes the raw $\theta_{\mathrm{signed}} = \mathrm{Vol}_{\log}(H_{\mathrm{fam}})$ field from this public source: the equality is derived through the Hodge–Arakelov theta-monoid degree carried by the same synchronization object that constructs the finite Hodge/SHE transport. The bridge audit and finite-divisor endpoint now expose this Hodge-calibrated possible-image route. A charted variant now derives the package theta/Hodge equality through the transported pre-ledger theta chart before entering the same possible-image Hodge/family-hull route, and the experiment surface exposes its bridge audit and finite-divisor C_Θ-dichotomy endpoint. The remaining task is to construct the underlying fields from the source-paper holomorphic-hull and determinant operations. The experiment surface now also exposes the record-native Step (xi) hull/determinant corridor directly. One endpoint presents the constructed IPL variant from finite Hodge/SHE transport plus the record possible-image union and canonical Ob4/tensor-power hull data. A stronger endpoint presents the history-separated, certificate-pinned IPL variant: finite Hodge/SHE transport is built from the source and target Hodge–Arakelov theta-evaluation sources and the record's history-separated descent permission, while the IPL boundary is the certificate link source. The constructed-IPL record-hull endpoint now also has a side-conditioned form: q-pilot positivity and source normalization are projected from the shared Stage 1 side-condition object instead of appearing as independent Step (xi) hull/determinant fields. The same side-conditioned record-hull endpoint now has a possible-image form: the q-pilot region is no longer supplied as an arbitrary set with a separate containment proof in the record possible-image union. It is fixed to the possible-image region of a chosen Theorem 3.11 output, and Lean derives the union containment internally. This possible-image source now also has its own holomorphic-hull/log-volume endpoint. Before any Hodge/SHE/IPL route wrapper is used, Lean projects certification, the SHE datum, equality with the record possible-image union, containment in the applied holomorphic hull, the θ_{signed} log-volume bound, q-pilot positivity, normalization, and the raw comparison $q_{\text{signed}} \leq \theta_{\text{signed}}$ from the single possible-image Step (xi) source object. The target-charted Hodge/determinant possible-image source now also has its own holomorphic-hull/log-volume endpoint and canonical hull/tensor-power bridge endpoint. The first projects certification, the SHE datum, equality with the record possible-image union, containment in the applied holomorphic hull, the θ_{signed} log-volume bound, and the raw comparison. The second identifies the package common-container hull/determinant bridge with the record-canonical Theorem 3.11 possible-image bridge while retaining the Hodge-calibrated θ_{signed} equality, tensor-power bound, side conditions, and raw comparison at the Step (xi) source boundary. The source-derived bridge itself now has a calibrated Hodge–Arakelov synchronization constructor. Given one calibrated source/target Hodge–Arakelov synchronization object, a constructed Theorem 3.11 IPL link, and a Step (xi) holomorphic-hull/determinant source, Lean constructs both the finite Hodge/SHE transport and the source theta/Hodge log-volume calibration internally before entering the finite-divisor vertical-IQ route. The experiment surface exposes this as the calibrated-synchronization IPL-construction hull finite-divisor C_Θ-dichotomy endpoint. The same calibrated route now has side-conditioned and possible-image side-conditioned Step (xi) variants. In the side-conditioned form, q-pilot positivity and source normalization are projected from the Step (xi) hull source; in the possible-image form, the q-pilot region is also fixed by the chosen Theorem 3.11 possible-image output. Thus the strongest experiment-facing endpoint now receives one calibrated Hodge–Arakelov synchronization object, one constructed IPL link source, and one possible-image side-conditioned hull/determinant source, rather than separate Hodge/SHE transport, source-calibration, profile, theta-root, q-region, and side-condition witnesses. The possible-image side-conditioned source now also has a direct bridge constructor and experiment-facing bridge audit endpoint. The constructor keeps the chosen Theorem 3.11 possible-image q-region visible at the Step (xi) boundary, then projects internally to the generic holomorphic-hull/determinant source before assembling the calibrated Hodge/SHE, constructed-IPL, and hull/determinant bridge used by the finite-divisor route. This strongest possible-image route now also exposes the raw signed Step (xi) comparison separately from the C_Θ dichotomy. The theorem named `...qSigned_le_thetaSigned` on the experiment surface constructs the source-derived Hodge/SHE/IPL/hull bridge from the calibrated Hodge–Arakelov synchronization, the constructed Theorem 3.11 IPL link, and the possible-image side-conditioned hull/determinant source, then derives $q_{\mathrm{Signed}} \le \theta_{\mathrm{Signed}}$ without assuming the later numeric bound $\theta_{\mathrm{Signed}} \le C_\Theta \cdot (-q_{\mathrm{Signed}})$. A companion audit endpoint records that the q-pilot region is the chosen Theorem 3.11 possible-image region, that it lies in the record possible-image union, and that q-positivity and source normalization are projected from the shared side-condition object. This possible-image side-conditioned boundary now has an obligations-backed refinement. The new source object still fixes the q-pilot region to the chosen Theorem 3.11 possible image, but the package hull/determinant bridge datum, q-pilot positivity, and source normalization are projected from the shared `IUTStage1SourceHullDetObligations` object. The experiment surface therefore exposes a raw Step (xi) comparison whose audit payload identifies the package bridge with the obligations bridge datum before deriving $q_{\mathrm{Signed}} \le \theta_{\mathrm{Signed}}$. The same obligations-backed source has now been lifted through the finite-divisor vertical-IQ C_Θ-dichotomy endpoint. Thus the current strongest calibrated possible-image finite-divisor route no longer takes q-positivity, source normalization, or the package hull/determinant bridge alignment as separate Step (xi) fields; these are read from the shared hull/determinant obligations object before applying the existing finite-divisor corridor. The later numeric input $\theta_{\mathrm{Signed}} \le C_\Theta \cdot (-q_{\mathrm{Signed}})$ remains an explicit hypothesis of the dichotomy theorem. There is now also an IPL-link variant of this same obligations-backed finite-divisor endpoint. It replaces the constructed-IPL source, and its certificate-equals-constructed-datum alignment, by the certificate-pinned Theorem 3.11 IPL link source. This does not construct the input-prime-strip link machinery; rather, it isolates the remaining IPL trust boundary to the two endpoint equalities identifying the certificate link source and target with the input and output prime strips, while the log-volume preservation is still read from the finite Hodge/SHE transport. A further Gaussian-to-Step (xi) audit starts this same comparison at the Hodge–Arakelov theta-evaluation source itself: canonical theta-root data, Gaussian zero/canonical/square-weight degree facts, and full-label log-volume calibration are projected before the constructed finite SHE transport, constructed IPL link, possible-image hull source, and raw $q_{\mathrm{Signed}} \le \theta_{\mathrm{Signed}}$ comparison are read from the assembled bridge. This Gaussian audit now also has the obligations-backed possible-image form. Its endpoint starts from the same Hodge–Arakelov theta-evaluation source and constructed IPL link, but the Step (xi) source object additionally records that the package hull/determinant bridge is the datum supplied by the shared hull/determinant obligations. Thus the audit chain from theta-root/Gaussian data to $q_{\mathrm{Signed}} \le \theta_{\mathrm{Signed}}$ now exposes the obligation-projected hull bridge alignment at the same boundary as the raw comparison and finite-divisor dichotomy. The obligations-backed constructed-IPL synchronized route now also has a named Gaussian-to-Step (xi) audit proposition. This packages the theta-root and Gaussian data from the calibrated synchronization's source evaluation, the factored finite Hodge/SHE transport, constructed IPL transport, obligations-backed Step (xi) hull data, and raw signed comparison before the later Hodge–Arakelov history-separated wrappers are applied. The IPL-link refinement has now been lifted to these audit layers as well. The raw comparison endpoint $q_{\mathrm{Signed}} \le \theta_{\mathrm{Signed}}$ and the Gaussian-to-Step (xi) endpoint can both use the certificate-pinned Theorem 3.11 IPL link source together with the obligations-backed possible-image hull source. Thus the audit chain from theta-root/Gaussian data through finite Hodge/SHE transport to the Step (xi) comparison no longer needs the constructed-IPL certificate-alignment field in this variant; it still keeps the two certificate link endpoint equalities explicit. This synchronized certificate-pinned IPL-link Gaussian payload now also has a named audit proposition, parallel to the synchronized constructed-IPL audit but without the constructed-datum equality. The two synchronized audit surfaces now mark exactly where the formal route switches from a certificate-pinned link to a constructed input-prime-strip link. The finite-divisor C_Θ-dichotomy route now has the same unbundled Hodge–Arakelov source surface. Instead of receiving a calibrated Hodge–Arakelov synchronization record, the endpoint takes the source and target theta-evaluation sources, the canonical-one preservation equality, the certificate-pinned IPL link source, the obligations-backed possible-image hull source, and the separate theta/Hodge log-volume calibration. The finite Hodge/SHE transport and source-derived Hodge/SHE/IPL/hull bridge are assembled internally before the existing finite-divisor vertical-IQ theorem is applied. This removes one synchronization-bundle boundary from the strongest possible-image finite-divisor route, but it does not derive the later numeric input $\theta_{\text{Signed}} \le C_\Theta \cdot (-q_{\text{Signed}})$. This same possible-image route now has a source-calibrated Hodge–Arakelov variant. The bridge constructor and experiment-facing finite-divisor theorem consume a calibrated Hodge–Arakelov theta-evaluation source, project the full-label theta/Hodge log-volume calibration internally, and then construct the finite Hodge/SHE transport, certificate-pinned IPL transport, and obligations-backed possible-image hull bridge. Thus the strongest possible-image finite-divisor surface no longer needs a standalone `IUTStage1SourceThetaHodgeLogVolumeCalibration` argument in this variant; the remaining external Step (xi) inputs are still the certificate-pinned IPL link source, the obligations-backed possible-image hull source, the finite divisor/log-Kummer data, and the later numeric C_Θ bound. The raw Step (xi) comparison has now been exposed at this same source-calibrated boundary. The new $q_{\text{Signed}} \le \theta_{\text{Signed}}$ theorem and its audit endpoint start from the calibrated Hodge–Arakelov theta-evaluation source, the target evaluation, canonical-one preservation, the certificate-pinned IPL link source, and the obligations-backed possible-image hull source. The audit records the projected theta/Hodge full-label calibration, the selected Theorem 3.11 possible-image q-region and containment in the record possible-image union, the obligations hull/determinant bridge alignment, the IPL certificate endpoint equalities, q-positivity, and source normalization before deriving the raw signed comparison, still without using the later C_Θ numeric hypothesis. This source-calibrated boundary now also has a Gaussian-to-Step (xi) audit endpoint. It exposes the canonical theta-root generator and full label, zero/canonical/square-weight Gaussian degree equations, full-label calibration, factored SHE comparison level and coordinate equivalence, the certificate-pinned IPL-link log-volume preservation, possible-image hull containment and obligations bridge alignment, q-positivity, normalization, and the same raw signed comparison in one payload. This is the most direct current audit chain from the Hodge–Arakelov theta-evaluation source to $q_{\text{Signed}} \le \theta_{\text{Signed}}$ for the finite nonarchimedean possible-image route. This certificate-pinned Gaussian-to-Step (xi) payload is now also promoted to a named audit proposition, and its finite-divisor route has a no-C_Θ boundary endpoint returning that audit together with the finite-divisor Step (x) packet equalities, vertical-IQ target equalities, and raw signed comparison. This gives a stable audit surface for the pre-construction IPL-link route, without asserting equality to a separately constructed IPL datum. The same source-calibrated possible-image route now also has the constructed IPL form. The bridge constructor, finite-divisor dichotomy, raw comparison, q-comparison audit, and Gaussian-to-Step (xi) audit can consume an input-prime-strip link construction source. The construction projects the link-source and link-target endpoint equalities internally; the remaining certificate-facing IPL condition at this boundary is the equality between the pre-ledger certificate datum and the datum generated by the construction. Thus this route removes another certificate-pinned IPL-link assumption from the finite nonarchimedean possible-image corridor, while still treating the existence of the input-prime-strip construction itself as Theorem 3.11 source data. The constructed-IPL Gaussian-to-Step (xi) payload is now also promoted to a named audit proposition, so later finite-divisor endpoints can cite this theta-root/Gaussian, history-separated Hodge/SHE, constructed IPL, and obligations-backed hull chain without restating the full conjunction. This named audit now also records the definitional provenance of the route components themselves: the bridge's finite Hodge/SHE transport is the one constructed from the history-separated Hodge–Arakelov evaluations, its IPL transport is the one obtained from the Theorem 3.11 construction source, and its hull constructor is the one induced by the possible-image obligations-backed Step (xi) source. The same audit now exposes the finite SHE preservation facts used by that transport: the allowed forgetting predicate is the history-separated permission attached to the record's Hodge-theater descent bridge, and the transported full-label log-volume equality and transported-average equality are projected from the constructed finite Hodge/SHE transport. The constructed-IPL log-volume equality is now similarly sourced: the audit records that the route's IPL source log-volume is the finite Hodge/SHE source average and that its target log-volume is the transported target average, before using transported-average preservation to identify them. On the Step (xi) side, the same audit now exposes the weighted tensor-power determinant bound and the equality identifying the route hull/determinant bridge with the record-canonical possible-image hull/tensor-power construction. The constructed-IPL audit also projects the induced theta-pilot hull endpoint itself: its possible-image union is the record's theta possible-image union, this union lies in the constructed holomorphic hull, and the resulting hull has log-volume at most θ_{Signed}. The synchronized calibrated constructed-route audit now exposes the same Step (xi) provenance one layer earlier: the record-canonical hull/tensor-power bridge, the weighted tensor-power determinant bound, and the induced theta-pilot hull endpoint are projected directly from the obligations-backed possible-image source before the history-separated source-calibrated wrapper is introduced. The certificate-pinned IPL-link version of this synchronized audit now exposes the same Step (xi) projections as well, so the pre-construction and constructed-IPL branches have the same visible possible-image hull/determinant provenance before the final finite-divisor boundary is applied. The same synchronized audit layer now also exposes the finite Hodge/SHE transport facts themselves: the target Hodge source is the synchronized target evaluation, the history-separated forgetful transport is allowed, and full-label log-volumes are preserved across the finite Hodge/SHE transport. Both synchronized audits now also expose the SHE-to-IPL log-volume handoff directly: the finite Hodge/SHE transported average is preserved, the route's IPL source log-volume is the finite Hodge/SHE source average, and the route's IPL target log-volume is the transported target average. The same finite-divisor route now has a no-C_Θ boundary endpoint returning this named audit together with the finite-divisor Step (x) packet equalities, vertical-IQ target equalities, and raw $q_{\mathrm{Signed}} \leq \theta_{\mathrm{Signed}}$ comparison, separating the source-derived Step (xi)/Step (x) provenance from the later numeric estimate. The corresponding finite-divisor C_Θ-dichotomy endpoint now returns this named audit proposition together with the dichotomy, keeping the external numeric C_Θ estimate as the only additional input at that boundary. The experiment surface now also has synchronized finite-divisor review wrappers: they consume the single Hodge–Arakelov synchronization object and return the synchronized Gaussian-to-Step (xi) audit directly, together with the same raw finite-divisor boundary or C_Θ-dichotomy conclusion. This makes the finite bad-local endpoint display the route theta-root/Gaussian $\to$ finite Hodge/SHE $\to$ IPL $\to$ possible-image hull before entering the later source-calibrated audit wrapper. The same bundled synchronized source now has the remaining-boundary ledger `SynchronizedRemainingSourceBoundaryAudit`, exposed by the experiment suffix `synchronizedRemainingBoundaryAudit`. It packages the Gaussian-to-Step (xi) audit, constructed IPL log-volume endpoint, possible-image hull/determinant endpoint, finite Step (x) packet equalities, vertical-IQ target facts, and raw signed comparison as projections from one source object. This gives the pre-matched synchronized corridor the same review shape as the later matched and side-conditioned exact corridors. The certificate-pinned IPL-link synchronized source now has the analogous remaining-boundary ledger `SynchronizedIPLLinkRemainingSourceBoundaryAudit`, exposed by the experiment suffix `synchronizedIPLLinkRemainingBoundaryAudit`. It packages the IPL-link Gaussian-to-Step (xi) audit, the certificate-pinned link source, possible-image hull/determinant endpoint, finite Step (x) packet equalities, vertical-IQ target facts, and raw signed comparison, while deliberately leaving the construction of the IPL datum outside this pre-construction review surface. For review, these strongest finite-divisor endpoints are also exposed under the compact milestone prefix `sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ`, with boundary and C_Θ-dichotomy variants carrying the `WithConstructedRouteAudit` suffix. These wrappers are definitionally just the constructed-route endpoints above; their purpose is to make the current review boundary easy to locate without renaming or hiding the longer source-calibrated audit theorem. The compact milestone prefix now also has all-in-one route variants. The no-C_Θ wrapper with suffix `boundaryEndpointWithAllInOneRouteBridgeAudit` consumes the target-charted Hodge/IPL/determinant possible-image route source directly and returns the route bridge audit together with the finite Step (x) packet and vertical-IQ target equalities. The corresponding C_Θ-boundary suffix `cThetaDichotomyWithAllInOneRouteGaussianStepXIAudit` returns the all-in-one Gaussian-to-Step (xi) audit before applying the same external numeric C_Θ estimate. Thus the compact review surface can now point to the stricter route object that packages the Hodge–Arakelov synchronization, finite Hodge/SHE transport, constructed IPL handoff, and record-native possible-image hull/determinant payload in one source-facing input. The same conditional boundary now also has the stronger compact suffix `cThetaDichotomyWithAllInOneRouteBridgeAudit`. This endpoint returns the full route bridge audit next to the dichotomy, rather than only the Gaussian-to-Step (xi) subaudit, while still keeping the numeric C_Θ-upper estimate as an explicit external hypothesis. The target-charted experiment surface has parallel suffixes `routeBridgeAuditEndpoint` and `matchedRouteBridgeAuditEndpoint`; the latter records the theta-monoid-matched route bridge audit beside the same conditional C_Θ conclusion. The all-in-one route bridge audit now also carries the named `TargetChartedHodgeSHEIPLConstructionAudit` as a field, and the compact experiment suffix `allInOneRouteHodgeSHEIPLAudit` exposes it directly. This means the milestone-facing route bridge reports the theta-root/Gaussian evaluation, factored SHE preservation, finite Hodge/SHE transport, constructed IPL log-volume transport, and finite-transport log-volume provenance before the Step (xi) hull/determinant payload or finite Step (x) boundary is used. At the bridge-constructor level, the calibrated Hodge–Arakelov constructed IPL bridge can now consume the constructor-built possible-image Step (xi) hull/determinant source directly. The core constructor `ofCalibratedHodgeSynchronizationT11IPLConstructionAndPossibleImage ConstructorBuiltHullSources` exposes, through the experiment suffix `calibratedSynchronizationIPLConstructionPossibleImage ConstructorBuiltHull_bridgeEndpoint`, the constructed IPL datum, finite Hodge/SHE synchronization, constructor-built Step (xi) source endpoint, record-canonical Step (xi) audit, compact boundary audit, and remaining-payload ledger from the same source object. The same constructor-built Step (xi) source now separates its Ob1/Ob2 holomorphic-hull absorption passage as `ConstructorBuiltOb1Ob2HullAbsorptionAudit`, exposed by the experiment suffix `possibleImageConstructorBuiltHullSource_ob1Ob2HullAbsorptionAudit` and carried by the bridge endpoint. This audit records that the selected q-pilot region is a Theorem 3.11 possible image, lies in the record possible-image union, and is absorbed by the canonical holomorphic hull before determinant/tensor-power data enters. The constructor-built Step (xi) source now also separates its Ob3/Ob4 determinant/tensor-power passage as `ConstructorBuiltOb3Ob4DeterminantAudit`, exposed by the experiment suffix `possibleImageConstructorBuiltHullSource_ob3Ob4DeterminantAudit` and carried by this bridge endpoint. The audit records weighted determinant summation, determinant normalization, hull-log-volume compatibility, the naive Frobenius tensor-power bound, and the determinant upper-ray comparison before the remaining-payload ledger adds q-choice, record-canonical bridge, positivity, and normalization fields. The same boundary now separates the Ob3/Ob5 determinant-compatibility chain as `ConstructorBuiltOb3Ob5DeterminantCompatibilityAudit`, exposed by the experiment suffix `possibleImageConstructorBuiltHullSource_ob3Ob5DeterminantCompatibilityAudit`. This audit derives the canonical hull-log-volume equality with the determinant log-volume by composing hull-log-volume compatibility with the normalized weighted determinant and the Ob3/Ob4 determinant normalization theorem, so this equality is no longer presented as an independent remaining-payload input. The theta-monoid-matched all-in-one route now has the same stronger no-C_Θ boundary surface with suffix `boundaryEndpointWithMatchedRouteBridgeAudit`. Its first component keeps the matched Gaussian audit together with the projected all-in-one route bridge audit, so the matched finite-divisor vertical-IQ endpoint no longer drops the Hodge/SHE/IPL construction payload when it records that canonical one-label preservation was reconstructed from target/source theta-monoid degree equality. The all-in-one Gaussian-to-Step (xi) audit has also been strengthened at the Step (xi) boundary. Its public experiment endpoint with suffix `gaussianToStepXIHullDetAuditEndpoint` now projects the record possible-image union, containment in the constructed holomorphic hull, the holomorphic-hull log-volume bound, the family-hull/determinant and tensor-power equalities, and the record-canonical hull/tensor-power bridge equality from the same route audit. Thus the finite-divisor wrappers that return this audit now carry more than the final raw comparison: they expose the holomorphic-hull/determinant provenance used to obtain it. The same all-in-one route now also exposes the narrower determinant-source boundary via `targetChartedHodgeIPLDeterminantPossibleImageRoute_determinantSourceBoundaryAuditEndpoint`. This audit separates the two determinant payload fields still supplied at that boundary—the package measure calibration and the Hodge theta/summand equality—from the record-native family-hull facts that Lean derives: the record possible-image union, the family-hull/determinant equality, the family-union bound by the hull, tensor-power normalization, and the resulting tensor-power bound against θ_{Signed}. This has now been tightened by the endpoint `targetChartedHodgeIPLDeterminantPossibleImageRoute_calibratedDeterminantSourceEndpoint`: the same route determinant payload is first viewed through a calibration-backed source object, so the Hodge theta/summand equality is projected from the target-charted summand calibration rather than exposed as a raw determinant field. The package measure identification remains the explicit determinant measure boundary. The stricter determinant payload has also been pushed through the possible-image Step (xi) source boundary. The endpoint `targetChartedHodgeIPLDeterminantPossibleImageRoute_calibratedPossibleImageSourceEndpoint` shows that the chosen q-region, record possible-image union, target-charted summand calibration, family-hull determinant facts, theta/family-hull equality, bridge alignment, and raw comparison are all projected from a calibration-backed possible-image source object. There is now also an all-in-one route object whose Step (xi) field is this calibration-backed possible-image source: `targetChartedHodgeIPLCalibratedPossibleImageRoute_sourceEndpoint` returns the projected Gaussian-to-Step (xi) audit together with the route's IPL handoff, calibrated q-region, determinant measure calibration, target-charted summand facts, bridge alignment, and raw comparison. The finite-divisor boundary can now consume this calibrated route object directly as well. The compact suffixes `boundaryEndpointWithCalibratedAllInOneRouteGaussianStepXIAudit` and `cThetaDichotomyWithCalibratedAllInOneRouteGaussianStepXIAudit` project to the existing finite Step (x) machinery only after the public route input has been tightened to the calibration-backed possible-image source. The calibrated route now also has matching review surfaces with `CalibratedGaussianStepXIAudit` suffixes; these return the calibrated possible-image audit directly, rather than only the projected audit for the older determinant possible-image route shape. The same boundary has now been refined once more by a measure-calibrated route object. The new source `IUTStage1MeasureCalibratedRecordBoundedFamilyHullDetLogVolumeSource` packages the record-native family-hull determinant source together with the identification of the package ledger measure with the induced hull log-volume measure. The all-in-one

`IUTStage1TargetChartedHodgeIPLMeasureCalibratedPossibleImageRouteSource` threads this measured family-hull payload through the calibrated possible-image Step (xi) source, so the determinant payload no longer exposes the package measure equality as a direct field. The experiment surface now exposes `targetChartedHodgeIPLMeasureCalibratedPossibleImageRoute_toMeasureCalibratedGaussianStepXIAudit` and a matching finite-divisor boundary wrapper returning the measure-calibrated Gaussian-to-Step (xi) audit; both project to the older calibrated route only internally. The measure-calibrated boundary now has a constructor-backed refinement as well. The new constructor-backed possible-image source constructs its hull/determinant obligations via `ofRecordCanonicalHullTensorPowerOfQSubsetUnion`, using the record-canonical possible-image hull/tensor-power bridge. The matching all-in-one route therefore no longer carries an arbitrary possible-image hull/determinant obligations object at this boundary; it projects to the measure-calibrated route only after the constructor has produced those obligations. The experiment surface exposes this as a constructor-backed Gaussian-to-Step (xi) audit, a matching no-C_Θ finite-divisor boundary wrapper, and now also a matching finite-divisor C_Θ dichotomy wrapper. Thus the final dichotomy surface can display the constructor-backed Step (xi) hull/determinant route before applying the still-external numeric C_Θ estimate. The same calibration-backed route has now been pushed into the strongest bundled finite exact vertical-IQ source as well. The experiment-surface calibrated finite-exact wrappers, with suffixes `boundaryEndpointWithCalibratedRouteAudit` and `cThetaDichotomyWithGaussianStepXIAudit`, consume one calibrated route/exact Step (x) package and project to the older exact finite-route theorem only internally. Thus the exact vertical-IQ review boundary now displays the calibration-backed possible-image payload before the final external numeric C_Θ estimate is supplied. The constructor-backed refinement has now been pushed through this bundled finite exact vertical-IQ boundary too. Its exact route package exposes a constructor-backed Gaussian-to-Step (xi) audit, a source-derived corridor audit linking the constructor-produced hull/determinant data to the exact Step (x) calibration, and a matching C_Θ dichotomy wrapper. The final dichotomy can therefore be reviewed from a single package whose Step (xi) payload is constructor-backed before the exact vertical-IQ projection and the still-external numeric estimate are applied. The adjusted Ob3/Ob4 version of this exact route is now explicit as well. The source

```
IUTStage1TargetChartedHodgeIPLConstructorBacked
  MeasureCalibratedAdjustedPossibleImage
  FiniteExactVerticalIQRouteSource
```

keeps the adjusted-summand Step (xi) equalities

$$\text{familyHullLogVolume} = \text{adjustedSummandLogVolume} = \text{determinantLogVolume}$$

visible at the same exact vertical-IQ boundary. Its public experiment wrappers expose the adjusted Gaussian audit, the source-derived corridor endpoint, and the corresponding C_Θ-dichotomy endpoint; projection to the older constructor-backed exact route is used only to reuse the established Step (x) vertical-IQ endpoint. The same adjusted exact route now has a theta-monoid-matched Hodge/IPL version. The source

```
IUTStage1TargetChartedThetaMonoidMatchedHodgeIPL
  ConstructionConstructorBackedMeasureCalibrated
  AdjustedPossibleImageFiniteExactVerticalIQSource
```

stores the matched Hodge/IPL construction source and the adjusted constructor-backed Step (xi) source as public fields, then projects to the ordinary adjusted exact source only internally. Its audit keeps the target/source theta-monoid degree equality, the adjusted exact-theta Step (xi) audit, and the exact vertical-IQ facts together before the still-external numeric C_Θ estimate is supplied. This matched adjusted source now also projects into the obligations-backed record-canonical corridor. The audit

`MatchedAdjustedObligationsCorridorAudit`

retains the matched adjusted Gaussian/exact audit and the adjusted exact-theta Step (xi) audit, then adds the projected matched obligations-backed record-canonical Gaussian corridor and the constructor-built finite-divisor route audit. Thus the strongest matched obligations boundary no longer loses the Ob3/Ob4 adjusted-summand provenance when it reviews the downstream finite-divisor route. This exact boundary now also has an assembled-source variant. Instead of storing the all-in-one constructor-backed route as a field, the package stores the target-charted Hodge/IPL construction source and the constructor-backed possible-image source separately, then builds the route only as an internal projection. The review surface therefore shows the constructed IPL datum, the constructor-produced Step (xi) obligations, the raw comparison, and the exact vertical-IQ calibration in one source-derived corridor audit before the C_Θ estimate is supplied. The exact boundary has now been tightened once more to a record-canonical constructor source. This source does not store the packaged constructor-backed possible-image object; it stores the measured determinant payload, q-choice, hull/determinant operation identifiers, record-canonical bridge equality, q-positivity, and normalization inputs from which the Theorem 3.11 hull/determinant constructor produces the Step (xi) source. Its experiment wrapper displays these constructor inputs together with the constructed IPL datum and exact vertical-IQ calibration before the external numeric C_Θ estimate is used. There is now an intermediate side-condition-backed form of the same record-canonical exact boundary. The side-conditioned source in `IUTStage1FrobenioidShift` keeps the measured determinant payload, q-choice, operation identifiers, and record-canonical bridge equality explicit, but replaces the separate q-positivity and source-normalization fields by one `IUTStage1SourceSideConditions` object. Its experiment-surface wrappers display the same corridor and C_Θ-dichotomy payload as the direct record-canonical constructor route, while making the side-condition source a single audited dependency before projection to the older endpoint. An obligations-backed version of this record-canonical exact boundary now projects q-positivity, source normalization, and the package bridge-data alignment from a single `IUTStage1SourceHullDetObligations` object. It still exposes the measured determinant payload, q-choice, and record-canonical hull/tensor bridge equality, but the final Step (xi) constructor no longer takes q-positivity and normalization as independent fields at this exact boundary. The finite-divisor C_Θ-route now has the same adjusted Ob3/Ob5 source form. The new source declaration with suffix `RecordOb3Ob5AdjustedPossibleImageFiniteDivisorVerticalIQ` starts from the record-canonical adjusted determinant log-volume source, derives the Ob4 tensor-power bound from the exact theta/family-hull equality, and projects the bridge equality, q-positivity, and normalization from the ambient hull/determinant obligations object before entering the constructor-built possible-image Step (xi) route. The experiment surface exposes this as `recordOb3Ob5AdjustedPossibleImageFiniteDivisorVerticalIQ_cThetaDichotomy`. This route now has a constructed Remark 3.9.5 source object at the finite-divisor boundary. The source `IUTStage1Remark395ConstructedHullDeterminantFiniteDivisorVerticalIQSource` bundles the measured adjusted Ob3/Ob5 determinant source, the exact θ_{signed}-as-family-hull equality, and the record-canonical obligations synchronization. Its endpoint proves that the package measure calibration, derived Ob4 tensor-power bound, obligations bridge equality, q-positivity, and source normalization are all projected from this single source object. The new finite-divisor theorem with suffix

```
remark395ConstructedOb3Ob5Adjusted
  HullDeterminantFiniteDivisorVerticalIQ
```

and experiment wrapper

```
remark395ConstructedOb3Ob5Adjusted
  HullDeterminantFiniteDivisorVerticalIQ_cThetaDichotomy
```

therefore remove the separate public measure-calibration and bridge synchronization arguments from this adjusted Ob3/Ob5 route. The constructed finite-divisor source now also projects the actual constructed Remark 3.9.5 Step (xi) object, rather than only the downstream constructor-built possible-image wrapper. The projection

```
IUTStage1Remark395ConstructedHullDeterminantFiniteDivisor
  VerticalIQSource.toPossibleImageConstructedHolomorphicHull
  DeterminantSource
```

fixes the q-region to the selected Theorem 3.11 possible image, derives its containment in the record possible-image union, exposes the package hull/determinant bridge equality, and records the two-step comparison

$$\mu_{\log}(qRegion) \leq \underset{\text{norm}}{\det} \leq \theta_{\text{signed}}.$$

The experiment theorem `remark395ConstructedHullDeterminantFiniteDivisorVerticalIQSource_possibleIm` is the review-facing form of this Step (xi) projection. At the same finite-divisor boundary, the source now also carries the Ob1–Ob5 bridge audit

```
IUTStage1Remark395ConstructedHullDeterminantFiniteDivisor
  VerticalIQSource.ConstructedFiniteDivisorOb1ToOb5BridgeAudit.
```

It is projected through the constructor-built possible-image Step (xi) source and records q-region absorption into the holomorphic hull, Ob3/Ob5 hull-log-volume-to-determinant compatibility, the record-canonical bridge equality, and $q_{\text{signed}} \leq \theta_{\text{signed}}$. The experiment surface exposes it as `remark395ConstructedHullDeterminantFiniteDivisorVerticalIQSource_ob1ToOb5BridgeAudit`. The same finite-divisor source now also has the Ob5-refined quotient bridge

```
IUTStage1Remark395ConstructedHullDeterminantFiniteDivisor
  VerticalIQSource.ConstructedFiniteDivisorOb1ToOb5QuotientBridgeAudit.
```

For a comparison possible-image choice, and nonemptiness witnesses for the selected q-choice image and the comparison image, Lean projects the constructor-built bounded-family quotient theorem to this boundary. The audit records containment of both possible images in the family hull, collapse of both quotient images to the distinguished upper-semi quotient point, equality of the quotient images, the quotient-collapse iff statement, and the same source-derived chain

$$\mu_{\log}(q\text{Region}) \leq \overset{\text{norm}}{\underset{\log}{\det}} \leq \theta_{\text{signed}}.$$

The experiment surface exposes this as `remark395ConstructedHullDeterminantFiniteDivisorVerticalIQSource`
This witness burden has been reduced at the same finite-divisor boundary:

```
IUTStage1Remark395ConstructedHullDeterminantFiniteDivisor
  VerticalIQSource.qChoiceRegion_nonempty_of_qPilotRegion_nonempty
```

derives nonemptiness of the selected Theorem 3.11 q-choice possible image from nonemptiness of the constructed q-pilot region, using the source equality $q\text{Region} = P_{q\text{Choice}}$. Consequently `remark395ConstructedHullDeterminantFiniteDivisorVerticalIQSource_ob1ToOb5QuotientBridgeAuditOfQP` requires only q-pilot-region nonemptiness plus a nonempty comparison possible image, and the canonical-qChoice specialization `remark395ConstructedHullDeterminantFiniteDivisorVerticalIQSource_ob1ToOb5QuotientBridgeAuditAtQC` uses a single q-pilot-region nonemptiness witness for both sides of the Ob5 quotient collapse. The same route now has a constructor-backed source form. The source

```
IUTStage1Remark395ConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQSource
```

does not take an ambient hull/determinant obligations object. Instead it carries the common-container equality with the record-canonical hull/tensor bridge, q-pilot positivity, and source normalization; Lean then constructs `IUTStage1SourceHullDetObligations` from the Theorem 3.11 hull/determinant constructor. The experiment wrapper

```
remark395ConstructorBackedConstructedOb3Ob5Adjusted
  HullDeterminantFiniteDivisorVerticalIQ_cThetaDichotomy
```

is therefore the current strongest adjusted Ob3/Ob5 finite-divisor route: it keeps the final numeric C_Θ estimate explicit, but no longer accepts an arbitrary obligations object together with a separate record-canonical bridge-data equality. This obligations-backed record-canonical exact corridor now also has a named audit proposition. The proposition packages the constructed IPL datum, the record-canonical Step (xi) bridge equality, the obligations-derived q-positivity and normalization witnesses, the raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison, and exact vertical-IQ Frobenioid precision. The experiment surface exposes this compact audit object directly, so dependency review can target the strongest constructor-obligations route without replaying the long corridor conjunction. The same exact corridor now has a combined Gaussian/exact audit. This audit joins the constructor-backed Gaussian-to-Step (xi) route audit with the obligations-backed exact vertical-IQ corridor audit for the same source package, and the C_Θ-dichotomy wrapper now has a suffix returning this combined audit as its first conjunct. Thus the public endpoint can display the theta-root/Hodge–Arakelov Gaussian data, factored SHE preservation, constructed IPL transport, record-canonical Step (xi) constructor, obligations-derived side conditions, raw comparison, and exact vertical-IQ precision through one named source object before the still-external numeric C_Θ estimate is applied. The Hodge/SHE/IPL portion of this combined audit is now itself a named pre-Step (xi) audit on the target-charted Hodge–Arakelov IPL construction source. It packages the theta-root Gaussian evaluation, factored SHE preservation, finite Hodge/SHE transport, constructed input-prime-strip IPL transport, and finite-transport log-volume provenance before any record-canonical hull/determinant constructor is invoked. Consequently the strongest Gaussian/exact corridor points back to the same `hodgeIPLSource` for both the Hodge/SHE/IPL construction facts and the later Step (xi) route audit. The Hodge synchronization boundary has also been sharpened. The new theta-monoid-matched synchronization source replaces the public canonical one-label preservation proof by an equality of source and target Hodge–Arakelov theta-monoid degrees. Lean then reconstructs canonical-one preservation from the finite Gaussian evaluation theorem that the canonical 1-label degree equals the theta-monoid degree on both sides. The compact finite-divisor wrappers with suffixes `WithThetaMonoidMatchedRouteAudit` expose this route before the external numeric C_Θ estimate is used. This matching has now been pushed one layer further into the target-charted Hodge/IPL construction source. The matched construction source stores the theta-monoid degree equality and the constructed Theorem 3.11 input-prime-strip link, projects to the ordinary Hodge/IPL source by reconstructing canonical-one preservation internally, and then feeds the same determinant possible-image Step (xi) payload. The experiment surface exposes the matched source endpoint, its Hodge/SHE/IPL audit projection, and a finite-divisor C_Θ-dichotomy wrapper. Thus the public determinant possible-image corridor can now consume a theta-degree-matched Hodge/IPL source directly, while the final numerical C_Θ bound remains an explicit external input. The matched Hodge/IPL source has also been threaded through the all-in-one target-charted route source. The new route object stores the theta-degree-matched Hodge/IPL construction source and the determinant possible-image Step (xi) payload, then projects to the older all-in-one route only internally. Its matched Gaussian-to-Step (xi) audit contains the projected Hodge/SHE/IPL/hull chain together with the explicit target/source theta-monoid degree match and the matched Hodge/SHE/IPL construction audit. The experiment surface now exposes this matched route endpoint and a finite-divisor C_Θ-dichotomy wrapper returning the matched audit before applying the still-external numerical estimate. There is now also a no-C_Θ finite-divisor boundary for this matched all-in-one route, exposed by the experiment suffix `boundaryEndpointWithMatchedGaussianStepXIAudit`. It returns the same matched Gaussian-to-Step (xi) audit together with the finite-divisor packet equalities, vertical-IQ target equalities, and raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison before any numeric C_Θ estimate is introduced. The matched route has now been carried into the assembled constructor-backed finite exact vertical-IQ source. This source stores the theta-degree-matched Hodge/IPL construction source, the constructor-backed Step (xi) possible-image source, and the exact Step (x) vertical-IQ package as separate public fields; the ordinary assembled constructor-backed exact source is only a projection. Its matched Gaussian/exact audit combines the matched all-in-one Gaussian-to-Step (xi) route audit, the constructor-backed hull/determinant audit, the exact vertical-IQ target calibration, and the raw comparison before the external numerical C_Θ estimate is supplied. The same theta-degree-matched Hodge/IPL input has now also reached the record-canonical constructor source. The matched record-canonical package stores the measured Hodge determinant source, the possible-image q-choice, the hull/tensor/determinant operations, the record-canonical bridge equality, the q-positivity and normalization side conditions, and the exact Step (x) vertical-IQ data, while the ordinary record-canonical source is recovered only by projection. Its public audit records the matched assembled Gaussian/exact route together with the record-canonical hull/tensor bridge, the raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison, and exact Frobenioid precision. The numerical C_Θ bound remains an explicit external input at this boundary. The matched record-canonical package has now been lifted through the obligations-backed record-canonical exact corridor as well. The new source keeps the theta-degree-matched Hodge/IPL construction public while projecting q-positivity, normalization, and bridge-data alignment from a single Step (xi) obligations object. Its review audit combines the matched record-canonical constructor audit with the existing obligations-backed Gaussian/exact corridor audit, so the strongest current endpoint records both the theta-monoid provenance of canonical-one preservation and the obligations-backed hull/determinant side conditions before the external numeric C_Θ estimate is applied. The same obligations-backed exact corridor now also exposes a named pre-exact Step (xi) hull/determinant audit. This audit records the chosen Theorem 3.11 possible-image q-region, its containment in the record possible-image union, the measured family-hull/determinant provenance, the tensor-power bound, the record-canonical hull/tensor bridge, the obligations-derived q-positivity and normalization witnesses, and the raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. The combined Gaussian/exact audit now contains this Step (xi) hull/determinant audit as a separate field, so the route can be reviewed between the Hodge/SHE/IPL construction and the exact vertical-IQ fields without unpacking the final C_Θ dichotomy theorem. The obligations-backed record-canonical source also now has a no-C_Θ raw-comparison endpoint at this same level. The endpoint returns the combined Gaussian/exact audit, the exact vertical-IQ target-calibration and Frobenioid-precision facts, the finite Step

(x) packet/product equalities, the q-pilot log-Kummer non-interference witness, the upper-semi inequality, and the raw $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison, all before the external numeric C_Θ estimate is supplied. The experiment surface now exposes this no-C_Θ target by the milestone-facing suffix `boundaryEndpointWithThetaMonoidMatchedRecordCanonicalGaussianExactCorridorAudit`. Thus the newest review target separates the source-derived Hodge/SHE/IPL/Step (xi) and exact Step (x) corridor from the final ordered-real dichotomy. At the same source level, the code now exposes a remaining-boundary ledger `ObligationsRecordCanonicalConstructorRemainingSourceBoundaryAudit`. This ledger does not add a new comparison theorem. Instead it records, in one kernel-checked proposition, the constructed Hodge/SHE/IPL audit, the determinant source-boundary audit, the pre-exact Step (xi) hull/determinant audit, the exact corridor audit, the finite-divisor packet equalities, and the vertical-IQ/log-Kummer boundary facts. Its purpose is to make the next paper-facing obligation precise: construct the still-public operation identifiers, q-choice, measure-calibrated determinant source, hull/determinant obligations object, upper-semi entry, and exact vertical-IQ source from IUT I–III data rather than supplying them as fields of the strongest corridor source. The same obligations-backed record-canonical source now also projects directly to the constructor-backed, measure-calibrated possible-image Step (xi) source. The new audit `ObligationsRecordCanonicalConstructorBackedStepXISourceAudit` records the constructor-backed Step (xi) endpoint, its measure-calibrated possible-image projection, and the assembled finite exact source endpoint. The combined Gaussian/exact audit and the remaining-boundary ledger include this constructor-backed Step (xi) audit, so the strongest exact corridor can now be reviewed either as the expanded record-canonical field list or as the source-facing Step (xi) object built from that list. The constructor-backed Step (xi) source itself now has a named audit, `ConstructorBackedStepXIHullDetAudit`. It records the chosen Theorem 3.11 possible-image q-region, the measure-calibrated determinant source, the constructed hull/determinant obligations object, the record-canonical hull/tensor bridge, q-positivity, normalization, and the raw comparison. The assembled exact source now also has `AssembledConstructorBackedRemainingSourceBoundaryAudit`, exposed on the experiment surface with suffix `remainingBoundaryAudit`. This is the direct review target corresponding to the Step (xi) passage in IUT III from the Theorem 3.11 multiradial output through holomorphic hulls, determinants/tensor powers, and log-volume, before the remaining finite Step (x) and vertical-IQ inputs are discharged. There is now also a side-conditioned constructor-backed Step (xi) source. Its experiment endpoint, with suffix `SideConditionedConstructorBackedMeasureCalibratedPossibleImage_sourceEndpoint`, projects q-pilot positivity and source normalization from one `IUTStage1SourceSideConditions` object before building the constructor-backed hull/determinant source. This removes two more flat public propositions at the low-level Step (xi) constructor boundary. The same low-level source now has the named audit `SideConditionedConstructorBackedStepXIHullDetAudit`. It packages the selected possible-image q-region, the constructor-backed hull/determinant provenance, the side-condition-derived q-positivity and normalization, and the raw signed comparison as one reusable Step (xi) proposition. The strongest matched side-conditioned exact corridor now consumes this audit directly before projecting to the constructor-backed and obligations-backed exact route. The theta-monoid-matched obligations-backed record-canonical corridor now has the same explicit source-facing exposure. Its new audit `MatchedObligationsRecordCanonicalConstructorBackedStepXISourceAudit` records the matched theta-degree equality, the projected constructor-backed Step (xi) source audit, and the matched assembled Gaussian/exact route audit. The matched Gaussian/exact corridor includes this audit as a separate field, and the experiment surface exposes it directly, so the matched route can be reviewed at the constructor-backed Step (xi) boundary without first unfolding the ordinary projection. It now also has the matched remaining-boundary ledger `MatchedObligationsRecordCanonicalConstructorRemainingSourceBoundaryAudit`, exposed by `sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ_thetaMonoidMatchedRecordCanonicalRemainingBoundaryAudit`. This ledger keeps the matched Gaussian/exact audit and matched constructor-backed Step (xi) audit next to the ordinary projected ledger of still-public source-paper inputs. The point is not to close those inputs, but to make the strongest theta-monoid-matched corridor display the exact remaining operation, q-choice, determinant, hull/determinant-obligation, upper-semi, and exact vertical-IQ construction boundary. Finally, this strongest source package now has a milestone-facing C_Θ-dichotomy alias using the existing source-derived finite-divisor endpoint prefix and the suffix `withRecordCanonicalGaussianExactCorridorAudit`. It specializes the source-derived endpoint to the obligations-backed record-canonical exact corridor and returns the combined Gaussian/exact audit. The experiment surface exposes the same theorem with the suffix `cThetaDichotomyWithRecordCanonicalGaussianExactCorridorAudit`. The record-native possible-image hull route has also gained an obligations-derived side-condition form. The new experiment wrapper uses the ambient `IUTStage1SourceHullDetObligations` object to supply q-pilot positivity and source normalization. The possible-image choice, record-canonical hull/tensor bridge equality, determinant source, and external numeric C_Θ estimate remain explicit, but the route no longer takes an independent `IUTStage1SourceSideConditions` record at this boundary. This same tightening now sits behind the history-separated Hodge–Arakelov surface: the finite Hodge/SHE transport is constructed from the source and target Hodge–Arakelov theta-evaluation sources and the record's separated descent permission, while the Step (xi) side conditions are still projected from the ambient hull/determinant obligations. This history-separated possible-image obligations route now also has an audited C_Θ-surface. The wrapper returns the Hodge–Arakelov canonical one-degree match, the containment of the chosen Theorem 3.11 possible image in the record possible-image union, the record-canonical hull/tensor bridge alignment, and the obligations-derived q-positivity and normalization witnesses alongside the final dichotomy. Thus the strongest currently exposed record-native finite-divisor endpoint can be reviewed at the source-data boundary without unpacking the long theorem statement. The certificate-pinned IPL-link record-hull path now has the same side-conditioned tightening. The new endpoint suffix `IPLLinkRecordHullSideConditionedFiniteDivisorVerticalIQ` projects q-pilot positivity and source normalization from `IUTStage1SourceSideConditions` before calling the older certificate-pinned record-hull route. Its history-separated Hodge–Arakelov wrapper uses the same side-condition object, so this public IPL-link boundary no longer needs those two witnesses as separate arguments. The experiment module now exposes both versions directly. The suffix `sideConditionedIPLLinkRecordHullFiniteDivisorVerticalIQ` is the finite Hodge/SHE transport form; the history-separated variant uses the `RecordHullSideConditionedFiniteDivisorVerticalIQ` suffix after constructing the transport from Hodge–Arakelov theta evaluations. These wrappers make the review-facing certificate-pinned IPL-link surface match the side-conditioned core theorem. It now also matches the possible-image obligations tightening available on the constructed-IPL side. The suffix `iplLinkPossibleImageObligationsRecordHullFiniteDivisorVerticalIQ` uses a chosen Theorem 3.11 possible-image q-choice, derives the q-region's containment in the record possible-image union internally, and projects q-pilot positivity and normalization from the ambient `IUTStage1SourceHullDetObligations` object. Its history-separated Hodge–Arakelov companion, `historySeparatedIPLLinkPossibleImageObligationsRecordHullFiniteDivisorVerticalIQ`, constructs the finite Hodge/SHE transport from the two theta-evaluation sources before applying the same certificate-pinned IPL-link possible-image obligations route. The compact synchronized route suffix now consumes one Hodge–Arakelov synchronization object at this boundary, so the source calibration, target evaluation, and canonical one-label preservation no longer appear as separate public inputs on the review-facing theorem. The no-C_Θ finite-divisor boundary now returns this strengthened audit as well, via the compact suffix `boundaryEndpointWithAllInOneRouteGaussianStepXIAudit`. This gives a raw Step (x) endpoint whose payload is the same all-in-one Gaussian-to-Step (xi) chain used by the later C_Θ-dichotomy wrapper, before any numeric C_Θ estimate is supplied. There is now also a C_Θ-boundary suffix `cThetaDichotomyWithAllInOneRouteHullDetGaussianStepXIAudit`; it displays the same holomorphic-hull and determinant bridge fields next to the final dichotomy. This keeps the finite-divisor review surface aligned between the raw and C_Θ endpoints. The synchronized Hodge/SHE/constructed-IPL/possible-image route now also has a finite-divisor vertical-IQ source object. This source object stores the calibrated Hodge–Arakelov synchronization, constructed Theorem 3.11 IPL source, obligations-backed possible-image Step (xi) hull source, finite-divisor packet, Kummer/forgetting compatibilities, and vertical-IQ target source together. Its experiment-facing boundary and C_Θ-dichotomy aliases therefore take one source package plus the external numeric estimate, while projecting the same Gaussian-to-Step (xi) audit and Step (x) finite-divisor/vertical-IQ endpoint facts internally. There is now also a bundled certificate-pinned IPL-link version of this finite-divisor source object. It packages the synchronized Hodge–Arakelov data, the Theorem 3.11 IPL-link source, the obligations-backed possible-image Step (xi) source, finite-divisor packet, Kummer/forgetting compatibilities, and vertical-IQ target source before the constructed IPL datum is introduced. The experiment suffixes `boundaryEndpointWithIPLLinkGaussianStepXIAudit` and `cThetaDichotomyWithIPLLinkGaussianStepXIAudit` expose the raw boundary and conditional C_Θ endpoint through this single source package. This keeps the certificate-pinned route auditable in the same form as the constructed-IPL route while preserving the intended distinction: the certificate-pinned link is still a pre-construction source surface, not a proof of the constructed input-prime-strip link machinery. There is now a theta-monoid-matched version of this bundled finite-divisor source as well. It stores a matched Hodge–Arakelov synchronization whose source/target theta-monoid degree equality derives the canonical one-label Gaussian preservation internally, then projects to the synchronized source object only inside the endpoint proof. The experiment aliases expose the theta-monoid equality, the derived canonical-one equality, the same Gaussian-to-Step (xi) audit, and either the no-C_Θ finite-divisor boundary facts or the final conditional C_Θ dichotomy. This matched source now also has a named pre-IPL Hodge/SHE transport audit: theta-monoid matching, source full-label calibration, the history-separated forgetful permission, pointwise representative comparison level, identity coordinate transport, and Hodge-theater side separation are packaged before the constructed IPL layer is invoked. The same matched bundled source now also exposes the constructed IPL/log-volume transport before the Step (xi) hull comparison is used. The new endpoint projects the constructed IPL datum, input/output prime-strip equalities, source/target Hodge-theater sides, log-volume preservation, allowed forgetting, and no-history-identification from the finite Hodge/SHE+IPL construction source. A separate finite-log-volume endpoint identifies the IPL source and target log-volumes with the finite Hodge/SHE source average and transported target average, making explicit that this preservation is inherited from the finite transport rather than supplied as an independent route equality. The suffix `thetaMonoidMatchedConstructedIPLTransportSourceEqConstructedEndpoint` also exposes the exact constructor identity for this IPL source at the finite-divisor review boundary. The matched bundled source now also exposes the Step (xi) possible-image holomorphic-hull/determinant endpoint from the same source object. This projection displays the selected Theorem 3.11 possible image, containment in the record possible-image union, obligations bridge alignment, tensor-power determinant bound, q-positivity, normalization, and the raw $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison. A companion record-canonical bridge endpoint identifies the package hull/determinant bridge with the canonical hull/tensor bridge constructed from the possible-image region, its record-union containment proof, and the weighted determinant data. These separate projections are now bundled into one named matched route audit, `MatchedSourceDerivedFiniteDivisorRouteAudit`, with experiment-surface wrapper `sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQSource_thetaMonoidMatchedFullRouteAudit`. This audit is the compact review target for the current one-source-object finite-divisor corridor: it contains the pre-IPL finite Hodge/SHE transport audit, the constructed IPL log-volume endpoints, the obligations-backed possible-image Step (xi) endpoint, the record-canonical hull/tensor bridge equality, and the no-C_Θ matched Gaussian-to-Step (xi) boundary. The target-charted route is stronger than this matched bundled source at the current review surface. The compact all-in-one target-charted Hodge/IPL/determinant possible-image route now exposes the target theta chart, the Hodge–Arakelov theta-monoid degree, the determinant summand sum, the record possible-image family-hull log-volume, the package hull/determinant bridge, and the raw signed comparison in one bridge audit. The same ordinary target-charted all-in-one route now also has a compact full-route audit, `TargetChartedFullRouteAudit`, with experiment wrapper `targetChartedHodgeIPLDeterminantPossibleImageRoute_fullRouteAudit`. It bundles the Gaussian-to-Step (xi) chain, determinant source-boundary audit, calibrated determinant endpoint, calibrated possible-image endpoint, and raw comparison endpoint without changing the underlying assumptions. This gives the stronger target-charted route a single review target parallel to the matched finite-divisor audit. Its theta-monoid-matched variant additionally records that canonical one-label preservation on the Hodge/IPL side is reconstructed from equality of the target and source theta-monoid degrees. Thus the strongest finite-divisor review endpoint no longer merely assumes a tensor-power upper bound at the public Step (xi) surface: it routes the equality θ_{signed} equals the record possible-image family-hull log-volume through the determinant payload before projecting the bound consumed by the older holomorphic-hull constructor. The exact vertical-IQ record-canonical corridor now also has a summand/family-hull source form. This source object stores the Hodge/IPL construction, the target-charted possible-image summand/family-hull Step (xi) source, the finite Step (x) source, and the exact vertical-IQ data. It projects internally to the older record-canonical obligations exact source, so the exact corridor no longer exposes the Step (xi) operation identifiers, family-hull source, measure equality, q-choice, and obligations bridge equality as separate public fields. The experiment suffixes `boundaryEndpointWithSummandFamilyHullGaussianExactCorridorAudit` and `cThetaDichotomyWithSummandFamilyHullGaussianExactCorridorAudit` expose the raw boundary and conditional C_Θ endpoint. This makes the exact vertical-IQ corridor line up more closely with the Step (xi) passage of IUT III, Remark 3.9.5 and Corollary 3.12, where possible images pass through holomorphic hulls, determinant/tensor-power data, and log-volume comparison before the final numerical estimate is invoked. The possible-image Step (xi) source has also gained a constructor-pinned form. This new source still stores the selected Theorem 3.11 possible image and the weighted determinant payload, but it additionally carries the Theorem 3.11 hull/determinant constructor whose hull data is identified with the ambient source obligations. Its endpoint exposes the constructor/obligation identification, the constructor bridge equality, q-positivity, normalization, the tensor-power bound, and the raw $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison together. The experiment surface now has theta-monoid-matched raw and C_Θ finite-divisor variants with the `ConstructorPinnedThetaMonoidMatchedRouteAudit` suffix. These variants project internally to the older obligations-backed possible-image source, but the public Step (xi) input is now pinned to a source constructor rather than only to a bridge-equality field. The next tightening removes even that ambient obligations object from the public Step (xi) source. The constructor-sourced hull source derives `IUTStage1SourceHullDetObligations` as `constructor.toHullDetObligations`; algorithm certification, SHE alignment, q-positivity, normalization, and hull/determinant data therefore enter through the same Theorem 3.11 constructor. The experiment suffix `ConstructorSourcedThetaMonoidMatchedRouteAudit` exposes the constructor-sourced endpoint, the projected constructor-pinned endpoint, and the same raw or conditional C_Θ finite-divisor boundary. The remaining unconstructed Step (xi) inputs are now concentrated in the constructor itself: the holomorphic hull shadow, weighted determinant payload, measure calibration, and tensor-power bound. There is now also a constructor-built source at the same review boundary. It does not receive a Theorem 3.11 hull/determinant constructor as public input. Instead, it applies the record-canonical constructor `ofRecordCanonicalHullTensorPowerOfQSubsetUnion` to the chosen possible image, the canonical hull/tensor bridge equation, q-positivity, and normalization. The experiment suffix `ConstructorBuiltThetaMonoidMatchedRouteAudit` exposes the built source endpoint, the projected constructor-sourced endpoint, the projected constructor-pinned endpoint, and then the same raw or conditional C_Θ finite-divisor boundary. Thus the remaining Step (xi) public payload has moved from a constructor object to the actual record-canonical hull/tensor inputs: holomorphic hull shadow, weighted determinant compatibility, measure calibration, tensor-power bound, and the bridge equality. The constructor-built possible-image source now also has the named audit `RecordCanonicalStepXIAudit`, exposed on the experiment surface by the suffix `recordCanonicalStepXIAudit`. This audit unfolds the record-canonical Step (xi) constructor itself: the selected q-pilot region lies in the Theorem 3.11 possible-image union and hence in the canonical holomorphic hull, that union agrees with both the Theorem 3.11 record and the package target union, the tensor-power determinant bound supplies the θ_{signed} bound, and the constructed hull/determinant source then provides certification, record-union hull containment, determinant-volume control, and the raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. This gives reviewers a Step (xi)-native audit target before the finite-divisor route projects it into the later vertical-IQ corridor. The same constructor-built source now exposes a sharper remaining-payload audit, `ConstructorBuiltRemainingPayloadAudit`, with experiment suffix `remainingPayloadAudit`. This ledger separates the still-supplied Remark 3.9.5/Step (xi) inputs from the constructor projections: selected possible-image q-region, holomorphic-hull absorption of the possible-image union and q-region, the named Ob1/Ob2 hull-absorption audit, the named Ob3/Ob4 determinant/tensor-power audit, the derived Ob3–Ob5 hull-log-volume compatibility chain, the record-canonical hull/tensor bridge equality, q-positivity, normalization, and the derived raw comparison. This constructor-built Step (xi) source is now also available as a single theta-monoid-matched finite-divisor route object, not merely as an external experiment wrapper. This source object packages the matched Hodge–Arakelov synchronization, the constructed Theorem 3.11 IPL source, the constructor-built possible-image hull source, finite-divisor packet data, Kummer/forgetting compatibilities, and vertical IQ target source. Its full-route, boundary, and conditional C_Θ audits now carry the `RecordCanonicalStepXIAudit` as an explicit component, and the experiment surface exposes it with suffix `ConstructorBuiltRecordCanonicalStepXIAudit`. Thus the route-level review target no longer hides the record-canonical Step (xi) constructor audit behind the compact constructor-built source endpoint. Its route audit first exposes the built Step (xi) endpoint, the projected constructor-sourced endpoint, and the projected constructor-pinned endpoint, then projects to the established matched finite-divisor audit. The experiment surface exposes this as the compact `ConstructorBuiltFullRouteAudit` review target. The same route now also has `ConstructorBuiltRemainingPayloadRouteAudit`, exposed with suffix `ConstructorBuiltRemainingPayloadRouteAudit`; it pairs the remaining-payload ledger with the full finite-divisor route audit so the source-paper Step (xi) obligations remain attached to the bundled source object. The same bundled source now also has no-C_Θ

and conditional C_Θ experiment endpoints with suffixes `ConstructorBuiltBoundaryAudit` and `ConstructorBuiltCThetaAudit`; these keep the external numeric estimate separate while ensuring that the Hodge/SHE, constructed IPL, constructor-built Step (xi), and finite vertical-IQ data all come from one route object. The conditional endpoint now also has the sharper suffix `ConstructorBuiltRemainingPayloadCThetaAudit`, which carries the same remaining-payload Step (xi) ledger next to the C_Θ dichotomy instead of exposing it only at the raw route-audit boundary. There is now also an IPL-link version of this constructor-built bundled route. The source object

```
IUTStage1ThetaMonoidMatchedHodgeSHEIPLLink
PossibleImageConstructorBuiltFiniteDivisorVerticalIQSource
```

keeps the same record-canonical Step (xi) constructor-built hull source, but uses only the certificate-pinned Theorem 3.11 IPL link source rather than an input-prime-strip link construction source. Its experiment suffixes

```
ConstructorBuiltIPLLinkFullRouteAudit,
ConstructorBuiltIPLLinkRemainingPayloadRouteAudit,
ConstructorBuiltIPLLinkBoundaryAudit,
ConstructorBuiltIPLLinkCThetaAudit,
ConstructorBuiltIPLLinkRemainingPayloadCThetaAudit
```

expose the route, no-C_Θ, conditional C_Θ, and remaining-payload conditional audits. The last suffix carries the same remaining Step (xi) payload ledger beside the certificate-pinned IPL-link dichotomy. This separates the Step (xi) constructor-built progress from the still-open construction of the actual Theorem 3.11 input-prime-strip link machinery. This constructor-built IPL-link route now also has a Hodge–Arakelov source endpoint. Instead of asking the public theorem for the bundled theta-monoid-matched synchronization record, the endpoint starts from a calibrated source theta evaluation, a target theta evaluation, and the equality of their theta-monoid degrees, constructs the matched synchronization internally, and then enters the same constructor-built Step (xi) finite-divisor route. The experiment surface now exposes both the no-C_Θ boundary, which includes the projected $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison, and the conditional endpoint, with suffixes

```
hodgeArakelovHistorySeparatedConstructorBuiltIPLLinkBoundaryAudit,
hodgeArakelovHistorySeparated
ConstructorBuiltIPLLinkCTheta.
```

This Hodge–Arakelov constructor-built IPL-link route now also has a canonical-one variant. Instead of receiving the theta-monoid degree equality directly, the endpoint assumes preservation of the Gaussian degree at the canonical 1-label and derives the theta-monoid equality from the Hodge–Arakelov evaluation laws on the source and target sides. Its experiment surface exposes the corresponding boundary and conditional endpoints with suffixes `hodgeArakelovCanonicalOneConstructorBuiltIPLLinkBoundaryAudit` and `hodgeArakelovCanonicalOneConstructorBuiltIPLLinkCTheta`. The canonical-one variant is now also available as a single source object,

```
IUTStage1CanonicalOneHodgeArakelovSHEIPLLink
PossibleImageConstructorBuiltFiniteDivisorVerticalIQSource.
```

This object stores the calibrated source Hodge–Arakelov evaluation, the target evaluation, the canonical 1-label Gaussian preservation equality, the certificate-pinned IPL link source, the constructor-built Step (xi) hull/determinant source, and the finite Step (x) vertical-IQ data. Its projection first reconstructs the theta-monoid-matched source object and then reuses the existing constructor-built IPL-link boundary. The experiment surface exposes this compact review target through the suffixes

`canonicalOneConstructorBuiltIPLLinkFullRouteAudit`,
`canonicalOneConstructorBuiltIPLLinkRemainingPayloadRouteAudit`,
`canonicalOneConstructorBuiltIPLLinkThetaEvaluationPairAudit`,
`canonicalOneConstructorBuiltIPLLinkBoundaryAudit`,
`canonicalOneConstructorBuiltIPLLinkCThetaAudit`,
`canonicalOneConstructorBuiltIPLLinkRemainingPayloadCThetaAudit`.

The theta-evaluation-pair audit records both source and target Hodge–Arakelov theta-evaluation endpoints, the source full-label calibration against the theta-source cusp-log-volume average, canonical 1-label Gaussian-degree preservation, and the derived theta-monoid degree equality. The latter is now exposed directly on the experiment surface as `canonicalOneConstructorBuiltIPLLinkThetaMonoidDegreeEq`; its proof uses the source and target Hodge–Arakelov theta-evaluation endpoints' `canonicalOneDegree_eq_thetaMonoidDegree` theorems rather than an inline route calculation. The same source object now exposes the raw Step (xi) comparison before the finite Step (x) boundary is unfolded. Its named audit `CanonicalOneConstructorBuiltIPLLinkQComparisonAudit` packages the constructor-built route audit, the pre-Step (xi) `CanonicalOneConstructorBuiltIPLLinkHodgeSHEIPLAudit`, and $q_{\text{signed}} \leq \theta_{\text{signed}}$. The pre-Step audit records the Hodge–Arakelov theta-root/Gaussian endpoint, finite Hodge/SHE transport, history-separated forgetful transport permission, and certificate-pinned IPL/log-volume transport before the possible-image hull comparison is used. The Step (xi) source-object handoff is now exposed as `CanonicalOneConstructorBuiltIPLLinkHolomorphicHullDeterminantSourceAudit`: the canonical-one route projects its constructor-built hull source to the existing obligations-backed possible-image holomorphic-hull/determinant source, where the q-pilot choice, record-union containment, obligations bridge, tensor-power bound, q-positivity, normalization, and raw signed comparison are read from that one source object. This is now followed by `CanonicalOneConstructorBuiltIPLLinkSourceDerivedBridgeAudit`, which constructs the source-derived Hodge/SHE/IPL/hull bridge from the canonical-one Hodge–Arakelov synchronization, the certificate-pinned Theorem 3.11 IPL link, and that projected possible-image hull source. Its endpoint records the IPL prime-strip equalities, source/target Hodge averages, possible-image q-region, obligations bridge, log-volume preservation, history separation, and $q_{\text{signed}} \leq \theta_{\text{signed}}$. The field-level Step (xi) hull side remains separated as `CanonicalOneConstructorBuiltIPLLinkStepXIHullAudit`, which records the q-pilot possible image, record-union containment, record-canonical hull/determinant bridge, tensor-power bound, theta-pilot hull containment and volume facts, q-positivity, and normalization before the raw signed comparison is projected. The same IPL-link Step (xi) source side now exposes the remaining payload in the same review buckets as the constructed-IPL route: Ob1/Ob2 hull absorption, Ob3/Ob4 determinant normalization and tensor-power control, Ob5 compatibility of the bounded-family hull passage, and the q-positivity/normalization side conditions. These buckets are then threaded into `CanonicalOneConstructorBuiltIPLLinkQComparisonAudit`, so the raw comparison no longer points only to a coarse Step (xi) hull audit. The experiment surface adds direct endpoints with suffixes

`canonicalOneConstructorBuiltIPLLinkThetaEvaluationPairAudit`,
`canonicalOneConstructorBuiltIPLLinkHodgeSHEIPLAudit`,
`canonicalOneConstructorBuiltIPLLinkHolomorphicHullDeterminantSourceAudit`,
`canonicalOneConstructorBuiltIPLLinkSourceDerivedBridgeAudit`,
`canonicalOneConstructorBuiltIPLLinkStepXIHullAudit`,
`canonicalOneConstructorBuiltIPLLinkOb1Ob2HullAbsorptionAudit`,
`canonicalOneConstructorBuiltIPLLinkOb3Ob4DeterminantAudit`,
`canonicalOneConstructorBuiltIPLLinkOb5CompatibilityAudit`,
`canonicalOneConstructorBuiltIPLLinkSideConditionAudit`,
`canonicalOneConstructorBuiltIPLLinkQComparisonAudit`,
`canonicalOneConstructorBuiltIPLLink_qComparison`,
`canonicalOneConstructorBuiltIPLLinkCTheta`,
`canonicalOneConstructorBuiltIPLLinkRemainingPayloadCThetaAudit`,
`canonicalOneConstructorBuiltIPLLinkMilestoneCThetaAudit`.

The no-C_Θ finite boundary is now staged through the named audit `CanonicalOneConstructorBuiltIPLLinkFiniteBoundaryAudit`: its first field is the raw q-comparison audit, followed by the Step (xi) hull audit, Ob1/Ob2 hull absorption, Ob3/Ob4 determinant, Ob5 compatibility, and side-condition audits, and then the finite-divisor packet equalities plus the vertical-IQ target equalities. The experiment surface exposes this as `canonicalOneConstructorBuiltIPLLinkFiniteBoundaryAudit`. The new `CanonicalOneConstructorBuiltIPLLinkMilestoneCThetaAudit` then packages the q-comparison audit, fine Step (xi) payload audits, finite-boundary audit, source-derived bridge audit, conditional C_Θ audit, the remaining-payload C_Θ audit, and final ordered-real dichotomy as one milestone-facing declaration. Thus this review target separates the raw source-derived Step (xi) comparison, the finite vertical-IQ boundary ledger, the still-supplied Step (xi) payload, and the later numeric C_Θ input while also offering a single audit surface for the canonical-one constructor-built route. There is now a stricter canonical-one variant whose middle route constructs IPL/log-volume transport rather than consuming only the certificate-pinned IPL-link source:

`IUTStage1CanonicalOneHodgeArakelovSHEIPLConstruction`
`PossibleImageConstructorBuiltFiniteDivisorVerticalIQSource`.

It carries the same calibrated source Hodge–Arakelov evaluation, target evaluation, canonical 1-label Gaussian preservation equality, constructor-built Step (xi) hull/determinant source, and finite Step (x) vertical-IQ data, but replaces the standalone IPL-link source by an `IUTStage1Theorem311IPLLinkConstructionSource`. Lean derives the theta-monoid match from canonical-one preservation, projects to the existing theta-monoid-matched constructed-IPL route, and exposes the finite Hodge/SHE transport audit, constructed IPL prime-strip equalities, and constructed IPL source/target log-volume provenance as fields of `CanonicalOneConstructorBuiltIPLConstructionMiddleRouteAudit`. The front Hodge–Arakelov Gaussian data is now also separated as `CanonicalOneConstructorBuiltIPLConstructionThetaEvaluationAudit`; it records the source and target theta-evaluation endpoints, the source theta/log-volume calibration, canonical 1-label degree preservation, and the derived theta-monoid degree equality before the finite Hodge/SHE transport is constructed. The equality itself is also available as the compact experiment target `canonicalOneConstructorBuiltIPLConstructionThetaMonoidDegreeEq`.

The transport-source handoff is now also named separately as `CanonicalOneConstructorBuiltIPLConstructionTransportSourceAudit`: it packages the theta-evaluation audit with the finite Hodge/SHE transport-source endpoint built from the source/target Hodge–Arakelov evaluations and history-separated forgetting. The next audit, `CanonicalOneConstructorBuiltIPLConstructionFactoredSHEAudit`, records the factored square/full-label SHE synchronization endpoint, the transport-source audit, and the finite Hodge/SHE transport endpoint: structured-SHE availability, identity-coordinate transport, full-label log-volume preservation, transported-average preservation, and history separation all appear before the constructed IPL log-volume transport is read. The constructed IPL segment now has two named subaudits: `CanonicalOneConstructorBuiltIPLConstructionLogVolumeTransportAudit` identifies the finite source's IPL/log-volume transport source with the construction from the Theorem 3.11 input-prime-strip link source, while `CanonicalOneConstructorBuiltIPLConstructionFiniteTransportLogVolumeAudit` reads the source and target IPL log-volumes from the finite Hodge/SHE source average and transported target average. The stronger chain audit `CanonicalOneConstructorBuiltIPLConstructionFiniteTransportLogVolumeChainAudit` packages the finite Hodge/SHE transport endpoint, the constructed Theorem 3.11 IPL source, the finite-transport log-volume endpoint, and the transported-average preservation theorem as a single provenance boundary. Thus the equality between the constructed IPL target and source log-volumes is read directly from finite transported-average preservation rather than from an independent route-level conjunction. The aggregate `CanonicalOneConstructorBuiltIPLConstructionConstructedIPLAudit` then packages these two subaudits together with the choice-wise prime-strip links. The named subaudit `CanonicalOneConstructorBuiltIPLConstructionChoiceLinkAudit` now exposes this choice-wise input-prime-strip link obligation directly and is threaded through the Hodge/SHE/IPL, middle-route, q-comparison, finite-boundary, and milestone C_Θ audits. The constructed-IPL route now also has its own bridge-construction audit, `CanonicalOneConstructorBuiltIPLConstructionSourceDerivedBridgeAudit`. This audit records the bridge built from the Hodge–Arakelov theta data, finite Hodge/SHE transport, the constructed Theorem 3.11 IPL source, and the constructor-built possible-image Step (xi) hull source, and then projects the generic source-derived bridge alignment endpoint from that bridge. The resulting route alignment is now also named as `CanonicalOneConstructorBuiltIPLConstructionRouteLogVolumeAlignmentAudit`. This audit exposes the concrete `IUTStage1HodgeSHEIPLHullRouteLogVolumeAlignment` constructed from the bridge, including IPL log-volume preservation, source theta/Hodge log-volume calibration, target SHE-transported log-volume equality, and q-pilot positivity. Its Step (xi) hull side is now separated as `CanonicalOneConstructorBuiltIPLConstructionStepXIHullAudit`, which records the constructor-built, constructor-sourced, and constructor-pinned possible-image hull endpoints, the projected possible-image hull/determinant endpoint, and the record-canonical hull/tensor bridge equality before the boundary audit is used. It now also carries the compact `CanonicalOneConstructorBuiltIPLConstructionOb1ToOb5BridgeAudit`, whose source-level core is `ConstructorBuiltOb1ToOb5BridgeAudit`. This audit packages the Remark 3.9.5(vii) chain from Ob1/Ob2 hull absorption through Ob3/Ob4 determinant and tensor-power normalization, Ob3/Ob5 determinant/log-volume compatibility, the remaining Step (xi) source payload, and the actual `hullDetSource_endpoint` produced by the Theorem 3.11 hull/determinant constructor. Thus the record-canonical bridge endpoint is now reviewable as the output of the Ob1–Ob5 source data, not only as a separate remaining-payload projection. The source-level audit is also exposed directly on the experiment surface as

`possibleImageConstructorBuiltHullSource_ob1ToOb5BridgeAudit`.

This lets the Remark 3.9.5(vii) bridge be reviewed before entering the finite-divisor vertical-IQ corridor. The source-level bridge is now also joined to the bounded-family Ob5 quotient collapse by

`possibleImageConstructorBuiltHullSource_ob1ToOb5QuotientBridgeAudit`.

This audit carries the Ob1–Ob5 bridge, the Ob5 quotient-independence audit, the record Ob3/Ob5 bridge, and the local canonical C_Θ chain in one object: two nonempty Theorem 3.11 possible images have the same quotient image, while the selected q-region still satisfies $\mu_{\log}(q\text{Region}) \leq \det_{\text{norm}} \leq \theta_{\text{signed}}$. Its flat-chain companion

`possibleImageConstructorBuiltHullSource_ob1ToOb5QuotientBridgeChain`

spells out the two family-hull inclusions, the quotient-image equality, $\mu_{\log}(q\text{Region}) \leq \det_{\text{norm}} \leq \theta_{\text{signed}}$, the canonical C_Θ upper bound, and the signed comparison $q_{\text{signed}} \leq \theta_{\text{signed}}$ without requiring reviewers to unfold the audit record. The same Step (xi) hull audit now exposes the constructor-built upper-ray log-volume audit

`ConstructorBuiltStepXIUpperRayAudit`,

which records the q-pilot region's holomorphic-hull containment, the equality between hull log-volume and normalized weighted determinant, membership of the q-pilot log-volume in the resulting upper ray, and the projected $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison. This upper-ray handoff is now named at the same route level as `CanonicalOneConstructorBuiltIPLConstructionUpperRayAudit`, which projects q-pilot upper-ray membership, theta-hull and determinant bounds, determinant normalization, and the raw signed comparison from the constructor-built Step (xi) source. Its Ob1/Ob2 hull-absorption payload is now isolated as `CanonicalOneConstructorBuiltIPLConstructionOb1Ob2HullAbsorptionAudit`: the selected q-region is identified with a Theorem 3.11 possible image, shown inside the record possible-image union, and both the union and selected q-region are absorbed by the canonical holomorphic hull before determinant data enters. Its Ob3/Ob4 determinant payload is now also isolated as `CanonicalOneConstructorBuiltIPLConstructionOb3Ob4DeterminantAudit`: the audit records weighted determinant summation, determinant normalization, hull-log-volume compatibility, tensor-power control, measure calibration, and the record-canonical hull/tensor bridge before the finite Step (x) boundary is used. The Ob5 independence-of-possibilities passage is now isolated as `CanonicalOneConstructorBuiltIPLConstructionOb5CompatibilityAudit`: the selected q-region is absorbed by the bounded-family hull, and the hull-log-volume, determinant normalization, package measure, and record-canonical bridge remain compatible with that hull-family passage. This route-level Ob5 audit now also carries the constructor-built Ob3/Ob5 determinant-compatibility chain, so the determinant log-volume equality is read through normalized weighted-determinant compatibility rather than a standalone remaining-payload projection. The same route now exposes the quotient form of this Ob5 passage as `CanonicalOneConstructorBuiltIPLConstructionOb5QuotientIndependenceAudit`. For any comparison possible-image choice, together with nonemptiness of the selected q-choice image and the comparison image, this audit imports the constructor-built Remark 3.9.5(v)–(vi) quotient collapse into the canonical-one constructed-IPL finite-divisor route: both possible images map to the same collapsed bounded-family quotient image, while the Ob1–Ob5 bridge, Ob5 compatibility, family-hull determinant/log-volume equality, family-union bound, tensor-power normalization, and raw signed comparison remain attached to the same source object. The remaining source side conditions are now split off as `CanonicalOneConstructorBuiltIPLConstructionSideConditionAudit`: the constructor-built hull source's q-pilot positivity and source normalization are repackaged through the existing `IUTStage1SourceSideConditions` interface before the finite Step (x) boundary consumes the raw signed comparison. The raw Step (xi) comparison is now also named directly as `CanonicalOneConstructorBuiltIPLConstructionQComparisonAudit`: it keeps the Hodge/SHE/constructed-IPL middle-route audit, Step (xi) hull audit, Step (xi) holomorphic-hull/determinant skeleton, named Ob1/Ob2 hull-absorption audit, named Ob3/Ob4 determinant audit, Ob5 compatibility audit, and source side-condition audit together before projecting $q_{\text{signed}} \leq \theta_{\text{signed}}$. The no-$C_\Theta$ finite boundary is now separated as `CanonicalOneConstructorBuiltIPLConstructionFiniteBoundaryAudit`; it keeps the middle-route audit, Step (xi) hull audit, the Step (xi) holomorphic-hull/determinant skeleton, compact boundary audit, finite-divisor packet equalities, exact vertical-IQ target equalities, and raw $q_{\text{signed}} \leq \theta_{\text{signed}}$ comparison attached to the same source object. The named skeleton `CanonicalOneConstructorBuiltIPLConstructionStepXIHolomorphicHullDeterminantSkeleton` records possible-image containment, holomorphic-hull absorption, determinant/tensor-power facts, the record-canonical bridge equality, side conditions, and the raw signed comparison as projections from the constructor-built hull source. The experiment surface exposes this stricter route through suffixes

```
canonicalOneConstructorBuiltIPLConstructionFullRouteAudit,
canonicalOneConstructorBuiltIPLConstructionRemainingPayloadRouteAudit,
canonicalOneConstructorBuiltIPLConstructionThetaEvaluationAudit,
canonicalOneConstructorBuiltIPLConstruction
  TransportSourceAudit,
canonicalOneConstructorBuiltIPLConstructionFactoredSHEAudit,
canonicalOneConstructorBuiltIPLConstruction
  LogVolumeTransportAudit,
canonicalOneConstructorBuiltIPLConstruction
  FiniteTransportLogVolumeAudit,
canonicalOneConstructorBuiltIPLConstruction
  FiniteTransportLogVolumeChainAudit,
canonicalOneConstructorBuiltIPLConstructionConstructedIPLAudit,
canonicalOneConstructorBuiltIPLConstructionChoiceLinkAudit,
canonicalOneConstructorBuiltIPLConstructionHodgeSHEIPLAudit,
canonicalOneConstructorBuiltIPLConstruction
  DirectHodgeSHEIPLAudit,
canonicalOneConstructorBuiltIPLConstructionDirectGaussianToStepXIAudit,
canonicalOneConstructorBuiltIPLConstructionMiddleRouteAudit,
canonicalOneConstructorBuiltIPLConstructionDirectMiddleRouteAudit,
canonicalOneConstructorBuiltIPLConstructionSourceDerivedBridgeAudit,
canonicalOneConstructorBuiltIPLConstruction
  RouteLogVolumeAlignmentAudit,
canonicalOneConstructorBuiltIPLConstructionStepXIHullAudit,
canonicalOneConstructorBuiltIPLConstruction
  Ob1ToOb5BridgeAudit,
canonicalOneConstructorBuiltIPLConstructionUpperRayAudit,
canonicalOneConstructorBuiltIPLConstruction
  StepXIHolomorphicHullDeterminantSkeleton,
canonicalOneConstructorBuiltIPLConstructionOb1Ob2HullAbsorptionAudit,
canonicalOneConstructorBuiltIPLConstructionOb3Ob4DeterminantAudit,
canonicalOneConstructorBuiltIPLConstructionOb5CompatibilityAudit,
canonicalOneConstructorBuiltIPLConstruction
  Ob5QuotientIndependenceAudit,
```

and

```
canonicalOneConstructorBuiltIPLConstructionSideConditionAudit,
canonicalOneConstructorBuiltIPLConstructionQComparisonAudit,
canonicalOneConstructorBuiltIPLConstruction
  DirectQComparisonAudit,
canonicalOneConstructorBuiltIPLConstruction_qComparison,
canonicalOneConstructorBuiltIPLConstructionPacketTargetAudit,
canonicalOneConstructorBuiltIPLConstructionDirectFiniteBoundaryAudit,
canonicalOneConstructorBuiltIPLConstruction
  DirectRouteFiniteBoundaryAudit,
canonicalOneConstructorBuiltIPLConstructionFiniteBoundaryAudit,
canonicalOneConstructorBuiltIPLConstructionBoundaryAudit,
canonicalOneConstructorBuiltIPLConstructionCTheta,
canonicalOneConstructorBuiltIPLConstructionDirectCThetaAudit,
canonicalOneConstructorBuiltIPLConstructionMilestoneCThetaAudit,
canonicalOneConstructorBuiltIPLConstruction
  DirectMilestoneCThetaAudit,
canonicalOneConstructorBuiltIPLConstruction
  Ob5QuotientMilestoneCThetaAudit.
```

The constructed-IPL route now also has compact milestone-facing aliases:

```
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ,
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ
  _withDirectMilestoneAudit,
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullConstructedIPLFiniteDivisorVerticalIQ,
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullConstructedIPLFiniteDivisorVerticalIQ
  _withMilestoneAudit,
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullConstructedIPLFiniteDivisorVerticalIQ
  _withDirectMilestoneAudit,
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullConstructedIPLFiniteDivisorVerticalIQ
  _withOb5QuotientMilestoneAudit.
```

The first pair is now the compact subgoal-6 public endpoint and its direct audit companion; the latter four names keep the more explicit constructed-IPL spelling. All six aliases return the same canonical-one constructor-built constructed IPL route: Hodge–Arakelov theta evaluation, factored SHE, constructed IPL/log-volume transport, Step (xi) hull/determinant audits, source side conditions, raw q-comparison, finite boundary, and conditional C_Θ audit, with the numeric C_Θ estimate still explicit. `CanonicalOneConstructorBuiltIPLConstructionMilestoneCThetaAudit` now carries the transport-source, log-volume transport, and finite-transport log-volume audits directly, together with the constructed source-derived bridge audit, the constructed route-log-volume alignment audit, a direct middle-route audit, constructor-built upper-ray audit, and a direct signed C_Θ audit. The direct middle-route audit bundles the canonical-one theta-evaluation, finite Hodge/SHE transport, factored SHE, constructed IPL log-volume transport, source-derived bridge, and route-log-volume alignment chain before the theta-monoid-matched finite boundary is used for compatibility. The direct C_Θ audit computes the ordered-real dichotomy from the constructed q-positivity side condition, the constructed $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ Step (xi) comparison, and the explicit numeric C_Θ estimate, rather than projecting that dichotomy only through the coarser theta-monoid-matched C_Θ package. The Hodge/SHE/IPL part of this direct path is now also exposed as `CanonicalOneConstructorBuiltIPLConstructionDirectHodgeSHEIPLAudit`. Unlike the broader compatibility audit, this direct audit does not carry the theta-monoid-matched finite-transport ledger; it records only the canonical-one theta evaluation, finite Hodge/SHE transport, factored SHE, constructed-IPL log-volume transport, finite-transport log-volume, and choice-link endpoints. The direct middle route, raw q-comparison audit, and direct milestone bundle consume this stricter audit. The raw comparison itself now also has a no-broad-middle-route form, `CanonicalOneConstructorBuiltIPLConstructionDirectQComparisonAudit`. It removes the older `MiddleRouteAudit` field from the comparison payload and reviews $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ through the direct Hodge/SHE/IPL audit, direct middle route, source-derived bridge, route-log-volume alignment, Step (xi) hull/determinant audits, Ob1–Ob5 payload, and source side conditions. The direct finite-boundary route, direct C_Θ audit, and direct milestone bundle consume this direct q-comparison audit. The public raw comparison suffix `canonicalOneConstructorBuiltIPLConstruction_qComparison` now projects through the direct comparison theorem as well. There is now also `CanonicalOneConstructorBuiltIPLConstructionDirectMilestoneCThetaAudit`, which removes the theta-monoid-matched route, remaining-payload, compact boundary, and matched C_Θ compatibility projections from the milestone bundle. Its load-bearing route is the direct middle-route audit, the direct route finite-boundary audit, and the direct signed C_Θ audit. The compact constructed-IPL alias with suffix `withDirectMilestoneAudit` returns this same direct milestone bundle. Thus the milestone-facing bundle exposes the pre-IPL and constructed-IPL transport chain, the Hodge/SHE/IPL/hull bridge constructor, the route alignment it constructs, and the Step (xi-f/g) upper-ray handoff instead of only the coarser constructed-IPL and hull audits. The quotient-aware milestone bundle, `CanonicalOneConstructorBuiltIPLConstructionOb5QuotientMilestoneCThetaAudit`, attaches the route-level Ob5 quotient-independence audit to this same direct C_Θ milestone surface. Its compact alias with suffix `withOb5QuotientMilestoneAudit` returns the final dichotomy together with the collapsed Ob5 quotient images for two nonempty possible-image choices, so the quotient-collapse payload is visible at the same review surface as the finite-divisor vertical-IQ C_Θ endpoint. The same route now has an unbundled canonical-one Hodge–Arakelov source endpoint,

```
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeArakelovCanonicalOneT11IPLConstruction
  ConstructorBuiltFiniteDivisorVerticalIQ,
```

exposed on the experiment surface with suffix

```
hodgeArakelovCanonicalOne
  ConstructorBuiltIPLConstructionCTheta.
```

It constructs the canonical-one source object internally from the calibrated source theta evaluation, target theta evaluation, canonical 1-label Gaussian preservation, Theorem 3.11 IPL-link construction source, constructor-built Step (xi) hull source, and finite Step (x) vertical-IQ data. Its audit-returning companion,

```
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeArakelovCanonicalOneT11IPLConstruction
  ConstructorBuiltFiniteDivisorVerticalIQ_withMilestoneAudit,
```

is exposed on the experiment surface with suffix

```
hodgeArakelovCanonicalOne
  ConstructorBuiltIPLConstructionMilestoneAudit.
```

It returns the full canonical-one constructor-built constructed-IPL milestone C_Θ audit for the internally assembled source object. The same canonical-one boundary now has a charted Hodge/family-hull Step (xi) variant,

```
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeArakelovCanonicalOneT11IPLConstruction
  ChartedFamilyHullFiniteDivisorVerticalIQ,
```

with audit companion ending in

```
ChartedFamilyHullFiniteDivisorVerticalIQ_withMilestoneAudit.
```

Its experiment-surface suffixes are

```
hodgeArakelovCanonicalOneChartedFamilyHull
  IPLConstructionCTheta,
hodgeArakelovCanonicalOneChartedFamilyHull
  IPLConstructionMilestoneAudit.
```

These endpoints no longer receive the constructor-built Step (xi) hull source directly: they project it from the possible-image charted Hodge-calibrated family-hull obligations-backed source before assembling the finite-divisor vertical-IQ corridor. The canonical-one boundary now also has a synchronized possible-image summand/family-hull refinement,

```
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeArakelovCanonicalOneT11IPLConstruction
  SynchronizedPossibleImageSummandFamilyHull
  FiniteDivisorVerticalIQ.
```

The experiment surface exposes this as

```
hodgeArakelovCanonicalOneSynchronizedPossibleImage
  SummandFamilyHullIPLConstructionCTheta.
```

Its no-C_Θ audit companion is

```
boundaryEndpoint_from
  _sourceDerivedHodgeArakelovCanonicalOneT11IPLConstruction
  SynchronizedPossibleImageSummandFamilyHull
  FiniteDivisorVerticalIQ_withQComparisonAudit,
```

with experiment suffix

`hodgeArakelovCanonicalOneSynchronizedPossibleImage`
`SummandFamilyHullIPLConstructionBoundaryAudit`.

Here the Hodge source and target evaluations are bundled into the synchronized Hodge–Arakelov source object, while the Step (xi) input still remembers the Theorem 3.11 possible-image q-choice and the determinant summand calibration. The audit companion exposes the summand/family-hull q-comparison audit and the finite-divisor packet and vertical-IQ target equalities before the numeric C_Θ estimate is supplied. Thus this public endpoint sits one stage upstream of the charted family-hull variant before projecting to the already verified finite-divisor corridor. The corresponding milestone C_Θ audit now also carries the remaining-payload route audit, the theta-evaluation audit, the factored SHE audit, and the constructed IPL audit as fields, so the final canonical-one constructed-IPL review target keeps the theta-root/Hodge–Arakelov Gaussian, factored Hodge/SHE, constructed IPL, direct Gaussian-to-Step (xi), Ob1/Ob2 hull absorption, Ob3/Ob4 determinant, Ob5 compatibility, finite Step (x), vertical-IQ, and Step (xi) source-paper payload ledgers together. At the same canonical-one boundary, the finite Hodge/SHE/IPL middle layer is now constructed directly from the route's source-calibrated and target Hodge–Arakelov evaluations plus canonical 1-label preservation. The new Hodge/SHE/IPL audit records that direct finite transport-source endpoint, the constructed IPL log-volume endpoint, and the finite-transport log-volume provenance before the route is projected through the theta-monoid-matched corridor. This keeps the history-separated forgetting permission and transported-average preservation visible at the canonical-one source surface. The same route now also carries a direct Gaussian-to-Step (xi) bridge audit: starting from the route-owned source-calibrated Hodge–Arakelov theta evaluation, target evaluation, canonical 1-label preservation, and constructed IPL source, it projects the constructor-built possible-image hull source to the side-conditioned obligations-backed source consumed by the source-calibrated Step (xi) bridge audit. This records the theta-root/Gaussian payload, constructed finite Hodge/SHE/IPL transport, hull containment, record-canonical bridge, determinant bound, side conditions, and $q_{\text{signed}} \le \theta_{\text{signed}}$ at the canonical-one milestone surface. The canonical-one constructed-IPL route now also has a separate Step (x) packet/vertical-IQ audit before the compact finite-boundary ledger. This audit is projected through the named finite-divisor packet-local source endpoint and the exact vertical-IQ target endpoint, so the packet-local equalities, upper-semi inequality, packet-normalized target equalities, and target (Ind3) equality are visible as source-endpoint facts in the milestone audit rather than only as loose components of the final boundary tuple. These pieces are now joined in two stages. The smaller direct finite-boundary audit projects the raw comparison from the source-calibrated Gaussian-to-Step (xi) bridge and the finite packet/vertical-IQ facts from the packet/target audit. The larger `CanonicalOneConstructorBuiltIPLConstructionDirectRouteFiniteBoundaryAudit` then carries the direct middle route, source-derived bridge, route-log-volume alignment, Step (xi) hull/determinant audits, side conditions, raw q-comparison, and finite packet/vertical-IQ target facts together before the matched compatibility boundary is used. The older compact matched boundary is still retained for compatibility with the existing C_Θ route, but the no-C_Θ review surface now has a direct source-derived boundary for the same canonical-one constructed-IPL source object. The constructor-built bundled route can now also be reached from the stronger theta-monoid-matched obligations-backed record-canonical exact corridor. The projection constructs the bundled route internally from the matched Hodge/IPL source, the measured determinant family, the selected possible image, the finite-divisor source, and the exact vertical-IQ source. In particular, the hull/determinant bridge equality, q-positivity, and normalization are first projected from the constructor-obligations object and then used to build the constructor-built Step (xi) source. The experiment surface exposes this review target through three suffixes:

`MatchedObligationsConstructorBuiltFullRouteAudit`,
`MatchedObligationsConstructorBuiltRemainingPayloadRouteAudit`,
`MatchedObligationsConstructorBuiltBoundaryAudit`,
`MatchedObligationsConstructorBuiltCThetaAudit`.

The remaining-payload variant keeps the obligations-to-record-canonical Step (xi) bridge together with the constructor-built Step (xi) payload ledger, so this intermediate corridor now reports the same unresolved hull/determinant source-paper payload that was previously visible only on the lower constructor-built route. The same obligations-backed record-canonical corridor is now also reachable from the earlier constructor-backed matched exact source. This projection sets the corridor's constructor-obligations object to `toHullDetSourceConstructor.toHullDetObligations`; consequently the obligations bridge equality with the record-canonical hull/tensor-power bridge is definitional at this boundary. The new review target therefore starts from constructor-backed Step (xi) data, projects to the obligations-backed record-canonical corridor, and only then uses the constructor-built finite-divisor route. The experiment surface exposes this as:

`MatchedConstructorBackedObligationsCorridorAudit`,
`MatchedConstructorBackedObligationsRemainingPayloadCorridorAudit`,
`MatchedConstructorBackedObligationsBoundaryAudit`,
`MatchedConstructorBackedObligationsCThetaAudit`.

The remaining-payload corridor threads the same constructor-built payload ledger through this projection, so the constructor-backed exact source no longer hides the unresolved Step (xi) hull operation, determinant source, bridge, positivity, and normalization payload behind the full-route audit. There is now a side-conditioned variant of this matched constructor-backed exact source as well. Its Step (xi) field is a `SideConditionedConstructorBacked` source, so q-pilot positivity and source normalization are projected from one `IUTStage1SourceSideConditions` object before the constructor-backed source is projected to the obligations-backed record-canonical corridor and the constructor-built finite-divisor route. The Hodge/SHE/IPL middle layer is also exposed separately by `MatchedSideConditionedHodgeSHEIPLAudit`: this audit records the theta-monoid degree match and then projects the theta-root evaluation, factored SHE preservation, history-separated finite Hodge/SHE transport, constructed IPL link, and finite-transport log-volume provenance from the target-charted Hodge–Arakelov/IPL construction source. The experiment surface exposes this boundary as:

`MatchedSideConditionedHodgeSHEIPLAudit`,
`MatchedSideConditionedConstructorBackedObligationsCorridorAudit`,
`MatchedSideConditionedConstructorBackedObligationsRemainingPayloadCorridorAudit`,
`MatchedSideConditionedConstructorBackedObligationsBoundaryAudit`,
`MatchedSideConditionedConstructorBackedObligationsCThetaAudit`.

The same strongest side-conditioned corridor is now exposed under the compact source-derived milestone names

`sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ_`
`boundaryEndpointWithMatchedSideConditionedConstructorBackedObligationsAudit`,
`sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ_`
`cThetaDichotomyWithMatchedSideConditionedConstructorBackedObligationsAudit`.

These are review-surface aliases rather than new mathematical assumptions: they return the existing side-conditioned Hodge/SHE/IPL audit, the named side-conditioned Step (xi) hull/determinant audit, and the projected constructor-backed obligations corridor under the milestone endpoint name. The matched-obligations conditional endpoint remains exposed under the explicit audit name

```
boundarySignedEqualityOrStrictCTheta_from
  _sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQ
  _withMatchedSideConditionedConstructorBackedObligationsAudit.
```

The compact theorem name without this suffix is now reserved for the constructed-IPL constructor-built route described above. The matched audit variant uses the strongest side-conditioned constructor-backed corridor, keeps the numeric C_Θ estimate explicit, and returns the same matched Hodge/SHE/IPL and Step (xi) audit package as the longer descriptive theorem. It now also has a matched side-conditioned remaining-boundary ledger, `MatchedSideConditionedConstructorBackedRemainingSourceBoundaryAudit`, exposed on the experiment surface with suffix

```
matchedSideConditionedConstructorBacked
  RemainingBoundaryAudit.
```

This ledger is the strongest current review target for what is still public at the source-paper boundary: the matched Hodge–Arakelov/IPL source, the side-conditioned constructor-backed Step (xi) source, the upper-semi finite-divisor Step (x) source, and the exact vertical-IQ source. It now also carries the projected constructor-backed remaining-payload corridor audit, so the route-level remaining-source ledger explicitly includes the constructor-built Step (xi) payload audit rather than only the older full obligations corridor and assembled remaining-boundary ledger. The matched side-conditioned Hodge/SHE/IPL audit now also carries the choice-wise input-prime-strip links directly. The experiment endpoint with suffix `matchedSideConditionedConstructedIPLChoiceLinkEndpoint` exposes that, for every possible output choice, the constructed IPL link has source equal to the route input prime strip and target equal to the choice-indexed output prime strip, before the Step (xi) hull/determinant source or exact vertical-IQ boundary is used. The constructor-backed Step (xi) audit has likewise been strengthened: it now records q-pilot containment in the Theorem 3.11 possible-image union, inclusion of that union in the holomorphic hull, the bounded-family determinant/log-volume equalities, the tensor-power bound, the record-canonical hull/tensor bridge, and the raw signed comparison as one source-facing skeleton. The experiment suffix `stepXIHolomorphicHullDeterminantEndpoint` exposes this skeleton before the finite-divisor vertical-IQ projection. The same skeleton is now carried by the strongest matched side-conditioned finite-exact source ledger itself, via `MatchedSideConditionedStepXIHolomorphicHullDeterminantSkeleton`; the experiment suffix `matchedSideConditionedStepXIHolomorphicHullDeterminantSkeleton` exposes it from the bundled source object that also contains the matched Hodge/SHE/IPL, finite-divisor, and exact vertical-IQ data. The no-C_Θ and conditional C_Θ milestone endpoints now also have remaining-boundary variants, `MatchedSideConditionedConstructorBackedRemainingBoundaryAudit` and `MatchedSideConditionedConstructorBackedRemainingCThetaAudit`. These variants return the projected obligations boundary together with the remaining-source ledger, so the public endpoint no longer drops the Step (xi) holomorphic-hull/determinant skeleton when it reaches the boundary or C_Θ audit surface.

The synchronized possible-image summand finite-divisor boundary now has a Step (xi) log-volume-chain audit endpoint as well. The new source declaration with suffix `withStepXILogVolumeChainAudit` refines the no-C_Θ q-comparison boundary by retaining the finite Step (x) packet and vertical-IQ equalities together with the summand-calibrated chain

$$\mu_{\log}(q\mathrm{Region}) \leq \det_{\log} \leq \theta_{\mathrm{signed}}.$$

This gives a sharper review surface for the current 3.11-to-3.12 corridor: the source-derived Step (xi) hull/determinant comparison and the finite Step (x) boundary are visible before the separate numerical C_Θ-estimate is supplied.

The constructor-built possible-image Step (xi) source can now also be constructed directly from an Ob3/Ob4 adjusted determinant package. The new constructor `ofOb3Ob4AdjustedDeterminantSource` takes the localization summands, structure-sheaf subtraction, and positive tensor-power data, projects the weighted determinant internally, and returns the existing constructor-built Ob3/Ob4 audit for the resulting possible-image hull source. Its experiment endpoint exposes the raw adjusted-localization identities, the weighted determinant summation, tensor-power normalization, and the $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison. Thus another piece of the Step (xi) determinant payload has moved from an already-weighted determinant input to a source-facing Remark 3.9.5(vii) Ob3/Ob4 construction.

There is now also a minimal-hull-system variant of this constructor,

```
possibleImageConstructorBuiltHullSource
  _ofRemark395HullSystemOb3Ob4AdjustedDeterminantSource
  _endpoint.
```

This route starts from the intersection-defined `IUTStage1Remark395HolomorphicHullSystem`, projects the older hull/log-volume shadow internally, and identifies the resulting Step (xi) hull with

$$\phi\Big(\bigcup_i P_i\Big)$$

for the Theorem 3.11 possible-image family. The endpoint proves that this object is a source-paper hull, that the chosen q-region is contained in it, and then returns the same Ob3/Ob4 determinant audit, bridge-inequality audit, and raw signed comparison. Thus this constructor form no longer treats the holomorphic-hull shadow as a supplied Step (xi) payload.

The constructor-built possible-image route has now been refined one step further to the product-hull formulation of Remark 3.9.5(i)–(ii). The new source constructor

```
ofRemark395ProductHullSystem
  Ob3Ob4AdjustedDeterminantSource
```

starts from an `IUTStage1Remark395ProductHullSystemSource`: a family of product hulls, a parameter selecting the intersection of all product hulls containing a region, and the corresponding log-volume calibration. The endpoint proves that the Step (xi) hull attached to $\bigcup_i P_i$ is exactly the product hull selected by this intersection parameter, that the chosen q-region is contained in it, and that it is minimal among product hulls containing $\bigcup_i P_i$. It then returns the same constructor-built Ob3/Ob4 determinant audit, bridge-inequality audit, and raw signed comparison. Thus the corridor now reaches the source-paper "smallest product hull" formulation before projecting to the older holomorphic-hull shadow. This product-hull provenance is now also available for any already constructed possible-image Step (xi) source whose hull/log-volume shadow is induced by a product-hull system:

```
possibleImageConstructorBuiltHullSource
  _productHullBackedConstructorBuiltAudit.
```

At the product-hull-backed finite-divisor boundary, the Ob5 quotient passage now also has a scalar quotient compatibility audit:

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _ob5ScalarQuotientCompatibilityAudit.
```

This theorem applies the same upper-semi quotient formalism to the real scalar $\mu_{\log}(\phi(P_B))$: the singleton images of $\det_{\log}^{\mathrm{norm}}$, $\det_{\log}$, the normalized $\det^{\otimes M}$ tensor-power scalar, and the family-hull log-volume all collapse to the distinguished quotient point, and hence have the same quotient image. Thus the product-hull route now records the compatibility with $\det^{\otimes M}(\phi(P_B))$ and $\mu_{\log}(\phi(P_B))$ that Remark 3.9.5(Ob5) states alongside the bounded-family hull quotient. The audit identifies the source's canonical hull of $\bigcup_i P_i$ with the selected smallest product hull, proves containment of both $\bigcup_i P_i$ and the q-pilot possible image in that hull, records minimality and product-hull log-volume, and keeps the record-canonical Step (xi) and bridge-inequality audits attached. The synchronized target-charted possible-image summand/family-hull source has a corresponding wrapper,

```
synchronizedTargetChartedPossibleImageSummand
  HodgeFamilyHull_productHullConstructorBuiltAudit,
```

which threads this product-hull audit through the Hodge–Arakelov chart, finite summand equality, exact theta/family-hull equality, and obligations-backed constructor reduction. Thus the strongest Hodge/summand Step (xi) source can now be reviewed together with the Remark 3.9.5 product-hull boundary, instead of reserving product-hull provenance only for the later finite-divisor wrapper.

The same constructor-built source now exposes the Step (xi) bridge inequality as a named audit `ConstructorBuiltStepXIBridgeInequalityAudit`. This audit records the selected q-pilot log-volume, the normalized determinant/tensor-power log-volume, and the package theta scalar in the direct chain

$$\mu_{\log}(q\mathrm{Region}) \le \det_{\mathrm{norm}} \le \theta_{\mathrm{signed}}.$$

The first inequality is derived from the upper-ray/hull absorption audit and the Ob3-3 determinant compatibility; the second is the Ob4 normalized tensor-power bound. The experiment surface exposes this as `possibleImageConstructorBuiltHullSource_bridgeInequalityAudit`, and the Ob3/Ob4-adjusted constructor endpoint now returns this bridge audit next to the determinant audit.

There is also an exact-theta constructor-built variant of the adjusted-summand route,

```
remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  _toPossibleImageConstructorBuiltHolomorphicHullDeterminantSource
  OfThetaEqFamilyHullLogVolume_endpoint.
```

Here the constructor no longer accepts the Ob4 tensor-power bound as a free input. It starts from the finite Ob3-3 summand identity, derives the record Ob3/Ob5 family-hull determinant source, assumes the source-facing exact-theta calibration

$$\theta_{\mathrm{signed}} = \mu_{\log}\left(\phi\left(\bigcup_i P_i\right)\right),$$

and then obtains

$$\det_{\mathrm{norm}} \le \theta_{\mathrm{signed}}$$

from the bounded-family theorem `tensorPower_bound_of_theta_eq_familyHullLogVolume`. The package bridge equality is still an explicit Step (xi) synchronization boundary, but the numerical Ob4 inequality is now constructed from the Remark 3.9.5 family-hull/log-volume source in this route.

The adjusted-summand source now also exposes the intermediate bridge itself, before assembling the final constructed Step (xi) object. The generic source constructor

```
IUTStage1Remark395HullDeterminantBridgeSource
  .ofAdjustedDeterminantLogVolumeSource
```

starts from the finite Ob3–3 identity

$$\mu_{\log}(\phi(\bigcup_i P_i)) = \sum_\beta \mathrm{AdjLoc}_\beta,$$

derives the Ob3/Ob5 determinant compatibility, and then proves the bridge chain

$$\mu_{\log}(q\mathrm{Region}) \leq \det_{\mathrm{norm}} = \det_{\log} \leq \theta_{\mathrm{signed}}.$$

The record-canonical companion `toRecordHullDeterminantBridgeSource` fixes the family to the Theorem 3.11 possible images, while the normalized experiment endpoints

`remark395HullDeterminantBridgeSource_normalizedBridge`,
`remark395RecordHullDeterminantBridgeSource_normalizedBridge`

make the direct $\det_{\mathrm{norm}}$-bridge reviewable without unpacking the longer endpoint. The experiment endpoint

```
remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  _toRecordHullDeterminantBridgeSource_endpoint
```

makes this bridge-level derivation reviewable without unfolding the later constructed-source route. The named finite-divisor milestone theorem

```
boundarySignedEqualityOrStrictCTheta
  _from_remark395ConstructedHullDeterminant
  FiniteDivisorVerticalIQ
```

now also has the audited companion with suffix

`_withRecordBridgeAudit`.

This companion returns the constructed record-canonical Remark 3.9.5 bridge audit together with the final conditional C_Θ dichotomy, so the public milestone name exposes the chain

$$\mu_{\log}(q\mathrm{Region}) \leq \det_{\log} \leq \theta_{\mathrm{signed}}$$

beside the finite-divisor vertical-IQ conclusion.

There is now also an exact-theta bridge-level variant with suffix

```
toRecordHullDeterminantBridgeSource
  OfThetaEqFamilyHullLogVolume
```

and experiment endpoint

```
remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  _toRecordHullDeterminantBridgeSource
  OfThetaEqFamilyHullLogVolume_endpoint.
```

This version no longer takes the Ob4 tensor-power bound as an argument at the bridge surface. The generic bounded-family theorem `IUTStage1BoundedFamilyHullDetLogVolumeSource .tensorPower_bound_of_theta_eq_familyHullLogVolume` derives the bound from

$$\theta_{\mathrm{signed}} = \mu_{\log}(\phi(\bigcup_i P_i)),$$

after the adjusted-summand source has already produced the Ob3/Ob5 determinant equality. Thus the intermediate bridge now has both forms: an ordinary form with an explicit Ob4 bound, and an exact-theta form in which that bound is constructed from the family-hull calibration.

The same exact-theta constructor-built route now has an obligations-backed form, with suffix

```
OfThetaEqFamilyHullLogVolumeFromObligations_endpoint.
```

This version consumes an `IUTStage1SourceHullDetObligations` object and a single synchronization statement identifying its hull/determinant bridge datum with the record-canonical Step (xi) construction. Lean then derives the package bridge equality from `hullDetData.hullDetBridge_eq_bridgeData`, and reads q-pilot positivity and source normalization from the same obligations object. Thus, at this constructor-built boundary, the remaining supplied Step (xi) data has been narrowed from four separate fields—tensor-power bound, bridge equality, q-positivity, and normalization—to the family-hull exact-theta calibration plus one obligations-backed record-canonical bridge synchronization.

The exact-theta adjusted-summand route has now also been tied to the local canonical C_Θ-scale audit. The source declaration

```
IUTStage1Remark395RecordOb3Ob5AdjustedDeterminantLogVolumeSource
  .toPossibleImageConstructorBuiltHolomorphicHullDeterminantSource
  OfThetaEqFamilyHullLogVolume_canonicalCThetaBridgeAudit
```

returns the constructor-built Remark 3.9.5 canonical-scale bridge audit together with the reviewable chain

$$\mu_{\log}(q\mathrm{Region}) \leq \det_{\mathrm{norm}} \leq \theta_{\mathrm{signed}} \leq C_{\Theta,\mathrm{can}} \cdot (-q_{\mathrm{signed}}).$$

Its obligations-backed companion, with suffix

```
OfThetaEqFamilyHullLogVolumeFromObligations_canonicalCThetaBridgeAudit,
```

adds the facts that the package hull/determinant bridge is the obligations bridge datum, $0 < -q_{\mathrm{signed}}$, and source normalization, all projected from the same obligations object. Thus the no-external-numeric-bound local scale is now attached directly to the finite adjusted-summand Ob3/Ob5, exact-theta Ob4, possible-image q-choice constructor, rather than only to the later constructed-source wrapper.

This canonical-scale bridge has now been lifted to the target-charted, measure-calibrated adjusted-summand sources as well. The audit

```
targetChartedHodgeMeasureCalibratedAdjustedPossibleImage
  HullDetObligationsBackedSource_canonicalCThetaBridgeAudit
```

projects the same lower constructor-built Remark 3.9.5 canonical-scale audit from the obligations-backed adjusted exact-theta source. Its constructor-backed companion

```
targetChartedHodgeConstructorBackedMeasureCalibrated
  AdjustedPossibleImageHullDetSource_canonicalCThetaBridgeAudit
```

does the same after replacing the obligations bridge by the Theorem 3.11 record-canonical hull/tensor-power constructor. Thus the charted Hodge calibration, finite Ob3–3 adjusted summands, measured family hull, possible image q-choice, and local no-external-bound $C_{\Theta,\mathrm{can}}$ chain now occur in one reviewable audit path.

There is now a named obligations-backed audit for the target-charted, measure-calibrated adjusted-summand possible-image source:

```
targetChartedHodgeMeasureCalibratedAdjustedPossibleImage
  HullDetObligationsBackedSource_adjustedExactTheta
  StepXIAudit.
```

This audit records the exact-theta source before passing to the constructor-backed form. The q-region is the chosen Theorem 3.11 possible image, the package measure is projected from the measured family hull, the transported target theta chart is calibrated to the Hodge–Arakelov theta-monoid degree, that degree is identified with the finite Ob3–3 adjusted summand sum, and the record family-hull/determinant source yields

$$\theta_{\mathrm{signed}} = \mu_{\log}\left(\phi\left(\bigcup_i P_i\right)\right).$$

The same audit keeps the obligations-backed hull/determinant bridge, q-positivity, source normalization, and the raw $q_{\mathrm{signed}} \le \theta_{\mathrm{signed}}$ comparison visible in one source-boundary object.

The constructor-backed adjusted-summand source now has a companion audit,

```
targetChartedHodgeConstructorBackedMeasureCalibrated
  AdjustedPossibleImageHullDetSource_adjustedExactTheta
  StepXIAudit.
```

It carries the same target-charted Hodge and Ob3–3 summand provenance for $\theta_{\mathrm{signed}} = \mu_{\log}(\phi(\bigcup_i P_i))$, but replaces the obligations-backed bridge boundary by the hull/determinant obligations constructed from the Theorem 3.11 record-canonical hull/tensor-power constructor. Thus the constructor-backed review surface no longer loses the source of the exact-theta equality when it records the constructed bridge, q-positivity, normalization, and raw signed comparison.

The same constructor-backed exact-theta source now projects directly into the current finite-divisor milestone source:

```
targetChartedHodgeConstructorBackedMeasureCalibrated
  AdjustedPossibleImageHullDetSource
  _toRemark395ConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQSource.
```

The projection fills the finite-divisor source's $\theta_{\mathrm{signed}} = \mu_{\log}(\phi(\bigcup_i P_i))$ field from the already audited Hodge theta-monoid and finite adjusted determinant summand chain, rather than accepting this equality again as a bare boundary input. Its companion audit

```
targetChartedHodgeConstructorBackedMeasureCalibrated
  AdjustedPossibleImageHullDetSource
  _toRemark395ConstructorBackedConstructedExactTheta
  FiniteDivisorAudit
```

keeps, in one object, the target-charted theta equality, the theta-monoid degree equality with the adjusted summand sum, the Ob3/Ob5 summand-to-determinant identities, and the constructed finite-divisor record-bridge audit proving the raw $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison. This is the current Lean bridge from the exact-theta Ob3/Ob5 implementation into the strongest constructor-backed finite-divisor corridor.

This exact-theta bridge now propagates one level further into the Ob5 product-hull hierarchy. The new internal constructors

```
toProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQSource,
toPrincipalProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQSource,
toPrincipalValuationBallBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQSource
```

start from the same target-charted exact-theta adjusted-summand source. They then add, respectively, the Remark 3.9.5 product hull, the principal $\lambda\mathcal{O}$ hull, and the valuation-ball direct-product cover identification. Their experiment-surface audits

```
_productHullBackedExactThetaFiniteDivisorAudit,
_principalProductHullBackedExactThetaFiniteDivisorAudit,
_principalValuationBallBackedExactThetaFiniteDivisorAudit
```

retain the Hodge theta-monoid/adjusted-summand equality while exposing the product-hull containment audit, the principal scalar-multiple hull identity, or the valuation-ball direct-product and calibrated-cell log-volume identities. Thus the product/principal/Haar corridor can now be reviewed as an extension of the exact-theta Step (xi) source rather than as a separate supplied constructor-backed payload. The principal valuation-ball projection has a tighter source-core variant as well. The named theorem

```
adjustedSource_eq_valuationBall
  _ofValuationBallRecordSource
```

derives the equality between the exact-theta measured adjusted source and the principal valuation-ball Ob3/Ob5 source from two narrower inputs: the valuation-ball possible-image family is synchronized with the Theorem 3.11 record family, and the measured adjusted source is the record-canonical source constructed from that valuation-ball cover. The companion constructor and audit,

```
toPrincipalValuationBallBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQSource
  OfValuationBallRecordSource,
toPrincipalValuationBallBackedExactThetaFiniteDivisorAudit
  OfValuationBallRecordSource,
_principalValuationBallBackedExactThetaFiniteDivisorAudit
  OfValuationBallRecordSource,
```

therefore no longer require the valuation-ball source-core equality as a standalone payload at this boundary.

The exact-theta source now reaches the final product-hull and principal valuation-ball dichotomy endpoints, not only the intermediate audits. The new experiment-surface wrappers

```
_productHullBackedCanonicalCThetaScaleDichotomy
  WithOb5Ob6Ob7SynchronizedAuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource,
_principalValuationBallCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource,
_principalValuationBallCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

take the target-charted exact-theta adjusted-summand source, construct the product-hull-backed or principal valuation-ball-backed finite-divisor source internally, and then invoke the strongest current finite product-formula Ob7 route. Consequently the returned C_Θ dichotomy now carries the exact-theta Step (xi) provenance through the Remark 3.9.5 product-hull, principal $\lambda\mathcal{O}$, valuation-ball topology/Haar, Ob5 quotient, Ob6 approximant, and Ob7 log-Kummer/product-formula layers. The record-canonical valuation-ball variant derives the principal valuation-ball source-core synchronization from the same possible-image family equality with the Theorem 3.11 record and the equality with the record-canonical valuation-ball adjusted source before invoking this final topology/Haar C_Θ endpoint.

The exact-theta route has also been pushed through the projected-factorwise local log-coordinate refinement. The constructor

```
_toValuationUnitBallNonzeroScalar
  LocalLogCoordinateProjectedFactorwiseTargetTransportSource,
_toValuationUnitBallNonzeroScalar
  LocalLogCoordinateProjectedFactorwiseTargetTransportSource
  OfValuationBallRecordSource
```

builds the projected-factorwise transport source from the target-charted Step (xi) source by first constructing the principal valuation-ball finite-divisor source and then attaching the literal dependent-product local factor data. The record-canonical variant derives the principal valuation-ball source-core equality from the Theorem 3.11 possible-image synchronization and the record-canonical valuation-ball adjusted source before adding these local coordinate factors. The corresponding final wrappers

```
_valuationUnitBallNonzeroScalarProjectedFactorwise
  CanonicalCThetaScaleDichotomyWithTopologyFiniteExtensionHaar
  Ob5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource,
_valuationUnitBallNonzeroScalarProjectedFactorwise
  CanonicalCThetaScaleDichotomyWithTopologyFiniteExtensionHaar
  Ob5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

returns the target-point transport audit together with the principal topology/Haar Ob5–Ob7 audit and the canonical-scale dichotomy. Thus the current strongest endpoint carries, simultaneously, exact-theta Step (xi) provenance, explicit product-index/valuation-place transport, local $\lambda_v O_v$ scalar-image regions, valuation-unit-ball provenance, and finite product-formula Ob7 provenance.

The same exact-theta final route now has a valuation-log-image form. The constructor

```
_toValuationUnitBallNonzeroScalarValuationLogImage
  ProjectedFactorwiseTargetTransportSource,
_toValuationUnitBallNonzeroScalarValuationLogImage
  ProjectedFactorwiseTargetTransportSource
  OfValuationBallRecordSource
```

builds the projected-factorwise source from the target-charted Step (xi) source while defining the local coordinate regions as $\ell_v(O_v)$ and $\ell_v(\lambda_v O_v)$. The record-canonical version derives the principal valuation-ball source-core equality from the Theorem 3.11 possible-image family synchronization and the record-canonical valuation-ball adjusted source. The corresponding final wrappers with suffix

```
_valuationUnitBallNonzeroScalarValuationLogImage
  ProjectedFactorwiseCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource,
_valuationUnitBallNonzeroScalarValuationLogImage
  ProjectedFactorwiseCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

returns the same target-point transport audit, principal topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy, but no longer accepts separate per-place image-equality hypotheses for the local coordinate regions. In the record-canonical variant it also no longer accepts the valuation-ball source-core equality as an independent input.

The finite-sum target-point form now has the same record-canonical strengthening. The constructor

```
_toValuationUnitBallNonzeroScalarValuationLog
  FiniteSumTargetPointSource
  OfValuationBallRecordSource
```

uses the actual point map

$$x \longmapsto \left\{ \mathrm{coord} = \sum_v \ell_v(x_v) \right\}$$

and asks only for the two transported target-point synchronizations with the principal local-integer and selected-hull regions. The valuation-ball source-core equality is derived from the same record-canonical valuation-ball source used above. The corresponding final endpoint

```
_valuationUnitBallNonzeroScalarValuationLog
  FiniteSumTargetPointCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

returns the target-point transport audit, the principal topology/Haar Ob5–Ob7 audit, and the canonical-scale dichotomy at this lower boundary. Thus the source-facing route has descended from per-place image regions to the finite local valuation/log sum as a point-level transport construction, while keeping the Theorem 3.11 possible-image synchronization as the source of the principal valuation-ball comparison.

This record-canonical finite-sum endpoint has also been pushed through the coordinate-place refinement. The new constructor

```
_toValuationUnitBallNonzeroScalarCoordinatePlace
  ValuationLogFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

uses the valuation cover's coordinate-place map as an equivalence $\delta \simeq \gamma_{\text{local}}$ and derives the older local-index compatibility from it. Its final endpoint

```
_valuationUnitBallNonzeroScalarCoordinatePlace
  ValuationLogFiniteSumTargetPointCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

therefore reaches the same target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy from the source-facing coordinate/place synchronization rather than from an arbitrary supplied index/place compatibility.

The same record-canonical route now descends one step further to normalized Haar log-coordinates. The constructor

```
_toValuationUnitBallNonzeroScalarHaarLogCoordinate
  FiniteSumTargetPointSource
  OfValuationBallRecordSource
```

replaces the raw local maps ℓ_v by local Haar-log-coordinate sources: each $\ell_v(x_v)$ is the normalized Haar log-volume of the source's local region attached to x_v. The transported target-point equalities remain the two synchronizations with the principal local-integer and selected-hull regions, and the principal valuation-ball source-core equality is still derived from the record-canonical valuation-ball source. The final endpoint

```
_valuationUnitBallNonzeroScalarHaarLogCoordinate
  FiniteSumTargetPointCanonicalCThetaScaleDichotomy
  WithTopologyFiniteExtensionHaarOb5Ob6Ob7AuditAtQChoice
  OfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

therefore reaches the target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy from a source boundary that records the Haar-log-volume origin of every local valuation/log coordinate.

The compact-open Haar finite-sum layer has now received the same record-canonical upgrade. The constructor

```
_toValuationUnitBallNonzeroScalarCompactOpen
  HaarLogCoordinateFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

starts from local compact-open Haar-log-coordinate sources, so every local region U_{v,x_v} carries finite-extension topological Haar provenance before being sent to the target point $\{\mathrm{coord} = \sum_v \ell_v(x_v)\}$. As in the preceding Haar layer, the transported target-point equalities are checked against the principal local-integer and selected-hull regions of the constructed valuation-ball source, while the source-core equality is derived from the record-canonical valuation-ball adjusted source. The corresponding final endpoint

```
_valuationUnitBallNonzeroScalarCompactOpen
  HaarLogCoordinateFiniteSumTargetPointCanonicalCThetaScale
  DichotomyWithTopologyFiniteExtensionHaarOb5Ob6Ob7
  AuditAtQChoiceOfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

therefore reaches the target-point transport audit, the principal topology/Haar Ob5–Ob7 audit, and the canonical-scale dichotomy from the compact-open local Haar boundary, without accepting the valuation-ball source-core equality as an independent hypothesis at this level.

The centered valuation-ball Haar finite-sum layer now has the same record-canonical form. The constructor

```
_toValuationUnitBallNonzeroScalarCenteredValuationBall
  HaarLogCoordinateFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

specializes each compact-open local factor to a centered closed valuation ball $\overline{B}(x_v, r_v)$ in a proper ultrametric local carrier. The local source supplies the centered-ball Haar log-coordinate and proves the compact-open provenance needed by the preceding layer. The record-canonical constructor still derives the principal valuation-ball source-core equality from the valuation-ball adjusted source, so the two transported target-point equalities are the only remaining local synchronization data at this boundary. The final endpoint

```
_valuationUnitBallNonzeroScalarCenteredValuationBall
  HaarLogCoordinateFiniteSumTargetPointCanonicalCThetaScale
  DichotomyWithTopologyFiniteExtensionHaarOb5Ob6Ob7
  AuditAtQChoiceOfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

returns the same target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy after projecting through the compact-open Haar finite-sum route.

The valued-field integer centered-ball layer has also been made record-canonical. The constructor

```
_toValuationUnitBallNonzeroScalarValuedFieldInteger
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

replaces the abstract proper-ultrametric local carrier by a nontrivially-normed valued field with its integer-source Haar normalization. Thus the local centered ball and its compact-open Haar provenance are read from valued-field integer data before the source projects to the centered-ball finite-sum corridor. The final endpoint

```
_valuationUnitBallNonzeroScalarValuedFieldInteger
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPoint
  CanonicalCThetaScaleDichotomyWithTopologyFiniteExtension
  HaarOb5Ob6Ob7AuditAtQChoiceOfCoricThetaMuProductFormula
  ConstructionSourceOfValuationBallRecordSource
```

returns the same target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy after deriving the valuation-ball source-core equality from the record-canonical valuation-ball adjusted source.

The concrete p-adic centered-ball layer now has the same record-canonical endpoint. The constructor

```
_toValuationUnitBallNonzeroScalarPadicCenteredValuationBall
  HaarLogCoordinateFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

fixes the local factor at a valuation place v to the actual field $\mathbb{Q}_{p_v}$, with the integer ring, residue prime, and Haar normalization supplied by the p-adic centered valuation-ball source. It still derives the principal valuation-ball source-core equality from the record-canonical valuation-ball adjusted source, and only retains the two transported target-point synchronizations with the constructed principal regions. The final endpoint

```
_valuationUnitBallNonzeroScalarPadicCenteredValuationBall
  HaarLogCoordinateFiniteSumTargetPointCanonicalCThetaScale
  DichotomyWithTopologyFiniteExtensionHaarOb5Ob6Ob7
  AuditAtQChoiceOfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

then reaches the target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy after projecting through the valued-field integer finite-sum route.

The concrete finite-extension p-adic centered-ball layer now has the same record-canonical strengthening. The constructor

```
_toValuationUnitBallNonzeroScalarPadicFiniteExtension
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

replaces the base local factor $\mathbb{Q}_{p_v}$ by an actual finite-dimensional local extension $K_v/\mathbb{Q}_{p_v}$. The local Haar coordinate records the finite-extension degree $\operatorname{finrank}_{\mathbb{Q}_{p_v}} K_v$, the residue prime p_v, and the uniformizer scaling map $x \mapsto p_v x$ through the algebra map into K_v. As in the base p-adic endpoint, the principal valuation-ball source-core equality is derived from the record-canonical valuation-ball adjusted source; the endpoint only retains the two target-point synchronizations identifying the transported local integer and selected-scalar images with the constructed principal regions. The final endpoint

```
_valuationUnitBallNonzeroScalarPadicFiniteExtension
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPoint
  CanonicalCThetaScaleDichotomyWithTopologyFiniteExtension
  HaarOb5Ob6Ob7AuditAtQChoiceOfCoricThetaMuProductFormula
  ConstructionSourceOfValuationBallRecordSource
```

then reaches the same target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy through the finite-extension valued-field integer projection.

The finite-extension p-adic endpoint has also been strengthened by base-prime dilation topology in record-canonical form. The constructor

```
_toValuationUnitBallNonzeroScalarPadicFiniteExtensionDilation
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPointSource
  OfValuationBallRecordSource
```

uses the same record-canonical valuation-ball adjusted source to construct the principal hull/determinant source, but each local factor now supplies a homeomorphism-backed Haar source for multiplication by the image of p_v in K_v. Thus the local source carries the finite-extension degree and residue-prime data of the previous layer together with the open/compact provenance of the base-prime dilation used by the finite-extension Haar normalization. The final endpoint

```
_valuationUnitBallNonzeroScalarPadicFiniteExtensionDilation
  CenteredValuationBallHaarLogCoordinateFiniteSumTargetPoint
  CanonicalCThetaScaleDichotomyWithTopologyFiniteExtension
  HaarOb5Ob6Ob7AuditAtQChoiceOfCoricThetaMuProductFormula
  ConstructionSourceOfValuationBallRecordSource
```

then projects through the finite-extension layer and returns the same target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy.

The same finite-extension local-field boundary now has a record-canonical constructed dilation-mass refinement. The constructor

```
_toValuationUnitBallNonzeroScalarPadicFiniteExtensionConstructed
  DilationMassCenteredValuationBallHaarLogCoordinateFiniteSum
  TargetPointSourceOfValuationBallRecordSource
```

keeps the valuation-ball record source as the origin of the principal hull/determinant object, while each local factor supplies the constructed nonzero base-prime dilation and the real-valued Haar scaling law

$$\mu_v(p_v S_v)_{\mathbb{R}} = \mu_v(S_v)_{\mathbb{R}} / p_v^{[K_v:\mathbb{Q}_{p_v}]}$$

for all local subsets S_v. The final endpoint

```
_valuationUnitBallNonzeroScalarPadicFiniteExtensionConstructed
  DilationMassCenteredValuationBallHaarLogCoordinateFiniteSum
  TargetPointCanonicalCThetaScaleDichotomyWithTopology
  FiniteExtensionHaarOb5Ob6Ob7AuditAtQChoiceOfCoricThetaMu
  ProductFormulaConstructionSourceOfValuationBallRecordSource
```

therefore exposes the local nonzero scalar facts, pointwise dilation formula, all-subset Haar mass formula, target-point transport audit, topology/Haar Ob5–Ob7 audit, and canonical-scale dichotomy in one record-canonical route.

The record-canonical finite-extension boundary has now been lowered one step further from real-valued mass scaling to the measure-level Haar modulus. The new constructor

```
_toValuationUnitBallNonzeroScalarPadicFiniteExtensionConstructed
  DilationHaarModulusCenteredValuationBallHaarLogCoordinate
  FiniteSumTargetPointSourceOfValuationBallRecordSource
```

uses the same valuation-ball record source for the principal hull/determinant object, but each local factor now supplies the identity in $\mathbb{R}_{\geq 0} \cup \{\infty\}$

$$\mu_v(p_v S_v) = \mathrm{ofReal}\Big(p_v^{-[K_v:\mathbb{Q}_{p_v}]}\Big) \cdot \mu_v(S_v)$$

for every subset $S_v \subseteq K_v$. The corresponding endpoint

```
_valuationUnitBallNonzeroScalarPadicFiniteExtensionConstructed
  DilationHaarModulusCenteredValuationBallHaarLogCoordinate
  FiniteSumTargetPointCanonicalCThetaScaleDichotomyWithTopology
  FiniteExtensionHaarOb5Ob6Ob7AuditAtQChoiceOfCoricThetaMu
  ProductFormulaConstructionSourceOfValuationBallRecordSource
```

first exposes this all-subset Haar-modulus formula and then projects through the constructed dilation-mass corridor to the canonical-scale dichotomy. Thus the Step (xi) route now distinguishes the genuine measure identity from its `toReal` mass-shadow at the record-canonical experiment surface.

The same record-canonical route now has the next lower Haar-character refinement. The constructor

```
_toValuationUnitBallNonzeroScalarPadicFiniteExtensionConstructed
  DilationHaarCharacterCenteredValuationBallHaarLogCoordinate
  FiniteSumTargetPointSourceOfValuationBallRecordSource
```

keeps the valuation-ball principal hull/determinant source fixed, but replaces the local measure identity by the additive Haar-character scalar for multiplication by p_v:

$$\mathrm{addEquivAddHaarChar}(x \mapsto p_v x) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}.$$

The final record-canonical endpoint first exposes this scalar, then the derived all-subset Haar-modulus formula, and finally the same canonical-scale dichotomy through the Haar-modulus route. In this layer the analytic source datum is therefore the Haar-character evaluation; the measure-level formula is a derived projection.

The record-canonical local Haar boundary has also been lowered to normalized valuation-unit-ball mass. The new constructor

```
_toValuationUnitBallNonzeroScalarPadicFiniteExtensionUnitBall
  HaarCharacterCenteredValuationBallHaarLogCoordinateFiniteSum
  TargetPointSourceOfValuationBallRecordSource
```

retains the same principal valuation-ball hull/determinant source, while each local factor supplies

$$\mu_v(\mathcal{O}_v) = 1, \qquad \mu_v(p_v\mathcal{O}_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}.$$

The corresponding endpoint exposes these two local mass facts, derives the additive Haar-character scalar, derives the all-subset Haar-modulus formula, and then reaches the canonical-scale dichotomy through the Haar-character route.

The record-canonical surface now also uses the core specialization theorem that derives the unit-ball Haar-character source from the constructed all-subset Haar-modulus source. The new endpoint

```
_valuationUnitBallNonzeroScalarPadicFiniteExtensionConstructed
  DilationHaarModulusDerivedUnitBallHaarCharacterCentered
  ValuationBallHaarLogCoordinateFiniteSumTargetPointCanonical
  CThetaScaleDichotomyWithTopologyFiniteExtensionHaarOb5Ob6Ob7
  AuditAtQChoiceOfCoricThetaMuProductFormulaConstructionSource
  OfValuationBallRecordSource
```

takes the constructed Haar-modulus local source and regularity of the Haar measure, then derives $\mu_v(\mathcal{O}_v) = 1$, $\mu_v(p_v\mathcal{O}_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}$, the additive Haar-character scalar, the all-subset Haar-modulus formula, and the same canonical-scale dichotomy. Thus the $p_v\mathcal{O}_v$ mass is no longer a separate record-canonical hypothesis. The remaining local analytic gap below this layer is now the source construction of the valuation ring $\mathcal{O}_v$, its normalized Haar mass, Haar regularity, and the all-subset base-prime Haar-modulus identity from the actual p-adic/Frobenioid input data.

The milestone-facing constructed source now also factors its log-volume comparison through the generic record-canonical Remark 3.9.5 bridge source. The projection `toRecordHullDeterminantBridgeSource` packages the same ϕ-hull operator, Theorem 3.11 possible-image family, q-region containment, Ob3/Ob5 determinant compatibility, and Ob4 tensor-power bound as the record bridge source. The new constructed-record bridge audit, exposed on the experiment surface, records that the constructed source's q-to-determinant-to-theta chain is read from this generic Remark 3.9.5 bridge theorem rather than re-proved as an isolated comparison. This audit now also has a leaner proof shape: it stores the coarse record-bridge endpoint and the constructed-source endpoint, while Lean derives the named projection theorems for q-hull containment, determinant-log comparison, normalized determinant comparison, and the final $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison. In particular, the normalized determinant chain

$$\mu_{\log}(q\text{Region}) \le \det_{\text{norm}} \le \theta_{\text{signed}}.$$

is a theorem projected from the record bridge endpoint, not an additional field of the audit object. The experiment surface exposes this two-step normalized form as

`remark395ConstructedHolomorphicHullDeterminantSource_normalizedRecordBridge`.

The public constructor endpoints for the record bridge, the exact-theta bridge, the constructed source, and the possible-image-choice constructor now include the same normalized inequalities directly in their returned conjunctions. Thus the milestone route displays $\mu_{\log}(q\text{Region}) \le \det_{\text{norm}} \le \theta_{\text{signed}}$ at the point where the Step (xi) source is built, not only after a later audit projection. The constructor-backed finite-divisor source now exposes the same bridge projection directly. Its experiment-surface audit,

```
remark395ConstructorBackedConstructedHullDeterminant
  FiniteDivisorVerticalIQ_recordBridgeAudit,
```

starts from the constructor-backed adjusted source, constructs the record Remark 3.9.5 bridge internally, and records

$$\mu_{\log}(q\text{Region}) \le \det_{\text{norm}} = \det_{\log} \le \theta_{\text{signed}}$$

together with the raw $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison projected from the same Theorem 3.11 hull/determinant constructor. Thus the newest finite-divisor endpoint no longer hides the Step (xi) q-to-normalized-determinant-to-theta comparison behind only the constructed obligations equality; the remaining external input at the C_Θ endpoint is still the global numeric estimate. The corresponding compact endpoint

```
remark395ConstructorBackedConstructedOb3Ob5AdjustedHull
  DeterminantFiniteDivisorVerticalIQ
  _cThetaDichotomyWithRecordBridgeAudit
```

returns this constructor-backed record bridge audit together with the finite-divisor vertical-IQ C_Θ dichotomy. Thus the public milestone surface now carries the normalized Step (xi) bridge and the conditional Corollary 3.12 dichotomy in one theorem. The finite-divisor constructor-backed source now also has a product-hull-backed form. This source records that the measured holomorphic-hull operator used by the constructed Step (xi) source is induced by an `IUTStage1Remark395ProductHullSystemSource`. Its audit identifies

$$\phi(P_{\mathrm{rec}})$$

with the selected product hull obtained as the intersection of all product hulls containing the Theorem 3.11 possible-image union P_{rec}, proves $q \subseteq \phi(P_{\mathrm{rec}})$, proves minimality among such product hulls, and identifies the hull log-volume with the product-hull log-volume. The endpoint

```
remark395ProductHullBackedConstructorBackedConstructed
  Ob3Ob5AdjustedHullDeterminantFiniteDivisorVerticalIQ
  _cThetaDichotomyWithProductHullAudit
```

returns this product-hull audit together with the finite-divisor vertical-IQ C_Θ dichotomy. Thus the current finite-divisor milestone no longer forgets the source-paper "smallest product hull" formulation when it enters the downstream Corollary 3.12 corridor. The product-hull source now also has a smaller possible-image containment audit,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _possibleImageContainmentAudit.
```

This audit separates the Ob1/Ob2 absorption facts from the later Ob5 and determinant payloads. It records that the source-facing possible-image family has union P_{rec}, that both $\phi(P_{\mathrm{rec}})$ and the canonical $\Phi(P_{\mathrm{rec}})$ approximant are the selected smallest product hull, and that

$$P_{\mathrm{rec}} \subseteq \phi(P_{\mathrm{rec}}), \qquad P_i \subseteq \phi(P_{\mathrm{rec}}), \qquad q \subseteq P_{\mathrm{rec}} \subseteq \phi(P_{\mathrm{rec}}).$$

These containments are proved from the product-hull intersection law and the constructor-backed q-choice possible image; they are no longer hidden inside the later quotient/determinant audits. The same finite-divisor surface now has a principal-product-hull refinement. The source

```
IUTStage1Remark395PrincipalProductHullBackedConstructorBacked
  ConstructedHullDeterminantFiniteDivisorVerticalIQSource
```

uses a principal $\lambda \cdot O$ hull system instead of a bare product-hull system, then forgets to the existing product-hull-backed source. Its audit records that the selected hull is the scalar image

$$\lambda_{\mathrm{sel}} \cdot O,$$

that λ_{sel} satisfies the nonzero-parameter predicate, that the Theorem 3.11 possible-image union and q-pilot region lie in this selected principal product hull, and that this selected hull is minimal among principal product hulls containing the possible-image union. The public finite-divisor endpoint

```
remark395PrincipalProductHullBackedConstructorBackedConstructed
  Ob3Ob5AdjustedHullDeterminantFiniteDivisorVerticalIQ
  _cThetaDichotomyWithPrincipalProductHullAudit
```

returns this principal-hull audit together with the usual conditional C_Θ dichotomy. This pushes the product-hull boundary one step closer to the source-paper formula of Remark 3.9.5(i), where hulls are smallest regions of the form $\lambda \cdot O$. The product-hull-backed source also exposes an Ob5 quotient/determinant audit,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _ob5QuotientDeterminantAudit.
```

Given a nonempty q-choice possible image and a nonempty comparison possible image, this audit proves that both images collapse to the distinguished upper-semi quotient point modulo the selected product hull and hence have the same quotient image. It also identifies $\mu_{\log}(\phi(P_{\mathrm{rec}}))$ with the product-hull log-volume, the normalized determinant, the determinant log-volume, and the normalized tensor-power log-volume, while retaining the q-pilot log-volume bound into the selected product hull and the raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison. This is the first finite-divisor product-hull surface where the source-paper "smallest product hull" formulation and the Remark 3.9.5(Ob5) quotient-compatible determinant formalism are visible in one audit object. The same Ob5 boundary now has a q-pilot-derived variant:

```
productHullBackedOb5QuotientDeterminantAudit
  _of_qPilotRegion_nonempty.
```

Here Lean derives nonemptiness of the selected q-choice possible image from nonemptiness of the q-pilot region and the constructor-backed inclusion $q \subseteq P_{q\text{-choice}}$. Thus the q-side quotient collapse no longer needs a separate selected-possible-image nonemptiness premise at this product-hull boundary. The same source now also has a Φ/Ξ-approximant audit,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _phiXiApproximantAudit.
```

It constructs the source-facing possible-image family $P_B = \{P_\beta\}_{\beta \in B}$ directly from the product-hull system and the Theorem 3.11 possible-image regions. Lean proves that the family union is P_{rec}, that the canonical hull $\phi(P_B)$ is the selected smallest product hull, and that the canonical $\Phi(P_B)$ approximant is this same hull. If an exact $\Xi(P_B)$ family is supplied, the audit proves

$$H_{\Xi(P_B)} = \phi(P_B)$$

at the selected product-hull boundary, records the exact Ξ-member log-volume equality, and proves that $H_{\Xi(P_B)}$ has the same collapsed upper-semi quotient image as any nonempty comparison possible image. This threads the paper's $\Phi(P)$, $\Xi(P)$, and $H_{\Xi(P)}$ formalism into the strongest current finite-divisor product-hull route instead of leaving it only at the generic source-core endpoint. The same product-hull source now exposes the Ob6 log-volume version of this audit,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _ob6HullApproximantLogVolumeAudit.
```

This audit identifies both $H_{\Phi(P_B)}$ and $H_{\Xi(P_B)}$ with the selected product hull, proves

$$\mu_{\log}(P_B) \leq \mu_{\log}(H_{\Phi(P_B)}), \qquad \mu_{\log}(P_B) \leq \mu_{\log}(H_{\Xi(P_B)}),$$

and records the corresponding comparison-region and q-pilot inequalities into the $H_{\Xi(P_B)}$ log-volume. The normalized determinant equality and the raw $q_{\mathrm{signed}} \leq \theta_{\mathrm{signed}}$ comparison remain in the same audit object, so the Ob6 passage is now visible directly on the finite-divisor product-hull route. The product-hull boundary also exposes an exact-theta bridge audit,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _exactThetaBridgeAudit.
```

This audit does not add a new comparison possible image. It packages the source-facing identities already needed by the determinant route:

$$\theta_{\text{signed}} = \mu_{\log}(\phi(P_B)), \qquad \mu_{\log}(\phi(P_B)) = \underset{\text{norm}}{\det} = \det,$$

with $\phi(P_B)$ the selected smallest product hull. From the exact theta/family-hull equality, Lean derives the normalized tensor-power upper bound into θ_{signed}, and the audit records the chain

$$\mu_{\log}(q) \leq \mu_{\log}(\phi(P_B)) = \underset{\text{norm}}{\det} \leq \theta_{\text{signed}}.$$

Thus the strongest current product-hull surface separates the remaining open source task–constructing the actual IUT product-hull/determinant data–from the ordered-real and determinant-log-volume consequences already checked by the kernel. The same surface now has an Ob7 log-Kummer compatibility audit,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _ob7LogKummerCompatibilityAudit.
```

Given a finite $F^{\times\mu}$-prime-strip lift whose global realified Frobenioid log-volume agrees with the determinant log-volume, this audit identifies the family-hull and normalized determinant log-volumes with that prime-strip global value and proves

$$\mu_{\log}(q) \leq \mu_{\log}(F^{\times\mu}_{\text{global}}).$$

For every environment prime it also returns the corresponding Gaussian prime-to-place equality, equality of local realified log-volumes at the evaluated prime, and equality of Gaussian and environment unit characters. Thus the product-hull route now retains the Ob7 log-Kummer/Frobenioid data at the precise boundary where the determinant comparison enters the 3.12 corridor. The same Ob7 boundary now has a source-object form. The declarations with suffixes

```
Ob7LogKummerCompatibilityAuditOfSource,
Ob5Ob6Ob7SynchronizedAuditAtQChoiceOfQPilotRegionNonemptyOfOb7Source
```

take the typed Remark 3.9.5(Ob7) compatibility source itself, rather than a loose prime-strip lift together with a separate determinant/global-log-volume equality. The only synchronization left at this boundary is the proof that the Ob7 source's hull/determinant bridge is the constructed product-hull Step (xi) bridge. Thus the determinant-to-prime-strip comparison, local prime/place square, local realified Frobenioid readings, and unit-character transport are reviewed as one source-paper object. There is now a further coric form of the same Ob7 boundary. The core source

```
IUTStage1Remark395Ob7CoricLogKummerCompatibilitySource
```

uses the finite $F^{\vdash\times\mu}$ invariant: a common unit character on the valuation set V whose pullbacks are the environment and Gaussian unit characters. It projects to the ordinary Ob7 source, while the product audit

`Ob7CoricLogKummerCompatibilityAudit`

keeps the projected Ob7 determinant comparison together with the coric unit equalities. The q-choice Ob5–Ob7 synchronized and canonical-scale audits now also have `OfCoricOb7Source` forms, so the unit part of Ob7 is no longer represented only by a transported equality between two prime-strip unit characters. This coric boundary has now been lowered one more step toward the source-paper Corollary 4.7(iv) construction. The declaration

`ProductHullBackedCoricThetaMuOb7ConstructionSource`

does not take a completed $F^{\vdash\times\mu}$ invariant as input. It takes an $F^{\times\mu}$-prime-strip lift, a common coric unit character on V, the environment pullback equality, and the determinant/global log-volume identification. Lean constructs the coric invariant by

```
IUTStage1CoricThetaMuPrimeStripInvariant.
  ofLiftAndCoricUnitCharacter,
```

then derives the Gaussian coric pullback equality from the prime-strip evaluation square. The product-hull and principal topology/Haar audit surfaces now have `OfCoricThetaMuConstructionSource` forms, so this finite Corollary 4.7(iv) construction is visible at the same Ob5–Ob7 boundary as the Remark 3.9.5 hull and determinant data. The product-hull final dichotomy now has the corresponding `OfCoricOb7Source` endpoint: the canonical-scale C_Θ dichotomy is returned together with the synchronized Ob5–Ob7 audit while consuming the single coric $F^{\vdash\times\mu}$ Ob7 source and its bridge equality with the constructed Remark 3.9.5 hull/determinant source, rather than a raw prime-strip lift and determinant-global equality. The product-hull boundary now also has a synchronized Ob5–Ob7 review object,

```
remark395ProductHullBackedConstructorBackedConstructed
  HullDeterminantFiniteDivisorVerticalIQ
  _ob5Ob6Ob7SynchronizedAudit.
```

It packages the Ob6 $H_{\Phi(P_B)}/H_{\Xi(P_B)}$ log-volume audit and the Ob7 finite $F^{\times\mu}$-prime-strip audit in one source object, while also recording that the Ob7 hull/determinant bridge uses exactly the same Theorem 3.11 possible-image family and the same selected product-hull operator as the Ob6 approximant boundary. Thus a prime-strip log-volume comparison can no longer be reviewed at this surface independently of the product-hull possible-image family whose Ob5 quotient collapse and Ob6 approximant inequalities are being used. This synchronized review object is now threaded through the finite-divisor vertical-IQ boundary as well. The experiment endpoint

```
remark395ProductHullBackedConstructorBackedConstructed
  Ob3Ob5AdjustedHullDeterminantFiniteDivisorVerticalIQ
  _cThetaDichotomyWithOb5Ob6Ob7SynchronizedAudit
```

returns the synchronized Ob5–Ob7 audit together with the usual conditional C_Θ dichotomy. Thus the strongest current product-hull finite-divisor route no longer reviews the final ordered-real conclusion after forgetting the Ob5 quotient, Ob6 hull-approximant, and Ob7 log-Kummer provenance attached to the same Remark 3.9.5 hull. This route now also has

```
_cThetaDichotomyWithOb5Ob6Ob7SynchronizedAudit
  OfQPilotRegionNonempty,
```

which replaces the direct q-choice possible-image nonemptiness hypothesis by a nonempty q-pilot region. The synchronized audit then obtains the selected possible-image witness internally from $q \subseteq P_{q\text{-choice}}$, so the public finite-divisor review target is closer to the source-paper q-pilot-object formulation. There is now a further q-choice-specialized version,

```
_cThetaDichotomyWithOb5Ob6Ob7SynchronizedAudit
  AtQChoiceOfQPilotRegionNonempty,
```

and its canonical-scale analogue. In this endpoint the Ob5 comparison possible image is not an arbitrary extra choice: it is the constructor-backed q-choice itself. Hence the single nonempty q-pilot region supplies both quotient-collapse nonemptiness witnesses through $q \subseteq P_{q\text{-choice}}$. The canonical-scale version therefore has neither an external numeric C_Θ estimate nor a separate comparison-choice nonemptiness hypothesis. The constructor-built source-level canonical-scale bridge is now exposed directly as

`possibleImageConstructorBuiltHullSource_remark395CanonicalCThetaBridgeAudit`.

It records the chain $\mu_{\log}(q\text{Region}) \le \det_{\text{norm}} \le \theta_{\text{signed}} \le C_{\Theta,\text{can}} \cdot (-q_{\text{signed}})$ at the Step (xi) source boundary, separate from the downstream finite-divisor wrappers.

The same milestone-facing constructed source now has a possible-image q-choice constructor. In this form the q-pilot region is definitionally the chosen Theorem 3.11 possible image P_i, so Lean derives $q \subseteq \bigcup_i P_i$ from the possible-image union rather than accepting that containment as a separate field. The endpoint then exposes containment in the record-canonical holomorphic hull, the Ob3/Ob4 weighted determinant payload, the q-to-determinant log-volume inequality, and the resulting $q_{\text{signed}} \le \theta_{\text{signed}}$ comparison. This is a small but important tightening of the current trust boundary: the generic Remark 3.9.5 bridge source can now be instantiated in the source-native possible-image case without an arbitrary q-region containment input.

The constructor-built Step (xi) source now also exposes the Remark 3.9.5(v)–(vi) upper-semi quotient passage as

```
possibleImageConstructorBuiltHullSource
  _ob5QuotientIndependenceAudit.
```

Given two nonempty Theorem 3.11 possible images, the audit proves that both images collapse to the same point under the quotient by the bounded-family holomorphic hull $\phi(\bigcup_i P_i)$. The same record carries the bounded-family determinant/log-volume equality, the family-union log-volume bound, the normalized tensor-power equality, and the raw Step (xi) signed comparison. Thus Ob5 is now visible on the constructor-built public surface as the paper's independence-of-possibilities quotient statement, not just as a background containment lemma.

The constructor-built Step (xi) source now has the same record-canonical Ob3/Ob5 projection. A shadow hull/log-volume object is first converted back to a Remark 3.9.5 ϕ-operator by `toRemark395Operator`; the constructor-built source then forms the record Ob3/Ob5 determinant-compatibility source and the record hull/determinant bridge source from its own q-region containment, determinant compatibility, and Ob4 tensor-power bound. The public `recordOb3Ob5BridgeAudit` endpoint records that the q-to-determinant and determinant-to-theta inequalities are obtained from the generic record-canonical bridge theorem. This does not construct the analytic determinant compatibility, but it removes another ad hoc audit boundary: the large constructor-built object now factors through the same source object as the milestone-facing constructed source.

The constructor-built Step (xi) source now also has a canonical local C_Θ-scale audit. From q-pilot positivity it defines

$$C_{\Theta,\text{can}} = \theta_{\text{signed}}/(-q_{\text{signed}})$$

and proves internally that

$$\theta_{\text{signed}} \leq C_{\Theta,\text{can}} \cdot (-q_{\text{signed}}).$$

The canonical-one constructed-IPL finite-divisor route therefore has a no-external-bound endpoint

```
sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQSource
  _canonicalOneConstructorBuiltIPLConstruction
  CanonicalCThetaScaleDichotomy.
```

The same source now exposes the stronger audit

```
sourceDerivedHodgeSHEIPLHullFiniteDivisorVerticalIQSource
  _canonicalOneConstructorBuiltIPLConstruction
  Remark395CanonicalCThetaBridgeAudit.
```

This audit keeps the canonical local scale attached to the constructed Remark 3.9.5 bridge chain

$$\mu_{\log}(q\text{Region}) \leq \det_{\text{norm}} \leq \theta_{\text{signed}} \leq C_{\Theta,\text{can}} \cdot (-q_{\text{signed}}).$$

Thus the local no-external-bound scale is no longer merely a final ordered-real specialization; it is recorded beside the Ob1–Ob5 holomorphic-hull, determinant, and record Ob3/Ob5 bridge payload that produces the signed comparison. The compact constructed-hull finite-divisor endpoint now has the same audited canonical-scale form,

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  ConstructedHullDeterminantFiniteDivisorVerticalIQ
  _canonicalCThetaScaleWithCanonicalScaleAudit.
```

It returns the constructed Remark 3.9.5 canonical-scale audit together with the finite-divisor vertical-IQ dichotomy, so the no-external-bound finite boundary exposes the normalized determinant bridge and the derived $\theta_{\text{signed}} \leq C_{\Theta,\text{can}}(-q_{\text{signed}})$ estimate instead of only the final ordered-real conclusion. This does not yet identify $C_{\Theta,\text{can}}$ with the global paper constant C_Θ. Its purpose is narrower and useful: the theta/log-q upper bound is no longer an opaque route hypothesis for this local scale, and the remaining mathematical task is now isolated as the comparison between this canonical source scale and the IUT constant from the paper. This comparison boundary is now a named Lean object rather than prose. The constructed source theorem

```
IUTStage1Remark395ConstructedHolomorphicHullDeterminantSource
  .constructedGlobalCThetaScaleComparisonAudit
```

assumes only

$$C_{\Theta,\text{can}} \leq C_\Theta$$

for the chosen paper/global constant C_Θ. From this domination and q-pilot positivity, Lean derives

$$\theta_{\text{signed}} \leq C_\Theta \cdot (-q_{\text{signed}})$$

and then applies the standard Corollary 3.12 ordered-real dichotomy. The experiment-surface endpoints

```
remark395ConstructedHolomorphicHullDeterminantSource
  _globalCThetaScaleComparisonAudit,
remark395ConstructedHolomorphicHullDeterminantSource
  _globalCThetaScaleChain
```

therefore replace the raw final numeric hypothesis by the sharper remaining source obligation "global C_Θ dominates the constructed local Remark 3.9.5 scale". This is still an obligation, but it is now the correct one to attack from the source-paper global estimate. This source-level comparison has now been threaded through the compact finite-divisor vertical-IQ boundary as

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  ConstructedHullDeterminantFiniteDivisorVerticalIQ
  _globalCThetaScaleWithGlobalScaleAudit.
```

The endpoint assumes $C_{\Theta,\mathrm{can}} \le C_\Theta$, constructs the global-scale Remark 3.9.5 audit, derives the numeric $\theta_{\mathrm{signed}} \le C_\Theta(-q_{\mathrm{signed}})$ estimate internally, and then returns the finite-divisor dichotomy. Thus the finite Step (x) boundary no longer needs to expose the raw final numeric hypothesis when the intended local-to-global C_Θ comparison is the available source obligation. The paper-side C_Θ expression from IUT IV is now connected to this same boundary. The experiment-surface audit

```
Remark395ConstructedIUTIVCThetaHandoffAudit
```

takes a constructed Remark 3.9.5 Step (xi) source and an `IUTStage1IUTIVThetaPilotLogVolumeEstimateShadow`. The latter supplies the displayed IUT IV expression

$$C_\Theta = \mathrm{arithmeticUpperTerm} - \mathrm{mainLogTerm} - 1$$

together with the existing IUT IV handoff endpoint. The only new comparison input is again

$$C_{\Theta,\mathrm{can}} \le C_\Theta.$$

The compact endpoint

```
remark395ConstructedHolomorphicHullDeterminantSource
  _iutIVCThetaHandoffChain
```

then records, in one proposition, the IUT IV expression for C_Θ, the constructed Step (xi) chain $\mu_{\log}(q\mathrm{Region}) \le \det_{\mathrm{norm}} \le \theta_{\mathrm{signed}}$, the derived paper-side bound $\theta_{\mathrm{signed}} \le C_\Theta(-q_{\mathrm{signed}})$, and the final Corollary 3.12 dichotomy. Thus the remaining global comparison is no longer split between the Step (xi) local scale audit and the separate IUT IV constant shadow. This IUT IV handoff has now been lifted to the compact finite-divisor vertical-IQ boundary:

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  ConstructedHullDeterminantFiniteDivisorVerticalIQ
  _iutIVCThetaWithHandoffAudit.
```

The endpoint consumes the constructed Remark 3.9.5 Step (xi) source, the IUT IV estimate object, and the single comparison $C_{\Theta,\mathrm{can}} \le C_\Theta$. It returns the IUT IV handoff inequalities, the constructed global-scale Remark 3.9.5 audit at `estimate.cTheta`, and the finite-divisor dichotomy. Thus the paper-side C_Θ expression is visible at the same boundary as the Step (x) vertical-IQ data. The first source-facing replacement of this last comparison is now in Lean. The structures

```
IUTStage1Theorem311Ind1Action,   IUTStage1Theorem311Ind2Action,
IUTStage1Theorem311Ind3UpperSemiRelation,   IUTStage1Theorem311TypedIndeterminacyCore
```

replace the earlier name/boolean indeterminacy shadow on the local-to-global critical path by typed relations on a Theorem 3.11 choice space. The formal payload matches the source-paper asymmetry needed in the Corollary 3.12 corridor: (Ind_1) and (Ind_2) preserve procession-normalized log-volume, while (Ind_3) supplies only an upper-semi log-volume inequality. The core constructs the generated quotient and quotient map, and the possible-image compatibility record states that (Ind_1) and (Ind_2) identify possible-image regions whereas (Ind_3) contributes the one-sided estimate. A finite one-point witness `finiteToyCore` checks joint satisfiability of this typed interface; it is deliberately not an IUT model.

The local-to-global comparison itself is now isolated as

`ConstructedTheorem311IndeterminacyLocalGlobalCThetaSource`.

This source object ties the typed Theorem 3.11 indeterminacy core to the constructed Remark 3.9.5 Step (xi) source and the IUT IV estimate shadow. It records the remaining paper-mathematical assembly obligations: local place log-volume assembly, finite-divisor valuation-ball Haar normalization, product/sum local-to-global assembly, and compatibility between the Theorem 3.11 indeterminacy construction and the Step (xi) determinant/hull source. Crucially, it no longer assumes the raw comparison $C_{\Theta,\mathrm{can}} \leq C_\Theta$. Instead Lean derives it from the paper-shaped arithmetic gap

$$C_{\Theta,\mathrm{can}} + 1 \leq \mathrm{arithmeticUpperTerm} - \mathrm{mainLogTerm}$$

together with the IUT IV identity

$$C_\Theta = \mathrm{arithmeticUpperTerm} - \mathrm{mainLogTerm} - 1.$$

The compact finite-divisor endpoint

```
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311IndeterminacyLocalGlobalCThetaFiniteDivisorVerticalIQ
```

therefore consumes this source object and returns both the source-obligation endpoint and the Corollary 3.12 dichotomy. This is progress on the disputed corridor because the last scalar comparison is now a typed Theorem 3.11/IUT IV local-to-global source obligation, not an exposed final numeric hypothesis. It remains an obligation until the local place assembly, Haar normalization, and product/sum comparison are constructed from the corresponding source-paper objects rather than recorded as proof-carrying fields. The Theorem 3.11 multiradial source has also been strengthened below this local-to-global handoff. The source object

`IUTStage1Theorem311QuotientMultiradialSource`

now ties the typed (Ind_1), (Ind_2), (Ind_3) core to an actual `IUTStage1MultiradialThetaImages` quotient package. It records that the multiradial quotient is the quotient generated by the typed indeterminacy core, that the possible-image family is the one carried by the Theorem 3.11 record, and that the selected q-region is literally one of these possible images. It also carries the finite F_l-label layer: a ΘNF gluing torsor, $|F_l| = l$, and procession-label transitions satisfying the zero and addition laws. Lean proves that every such label transition has the same quotient-map class and hence the same multiradial possible-image region.

The source is no longer constructed only by supplying this whole package as a raw payload. The finite procession layer is separated as

`IUTStage1Theorem311FLProcessionAction`, which records the ΘNF gluing torsor, the Z/lZ-indexed procession transition on choices, the zero/addition action laws, and the fact that each transition stays inside the quotient generated by the typed indeterminacy core. The constructor

`IUTStage1Theorem311QuotientMultiradialConstructionSource`

then builds the quotient-aware source: the multiradial image object is formed from the typed quotient and the record's Theorem 3.11 theta possible images, the selected q-region is defined to be the possible image at the selected choice, and $|F_l| = l$ is proved by `ZMod.card`. Thus the remaining paper-side payload at this layer has been narrowed to two mathematical assertions: possible-image invariance along the typed quotient and the actual finite procession action on the source choice space.

There is now a source-faithful refinement of this last statement. The typed core exposes an equality quotient

`IUTStage1Theorem311TypedIndeterminacyCore.equalityQuotient`

generated only by (Ind_1) and (Ind_2); the (Ind_3) generator is not part of the possible-image equality quotient. From a `PossibleImageQuotientCompatibility` object, Lean proves

$$i \sim_{\mathrm{Ind}_1,\mathrm{Ind}_2} j \Longrightarrow \Theta\mathrm{Image}_i = \Theta\mathrm{Image}_j$$

as `PossibleImageQuotientCompatibility.equalityQuotient_image_invariant`, while retaining the separate theorem that an (Ind_3)-step gives only the upper-semi log-volume inequality. The Step (xi) source

`IUTStage1Theorem311OneSidedMultiradialConstructionSource`

is the preferred version of the multiradial source boundary: it constructs the `IUTStage1MultiradialThetaImages` object from this equality quotient, uses an `IUTStage1Theorem311EqualityFLProcessionAction` for the F_l-procession transitions, and returns the selected q-region as a genuine Theorem 3.11 possible image. Its endpoint explicitly reports the one-sided (Ind_3) inequality instead of assuming (Ind_3)-invariance of possible-image regions.

The bridge to the constructed Remark 3.9.5 source is now named as

`IUTStage1Theorem311QuotientMultiradialRemark395Compatibility`.

Its essential field identifies the constructed Step (xi) q-pilot region with the selected Theorem 3.11 possible image. From this, Lean re-derives

$$q\mathrm{Region} \subseteq \bigcup_i \Theta\mathrm{Image}_i$$

through the quotient-aware Theorem 3.11 source, instead of treating the record-union containment as an unrelated payload. The construction-source endpoint also produces this compatibility once the constructed Remark 3.9.5 q-pilot region is identified with the selected possible image. The next refinement removes the single opaque arithmetic-gap field from the preferred source surface. The Lean structure

`ConstructedTheorem311FinitePlaceLocalGlobalCThetaSource` introduces a finite set of local places and records four local functions: the local canonical Step (xi) scale, the local IUT IV arithmetic upper contribution, the local main-log contribution, and the local Haar-normalization defect. Its load-bearing local assertion is the pointwise inequality

$$C_{\Theta,\mathrm{can},v} + \delta_v \leq A_v - M_v.$$

Lean proves

$$C_{\Theta,\mathrm{can}} + 1 \leq \mathrm{arithmeticUpperTerm} - \mathrm{mainLogTerm}$$

by summing these local inequalities with `Finset.sum_le_sum`, using the equalities

$$C_{\Theta,\mathrm{can}} = \sum_v C_{\Theta,\mathrm{can},v}, \qquad \mathrm{arithmeticUpperTerm} = \sum_v A_v, \qquad \mathrm{mainLogTerm} = \sum_v M_v,$$

and the finite-divisor/Haar normalization condition

$$1 \leq \sum_v \delta_v.$$

The resulting conversion theorem

```
ConstructedTheorem311FinitePlaceLocalGlobalCThetaSource
  .toConstructedTheorem311IndeterminacyLocalGlobalCThetaSource
```

constructs the coarser source object automatically. The public finite-divisor endpoint

```
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311FinitePlaceLocalGlobalCThetaFiniteDivisorVerticalIQ
```

therefore derives the Corollary 3.12 dichotomy from a finite local-place assembly source, rather than from a raw numerical bound, a raw $C_{\Theta,\mathrm{can}} \leq C_\Theta$ comparison, or a single global arithmetic-gap hypothesis. The remaining mathematical work has been pushed down to constructing the local terms $C_{\Theta,\mathrm{can},v}$, A_v, M_v, and δ_v from the actual Theorem 3.11/Step (xi)/IUT IV objects and proving the pointwise local inequalities.

The preferred local-to-global surface has also been tightened to use the one-sided Theorem 3.11

construction directly:

```
ConstructedTheorem311OneSidedLocalGlobalCThetaSource,
ConstructedTheorem311OneSidedFinitePlaceLocalGlobalCThetaSource,
ConstructedTheorem311OneSidedStepXILocalTermCThetaSource,
ConstructedTheorem311OneSidedNormalizedHaarStepXILocalTermCThetaSource,
ConstructedTheorem311OneSidedPadicUnitBallHaarStepXILocalTermCThetaSource,
ConstructedTheorem311OneSidedIUTIVArithmeticDefectStepXILocalTermCThetaSource,
ConstructedTheorem311OneSidedIUTIVArithmeticDivisorEvaluationStepXILocalTermCThetaSource,
ConstructedTheorem311OneSidedIUTIVTheorem110ArithmeticDivisorStepXILocalTermCThetaSource,
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311OneSidedFinitePlaceLocalGlobalCThetaFiniteDivisorVerticalIQ,
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311OneSidedNormalizedHaarStepXILocalTermCThetaFiniteDivisorVerticalIQ,
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311OneSidedPadicUnitBallHaarStepXILocalTermCThetaFiniteDivisorVerticalIQ,
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311OneSidedIUTIVArithmeticDefectStepXILocalTermCThetaFiniteDivisorVerticalIQ,
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311OneSidedIUTIVArithmeticDivisorEvaluationStepXILocalTermCThetaFiniteDivisorVerticalI
boundarySignedEqualityOrStrictCTheta_from_constructed
  Theorem311OneSidedIUTIVTheorem110ArithmeticDivisorStepXILocalTermCThetaFiniteDivisorVerticalI
```

The source records carry an `IUTStage1Theorem311OneSidedMultiradialConstructionSource`, identify the constructed Remark 3.9.5 q-pilot region with its selected possible image, and then feed the same finite-place Step (xi)/Haar summation into the $C_{\Theta,\mathrm{can}} \leq C_\Theta$ comparison. The public endpoint returns the Corollary 3.12 dichotomy while also reporting the equality quotient generated only by (Ind$_1$) and (Ind$_2$), the separate one-sided (Ind$_3$) log-volume inequality, and the selected q-region containment in the Theorem 3.11 possible-image union. Thus the current preferred route no longer exposes the raw numeric C_Θ comparison and no longer uses a bare typed indeterminacy core at the local-to-global boundary.

The newest constructor, `ConstructedTheorem311OneSidedStepXILocalTermCThetaSource`, removes one more arbitrary layer from the finite-place comparison. Its local canonical terms are the weighted adjusted localization contributions of an `IUTStage1Remark395Ob3Ob4LocalizedVectorBundleDeterminantSource`; the local arithmetic upper contribution is then definitionally this Step (xi) term plus the local Haar defect and local main-log term. Lean derives the pointwise Step (xi)/Haar inequality and constructs the finite-place local-to-global source from these data. The remaining source obligations at this boundary are now explicit: identify $C_{\Theta,\mathrm{can}}$ with the sum of the localized Step (xi) terms, identify the IUT IV arithmetic and main-log terms with their local sums, and identify the local Haar-defect terms with the precise IUT IV local arithmetic defects. The Haar-defect lower-bound input is no longer consumed as a raw sum inequality: `IUTStage1FinitePlaceHaarDefectLowerBoundSource` records nonnegative local Haar defects and a distinguished normalized valuation-ball place whose defect is at least one, and Lean proves $\sum_v \delta_v \geq 1$ from those facts. The stricter constructor `ConstructedTheorem311OneSidedNormalizedHaarStepXILocalTermCThetaSource` threads this derived bound into the Step (xi) local-to-global comparison and now has its own finite-divisor vertical-IQ endpoint. The next constructor `IUTStage1FinitePlacePadicUnitBallHaarIndexDefectSource` derives this finite-place lower-bound source from placewise p-adic unit-ball Haar normalization. For each place v, the local defect is the dilation index

$$\delta_v = p_v^{[K_v:\mathbb{Q}_{p_v}]};$$

Lean proves $\delta_v \geq 0$ and $\delta_{v_0} \geq 1$ for a distinguished normalized place from primality of p_v and the natural finrank exponent. The wrapper `ConstructedTheorem311OneSidedPadicUnitBallHaarStepXILocalTermCThetaSource` threads this p-adic Haar-index source into the normalized Step (xi) local-to-global surface, and the corresponding finite-divisor vertical-IQ endpoint projects through the normalized-Haar route to the same Corollary 3.12 dichotomy. The new source `IUTStage1FinitePlaceIUTIVLocalArithmeticDefectSource` names the paper-side local arithmetic-defect term in the IUT IV C_Θ estimate and records its placewise equality with the p-adic unit-ball Haar index. It therefore derives the same nonnegativity, distinguished-place lower bound, and total $\sum_v \delta_v \geq 1$ statement for the IUT IV local defects. The corresponding Step (xi) wrapper `ConstructedTheorem311OneSidedIUTIVArithmeticDefectStepXILocalTermCThetaSource` expresses the arithmetic upper sum using these named IUT IV local defects and projects to the p-adic Haar-index route by the recorded equality. Its finite-divisor vertical-IQ endpoint is now the preferred local-to-global C_Θ surface for this part of the corridor. The divisor-evaluation refinement `IUTStage1IUTIVThetaPilotArithmeticDivisorLocalEvaluationSource` now records the local Definition 1.9/Theorem 1.10 arithmetic-divisor degree evaluation: local different degrees, conductor degrees, absorbed prime/error contributions, local main-log terms, and finite-sum identities recovering $\log \mathrm{Diff}_{F_{\mathrm{tpd}}}$, $\log \mathrm{Cond}_{F_{\mathrm{tpd}}}$, the absorbed $10(e^*_{\mathrm{mod}} l + \eta_{\mathrm{prm}})$ term, the global arithmetic upper term, and the global main-log term. The wrapper `ConstructedTheorem311OneSidedIUTIVArithmeticDivisorEvaluationStepXILocalTermCThetaSource` threads this source into Step (xi), leaving the local matching condition that the IUT IV arithmetic upper contribution is the Step (xi) weighted determinant contribution plus the local IUT IV defect plus the local main-log contribution. The next source layer now makes this more faithful to Definition 1.9 itself. `IUTStage1IUTIVArithmeticDivisorSource` formalizes a finite arithmetic divisor as coefficients c_v, degree weights, a positive normalizing degree, support, effectivity, local normalized degree summands, and the theorem that the global normalized degree is the finite sum of the local summands. `IUTStage1IUTIVTheorem110ArithmeticDivisorSource` packages the three paper divisors d_{ADiv}, q_{ADiv}, and f_{ADiv}, including effectivity, equality of the conductor and q-parameter supports, and recovery of $\log \mathrm{Diff}_{F_{\mathrm{tpd}}}$ and $\log \mathrm{Cond}_{F_{\mathrm{tpd}}}$ from normalized degrees. `IUTStage1IUTIVTheorem110LocalEstimateGateSource` names the three local cases of Theorem 1.10, Steps (v)–(vii): distinguished nonarchimedean, nondistinguished nonarchimedean with zero upper bound, and archimedean. Its endpoint proves the local upper-semi bound into the arithmetic-minus-main-log gap and supplies the finite-sum identities for the absorbed prime/error and main-log terms. The wrapper `ConstructedTheorem311OneSidedIUTIVTheorem110ArithmeticDivisorStepXILocalTermCThetaSource` threads this typed Theorem 1.10 layer into the Step (xi) route. Thus the preferred public route no longer exposes the raw total Haar-defect sum inequality, an unnamed Haar-defect summand, the global arithmetic/main-log sum identities, or bare local degree functions as separate endpoint hypotheses. The remaining source-paper work is now to instantiate these typed finite sources from actual number-field valuations, the different ideal, q-parameters, the conductor support divisor, and the Proposition 1.4/1.5 local log-shell estimates, and to prove the Step (xi) matching equality from those constructions rather than taking it as source data.

The practical priority order is:

1. instantiate the typed IUT IV arithmetic-divisor source from actual number-field valuations, different/q/conductor divisors, and Proposition 1.4 and Proposition 1.5 local log-shell esti-

mates, then derive the current Step (xi) matching equality from those objects;

2. extend the audited Hodge–Arakelov theta evaluation wrapper beyond the finite degree-evaluation, bad-local quotient, and canonical-label shadows toward actual Gaussian/Frobenioid material from IUT II;

3. continue replacing the finite Hodge/SHE transport source boundary by source-paper constructions of the Hodge-theater descent bridge, the actual forgetting operation beneath the current history-separated permission predicate, and deeper Gaussian/Frobenioid preservation data beneath the current Hodge–Arakelov theta-evaluation wrapper; the canonical-one constructed-IPL route now also constructs its finite Hodge/SHE/IPL middle source directly from the canonical-one Hodge–Arakelov evaluation pair, but the history-separated forgetting permission itself still comes from the current descent-bridge predicate;

4. continue replacing certificate-pinned IPL endpoint-equality variants by bundled and then construction-source forms; the certificate-pinned finite-divisor route now has a one-source-object review surface, so the next mathematical replacement is the actual Theorem 3.11 input-prime-strip link machinery beneath both the certificate-pinned and constructed-IPL surfaces;

5. construct the packet-correspondence source from the actual log-Kummer/Frobenioid machinery of IUT III, Step (x), rather than treating this finite correspondence object itself as source-facing input; the canonical-one constructed-IPL route now exposes the finite packet source endpoint and exact vertical-IQ target endpoint as one packet/target audit, and joins it to the direct Gaussian-to-Step (xi) audit in a no-C_Θ finite-boundary audit, but the finite correspondence object is still supplied at that boundary;

6. construct the remaining fields of the Step (xi) holomorphic-hull/determinant source from the actual holomorphic-hull and determinant operations of Remark 3.9.5 and Step (xi): the target-charted route now factors θ_{signed} through the record possible-image family hull and determinant summand payload, and the canonical-one constructed-IPL route now carries a direct Gaussian-to-Step (xi) audit through the constructor-built hull projection, so the named constructor-built Step (xi) boundary audit, and now its route-carried remaining-payload audit, now threaded through the obligations-backed, constructor-backed, and side-conditioned corridors, should be treated as the current ledgers of remaining source-paper obligations; the charted Hodge-calibrated family-hull source now constructs this constructor-built boundary from a stronger source object, and the synchronized target-charted possible-image summand source now reduces to that same boundary through a named audit; the synchronized finite-divisor route now also attaches the constructor-built remaining-payload ledger to the Ob5 quotient boundary, and the source-derived bridge constructor can consume the constructor-built possible-image Step (xi) source directly and expose its remaining-payload ledger at the bridge boundary; product-hull provenance is now also threaded through the synchronized target-charted possible-image summand source when its family-hull shadow is product-hull-induced, so the lower-level record-hull finite-divisor endpoint can also consume this constructor-built possible-image source directly, and the adjusted Ob3/Ob5 finite-divisor endpoint now constructs that source from the record-canonical adjusted determinant log-volume data plus obligations; the product-hull route now also retains the Ob7 $F^{\times\mu}$-prime-strip log-Kummer compatibility at the determinant boundary, and the typed Theorem 3.11 possible-image quotient now has a combined Ob5–Ob7 audit tying related possible images, hull-approximant log-volume

estimates, and retained prime-strip data to the same hull operator; the newest calibrated local-ring charted vector-bundle source pushes the factor regions, local-ring matching, and direct-summand volumes below the previous vector-bundle local-factor interface, the product-log-volume variant now factors the per-cell direct-product equality through a reusable single-cell tensor/product log-volume source, the tensor-packet variant now derives that single-cell law from a finite tensor-region/direct-sum log-volume datum for each actual charted localization cell, and the compact-open tensor-packet variant now constructs that finite tensor datum from compact-open local factors plus a direct-sum tensor-region source, and the nonarchimedean local tensor-packet variant now constructs the compact-open factors from local compact-open log-volume data with integral-structure, degree-normalization, and uniformizer-shift laws, and the finite-extension Haar tensor-packet variant now constructs those local sources from raw finite-extension Haar log-volume functions plus the tensor-product direct-sum decomposition law, and the compact-open topology/Haar tensor-packet variant now constructs the finite-extension Haar source from compact-open provenance, raw Haar measure, raw log-volume, degree normalization, and uniformizer compact-open preservation, and the additive Haar variant now constructs the compact-open topology source from a mathlib additive Haar measure together with Haar mass-one and uniformizer-scaling laws, and the valuation-ball variant now defines the local ring of integers and selected compact-open subset as valuation balls before constructing the additive Haar source, with the newest proper-ultrametric variant deriving openness from the nonarchimedean metric inequality and compactness from properness of the local metric topology. The valuation-ball factor-calibrated route now also projects through the local-ring charted cover, direct-product localized cover, calibrated localized cover, and localized hull decomposition to the adjusted Ob3/Ob5 determinant source. The experiment endpoint

```
remark395ValuationBallFactorCalibratedHaarTensorPacket
  FiniteAdditiveCalibratedLocalRingChartedVectorBundle
  HullCoverSource_adjustedDeterminantEndpoint
```

exposes the resulting identities $\mu_{\log}(\phi(P_B)) = \sum_\beta \mathrm{AdjLoc}_\beta = \det_{\log} = \det_{\log}^{\mathrm{norm}}$ and the induced Ob3/Ob5 determinant compatibility. Thus the lowest current valuation-ball/Haar local-ring route is now connected directly to the determinant payload used by the Step (xi) bridge, not only to its own cover-log-volume endpoint. This valuation-ball determinant payload now also has a record-canonical constructor

```
IUTStage1Remark395RecordOb3Ob5Adjusted
  DeterminantLogVolumeSource.ofValuationBallFactor
  CalibratedHaarTensorPacketFiniteAdditive
  CalibratedLocalRingChartedVectorBundleHullCoverSource
```

together with an endpoint audit. Its only extra synchronization input is that the valuation-ball cover's possible-region family is the Theorem 3.11 record possible-image family; after this identification, Lean transports the valuation-ball/Haar/local-ring finite-cover determinant equality into the record Ob3/Ob5 source used by the constructed Step (xi) bridge. A narrower cover-additive variant now removes the global `IUTStage1FiniteAdditiveHullLogVolumeSource` obligation from this valuation-ball branch: it assumes only the direct-product cover equality $\mu_{\log}(\bigcup_\beta H_\beta) = \sum_\beta \mu_{\log}(H_\beta)$, derives the calibrated cell sum, and then projects through the same localized cover and adjusted Ob3/Ob5 determinant source. The valuation-ball route now has a further finite-cover-additive source form: instead of carrying this equality as a raw field, it carries a family-local finite-cover additivity certificate for the selected cells H_β; Lean derives pairwise disjointness from local valuation-ball factor separation and then obtains $\mu_{\log}(\bigcup_\beta H_\beta) = \sum_\beta \mu_{\log}(H_\beta)$ before projecting to the adjusted Ob3/Ob5 determinant source. This keeps the trust boundary below global finite additivity while removing the bare cover equality from the public valuation-ball determinant route. The newest measure-backed variant now lowers this boundary one more level. The source object

`IUTStage1FiniteCoverLogVolumeMeasureModelSource`

records a real-valued log-volume measure model for the selected finite cover, proves that the hull log-volume agrees with this model, and proves finite additivity for the measure model on pairwise-disjoint cover families. Lean then constructs the family-local `IUTStage1FiniteCoverAdditiveHullLogVolumeSource`. The valuation-ball specialization

```
IUTStage1Remark395ValuationBallMeasureBacked
  FactorCalibratedHaarTensorPacketHullCoverSource
```

uses this measure model for the valuation-ball direct-product cells and projects through the same adjusted Ob3/Ob5 determinant endpoint. Thus the public valuation-ball determinant route no longer takes even the cover-local hull additivity certificate as primitive; it is produced from a named measure/log-volume layer. A tensor-measure-backed refinement now constructs this measure/log-volume layer from the valuation-ball tensor packet cells themselves. The source object

```
IUTStage1Remark395ValuationBallTensorMeasureBacked
  FactorCalibratedHaarTensorPacketHullCoverSource
```

replaces the supplied measure model by the Step (xi)-shaped assertion

$$\mu_{\log}\Big(\bigcup_\beta H_\beta\Big) = \sum_\beta \mu_{\log}^{\mathrm{tensor}}(H_\beta),$$

where each tensor-normalized cell value is identified in Lean with the hull log-volume of the corresponding valuation-ball direct-product cell. It then constructs the previous measure-backed valuation-ball source and projects to the same adjusted Ob3/Ob5 determinant endpoint. Thus the remaining cover input is no longer an arbitrary finite-cover measure model; it is localized to the tensor-packet log-volume sum for the selected valuation-ball cells. This projection now has a named measure-model audit,

```
remark395ValuationBallTensorMeasureBacked
  FactorCalibratedHaarTensorPacketHullCoverSource
  _measureModelAudit.
```

The audit records that the constructed finite-cover measure model is the hull log-volume itself on the selected cover, that each direct-product valuation-ball cell has the corresponding tensor-normalized log-volume, that the finite-cover additive equality is derived from this model and the pairwise-disjoint valuation-ball cover, and that the resulting measure-backed source is the one used by the adjusted Ob3/Ob5 determinant handoff. In the principal valuation-ball branch this tensor-measure audit is now lifted to the Remark 3.9.5(i) $\lambda \cdot O$ surface as

```
remark395PrincipalValuationBallProductHullCoverSource
  _tensorMeasureAudit.
```

This audit records that the selected principal hull is the same valuation-ball direct-product cover used by the tensor-measure-backed source, that its log-volume is the sum of the tensor-normalized valuation-ball cell log-volumes, that the cover hull system is the principal product-hull system, and that the adjusted Ob3/Ob5 determinant handoff retains the principal holomorphic-hull operator. Thus the finite-cover measure model is now visible at the principal $\lambda \cdot O$ hull boundary, not only at the generic valuation-ball cover boundary. In the finite-cover-additive branch this tensor-cell cover sum is no longer a separate primitive: Lean derives it from the family-local finite-cover additivity certificate together with the per-cell identification $\mu_{\log}(H_\beta) = \mu_{\log}^{\mathrm{tensor}}(H_\beta)$, and then constructs the tensor-measure-backed source. Thus the public audit path can view the tensor-cover theorem as a derived consequence of the finite-cover valuation-ball source whenever that source has been supplied; the older global finite-additive valuation-ball/Haar factor route now first narrows to this finite-cover source and then enters the same tensor-measure-backed determinant corridor. The valuation-ball Haar layer itself now has a named normalization audit,

```
remark395ValuationBallAdditiveHaarNormalizationSource
  _normalizationAudit,
```

which records the unit-ball and selected positive-radius valuation-ball definitions, compact-openness, the raw Haar log-volume shift under uniformizer scaling, the normalized $\mu_{\log}(\mathcal{O}) = 0$ law, the normalized $\mu_{\log}(\varpi U) = \mu_{\log}(U) - \log(q)$ law, and the projected nonarchimedean compact-open log-volume source. The per-factor vector-bundle calibration surface now carries this same local Haar audit as

```
valuationBallVectorBundleFactorCalibrationSource_haarCalibrationAudit,
```

together with the chart/local-ring equality, region equality, additive normalized log-volume equality, and direct-summand calibration. Thus the local Step (xi-d) vector-bundle factor no longer exposes only the final log-volume equality; it retains the valuation-ball Haar normalization chain that produced it. This local Haar audit is now also visible at the principal valuation-ball finite-divisor boundary. The source object

```
PrincipalValuationBallHaarNormalizationBridgeAudit
```

records, for every cover cell and place, the valuation-ball/vector-bundle Haar calibration audit, the unit-ball and selected compact-open valuation-ball definitions, compact-openness of the uniformizer-scaled subset, the normalized $\mu_{\log}(\mathcal{O}) = 0$ and uniformizer-shift laws, the realized local-factor region equality, and the direct-summand log-volume equality. The finite-divisor endpoint

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  PrincipalValuationBallHaarNormalizedConstructorBacked
  ConstructedOb3Ob5AdjustedHullDeterminant
  FiniteDivisorVerticalIQ_canonicalCThetaScaleWithHaarAudit
```

returns this Haar bridge audit together with the principal valuation-ball canonical-C_Θ audit and the Corollary 3.12 dichotomy. Thus the current strongest principal $\lambda \cdot O$ boundary keeps the local additive Haar normalization data attached all the way to the finite vertical-IQ route. This has now been strengthened by the finite-cover/tensor-measure bridge

```
PrincipalValuationBallFiniteCoverHaarBridgeAudit
```

and the endpoint

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  PrincipalValuationBallFiniteCoverHaarConstructorBacked
  ConstructedOb3Ob5AdjustedHullDeterminant
  FiniteDivisorVerticalIQ
  _canonicalCThetaScaleWithFiniteCoverHaarAudit.
```

The new audit returns the local Haar normalization bridge together with the principal tensor-measure audit. It records that the tensor-measure-backed hull system is the principal product-hull system, that the selected principal hull is the tensor-measure direct-product cell union, that its log-volume is both the calibrated cell sum and the tensor-normalized cell sum, that the finite-cover additivity equality is derived at this cover, and that the adjusted Ob3/Ob5 determinant source still carries the principal hull operator and normalized determinant equality. Thus the finite-cover additivity and tensor-cell measure model are no longer visible only in the lower source-core endpoints; they are retained at the current principal finite-divisor C_Θ boundary. The same boundary now has a finite-extension/Haar refinement

```
PrincipalValuationBallFiniteExtensionHaarBridgeAudit
```

and the endpoint

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  PrincipalValuationBallFiniteExtensionHaarConstructorBacked
  ConstructedOb3Ob5AdjustedHullDeterminant
  FiniteDivisorVerticalIQ
  _canonicalCThetaScaleWithFiniteExtensionHaarAudit.
```

This audit keeps the finite-cover/Haar bridge and exposes, for every local valuation-ball factor, the finite-extension degree positivity, raw Haar log-volume as the logarithm of the Haar measure, normalized log-volume as the degree-normalized raw log-volume, Haar mass one on the valuation unit ball, positivity on the selected compact-open valuation ball, the raw uniformizer-scaling identity, the normalized $-\log(p)$ shift, and the projection to the nonarchimedean compact-open log-volume source. Thus the public principal $\lambda \cdot O$ finite-divisor route now displays the finite-extension compact-open Haar normalization layer, not merely the tensor-cover additivity layer. This has now been sharpened by the topology/finite-extension refinement

```
PrincipalValuationBallTopologyFiniteExtensionHaarBridgeAudit
```

and the endpoint

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  PrincipalValuationBallTopologyFiniteExtensionHaarConstructorBacked
  ConstructedOb3Ob5AdjustedHullDeterminant
  FiniteDivisorVerticalIQ
  _canonicalCThetaScaleWithTopologyFiniteExtensionHaarAudit.
```

The new audit keeps the finite-extension/Haar bridge and additionally records the valuation-ball topology provenance of every local factor: the ring of integers is the radius-one valuation ball, the selected compact-open is a positive-radius valuation ball, the ring of integers, selected ball, and uniformizer-scaled selected ball are compact-open, the additive Haar source uses the same unit and selected compact-open subsets, and the realized local factor still agrees with the charted direct-summand log-volume. This makes the current principal finite-divisor boundary reviewable through the actual valuation-ball layer before the finite-extension Haar normalization layer. The same boundary now also retains the synchronized Ob5–Ob7 review object. Lean exposes

`PrincipalValuationBallTopologyOb5Ob6Ob7CanonicalCThetaScaleAudit`

and the public endpoint

```
boundarySignedEqualityOrStrictCTheta_from_remark395
  PrincipalValuationBallTopologyFiniteExtensionHaarConstructorBacked
  ConstructedOb3Ob5AdjustedHullDeterminant
  FiniteDivisorVerticalIQ
  _canonicalCThetaScaleWithTopologyFiniteExtensionHaarOb5Ob6Ob7Audit.
```

This audit identifies the product-hull selected Ob5 quotient hull and the Ob6 $H_\Phi(P_B)$, $H_\Xi(P_B)$ approximant hulls with the same selected principal valuation-ball hull used by the topology/Haar route. It also keeps the Ob7 finite $F^{\times\mu}$-prime-strip determinant comparison at that boundary, including the q-region-to-prime-strip log-volume inequality, normalized determinant equality, and family-hull log-volume equality. Thus the strongest principal $\lambda \cdot O$ finite-divisor endpoint no longer separates the local valuation-ball topology/Haar provenance from the synchronized Ob5–Ob7 determinant comparison. There is now also a q-choice-specialized version of this endpoint, with suffix

`AtQChoiceOfQPilotRegionNonempty`.

It fixes the Ob5 comparison possible image to the constructor-backed q-choice and derives both quotient-collapse nonemptiness witnesses from the single nonempty q-pilot region. Thus the current strongest principal valuation-ball/topology/Haar route has no external C_Θ estimate and no arbitrary comparison-choice nonemptiness hypothesis. Its Ob7 input also has a typed-source variant,

```
Ob5Ob6Ob7CanonicalCThetaScaleAudit
  AtQChoiceOfQPilotRegionNonemptyOfOb7Source.
```

This variant keeps the same topology/Haar provenance and q-choice Ob5 specialization, but obtains all prime-strip and determinant-normalization facts from a single Remark 3.9.5(Ob7) log-Kummer compatibility object synchronized with the constructed hull/determinant bridge. The principal audit also has the corresponding `OfCoricOb7Source` form, which replaces that Ob7 object by the coric $F^{\vdash\times\mu}$ source and hence exposes the common valuation-indexed unit character at the strongest current topology/Haar boundary. The final principal finite-divisor dichotomy now has the same coric-source form: its topology/Haar Ob5–Ob7 audit and canonical-scale dichotomy are produced from the coric Ob7 source plus the bridge synchronization with the constructed hull/determinant source, not from separate prime-strip and determinant-normalization payloads. At the product-hull boundary, the coric Ob7 synchronization has been lowered one step further. The source

`ProductHullBackedConstructedCoricOb7Source` stores the coric $F^{\vdash \times \mu}$ invariant and the determinant/global log-volume equality while fixing its Ob7 `bridgeSource` to the constructed product-hull Step (xi) bridge by definition. Consequently the experiment endpoint with suffix

`OfConstructedCoricOb7Source`

reaches the same canonical-scale product-hull Ob5–Ob7 finite-divisor dichotomy without a separate coric bridge-equality argument; the remaining Ob7 payload is now the coric invariant plus the determinant/global log-volume identification. The same synchronization has now been pushed through the principal valuation-ball topology/Haar boundary. Lean exposes the abbreviation

`PrincipalValuationBallConstructedCoricOb7Source`

as the product-hull constructed coric source attached to the product-hull bridge forgotten from the principal valuation-ball Step (xi) source. Its audit and endpoint forms with suffix

`OfConstructedCoricOb7Source`

therefore return the strongest current principal topology/Haar Ob5–Ob7 audit together with the canonical-scale finite-divisor dichotomy without an external Ob7 bridge-equality premise. At this boundary the remaining source data is the coric $F^{\vdash \times \mu}$ invariant, the determinant versus prime-strip global log-volume equality, and the already constructed principal valuation-ball topology/Haar bridge. The product-hull route now lowers the determinant part of this Ob7 boundary: Lean defines

`IUTStage1WeightedDeterminantPrimeStripProductFormulaSource`

whose fields are pointwise identifications of each adjusted Ob3 determinant summand with a converted local Gaussian realified Frobenioid log-volume, together with the finite product-formula equality identifying the resulting sum with the Gaussian prime-strip global log-volume. The theorem

`determinantLogVolume_eq_primeStripGlobal`

then derives the Ob7 determinant/global equality from these local data. The experiment endpoint with suffix

`OfCoricThetaMuProductFormulaConstructionSource`

threads this product-formula determinant source through the coric $F^{\vdash \times \mu}$ unit-character construction and reaches the same canonical-scale finite-divisor dichotomy. The principal valuation-ball topology/Haar route now has the same product-formula refinement: Lean defines

`PrincipalValuationBallCoricThetaMuProductFormulaOb7ConstructionSource`

by forgetting the principal Step (xi) source to its product-hull bridge, and the endpoint with suffix

`OfCoricThetaMuProductFormulaConstructionSource`

returns the principal topology/Haar Ob5–Ob7 audit together with the canonical-scale finite-divisor dichotomy. Thus the strongest principal endpoint no longer treats the Ob7 determinant/global comparison as a bare equality premise; it is supplied by local adjusted Ob3 summand calibrations, the finite product-formula sum, and the coric $F^{\vdash\times\mu}$ unit-character construction. The valuation-unit-ball/nonzero-scalar product-cover route is now threaded to this finite-divisor boundary as well: Lean exposes a transport-backed source whose explicit carrier equivalence identifies the literal local product scalar hull and valuation-unit-ball direct-product cover with the principal valuation-ball hull before applying the canonical-scale vertical-IQ dichotomy. A reduced synchronization-source endpoint now derives the transported valuation-cover equality from the selected-scalar/product-cover theorem, so this boundary no longer lists the local-integer, selected-scalar, and valuation-cover transport equalities as three independent hypotheses. A charted synchronization-source endpoint now removes the bare product carrier equivalence as well, replacing it by an explicit real-coordinate chart whose inverse laws generate the equivalence to the target real-line copy. A forward target-point transport endpoint now lowers the finite-divisor boundary further: it uses only a map from the product carrier to the target real-line copy and audited image equalities, not inverse chart laws. A coordinate-derived target transport endpoint now constructs that point map from a real coordinate map and proves the point-image equalities from coordinate-region image equalities. A selected-image coordinate transport endpoint now rewrites the selected hull coordinate image through the source theorem `selectedScalarImageHull` $= \lambda \cdot O$, so this half of the transport boundary is stated for the explicit selected nonzero-scalar multiplication image of the local integer product region. A valuation-anchor coordinate transport endpoint now rewrites the local-integer coordinate image through the source theorem $O = \prod_v O_v$, so both coordinate-image obligations are stated for the explicit source regions O and λO. A finite local log-coordinate endpoint now replaces the arbitrary target real-coordinate map at this boundary by the assembled finite sum $x \mapsto \sum_v \ell_v(x)$ of local log-coordinate readings. A product-image local log-coordinate endpoint now derives those finite-sum image laws from the vector-valued image laws for $x \mapsto (v \mapsto \ell_v(x))$ and the local coordinate product regions. A preimage-form local log-coordinate endpoint now derives those vector-valued product-image laws from coordinatewise landing and simultaneous realization of compatible local log-coordinate tuples on both O and λO. A placewise-preimage endpoint now derives the simultaneous realization laws from local factor preimages plus a direct-product assembly map whose assembled point has the prescribed local log-coordinate tuple. A factorwise local log-coordinate endpoint now derives coordinatewise landing from point-to-factor projections, local factor membership, coordinate agreement, and local factor landing. A projected-factorwise endpoint now specializes this factorwise source to the literal dependent-product carrier: a valuation-place/product-index equivalence makes point-to-factor projection ordinary coordinate evaluation, and direct-product assembly is supplied by the dependent-product equivalence. This projected-factorwise source now also reaches the strongest principal topology/Haar Ob5–Ob7 finite product-formula endpoint, and the target-charted exact-theta Step (xi) source constructs this projected endpoint internally. A valuation-log-image projected endpoint now defines the local coordinate regions as the images $\ell_v(O_v)$ and $\ell_v(\lambda_v O_v)$, then projects to the same final Ob5–Ob7 route. The exact-theta Step (xi) source also constructs this valuation-log-image endpoint internally. A finite-sum target-point endpoint now goes below this: the target map is definitionally $x \mapsto \{\text{coord} = \sum_v \ell_v(x_v)\}$, the coordinate-place map is required to agree with the valuation-place/product equivalence, and Lean proves the finite-sum real-line laws from the dependent-product source. A coordinate-place finite-sum endpoint lowers this further: the valuation-cover coordinate-place map is carried as the source equivalence $e : \delta \simeq \gamma_{\text{local}}$, and the old compatibility `coordinatePlace`$(e^{-1}(v)) = v$ is derived internally. The exact-theta Step (xi) source now constructs both the finite-sum target-point source and this coordinate-place finite-sum source from point-level transport of O and λO, rather than from supplied finite-sum real-line equalities. A Haar-log-coordinate finite-sum endpoint lowers the local maps themselves: each $\ell_v(x_v)$ is the normalized Haar log-volume of the compact-open local region attached to x_v, and the Step (xi) constructor can build this Haar-backed source. A compact-open Haar refinement now carries the finite-extension compact-open topology/Haar normalization source at each place and proves that every assigned U_{v,x_v} is compact-open before projecting to the Haar finite-sum endpoint. The newest centered valuation-ball refinement removes this compact-open proof as external data: it sets $U_{v,x_v} = \overline{B}(x_v, r_v)$ inside a proper ultrametric local factor, proves openness by the ultrametric inequality, proves compactness by properness, and then projects through the compact-open Haar source. This has now been lowered further to a valued-field integer local source: each proper ultrametric centered-ball Haar source is constructed from a valued-field-integer Haar normalization object whose local ring is $\mathcal{O}_K$. A p-adic refinement specializes this once more to the base local field $\mathbb{Q}_{p_v}$ at each valuation place: Lean identifies the integer ring with $\mathbb{Z}_{p_v}$, records residue prime p_v, degree 1, and uniformizer action $x \mapsto p_v x$, and threads the resulting centered-ball Haar coordinates through the finite-sum target map. The newest finite-extension p-adic refinement replaces those base factors by actual finite-dimensional local fields $K_v/\mathbb{Q}_{p_v}$: the normalization degree is the Lean $\mathrm{finrank}_{\mathbb{Q}_{p_v}} K_v$, the distinguished scale is multiplication by the image of p_v in K_v, and the local centered-ball Haar coordinates are threaded through the same Step (xi) constructor and C_Θ-dichotomy endpoint. The newest dilation refinement lowers one of the finite-extension topology assumptions: instead of separately supplying open and compact preservation for $x \mapsto p_v x$, each local factor carries a homeomorphism realizing this base-prime dilation, and Lean derives the open/compact image laws used by the finite-extension Haar source. The finite-product and experiment-surface route now expose this homeomorphism-backed provenance before projecting to the same C_Θ-dichotomy endpoint. The next refinement lowers the Haar-mass input itself one step. Instead of assuming only the selected compact-open equality

$$\mu_v(p_v U_v) = \mu_v(U_v)/p_v^{[K_v:\mathbb{Q}_{p_v}]},$$

the new source object carries the uniform dilation-modulus law

$$\mu_v(p_v S) = \mu_v(S)/p_v^{[K_v:\mathbb{Q}_{p_v}]}$$

for every local subset S. Lean specializes this theorem to the selected valuation ball, projects to the existing finite-extension Haar source, and exposes the per-place all-subset mass-scaling theorem at the experiment-surface C_Θ route. This boundary has now been lowered once more from a real-valued `toReal` statement to the measure codomain itself. The Haar-modulus source carries the identity in $\mathbb{R}_{\geq 0} \cup \{\infty\}$,

$$\mu_v(p_v S) = \mathrm{ofReal}\Big(p_v^{-[K_v:\mathbb{Q}_{p_v}]}\Big) \cdot \mu_v(S),$$

and Lean derives the older real-valued division formula only after applying `ENNReal.toReal`. Thus the finite-extension route now separates three layers: construction of the dilation homeomorphism, the genuine measure-level Haar modulus law, and the real log-volume normalization used by the determinant corridor. The dilation homeomorphism itself is now also lowered out of the source data. Since $p_v \neq 0$ in $\mathbb{Q}_{p_v}$, Lean proves that its image in the finite-extension field K_v is nonzero and constructs the homeomorphism $x \mapsto p_v x$ by `Homeomorph.smulOfNeZero`. The constructed source records the pointwise formula for this homeomorphism, specializes the uniform Haar-mass law through it, and then projects to the previous dilation-mass route. At that stage the analytic Haar input was the uniform local modulus theorem itself, not the invertibility or topological automorphism of base-prime multiplication. This boundary has now been lowered below the all-subset modulus theorem. Lean constructs the continuous additive equivalence $x \mapsto p_v x$ on the finite-extension local field and applies mathlib's additive Haar-character theorem:

$$\mathrm{addEquivAddHaarChar}(x \mapsto p_v x) \cdot \mu_v((x \mapsto p_v x)^{-1} X) = \mu_v(X).$$

Injectivity of the constructed equivalence converts this preimage statement into the image law for $p_v S$. The source input has therefore moved from the uniform measure identity itself to the local arithmetic Haar-character evaluation

$$\mathrm{addEquivAddHaarChar}(x \mapsto p_v x) = p_v^{-[K_v:\mathbb{Q}_{p_v}]},$$

recorded in Lean by the new finite-extension Haar-character normalization source. The local, finite-product, and experiment-surface Remark 3.9.5 routes now expose this Haar-character scalar before projecting to the previous measure-level Haar-modulus endpoint. The Haar-character boundary has now been lowered once more to the normalized valuation unit ball. The new unit-ball source carries the two local mass facts

$$\mu_v(\mathcal{O}_v) = 1, \qquad \mu_v(p_v \mathcal{O}_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}.$$

Applying the same additive Haar-character theorem with $X = p_v \mathcal{O}_v$ gives

$$\mathrm{addEquivAddHaarChar}(x \mapsto p_v x) \cdot \mu_v(\mathcal{O}_v) = \mu_v(p_v \mathcal{O}_v),$$

and Lean cancels the normalized unit-ball mass to derive the Haar-character scalar. Thus the local, product, and experiment surfaces now expose valuation-unit-ball mass provenance before projecting to the Haar-character and all-subset Haar-modulus routes. A further core bridge now derives the second displayed mass equality from the all-subset Haar-modulus identity itself: specializing

$$\mu_v(p_v S) = p_v^{-[K_v:\mathbb{Q}_{p_v}]} \mu_v(S)$$

at $S = \mathcal{O}_v$, and using the normalized equality $\mu_v(\mathcal{O}_v) = 1$, Lean proves $\mu_v(p_v \mathcal{O}_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}$. The constructed Haar-modulus source therefore projects to the unit-ball Haar-character source without taking the $p_v \mathcal{O}_v$ mass as an independent assumption. This projection is now exposed not only in the source core, but also at the local Haar-log-coordinate factor, finite-product target-point source, and record-canonical Step (xi) experiment endpoint: the final route consumes the constructed all-subset Haar-modulus source plus Haar regularity, derives the unit-ball Haar-character data, and continues to the same C_Θ dichotomy. The finite-extension unit-ball boundary has now also been lowered in the other direction, from a supplied $p_v \mathcal{O}_v$ mass to a residue-coset calculation. The first algebraic layer below this boundary is the generic finite additive quotient partition source: for additive subgroups $H \subseteq U$, an equivalence

$$\mathrm{Fin}(n) \simeq U/H$$

and representatives $a_i \in U$, Lean proves $U = \bigcup_i (a_i + H)$, proves pairwise disjointness of the translated H-cosets from quotient injectivity, and proves coset measurability from measurability of H using the measurable equivalence given by additive translation. This is the formal algebraic core of the source-paper decomposition $O_v/p_v O_v$. The next refinement constructs the scaled subgroup itself. Lean defines the additive homomorphism $x \mapsto p_v x$, proves that the subgroup image of $\mathcal{O}_v$ under this homomorphism has underlying set $p_v\mathcal{O}_v$, proves $p_v\mathcal{O}_v \subseteq \mathcal{O}_v$ from the valuation-style closure input, and derives measurability of $p_v\mathcal{O}_v$ from measurability of $\mathcal{O}_v$ via the continuous additive equivalence $x \mapsto p_v x$. Thus the stricter dilation-quotient source no longer treats $p_v\mathcal{O}_v$ as an arbitrary subgroup. The next refinement constructs the upper subgroup $\mathcal{O}_v$ itself. The valued-field integer source now exposes the additive subgroup of the mathlib valued integer subring $\mathcal{O}[K]$, proves that its underlying set is the valuation ring/unit ball already used in the local Haar normalization, and proves its measurability from the closed metric unit ball. The valued-integer dilation-quotient source therefore no longer carries an arbitrary subgroup representing $\mathcal{O}_v$, nor a separate equality with the valuation unit ball, nor a separate measurability proof. It keeps only the still-genuine local valuation input $p_v\mathcal{O}_v \subseteq \mathcal{O}_v$, together with the finite quotient equivalence and representatives. That closure input has now been removed at the norm-compatible finite-extension boundary. The normed valued-integer source assumes the finite extension norm is compatible with the base $\mathbb{Q}_{p_v}$-norm as a $\mathbb{Q}_{p_v}$-normed algebra, proves

$$\|\,\mathrm{algebraMap}(p_v)\| = \|p_v\| = p_v^{-1} \le 1,$$

and derives $p_v\mathcal{O}[K_v] \subseteq \mathcal{O}[K_v]$ from the valued-integer unit-ball condition. It projects to the valued-integer dilation-quotient source while supplying the closure proof internally. The quotient-equivalence and representative inputs have now also been lowered by one mathematical layer. A generic finite additive quotient cardinality source carries only a finite quotient U/S, its cardinality, and measurability of S; Lean constructs $\mathrm{Fin}(\#(U/S)) \simeq U/S$ via `Fintype.equivFin`, chooses representatives via `Quot.exists_rep`, proves their compatibility with `QuotientAddGroup.mk'`, and then applies the quotient-coset partition theorem. The normed finite-extension valued-integer cardinality source specializes this to

$$\mathcal{O}[K_v]/p_v\mathcal{O}[K_v], \qquad \#(\mathcal{O}[K_v]/p_v\mathcal{O}[K_v]) = p_v^{[K_v:\mathbb{Q}_{p_v}]},$$

so the downstream finite-index equivalence and residue representatives are no longer independent payloads. This cardinality boundary has now been refined again through a residue vector-space model. The generic residue vector-space source carries a finite residue field κ, a finite κ-vector space V, and an additive equivalence $V \simeq U/S$; Lean proves

$$\#(U/S) = \#V = \#\kappa^{\dim_\kappa V}$$

by applying `Module.card_eq_pow_finrank`. The finite-extension specialization applies this with $U = \mathcal{O}[K_v]$, $S = p_v\mathcal{O}[K_v]$, together with $\#\kappa = p_v$ and $\dim_\kappa V = [K_v : \mathbb{Q}_{p_v}]$, and therefore derives $\#(\mathcal{O}[K_v]/p_v\mathcal{O}[K_v]) = p_v^{[K_v:\mathbb{Q}_{p_v}]}$ in Lean. This has been lowered once more to a quotient-native residue module source: the quotient $\mathcal{O}[K_v]/p_v\mathcal{O}[K_v]$ itself carries the finite κ-vector-space structure, so no separate carrier V nor equivalence $V \simeq \mathcal{O}[K_v]/p_v\mathcal{O}[K_v]$ is needed at this boundary. Lean again applies `Module.card_eq_pow_finrank` directly to the quotient and then uses $\#\kappa = p_v$ and the quotient finrank identity to obtain the required cardinality. This quotient-native residue module source now projects all the way to the valued-integer dilation-quotient/coset/Haar-character endpoint: once the quotient module cardinality, quotient finrank identity, residue-field cardinality, and normalized Haar data are present, Lean derives the coset cover, pairwise disjointness, coset measurability, $\mu_v(p_v\mathcal{O}_v)$, the additive Haar-character scalar, and the all-subset base-prime dilation law from the source record. The finite-extension residue boundary has now been lowered one step below this quotient-native source. The new submodule-level source carries the residue-field action on the additive group of $\mathcal{O}[K_v]$ before quotienting, together with a κ-submodule whose underlying additive subgroup is $p_v\mathcal{O}[K_v]$. Lean constructs the quotient module from this submodule, computes

$$\#(\mathcal{O}[K_v]/(p_v\mathcal{O}[K_v])) = p_v^{[K_v:\mathbb{Q}_{p_v}]}$$

on the submodule quotient by `Module.card_eq_pow_finrank`, and also rederives $p_v\mathcal{O}[K_v] \subseteq \mathcal{O}[K_v]$ from the normed-algebra inequality $\|p_v\| \leq 1$. This boundary now includes the quotient transport itself: Lean forms the additive quotient equivalence induced by the equality between the submodule's underlying additive subgroup and $p_v\mathcal{O}[K_v]$, transports the κ-module structure across this equivalence, obtains a quotient linear equivalence, transports the finrank identity to the additive quotient used by the coset/Haar partition, and then reaches the full local coset/Haar-character endpoint from the submodule-level source. For the base field itself, this residue quotient is no longer an obligation: Lean now uses `PadicInt.toZMod`, its kernel/maximal-ideal theorem, and the additive first isomorphism theorem to construct

$$\mathbb{Z}_{p_v}/\ker(\texttt{toZMod}) \simeq \mathbb{Z}/p_v\mathbb{Z}$$

and prove both the cardinality p_v and the one-dimensional residue vector-space statement matching $\mathrm{finrank}_{\mathbb{Q}_{p_v}} \mathbb{Q}_{p_v} = 1$. In the base local field $\mathbb{Q}_{p_v}$, this closure input is now also constructed: the base-field valued-integer source proves $\|p_v\| = p_v^{-1} \leq 1$, uses the characterization $x \in \mathcal{O}[\mathbb{Q}_{p_v}] \iff \|x\| \leq 1$, and derives $p_v\mathcal{O}[\mathbb{Q}_{p_v}] \subseteq \mathcal{O}[\mathbb{Q}_{p_v}]$. It then projects to the valued-integer dilation-quotient source without carrying a closure field. The p-adic quotient-coset source specializes the generic quotient theorem to the finite-extension normalization boundary: it carries the additive subgroup $\mathcal{O}_v = \mathcal{O}[K]$, constructs $p_v\mathcal{O}_v$ as this image, supplies the equivalence

$$\mathrm{Fin}(p_v^{[K_v:\mathbb{Q}_{p_v}]}) \simeq \mathcal{O}_v/(p_v\mathcal{O}_v),$$

and then derives the cover

$$\mathcal{O}_v = \bigcup_i (a_i + p_v\mathcal{O}_v),$$

pairwise disjointness of the cosets, and measurability of each coset from the quotient partition theorem. It projects to the older coset Haar source, so Lean derives the finite-additivity equality for this cover by applying `measure_iUnion`, and derives equality of each translated coset measure with $\mu_v(p_v\mathcal{O}_v)$ from additive Haar invariance via `measure_preimage_add`. Lean then proves

$$\mu_v(p_v\mathcal{O}_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}$$

from $\mu_v(\mathcal{O}_v) = 1$ by finite-sum algebra in $\mathbb{R}_{\geq 0} \cup \{\infty\}$. This coset-derived source now projects to the unit-ball Haar-character source and is threaded through the local centered-ball Haar-log-coordinate and finite-product target-point layers. Remark 3.9.5(v)–(vii) quotient, approximant, and log-Kummer comparison audit. The next work is now below the finite-extension boundary: instantiate the submodule-level residue source for genuine finite extensions by constructing the residue field action on $\mathcal{O}[K_v]$ and proving that $p_v\mathcal{O}[K_v]$ is a κ-submodule. Once supplied, the existing Lean transport now feeds the resulting coset/Haar-character endpoint into the determinant and hull comparison source. In parallel, one still has to derive the positive valuation-ball radius, the coordinate-place equivalence, and the point-level target transport identifying O and λO with the principal finite-divisor target regions. Below that, the local directional scalar-image laws for each local factor $\lambda_v \cdot O_v$ still have to be derived from the valuation/Kummer model: landing in the valuation/direct-product factor $H_v(\lambda)$, existence of local scalar preimages for all points of $H_v(\lambda)$, direct-product localization cover construction, family-local finite-cover additivity theorem, construction of the compact-open valuation-norm from actual finite extensions of $\mathbb{Q}_p$, derivation of the proper ultrametric local topology from the actual finite-extension valuation, normalized Haar measure choice, and uniformizer-scaling theorem, per-factor chart/direct-summand calibration construction, local region/vector-bundle log-volume calibration, measure calibration, record-canonical synchronization, and chart synchronization witnesses by source-paper constructions;

7. only then evaluate whether the Stage 1 interface has enough source content to bear on the Mochizuki–Scholze–Stix disagreement.

References

[1] S. Mochizuki, *Inter-universal Teichmuller Theory I: Construction of Hodge Theaters.*

[2] S. Mochizuki, *Inter-universal Teichmuller Theory II: Hodge-Arakelov-theoretic Evaluation.*

[3] S. Mochizuki, *Inter-universal Teichmuller Theory III: Canonical Splittings of the Log-theta-lattice.*

[4] S. Mochizuki, *Inter-universal Teichmuller Theory IV: Log-volume Computations and Set-theoretic Foundations.*

[5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.

[6] P. Scholze and J. Stix, *Why abc is still a conjecture.*